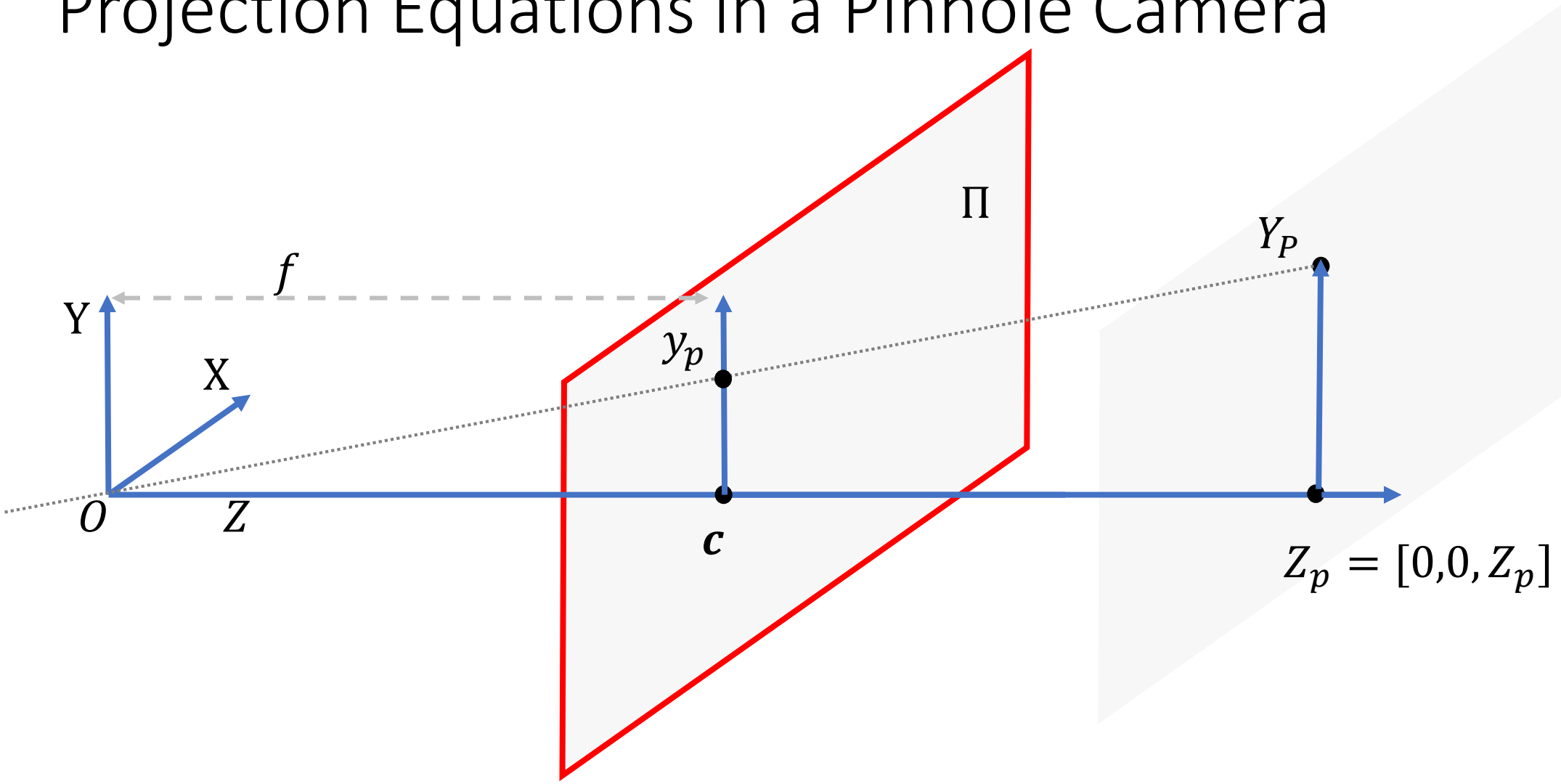
Projective 2D (Planar) Geometry and 2D Reconstruction

Topics

- **Introduction**
- Planar (2D) Projective Geometry
- Spatial (3D) Projective Geometry
- Camera Geometry (3D → 2D Projection)

The image of a point P in the scene

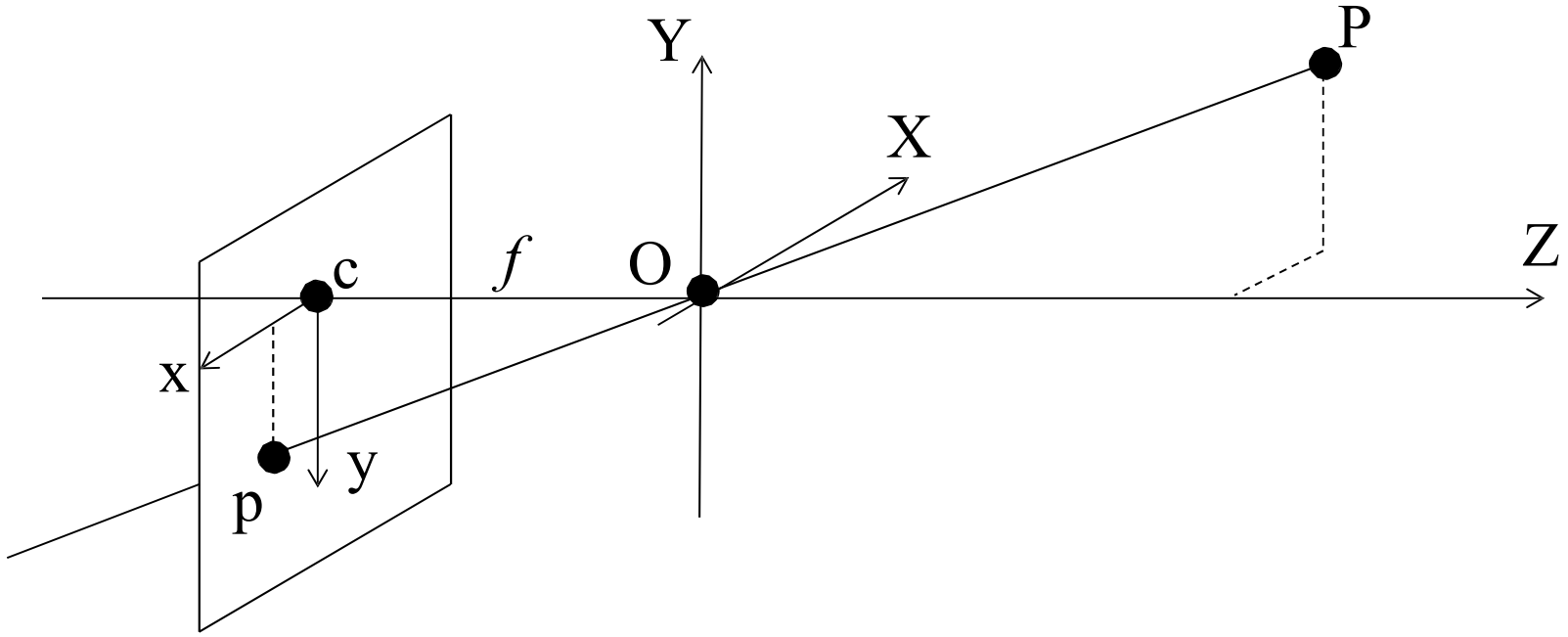
Projection Equations in a Pinhole Camera



from triangle similarity follows Oc: $Oy_p = OZ_P : Z_P Y_P$,

Thus $f : y_p = Z_P : Y_P$ (similar relation for x_p)

The equations of Pinhole Camera Projection



$$P = [X_P, Y_P, Z_P] \rightarrow \boldsymbol{p} = [x_p, y_p]$$

where

$$\begin{cases} x_p = f * X_P / Z_P \\ y_p = f * Y_P / Z_P \end{cases}$$

central projection

-nonlinear
-not shape-preserving
-not length-ratio preserving

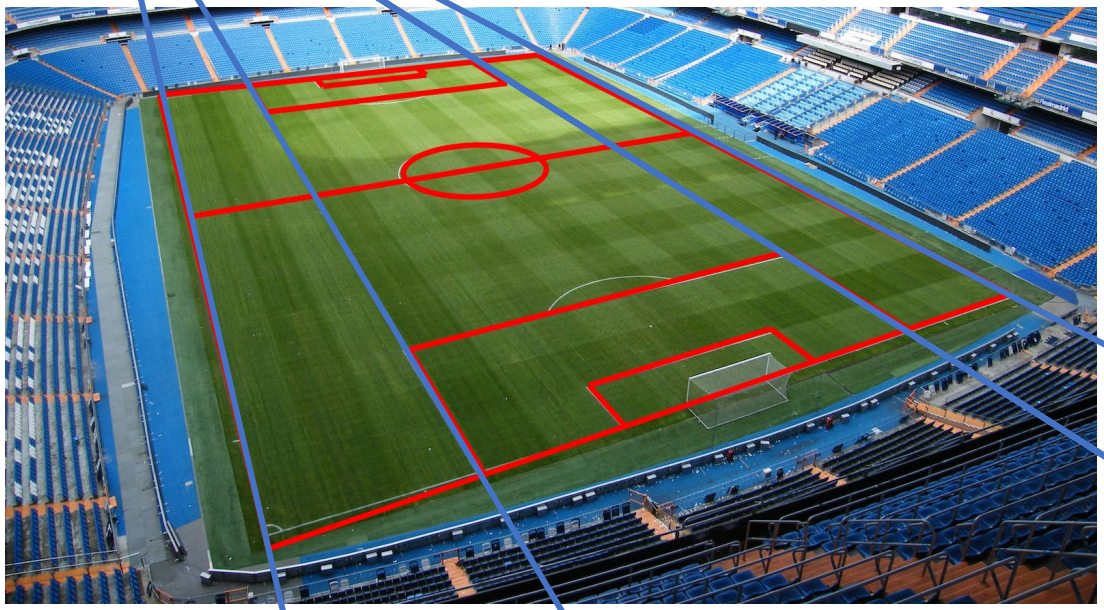
TO UNDERSTAND WHAT WE GET FROM WHAT WE SEE,
WE NEED TO STUDY THE GEOMETRY OF CENTRAL PROJECTION

- Introduction
- **Planar (2D) Projective Geometry**
- Spatial (3D) Projective Geometry
- Camera Geometry (3D → 2D Projection)

Preview of Geometry:
useful properties
1. The vanishing point of a direction

Parallel lines project onto **concurrent** lines

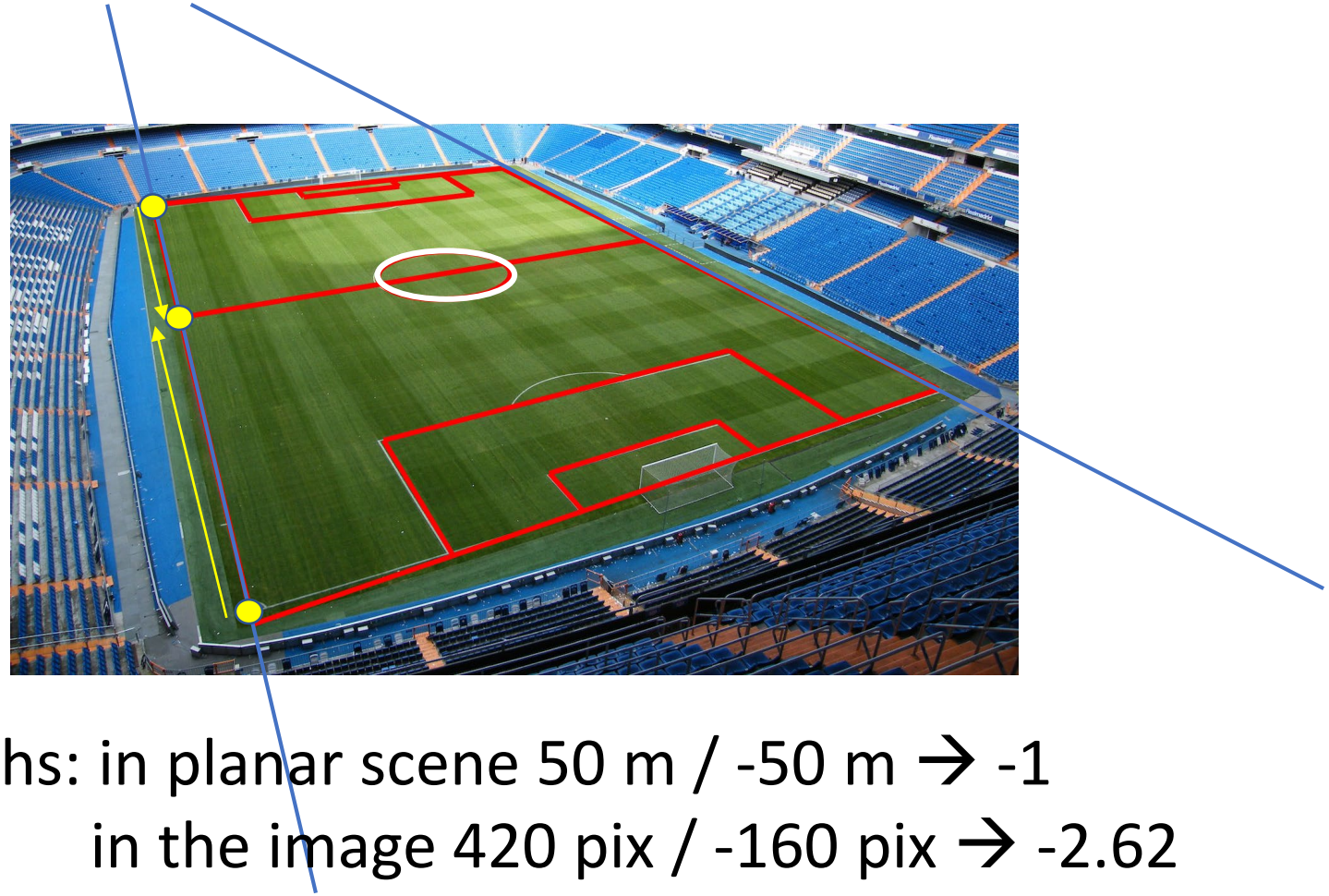
vanishing point
of a direction



Preview of Geometry:
useful properties
2. The cross ratio of four colinear points

ratio of lengths: NOT INVARIANT

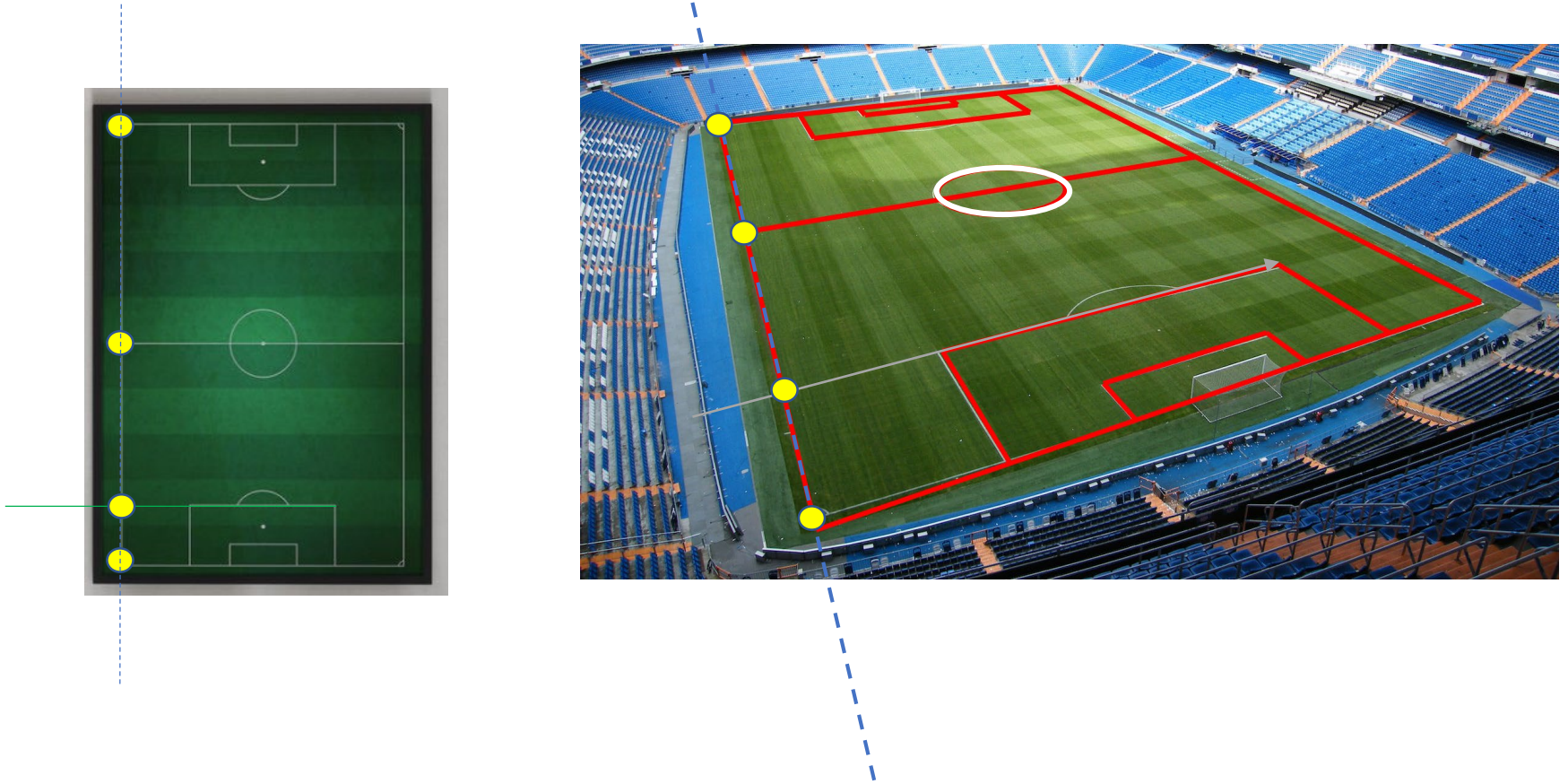




ratio of lengths: in planar scene 50 m / -50 m → -1

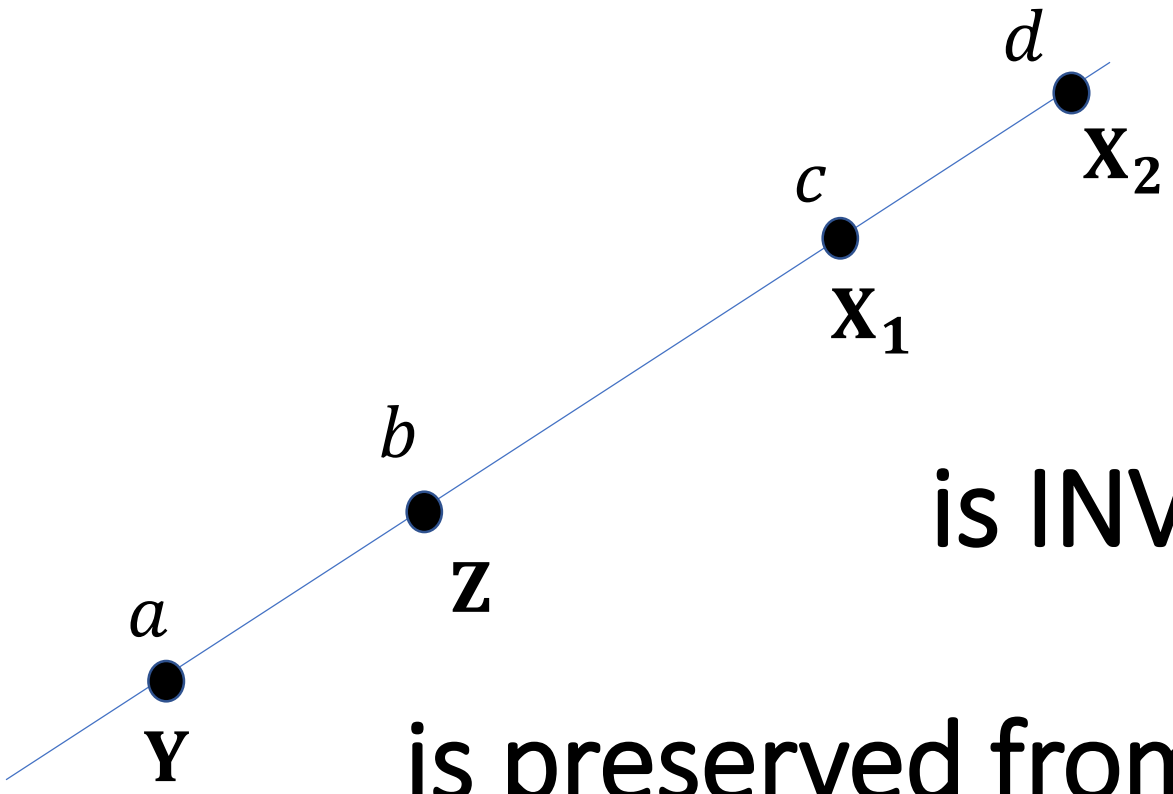
in the image 420 pix / -160 pix → -2.62

the cross ratio, i.e., the ratio between the two ratios, is INVARIANT: it is preserved from real word to image



Cross ratio of a 4-tuple of colinear points

$$CR_{Y,Z,X_1,X_2} = \frac{c-a}{c-b} \Big/ \frac{d-a}{d-b}$$



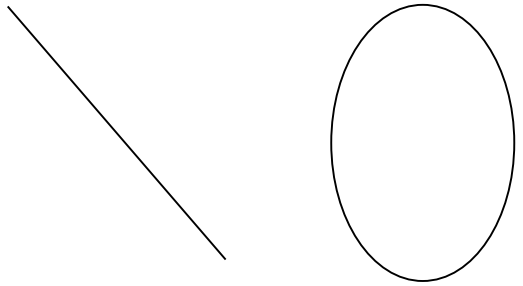
is INVARIANT:

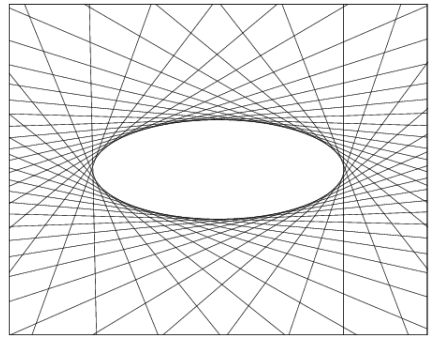
is preserved from real world to image

Planar (2D) Projective Geometry

Planar Projective Geometry

- **Elements**
- **Points**
- Lines
- Conics
- Dual conics

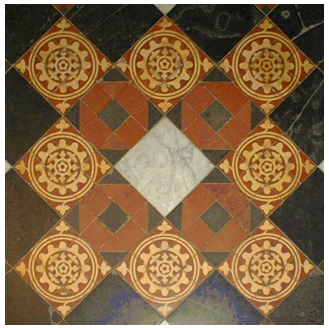




- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities

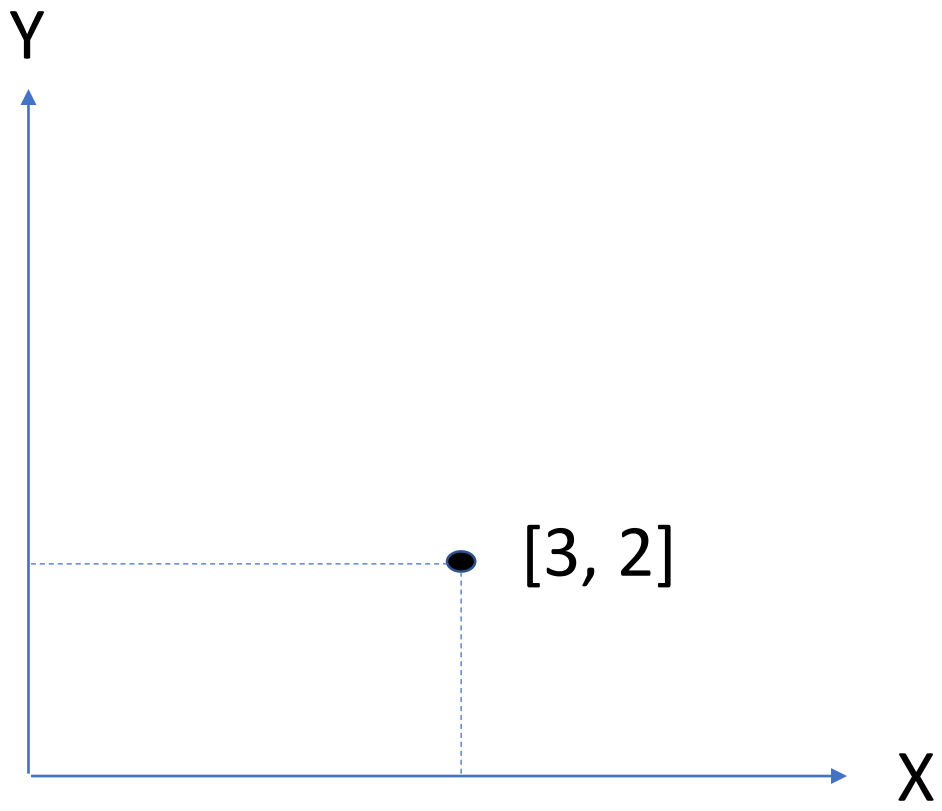






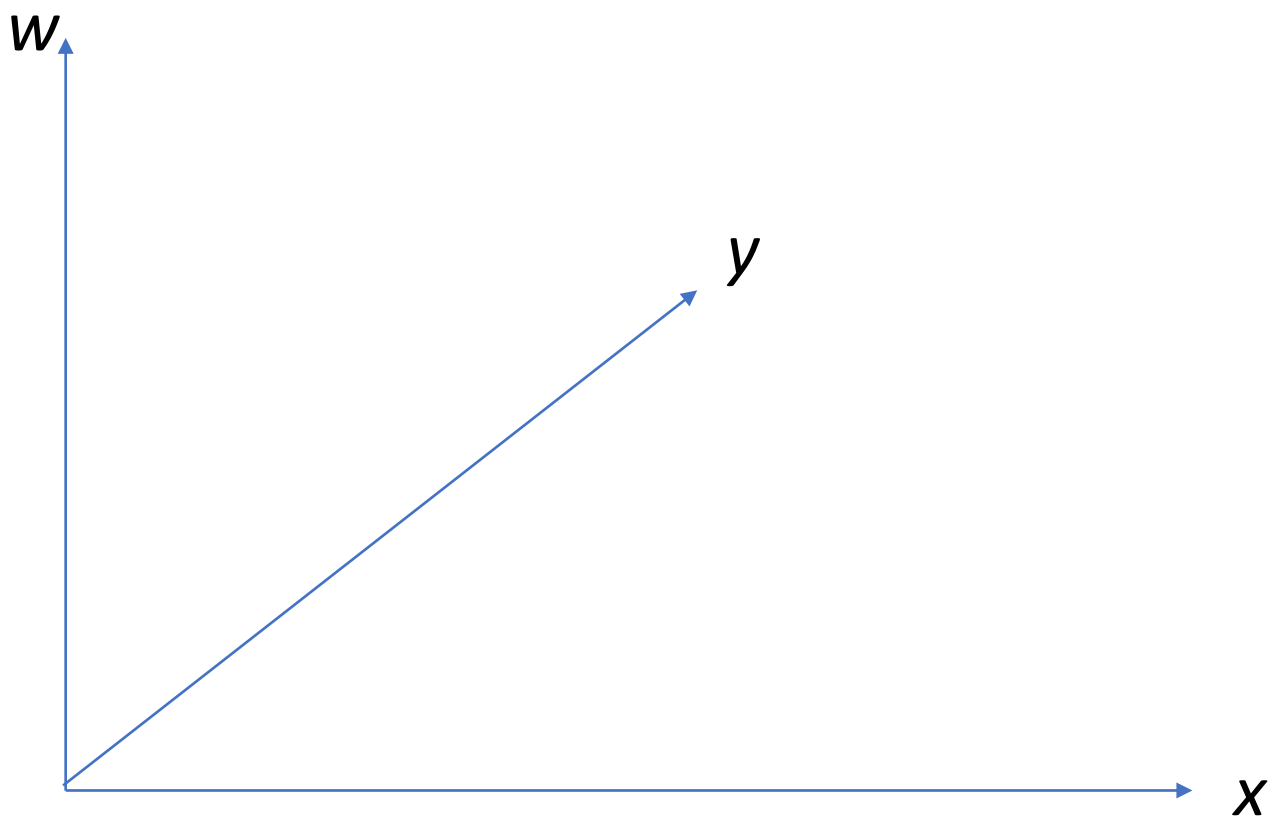
Points in 2D Projective Geometry

Euclidean plane – cartesian coordinates

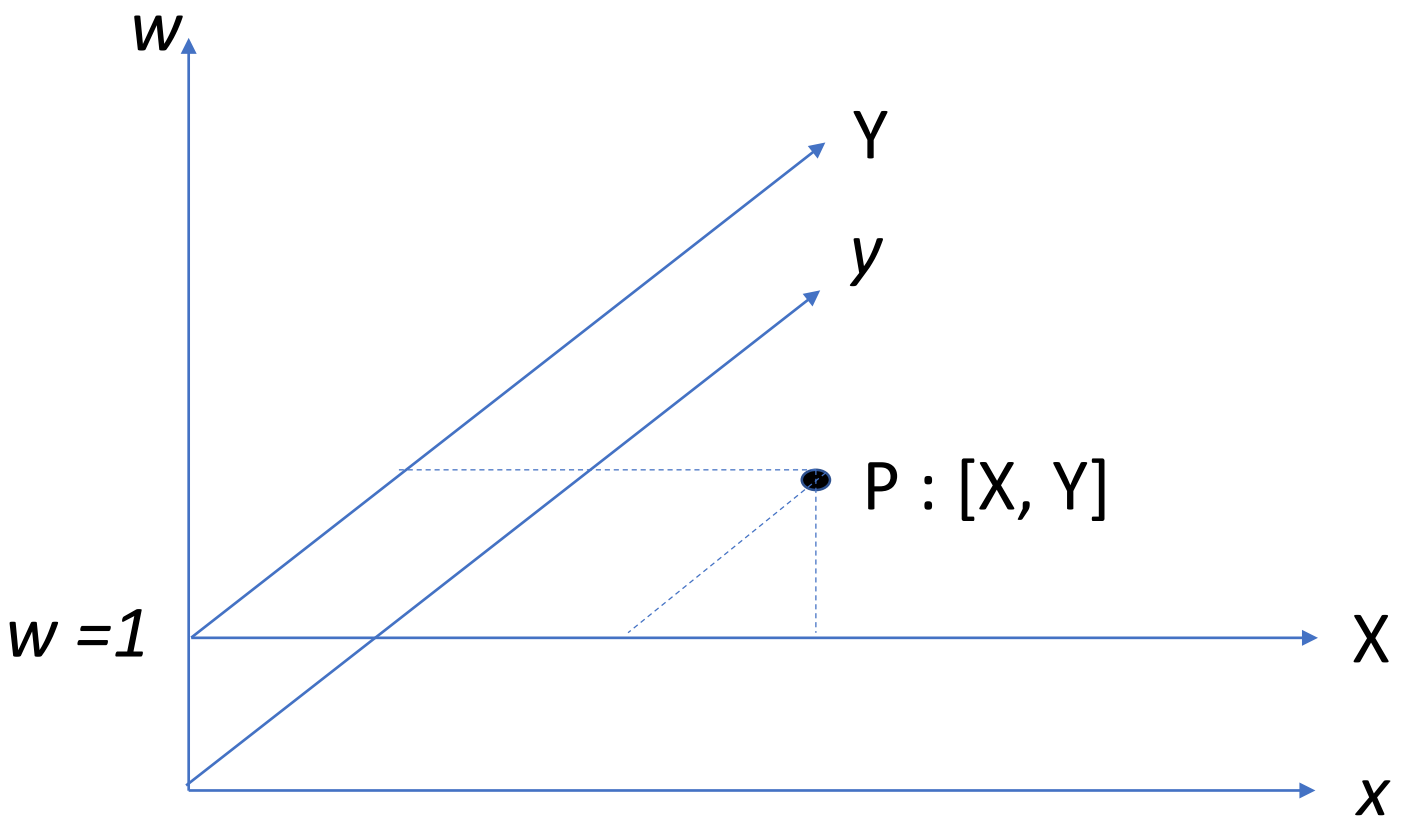


Homogeneous coordinates

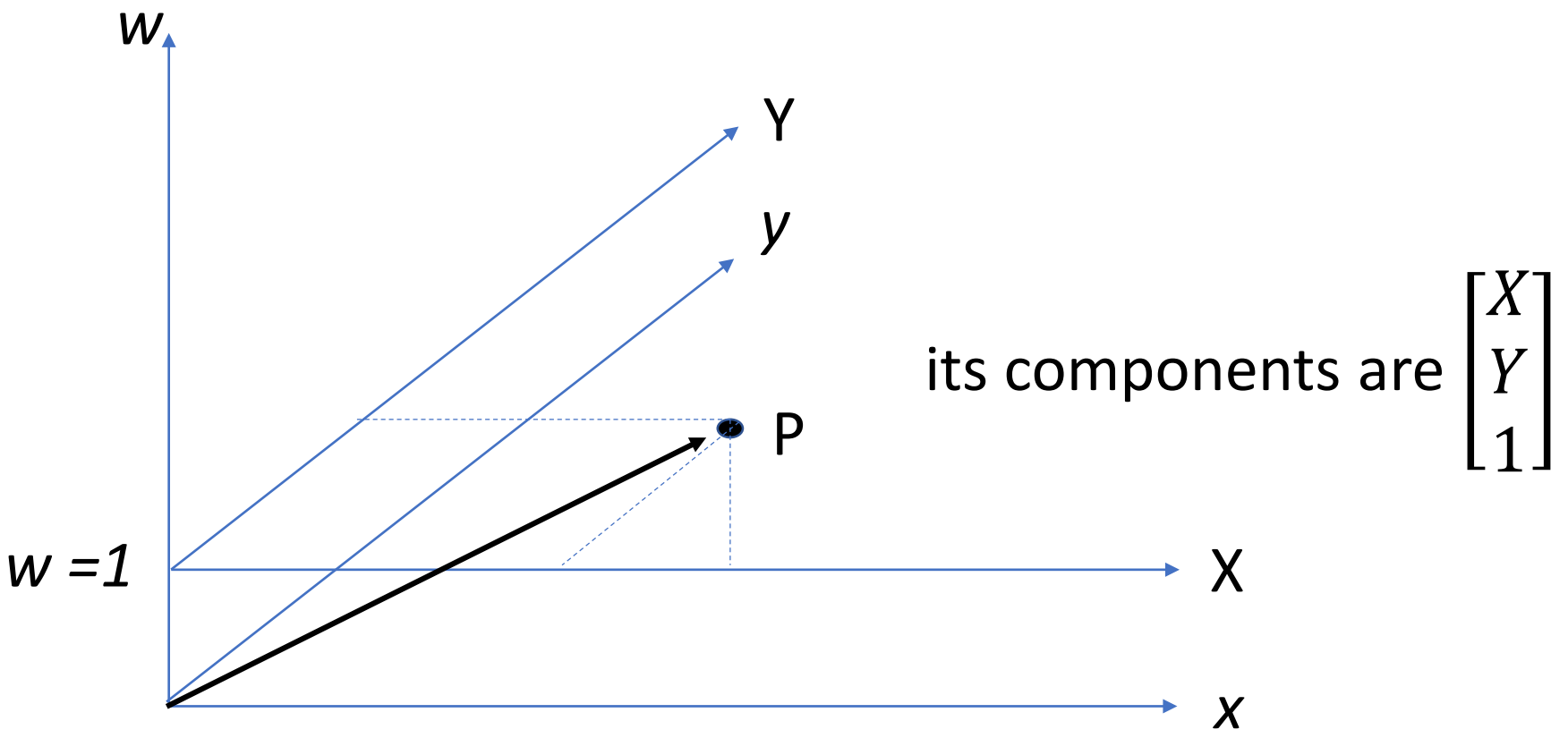
Consider a (3D) space of coordinate vectors



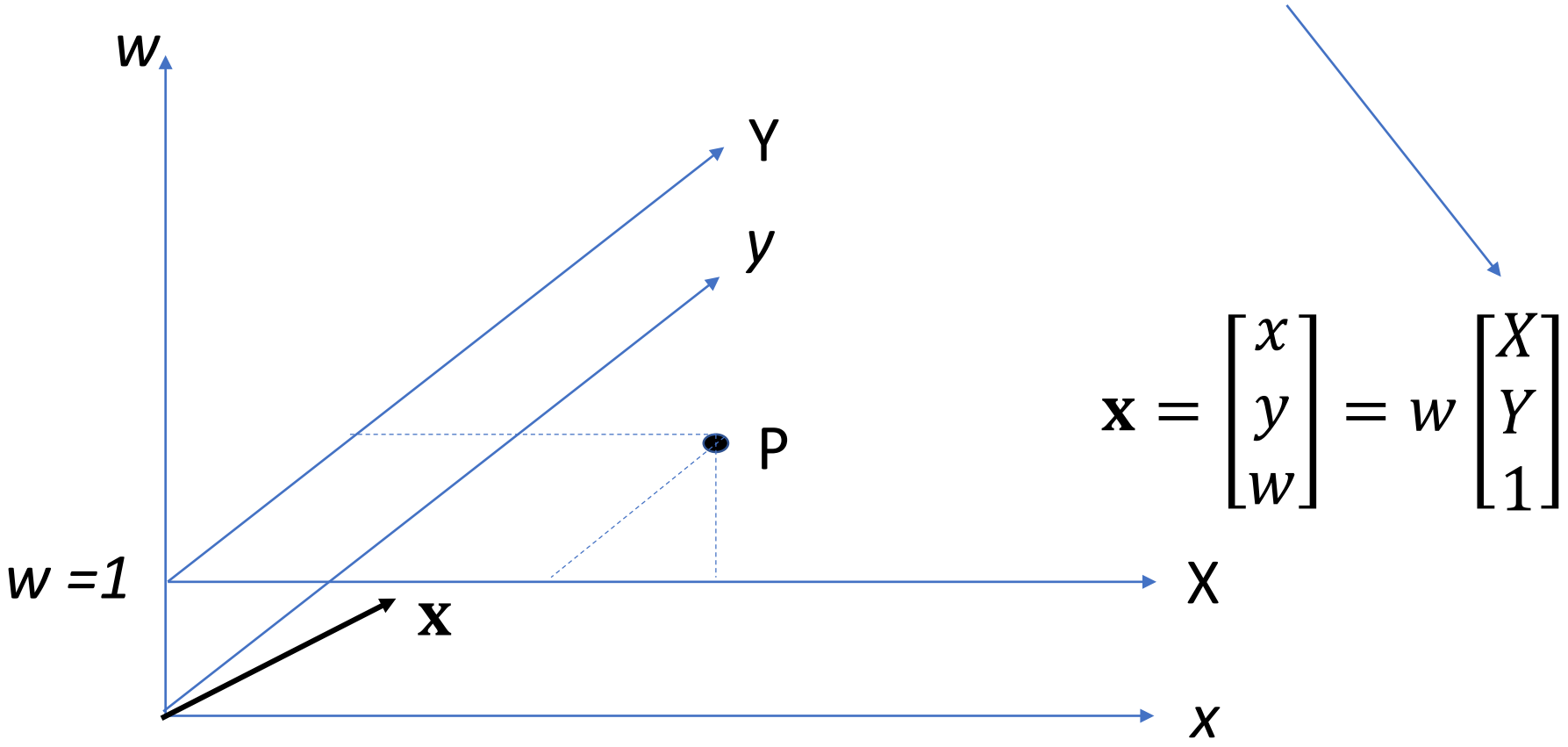
Embed the Euclidean plane into the (3D) space of coordinate vectors as the plane $w = 1$



Consider the vector from the origin of the space to the point P

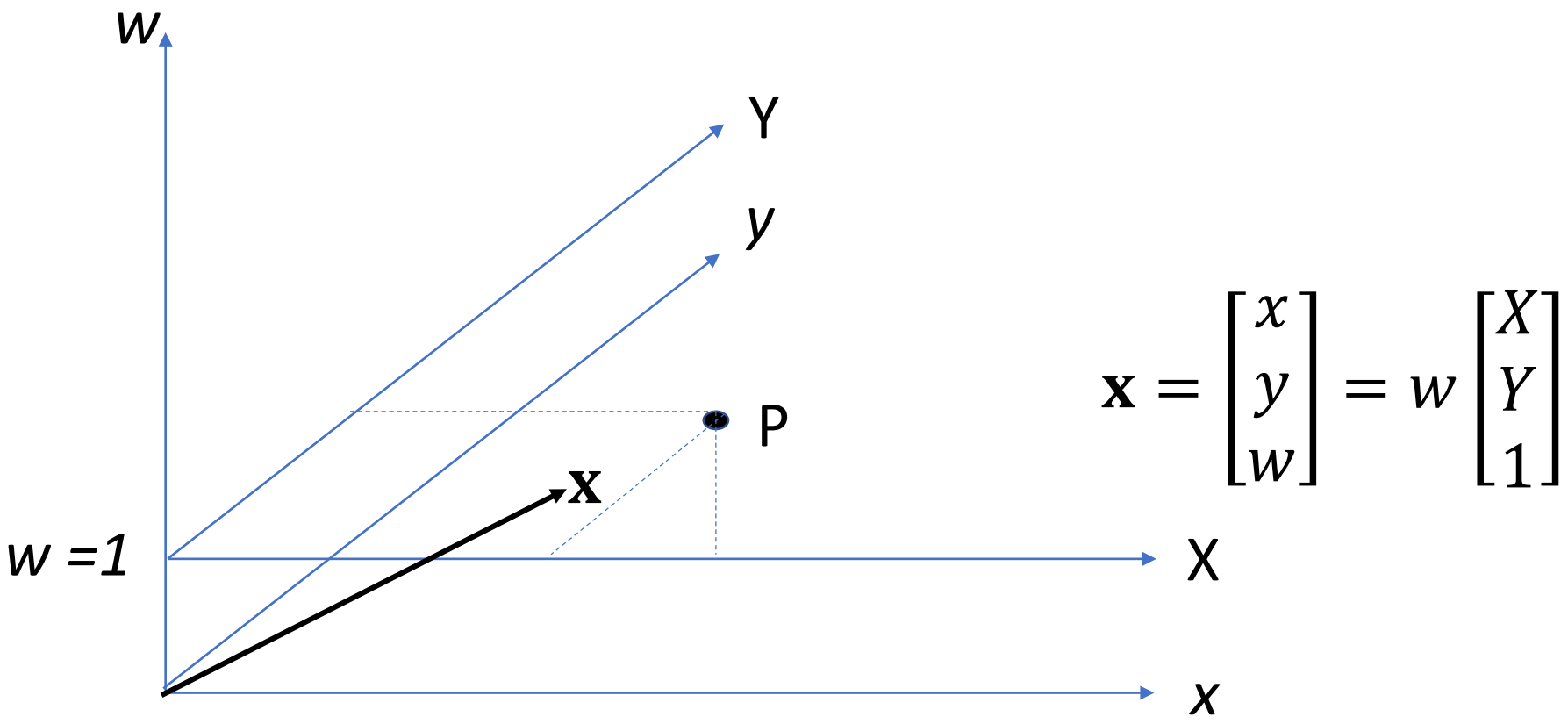


the point P is represented by any vector **x** , that is a nonzero multiple of it



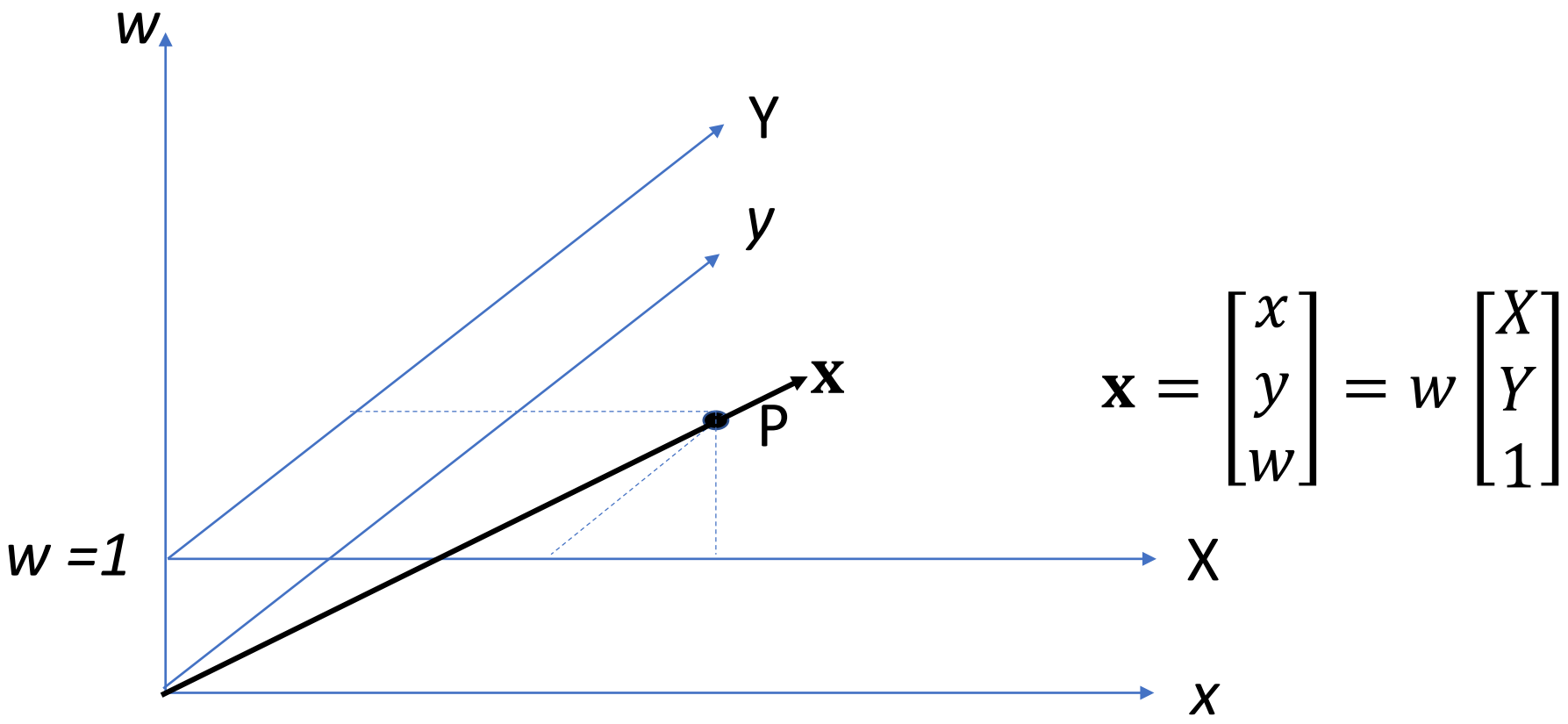
$$\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} = w \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

the point P is represented by any vector **x** , that is a nonzero multiple of it



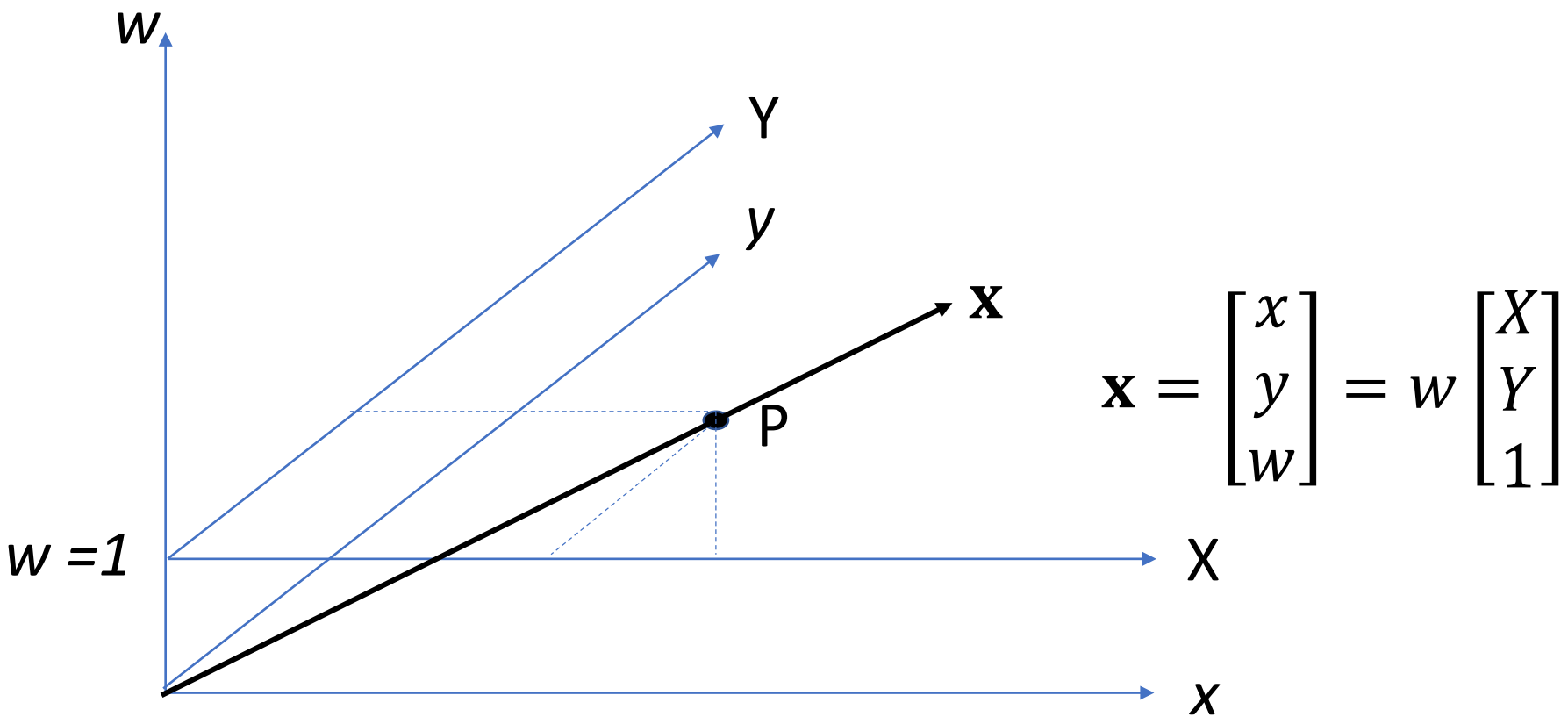
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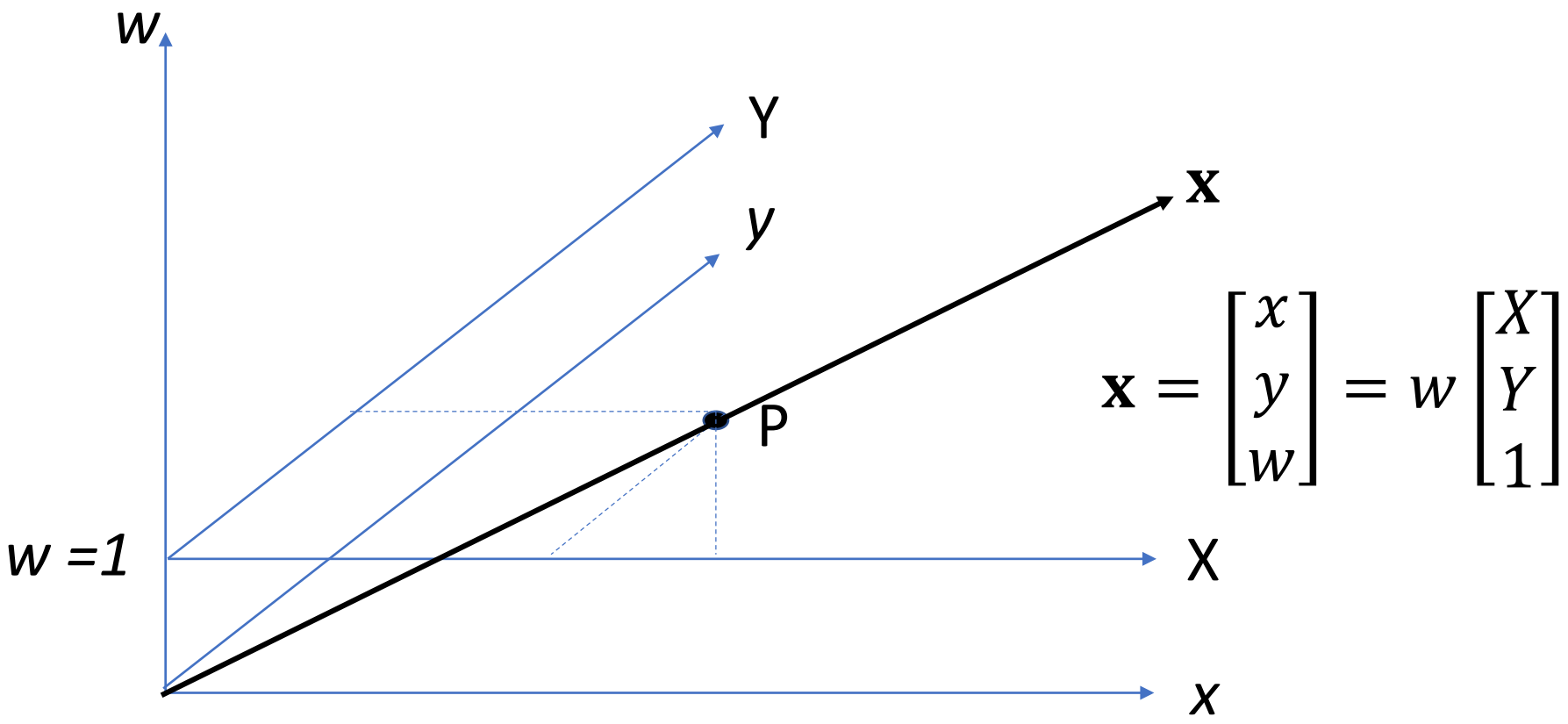
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$$\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} = w \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

the point P is represented by any vector **x** , that is a nonzero multiple of it



$$\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} = w \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

A vector $\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix}$ and all its nonzero multiples $\lambda \begin{bmatrix} x \\ y \\ w \end{bmatrix}$, including $\begin{bmatrix} x/w \\ y/w \\ 1 \end{bmatrix}$,
represent the same point of cartesian coordinates $[X \quad Y] = [x/w \quad y/w]$
on the Euclidean plane

→ **homogeneity**: any vector $\mathbf{x}$ is equivalent to all its nonzero multiples $\lambda\mathbf{x}$, $\lambda \neq \mathbf{0}$
since they represent the same point

→$[x \quad y \quad w]$ are ***homogeneous*** coordinates of the point on the plane

its **two** degrees of freedom are the two independent ratios between the three coordinates $x : y : w$

doesn't represent any point

Projective plane $\mathbb{P}^2 = \{[x \quad y \quad w]^T \in \mathbb{R}^3\} - \{[0 \quad 0 \quad 0]^T\}$

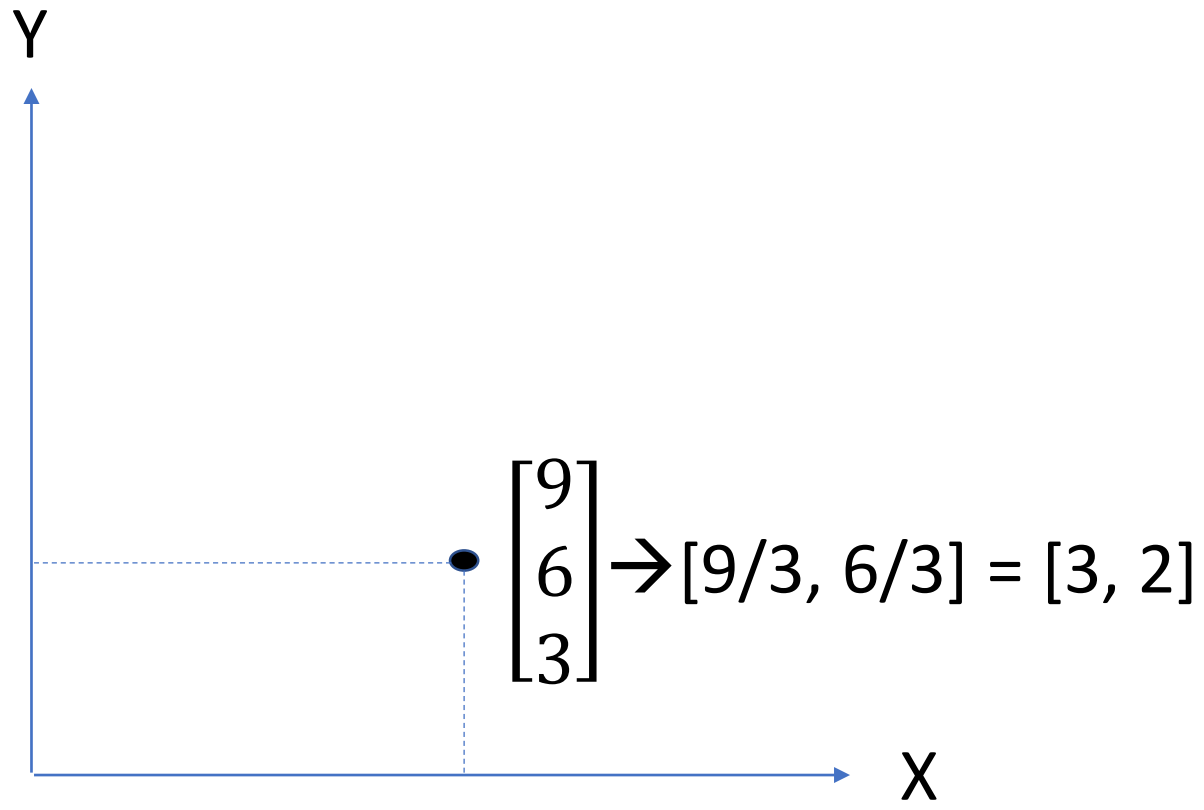
redundancy

3 homogeneous coordinates to represent points in the 2D plane (2 dof)

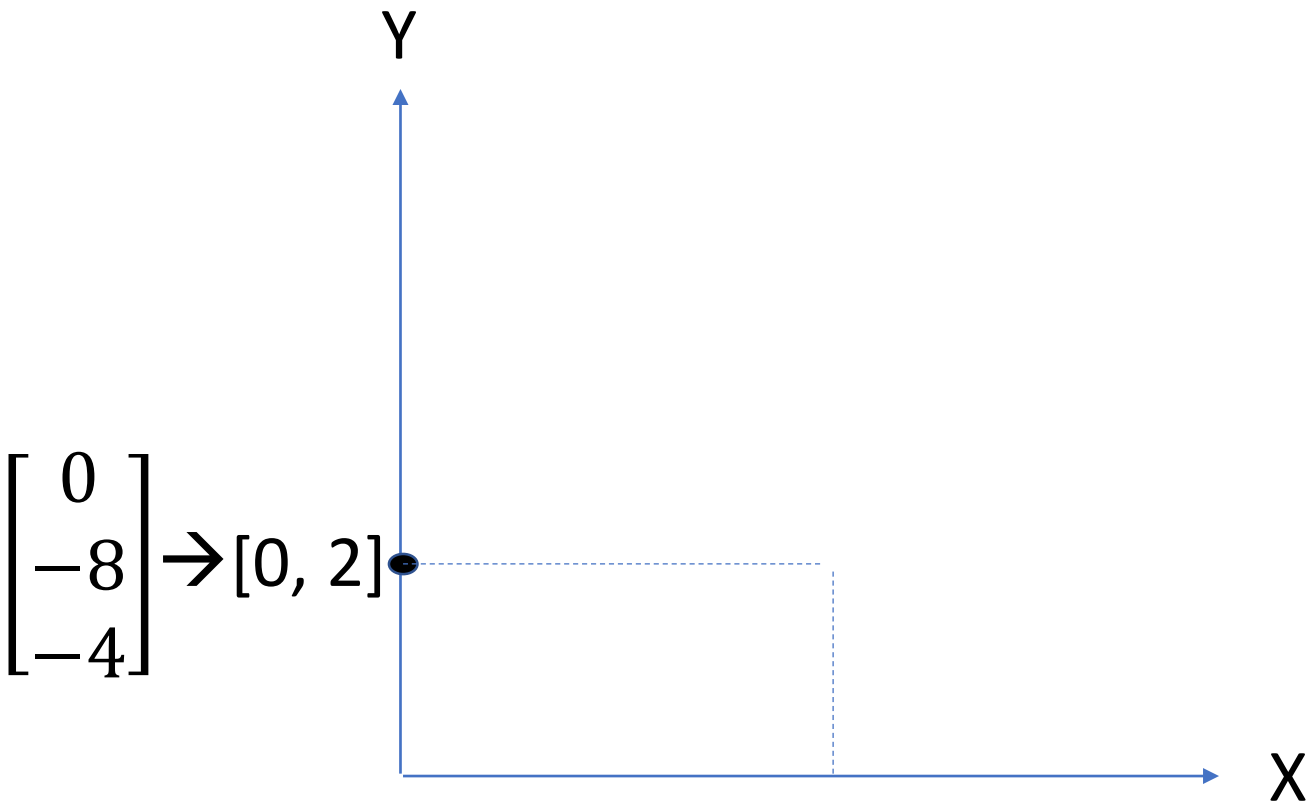
an infinite number of equivalent representations for a single point, namely all nonzero multiples of the vector $[X \quad Y \quad 1]^T$

the null vector $[0 \quad 0 \quad 0]^T$ **does not** represent any point

Example: cartesian coordinates vs homogeneous coordinates $[X \quad Y] = [x/w \quad y/w]$

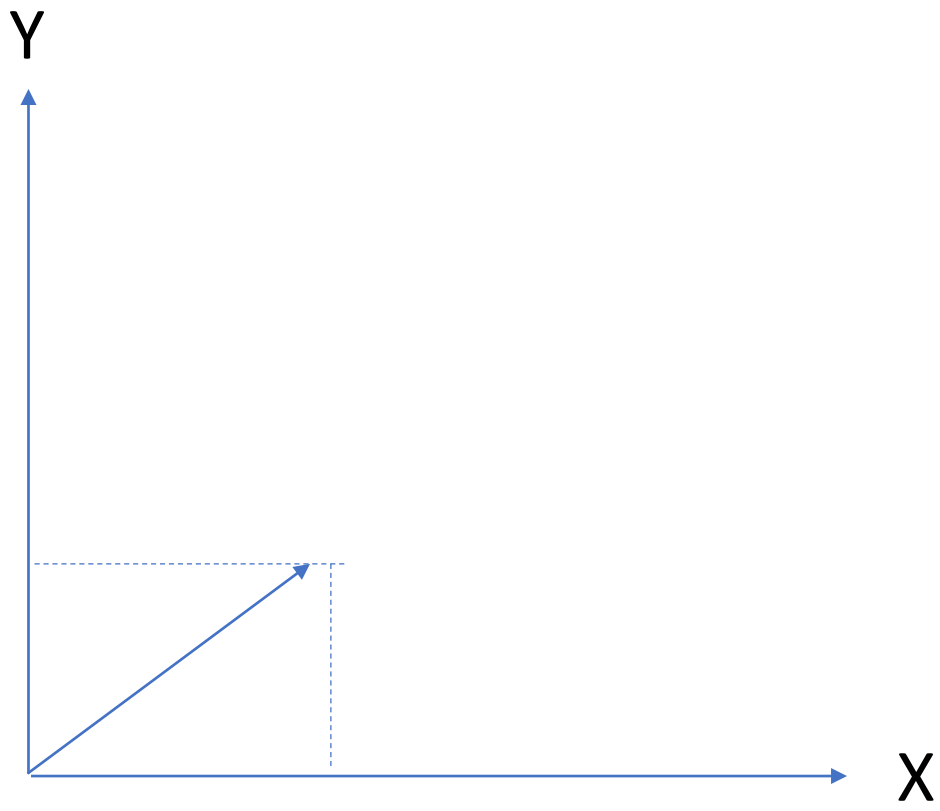


Example: cartesian coordinates vs homogeneous coordinates $[X \quad Y] = [x/w \quad y/w]$



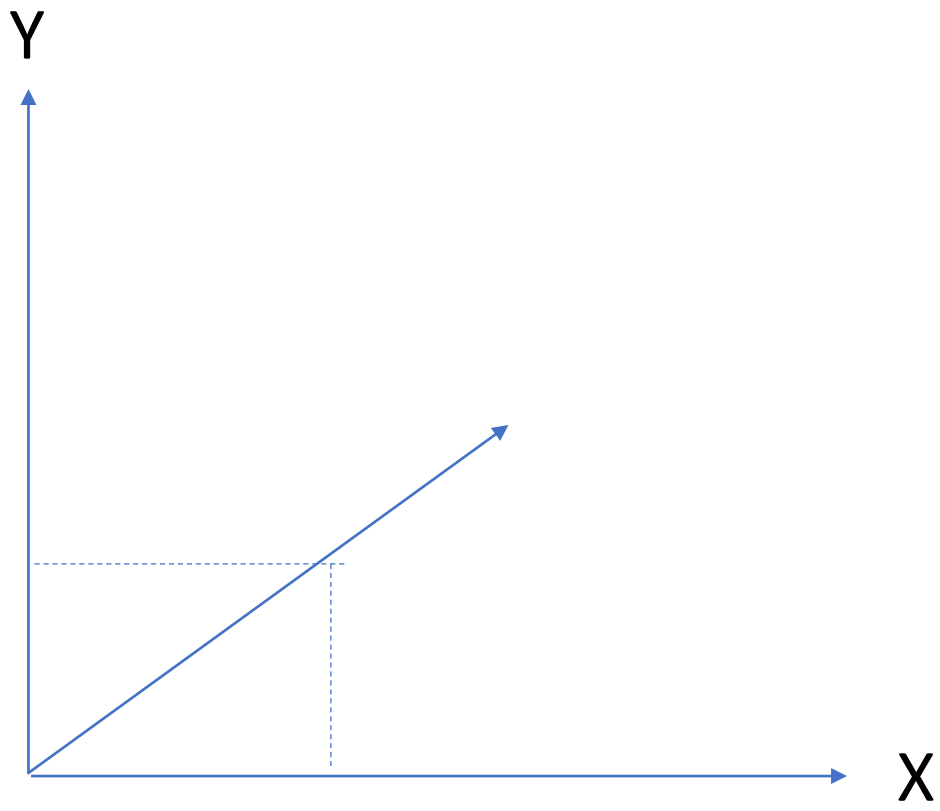
Points at the infinity

points at the infinity



$[3 \quad 2 \quad 0]^T \rightarrow ?$

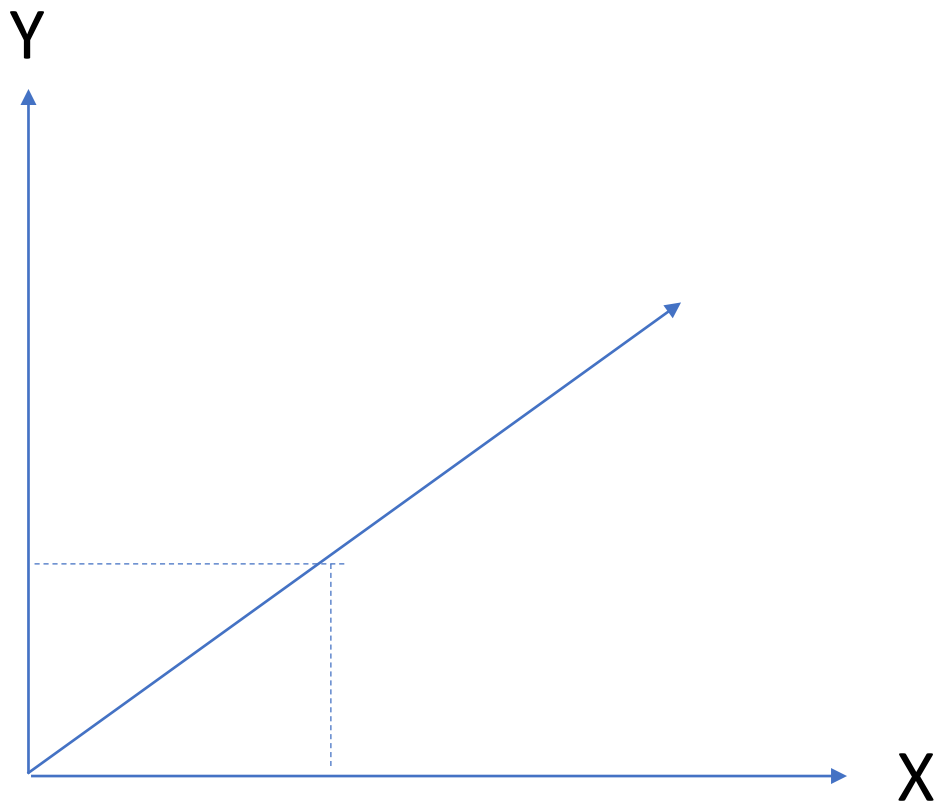
points at the infinity



$$[3 \quad 2 \quad 0]^T$$

$$\rightarrow \lim_{w \to 0} [3/w \quad 2/w]$$

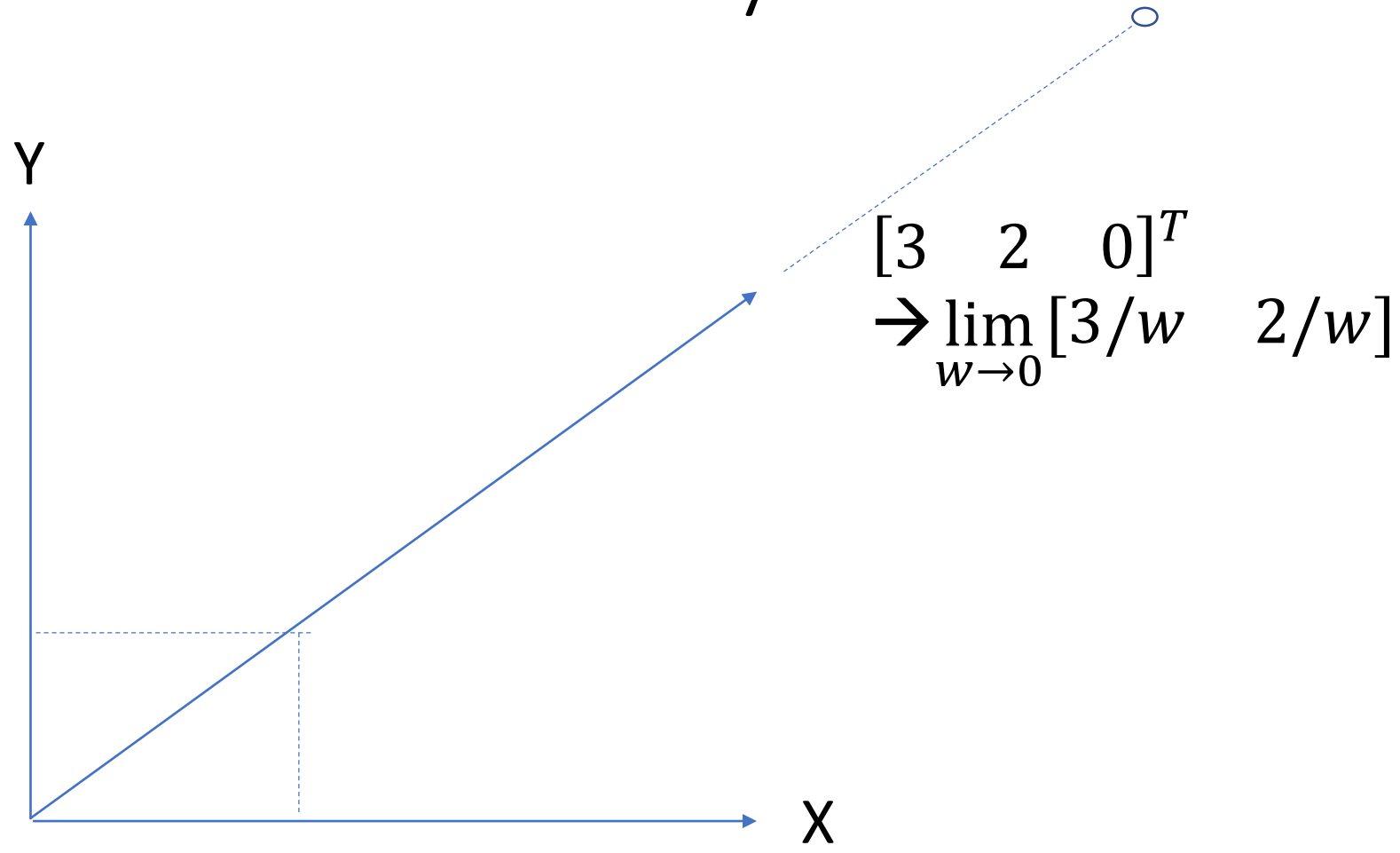
points at the infinity



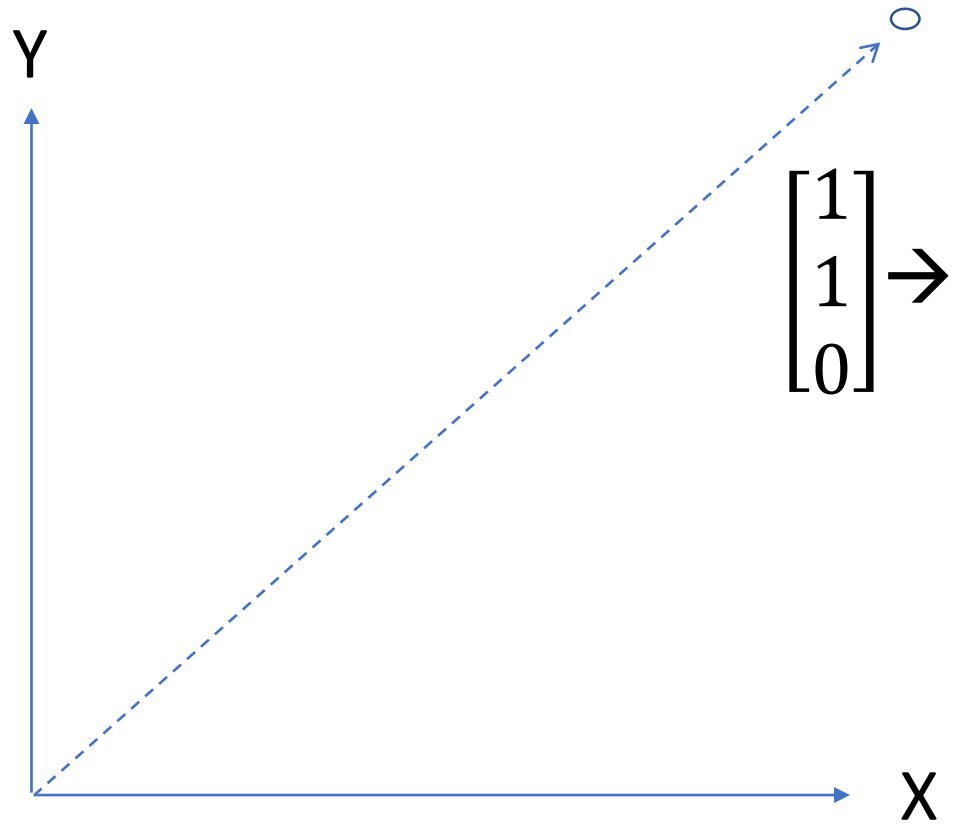
$$[3 \quad 2 \quad 0]^T$$

$$\rightarrow \lim_{w \to 0} [3/w \quad 2/w]$$

points at the infinity



Points at the infinity, who represent directions, are not part of the Euclidean plane: they are extra points, well defined within the Projective plane.



$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$ → point at the infinity

along direction [1, 1]

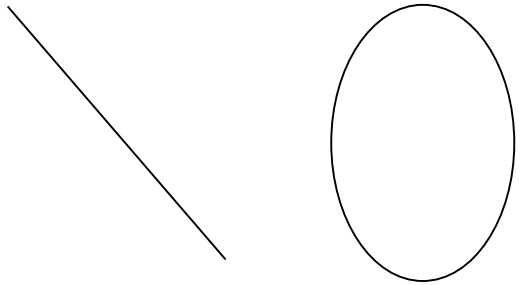
Euclidean plane and Projective plane

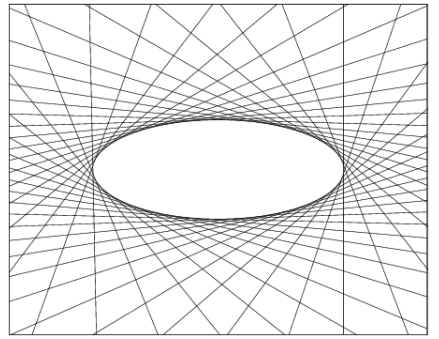
Projective plane $\mathbb{P}^2 = \{[x \quad y \quad w]^T \in \mathbb{R}^3\} - \{[0 \quad 0 \quad 0]^T\}$

= Euclidean plane ∪ set of the points at the infinity

Planar Projective Geometry

- **Elements**
- Points
- **Lines**
- Conics
- Dual conics

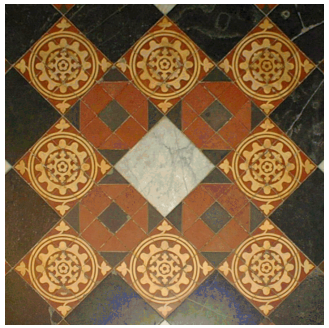




- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities

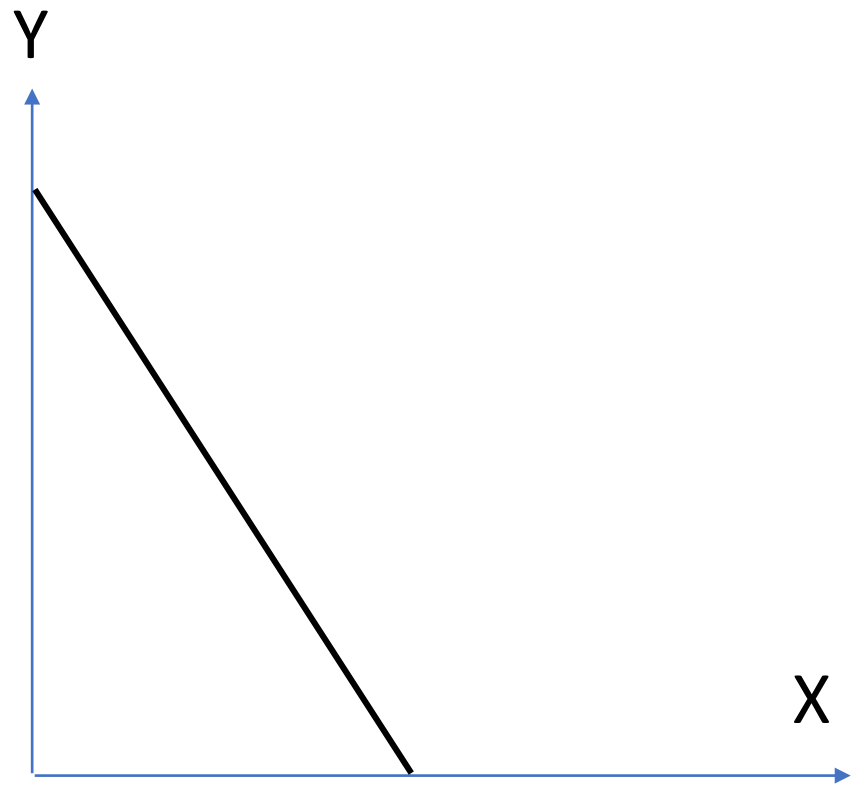






Lines in 2D Projective Geometry

Consider a line on the Euclidean plane



$$aX + bY + c = 0$$

Since $[X \quad Y] = [x/w \quad y/w]$

$$a\frac{x}{w} + b\frac{y}{w} + c = 0$$

$$ax + by + cw = 0$$

$$ax + by + cw = 0 \rightarrow [a \quad b \quad c]\begin{bmatrix} x \\ y \\ w \end{bmatrix} = 0$$

$$\boldsymbol{l}^T \mathbf{x} = 0$$

$$ax + by + cw = 0 \rightarrow \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix} = 0$$

homogeneous, **linear** equation in $\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix}$: $\boldsymbol{l}^T \mathbf{x} = 0$,

the vector $\boldsymbol{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ and all its nonzero multiples $\lambda \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ are equivalent:

they represent the same line

→ **homogeneity**: $\begin{bmatrix} a & b & c \end{bmatrix}$ are ***homogeneous*** parameters of the line

its **two** degrees of freedom are the two independent ratios between the three parameters $a : b : c$

doesn't represent any line

Projective «dual» plane $\mathbb{P}^2 = \{[a \quad b \quad c]^T \in \mathbb{R}^3\} - \{[0 \quad 0 \quad 0]^T\}$

redundancy

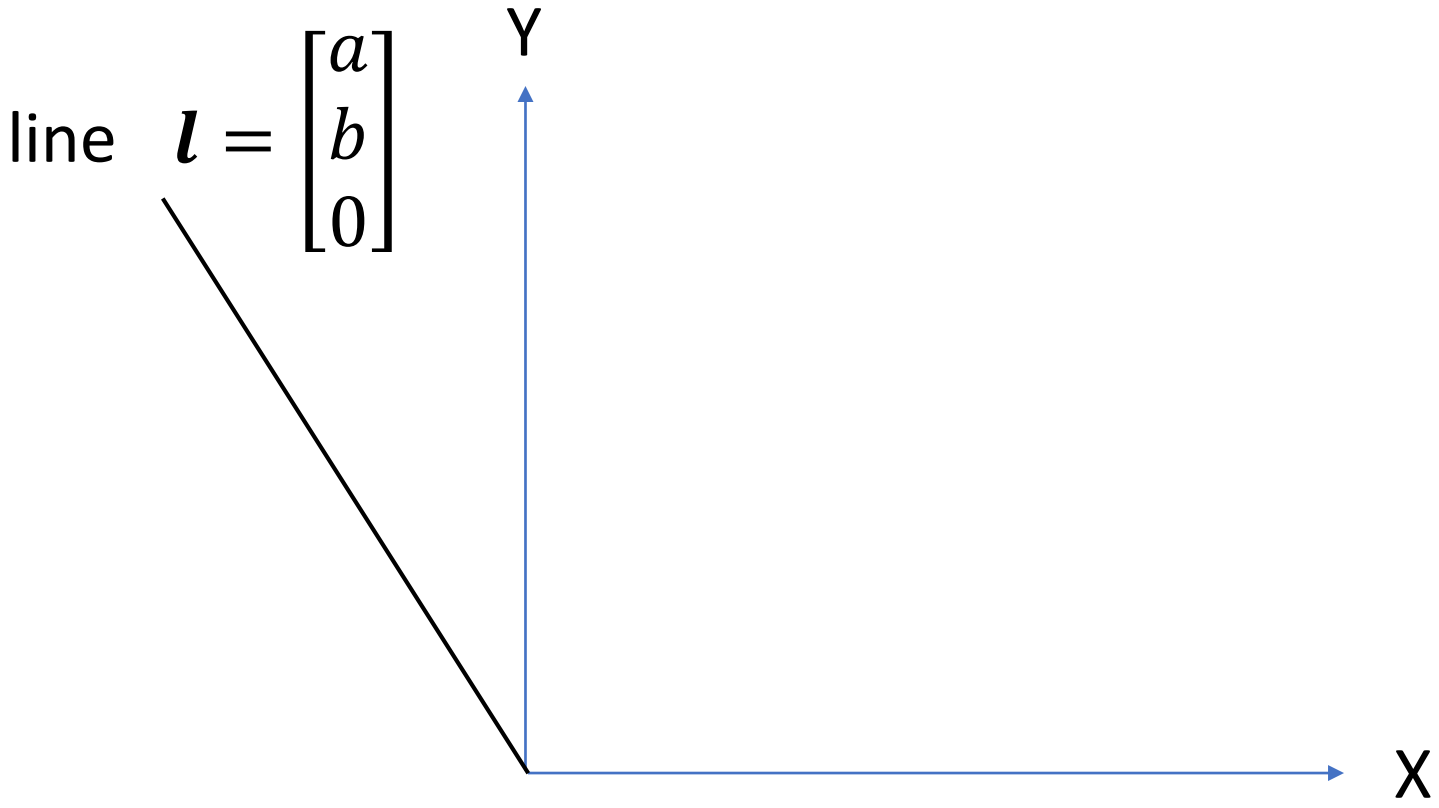
3 homogeneous parameters to represent lines in the 2D plane (2 dof)

an infinite number of equivalent representations for a single line, namely all nonzero multiples of the vector $[a \quad b \quad c]^T$

the null vector $[0 \quad 0 \quad 0]^T$ **does not** represent any line

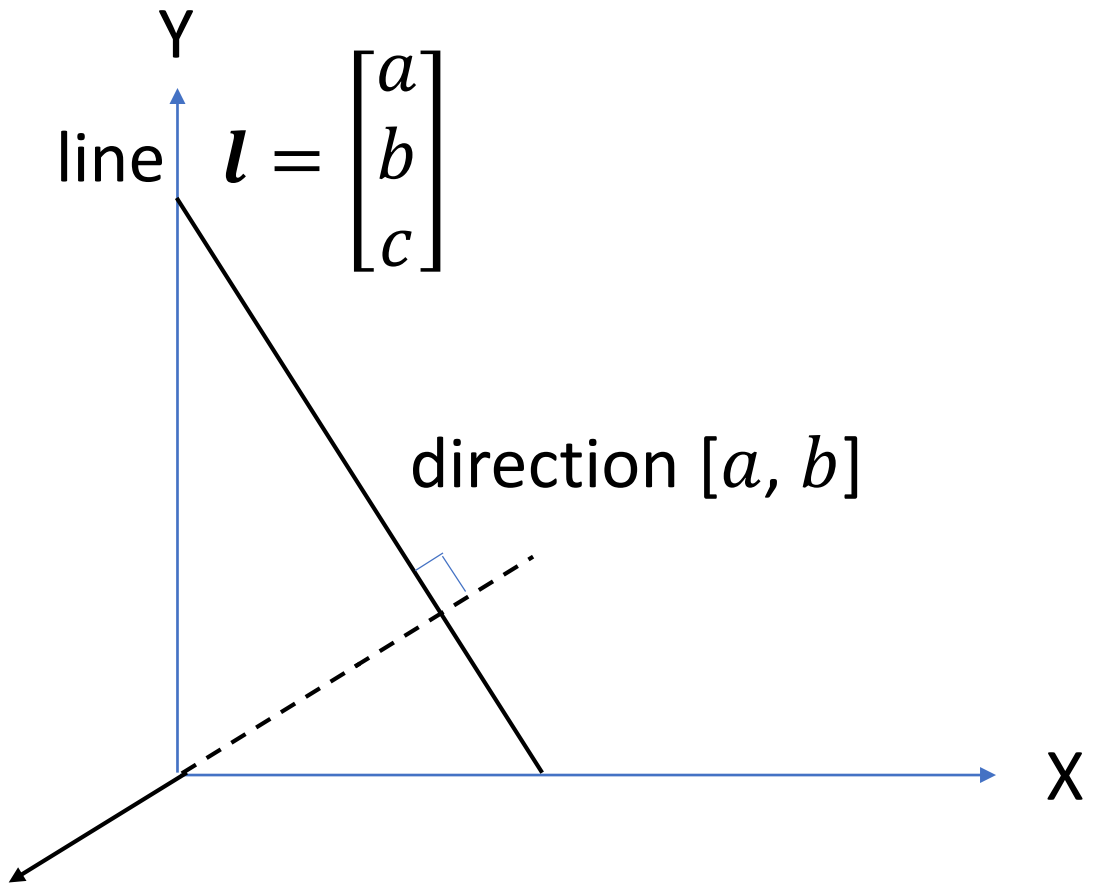
Three remarks

1. If the third parameter is null, $\boldsymbol{l} = \begin{bmatrix} a \\ b \\ 0 \end{bmatrix}$, then $\boldsymbol{l}$ goes through point $[0,0]$



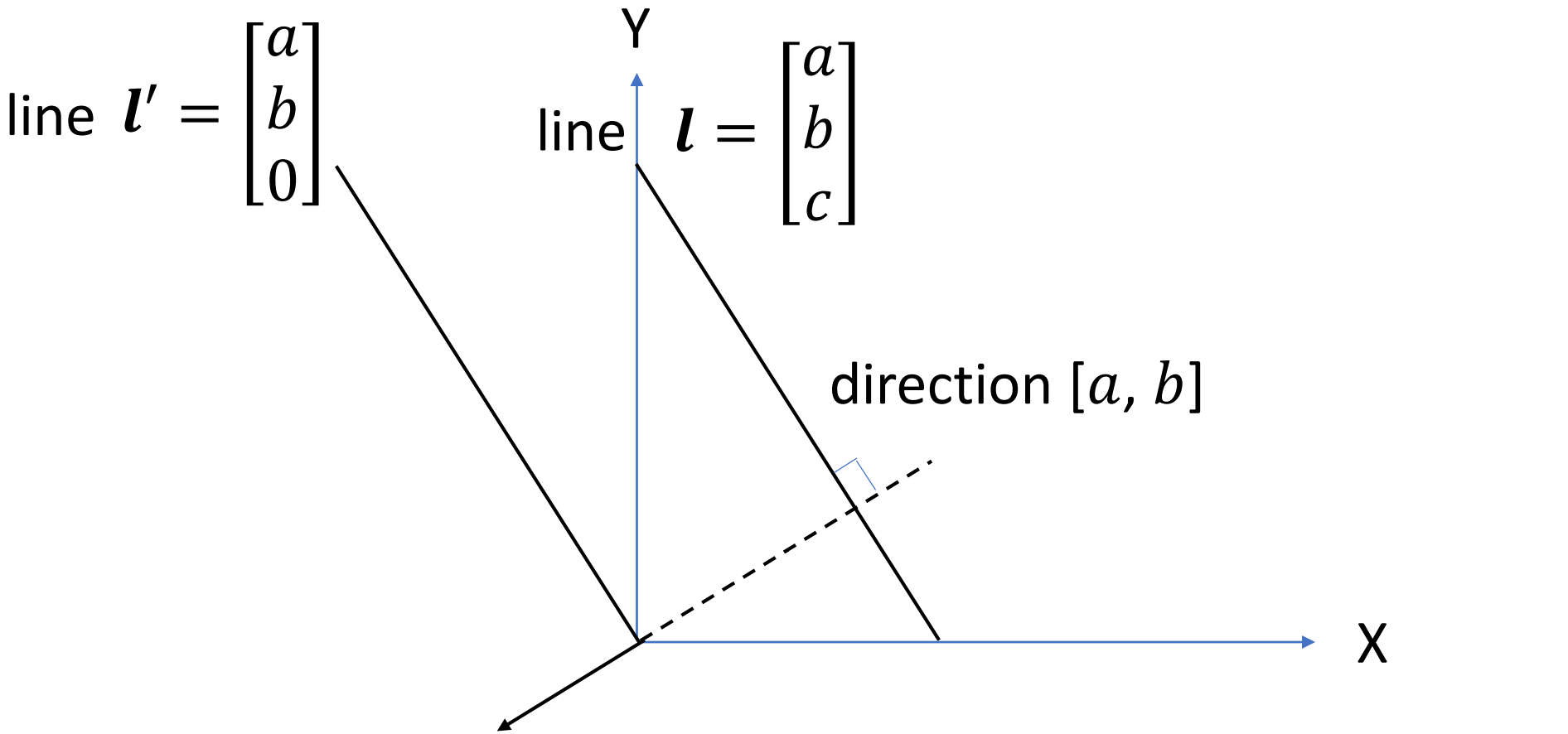
A line $\boldsymbol{l} = [\boldsymbol{a} \quad \boldsymbol{b} \quad \boldsymbol{0}]^{\boldsymbol{T}}$, whose third parameter is zero, goes through the origin of the plane

2. the direction $[a, b]$ is normal to the line $\boldsymbol{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$,



direction $[a, b]$ is normal to the line $\boldsymbol{l} = [\boldsymbol{a} \quad \boldsymbol{b} \quad \boldsymbol{c}]^{\boldsymbol{T}}$

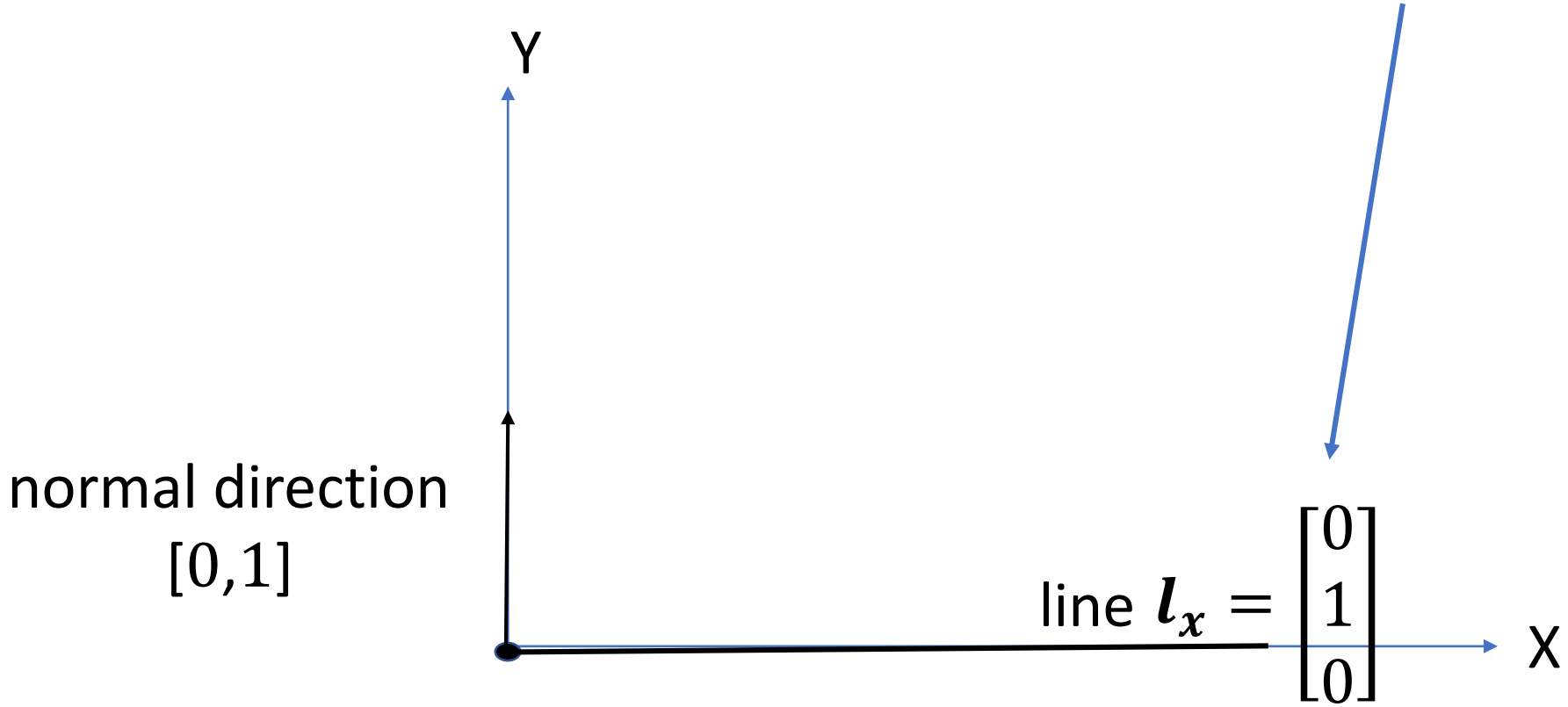
3. the lines $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ and $\begin{bmatrix} a \\ b \\ c' \end{bmatrix}$ are parallel: their common direction is $[b,\text{-}a]$



direction $[a, b]$ is normal both to the line $\boldsymbol{l} = [a \quad b \quad c]^{\boldsymbol{T}}$
and to the line $\boldsymbol{l} = [a \quad b \quad 0]^{\boldsymbol{T}}$

Example: the X-axis

(Y = 0 → y = 0 → 0 x + 1 y + 0 w = 0)



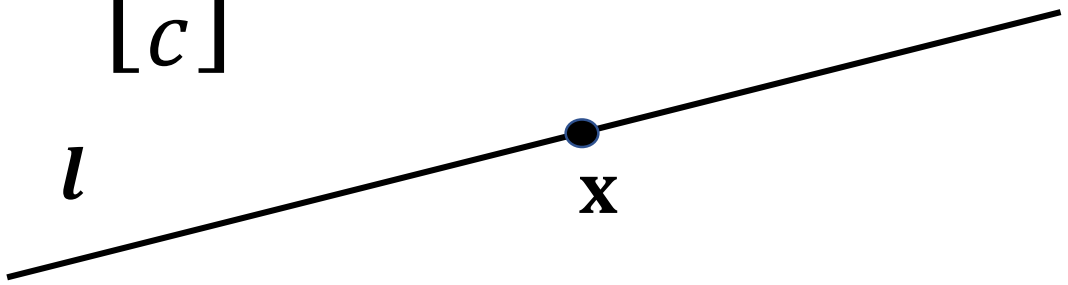
The incidence relation

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \boldsymbol{l}^T \mathbf{x} = 0$$

the point $\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix}$ **is on** the line $\boldsymbol{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

or

the line $\boldsymbol{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ **goes through** the point $\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix}$



The line at the infinity:
the locus of the points at the infinity

As there are infinite points at the infinity (one for each direction), what is the aspect of the set of these points at the infinity?

Simply:

$$w = 0 = 0x + 0y + 1w$$

This set is a line: $[\boldsymbol{a} \quad \boldsymbol{b} \quad \boldsymbol{c}]\begin{bmatrix}\boldsymbol{x}\\ \boldsymbol{y}\\ \boldsymbol{w}\end{bmatrix} = 0$, actually $[\mathbf{0} \quad \mathbf{0} \quad \mathbf{1}]\begin{bmatrix}\boldsymbol{x}\\ \boldsymbol{y}\\ \boldsymbol{w}\end{bmatrix} = \boldsymbol{w} = 0$

namely, **the line at the infinity** $\boldsymbol{l}_\infty = \begin{bmatrix}0\\0\\1\end{bmatrix}$, or equivalently $\boldsymbol{l}_\infty = \begin{bmatrix}0\\0\\\lambda\end{bmatrix}$

The duality principle between points and lines

Since dot product is commutative

→ incidence relation is commutative

$$\boldsymbol{l}^T\mathbf{x} = [\boldsymbol{a} \quad \boldsymbol{b} \quad \boldsymbol{c}]\begin{bmatrix}\boldsymbol{x}\\ \boldsymbol{y}\\ \boldsymbol{w}\end{bmatrix} = [\boldsymbol{x} \quad \boldsymbol{y} \quad \boldsymbol{w}]\begin{bmatrix}\boldsymbol{a}\\ \boldsymbol{b}\\ \boldsymbol{c}\end{bmatrix} = \mathbf{x}^T\boldsymbol{l} = \mathbf{0}$$

point $\mathbf{x}$ is on line $\boldsymbol{l}$ ⇔ **point $\boldsymbol{l}$ is on line $\mathbf{x}$**

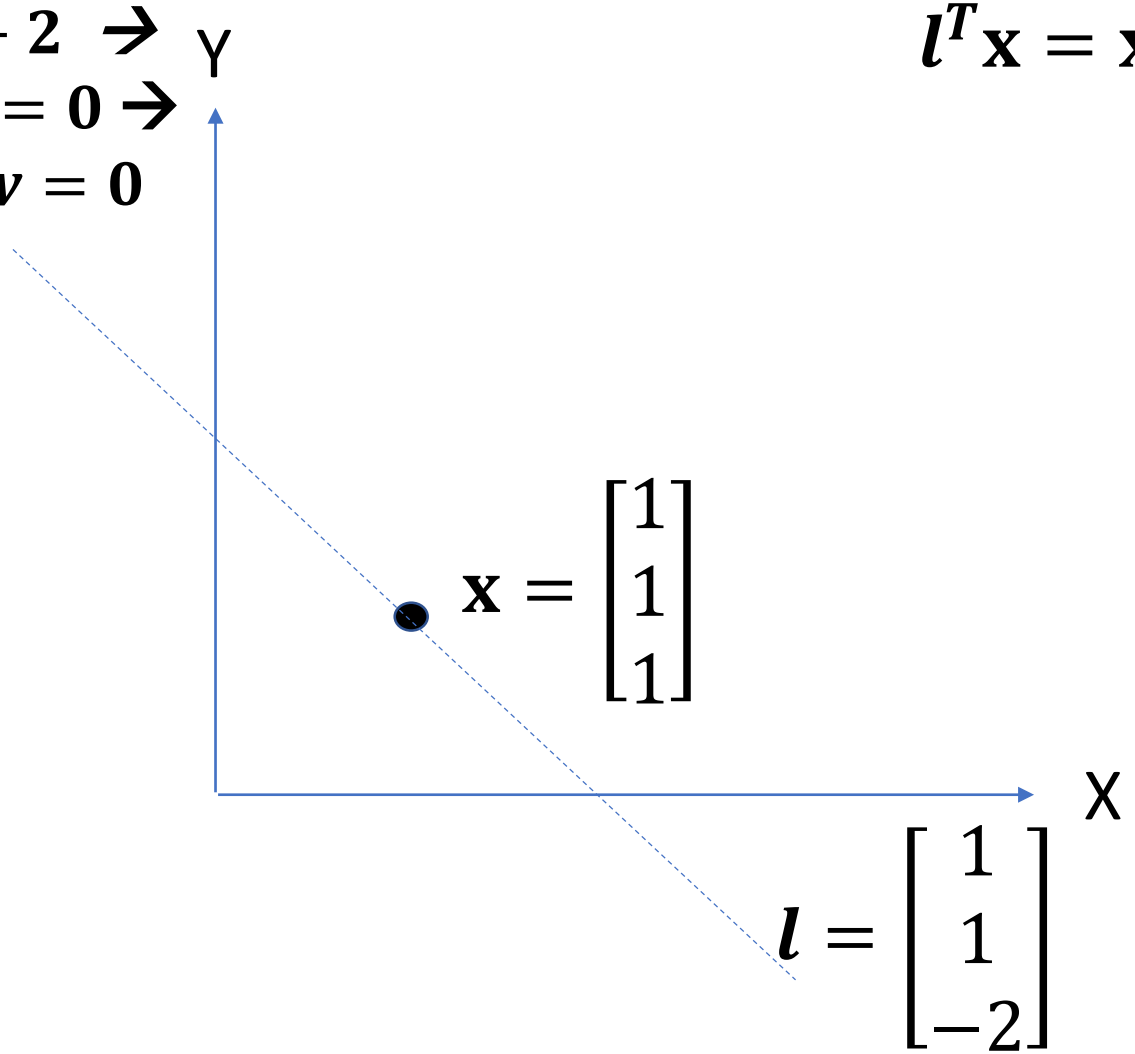
Principle of **duality** between points and lines

(50% discount principle)

point $\mathbf{x}$ is on line $\boldsymbol{l}$ (i.e. line $\boldsymbol{l}$ goes through point $\mathbf{x}$)

$\boldsymbol{Y = -X + 2} \rightarrow$
$\boldsymbol{X + Y - 2 = 0} \rightarrow$
$\mathbf{x} + \boldsymbol{y} - \mathbf{2}\boldsymbol{w} = \mathbf{0}$

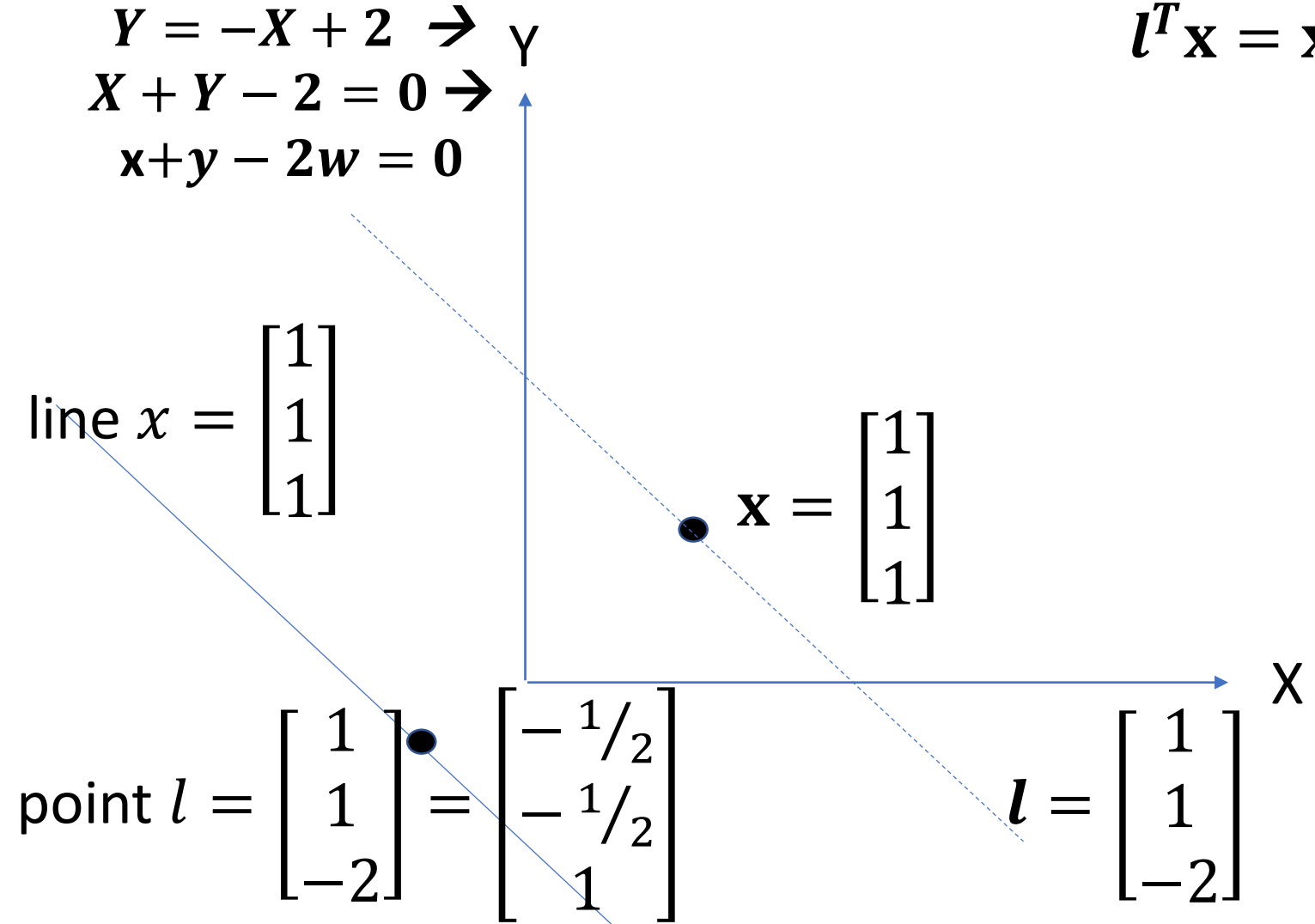
$$\boldsymbol{l}^T\mathbf{x} = \mathbf{x}^T\boldsymbol{l} = \mathbf{0}$$



point $\boldsymbol{l}$ is on line $\boldsymbol{x}$ (i.e. line $\boldsymbol{x}$ goes through point $\boldsymbol{l}$)

$\boldsymbol{Y = -X + 2}$ → $\boldsymbol{X + Y - 2 = 0}$ → $\mathbf{x} + \boldsymbol{y} - \mathbf{2}\boldsymbol{w} = \mathbf{0}$

$\boldsymbol{l}^T\mathbf{x} = \mathbf{x}^T\boldsymbol{l} = \mathbf{0}$



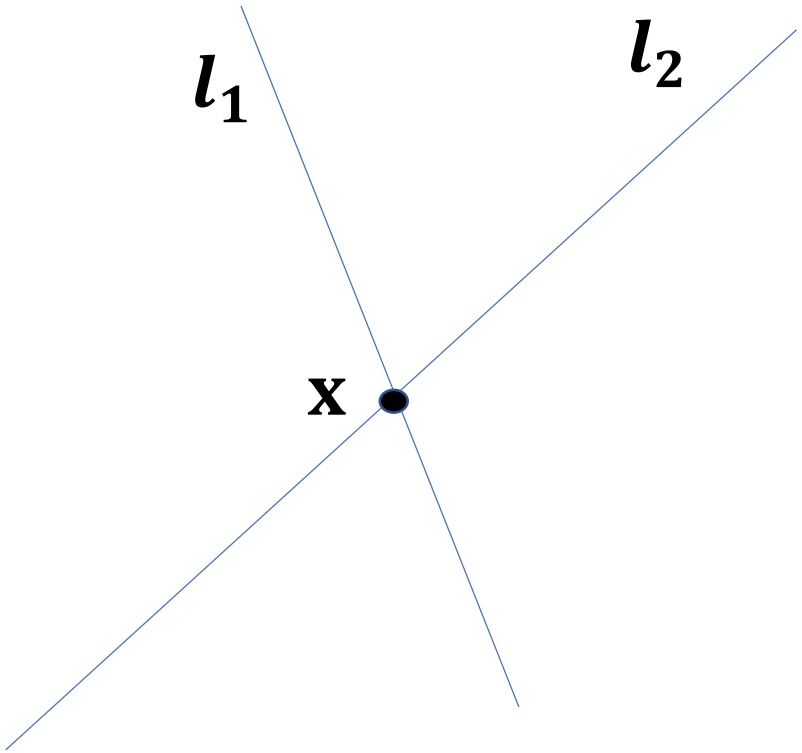
For any true sentence containing the items

- point
- line
- is on
- goes through

there is a DUAL sentence -also true- obtained by substituting, in the previous one, each occurence of

- point	by	- line
- line	by	- point
- is on	by	- goes through
- goes through	by	- is on

The point on two lines



$$\begin{cases} {l_1}^T \mathbf{x} = 0 \\ {l_2}^T \mathbf{x} = 0 \end{cases}$$

$$\begin{bmatrix} {l_1}^T \\ {l_2}^T \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\mathbf{x} = \mathbf{RNS}(\begin{bmatrix} {l_1}^T \\ {l_2}^T \end{bmatrix})$$

$\mathbf{x}$ is a vector orthogonal to both l_1 and l_2 vectors

→

$\mathbf{x}$ is (a multiple of) the cross product of l_1 and l_2 : $\mathbf{x} = l_1 \times l_2$

Example: intersection of two parallel lines

Suppose that lines $\boldsymbol{l_1}$ and $\boldsymbol{l_2}$ are parallel: this means that

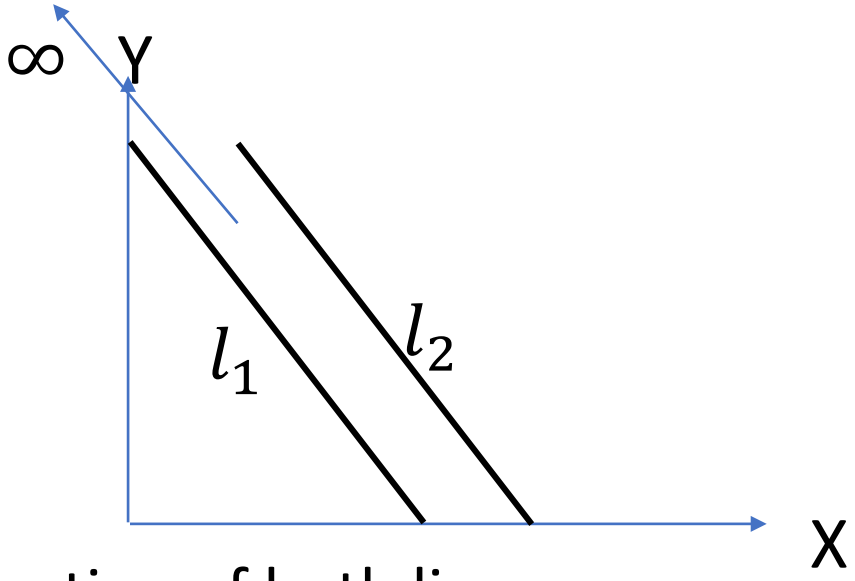
$$l_1 = [a \quad b \quad c_1]^T \text{ and}$$
$$l_2 = [a \quad b \quad c_2]^T$$

The point $\mathbf{x} = [\boldsymbol{x} \quad \boldsymbol{y} \quad \boldsymbol{w}]^T$ common two these two lines satisfies

$$\begin{cases} ax + by + c_1 w = 0 \\ ax + by + c_2 w = 0 \end{cases}$$

$$\downarrow$$

$$\mathbf{x} = [b \quad -a \quad 0]^T$$



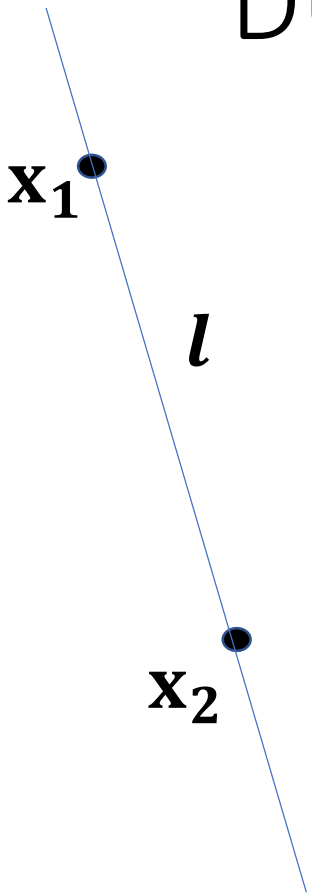
Namely, the point at the ∞ along the direction of both lines

(remember: [b, -a] is the direction of both lines)

PREVIEW: the *vanishing point* of a direction d is the image of the point at the ∞ along d

The line through two points

previous: the *point on* two *lines*
DUAL: the *line through* two *points*



$$\boldsymbol{l} = \mathbf{RNS}\left(\begin{bmatrix} \mathbf{x_1}^T \\ \mathbf{x_2}^T \end{bmatrix}\right)$$

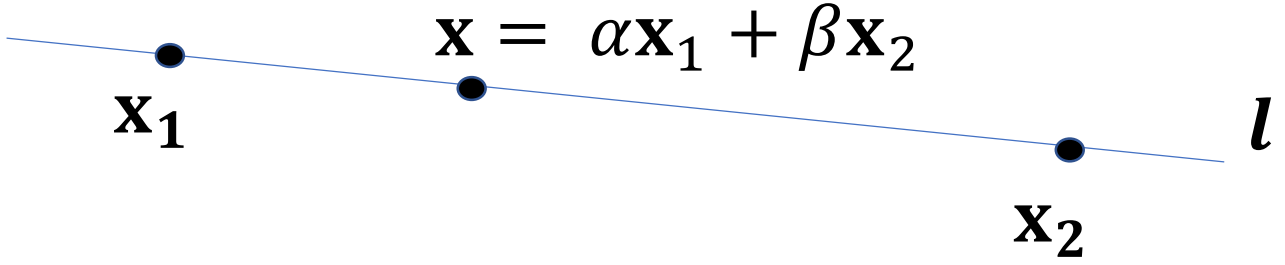
or, just in 2D Proj. Geo.

$$\boldsymbol{l} = \mathbf{x_1} \times \mathbf{x_2}$$

A useful property and its dual

The linear combination of two points

Property: the point $\mathbf{x}$ given by the linear combination $\mathbf{x} = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2$ of two points $\mathbf{x}_1$ and $\mathbf{x}_2$ is on the line $\boldsymbol{l}$ through $\mathbf{x}_1$ and $\mathbf{x}_2$ (i.e. on the line joining $\mathbf{x}_1$ and $\mathbf{x}_2$) . In less words: $\mathbf{x}$ is COLINEAR to $\mathbf{x}_1$ and $\mathbf{x}_2$

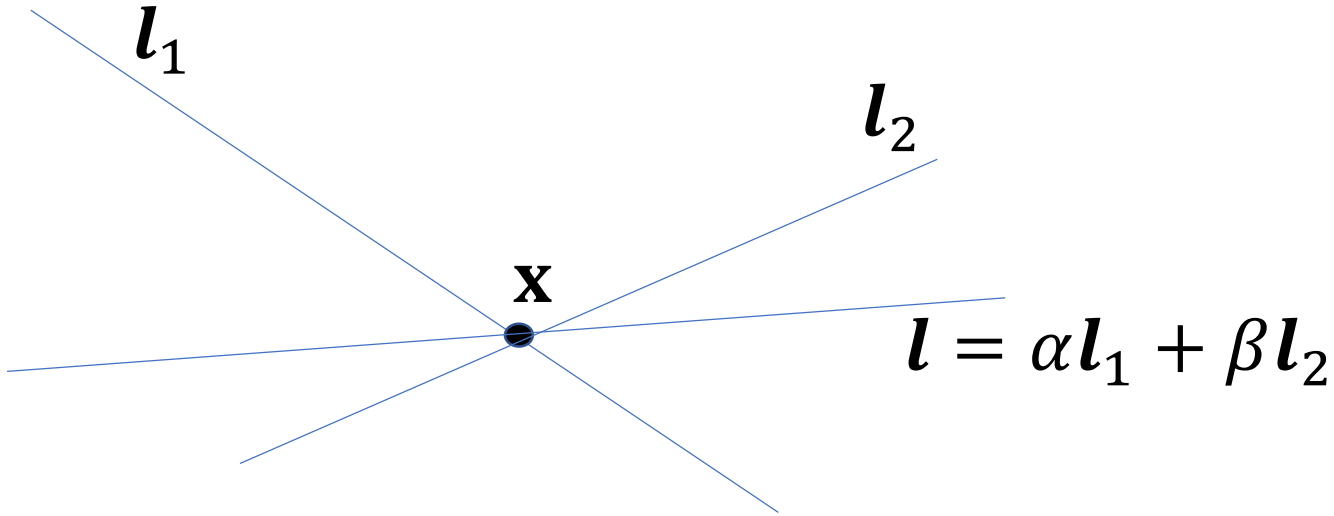


Proof: the line $\boldsymbol{l}$ through both points satisfies $\boldsymbol{l}^T\mathbf{x_1} = 0$ and $\boldsymbol{l}^T\mathbf{x_2} = 0$.
By adding α times the first eqn. to β times the second eqn., we obtain
$0 = \boldsymbol{l}^T(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = \boldsymbol{l}^T\mathbf{x} = 0$, i.e. $\mathbf{x}$ is on the same line joining $\mathbf{x}_1$ and $\mathbf{x}_2$.

NOTE: this property is only valid with homogeneous coordinates

DUAL: **The linear combination of two lines**

Property: the point $\mathbf{x}$, given by the linear combination $\mathbf{x} = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2$ of two points $\mathbf{x}_1$ and $\mathbf{x}_2$, is on the line $\boldsymbol{l}$ through $\mathbf{x}_1$ and $\mathbf{x}_2$

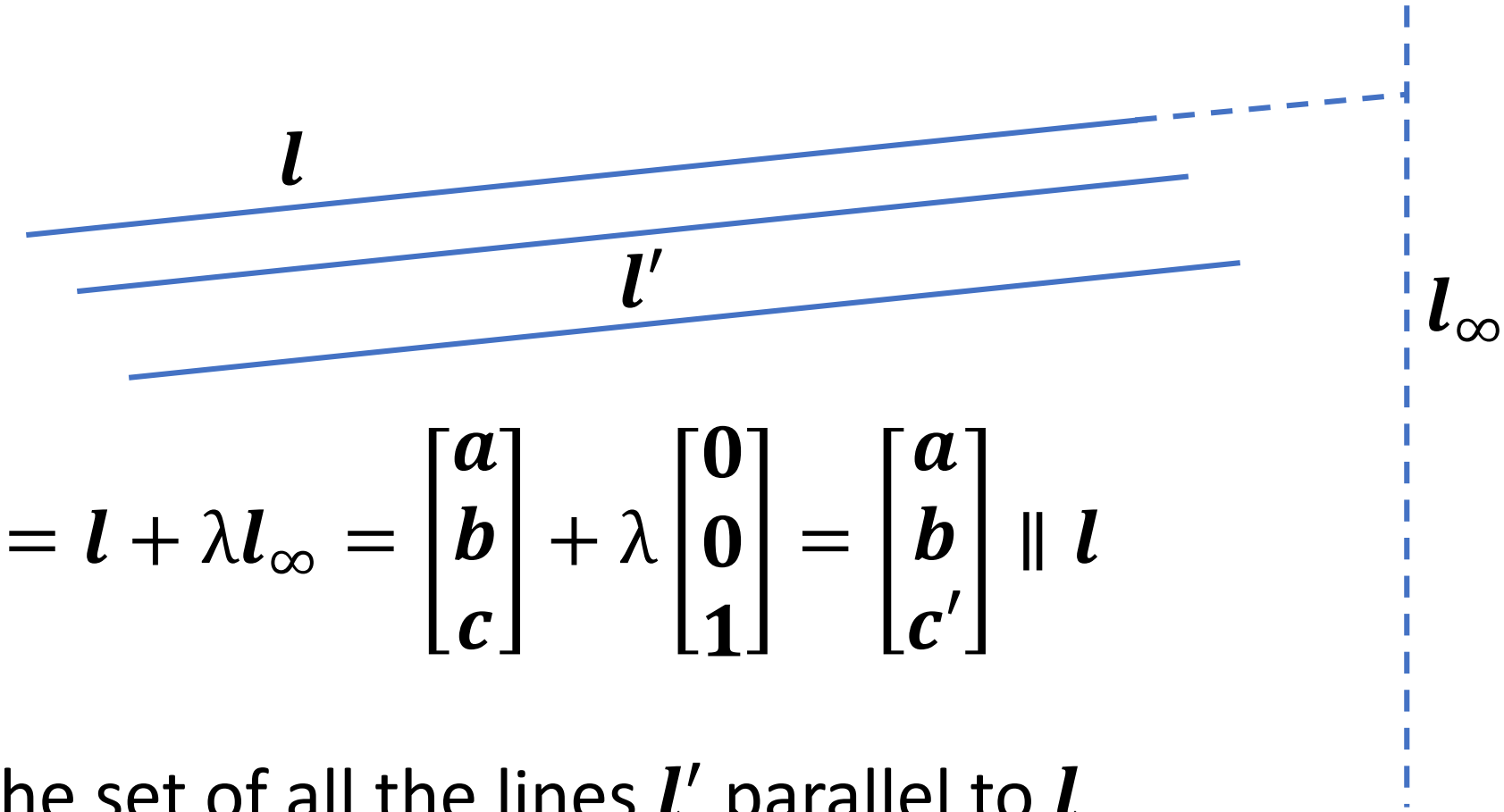


Dual: the line $\boldsymbol{l}$, given by the linear combination $\boldsymbol{l} = \alpha\boldsymbol{l}_1 + \beta\boldsymbol{l}_2$ of two lines $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$, goes through the point $\mathbf{x}$ on $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$.
In less words: line $\boldsymbol{l} = \alpha\boldsymbol{l}_1 + \beta\boldsymbol{l}_2$ is CONCURRENT to lines $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$

New dual pair

- point	by	- line
- line	by	- point
- is on	by	- goes through
- goes through	by	- is on
- colinear	by	- concurrent
- concurrent	by	- colinear

A special case:
linear combinations of a line $\boldsymbol{l}$ and the line $\boldsymbol{l}_\infty$



$$\boldsymbol{l}' = \boldsymbol{l} + \lambda \boldsymbol{l}_\infty = \begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \\ \boldsymbol{c} \end{bmatrix} + \lambda \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ \boldsymbol{1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \\ \boldsymbol{c}' \end{bmatrix} \parallel \boldsymbol{l}$$

→ the set of all the lines $\boldsymbol{l}'$ parallel to $\boldsymbol{l}$

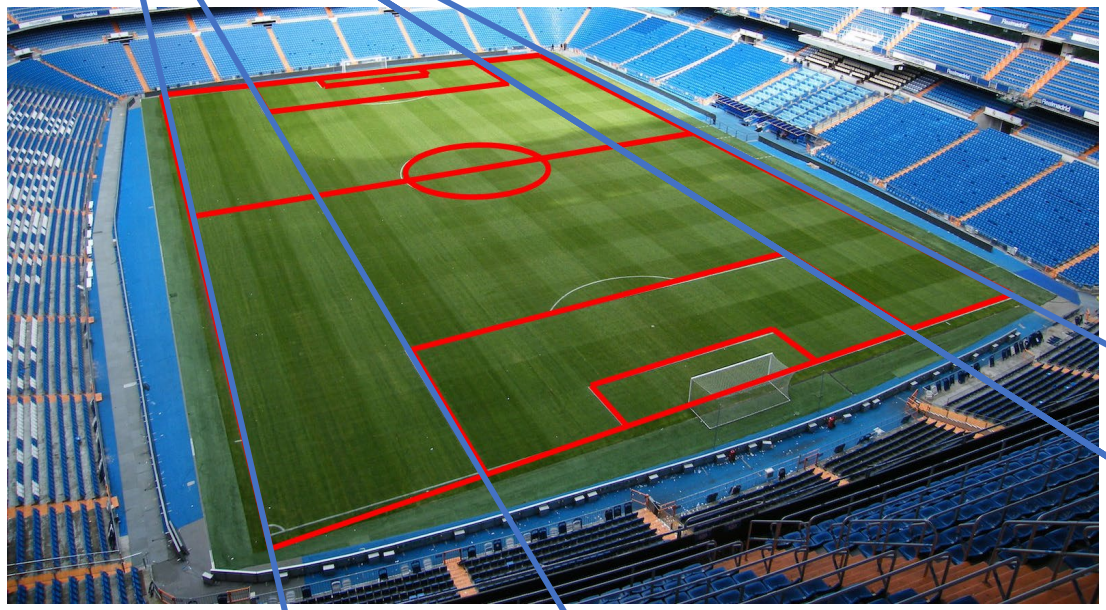
A preview:
time to impact for a translating camera
e.g., mounted on a moving robot or vehicle

Preview of Geometry (preview):

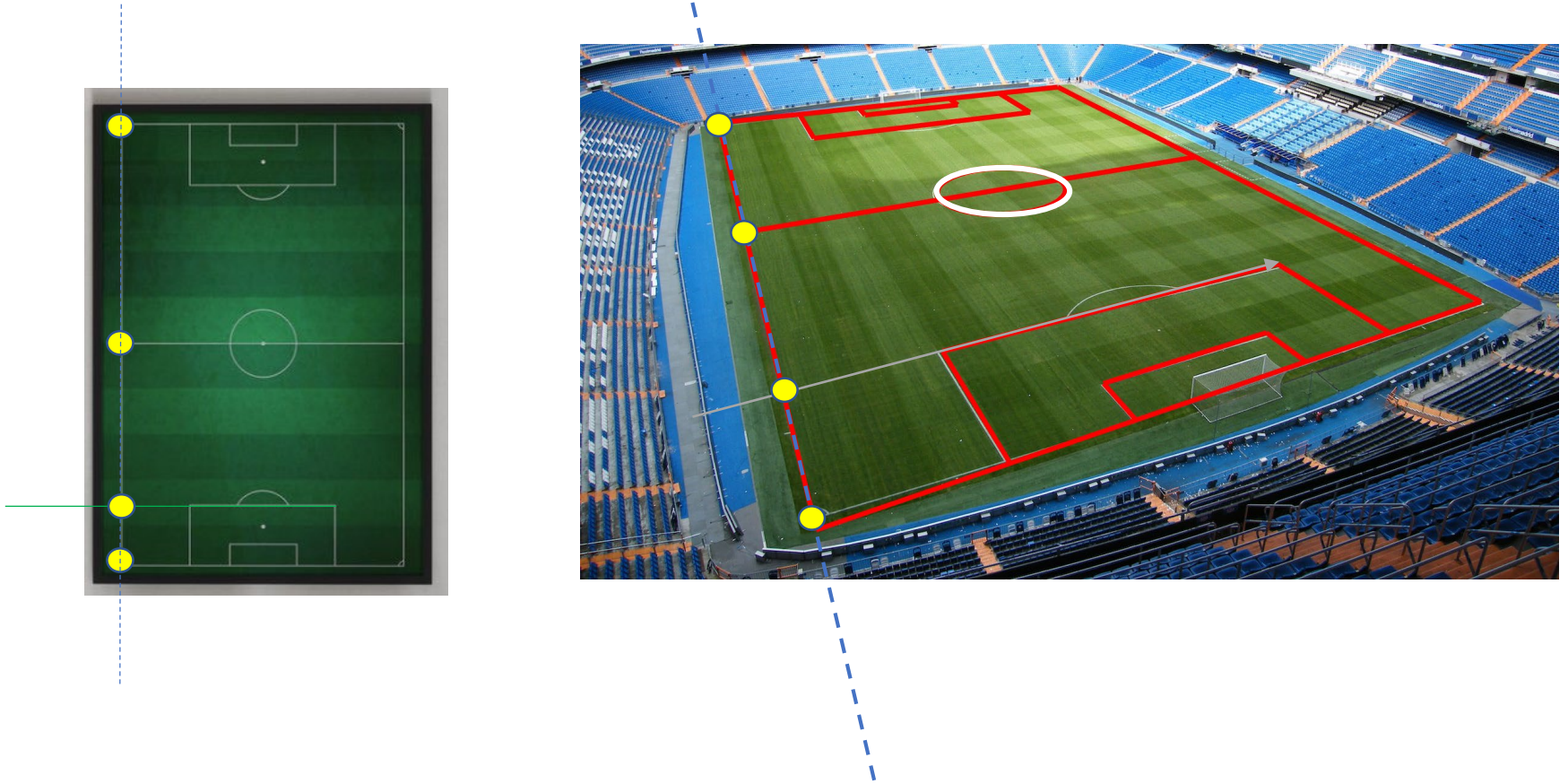
1. The vanishing point of a direction
2. The cross ratio of four colinear points is INVARIANT between real world and image

Parallel lines project onto **concurrent** lines

vanishing point
of the direction
of the // lines

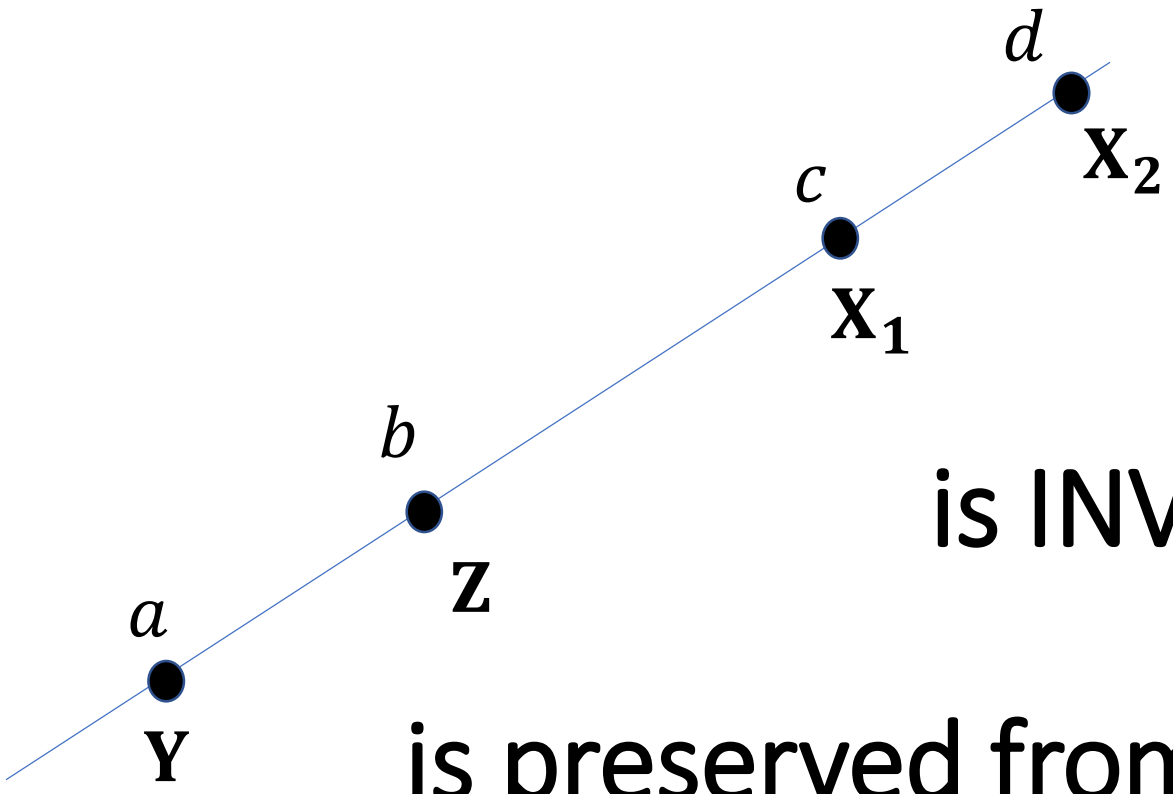


the cross ratio, i.e., the ratio between the two ratios, is INVARIANT: it is preserved from real word to image



Cross ratio of a 4-tuple of colinear points

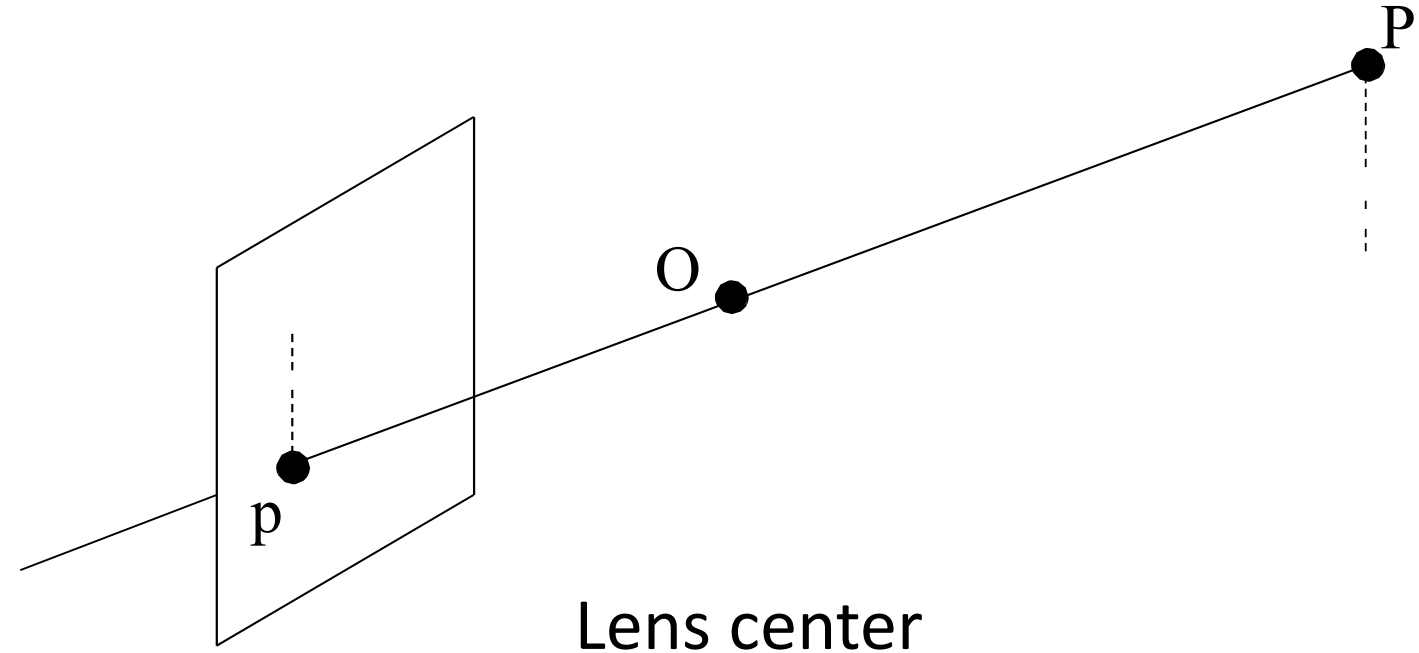
$$CR_{Y,Z,X_1,X_2} = \frac{c-a}{c-b} \Big/ \frac{d-a}{d-b}$$



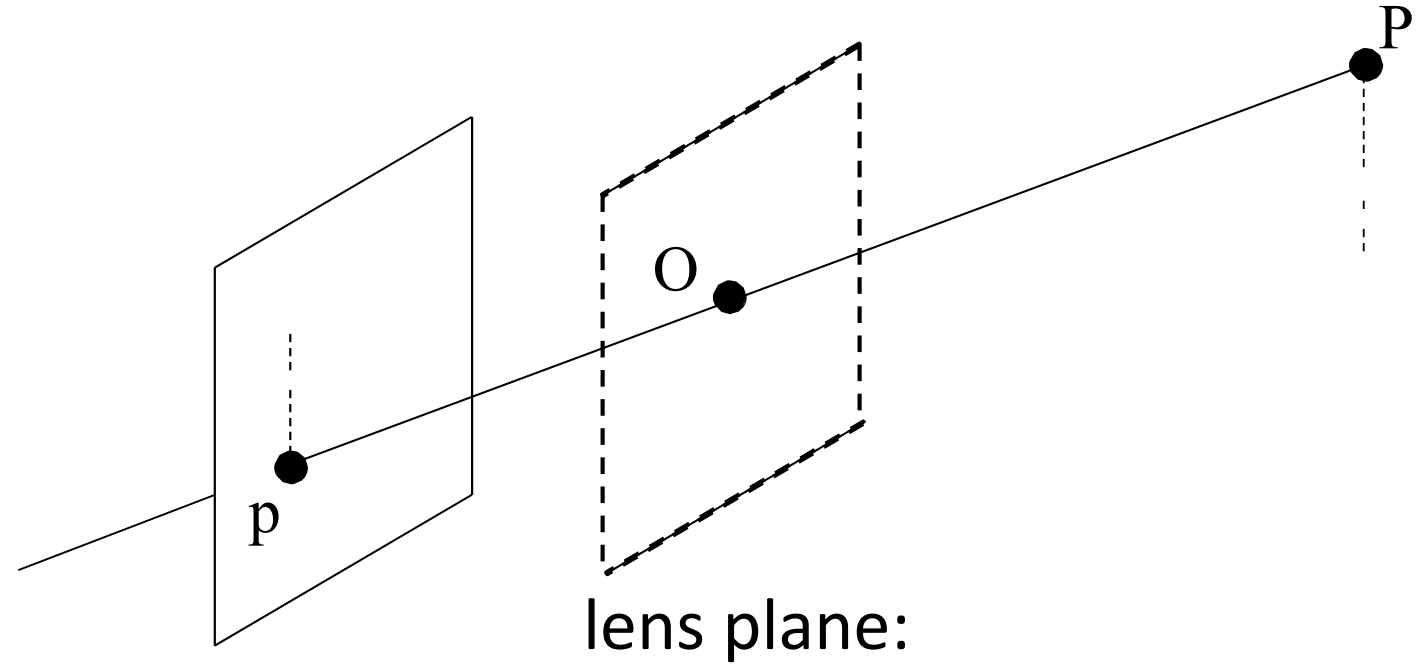
is INVARIANT:

is preserved from real world to image

camera projection

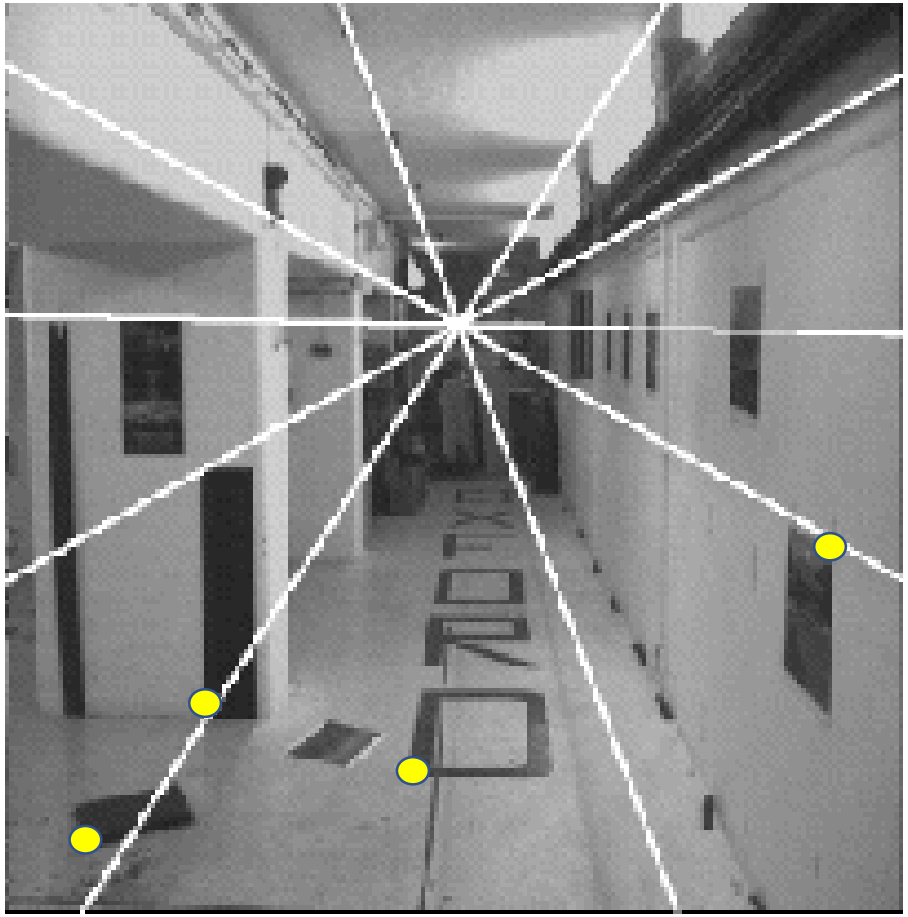


camera projection

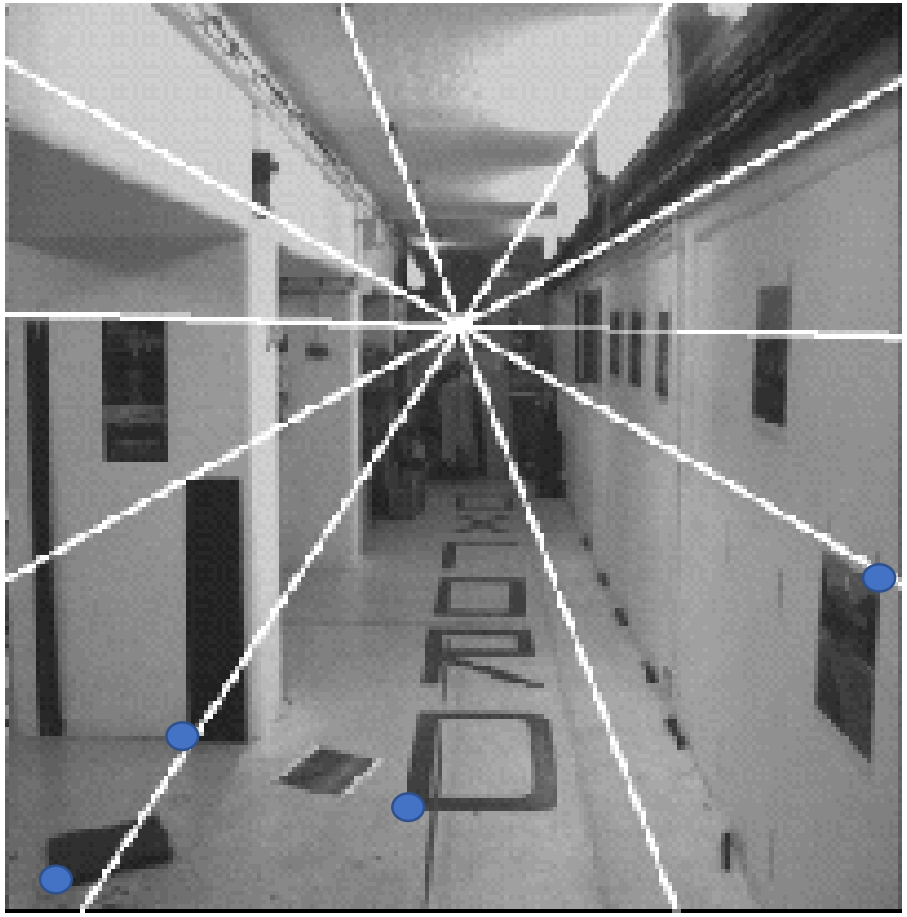


Time to impact: translating camera

t = 0



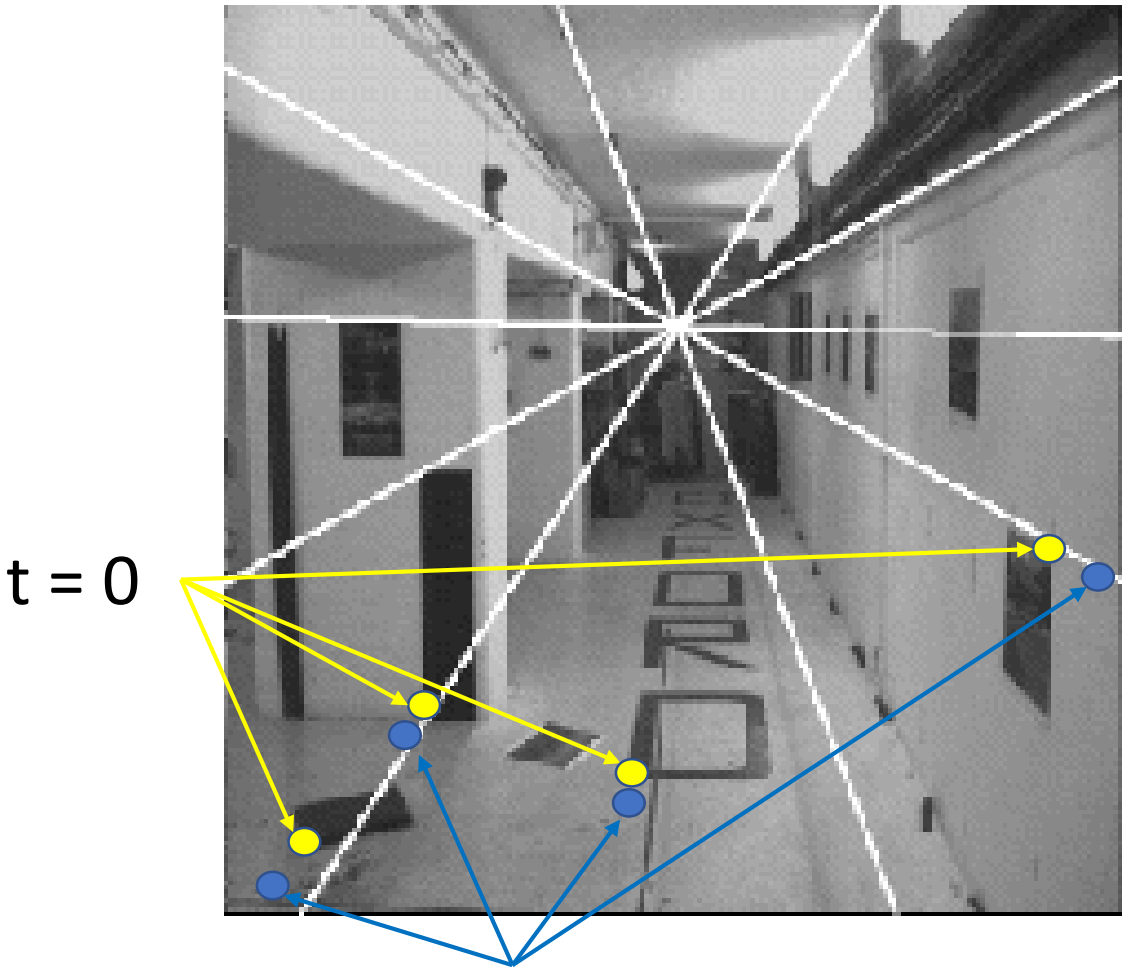
t = Δt

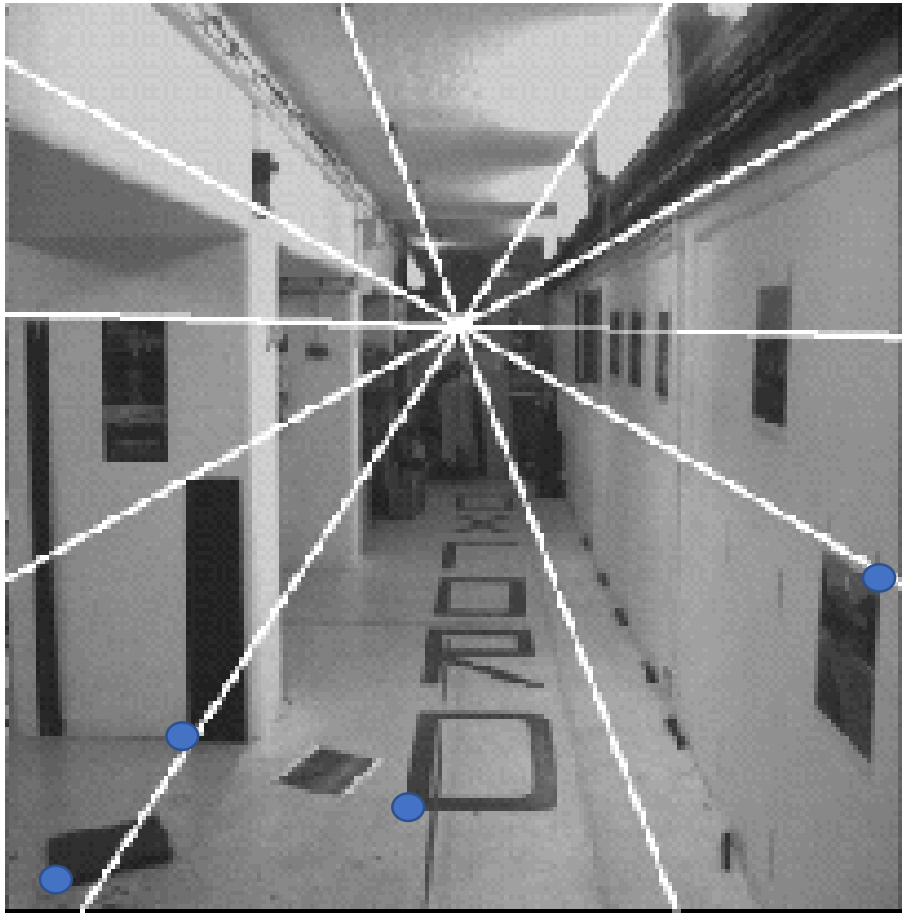


Apparent motion segments “yellow – blue” are PARALLEL in real world

→ their images are CONCURRENT at the VANISHING POINT of motion direction

= image of the point at the ∞ along motion direction corresponds to ∞ distance

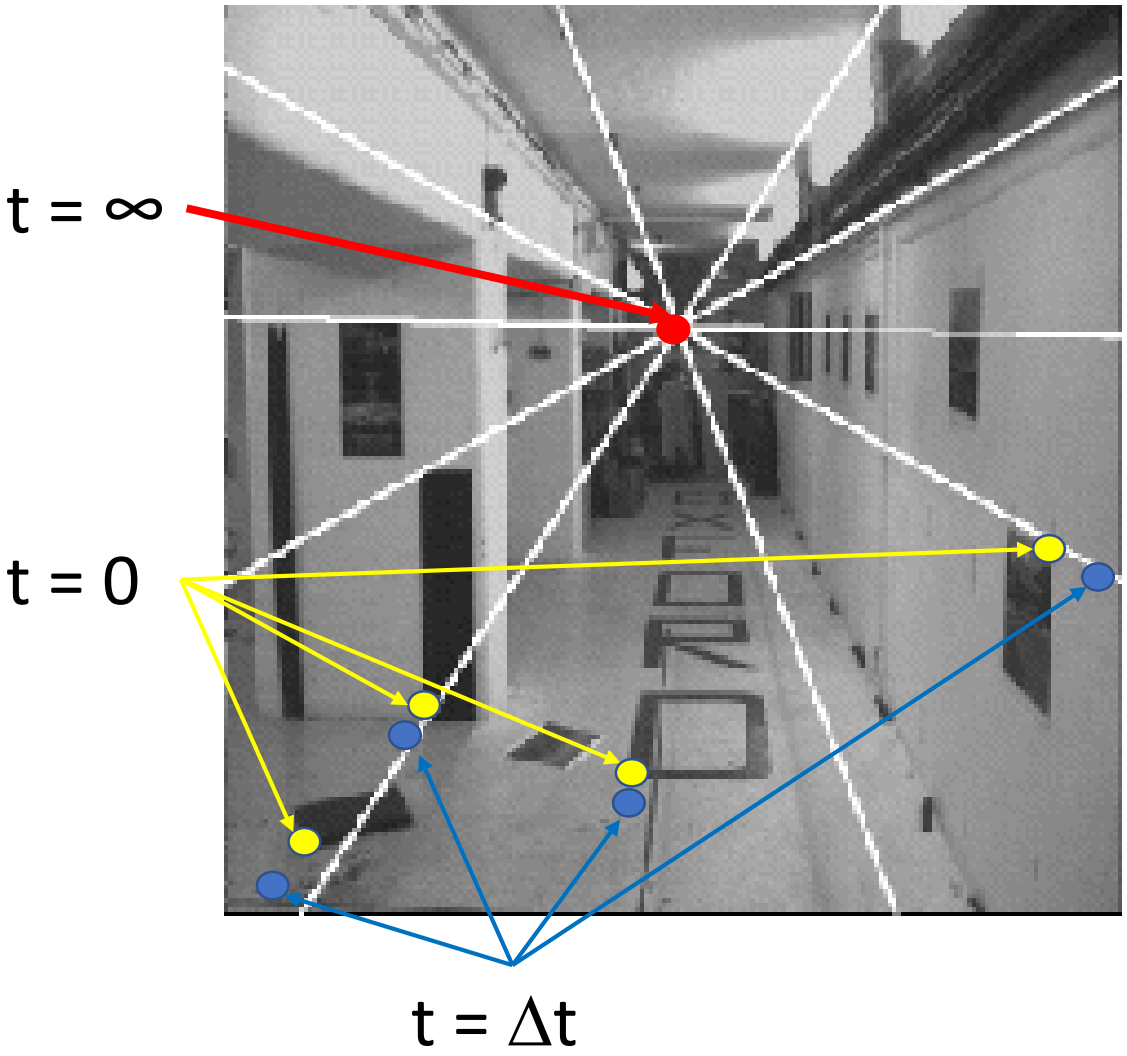


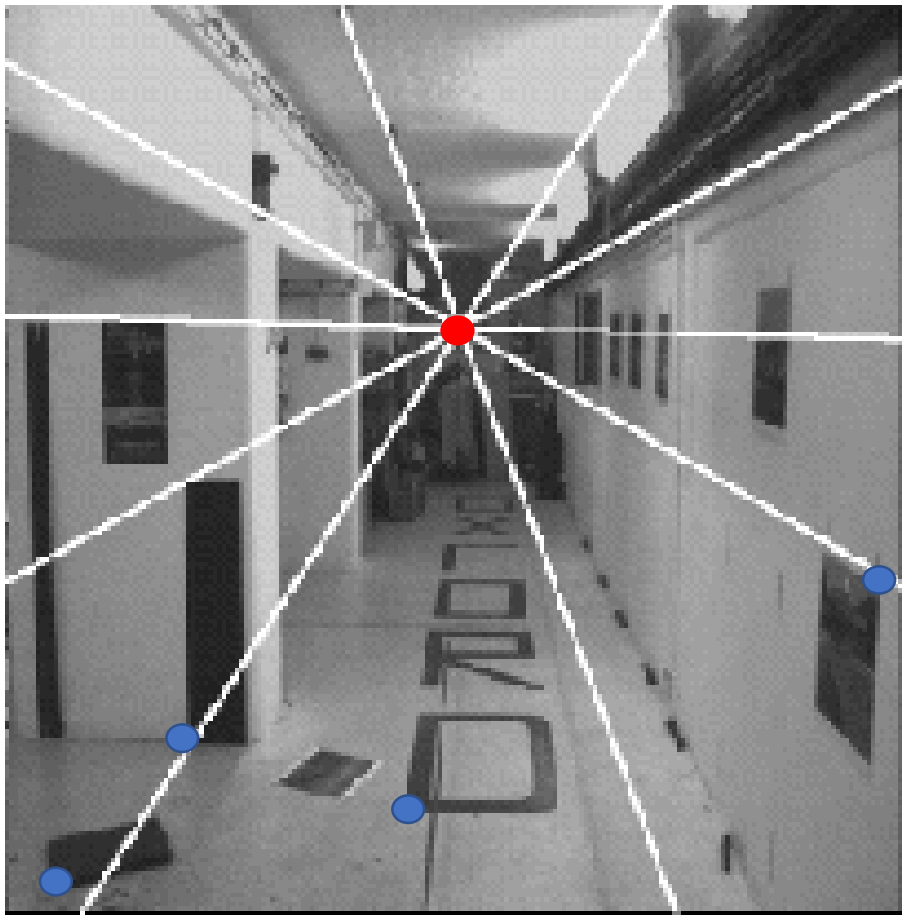


Constant speed: space proportional to time → ∞ distance corresponds to t = ∞

In addition (2.): cross ratio of times = cross ratio of real distances

= cross ratio of image coordinates

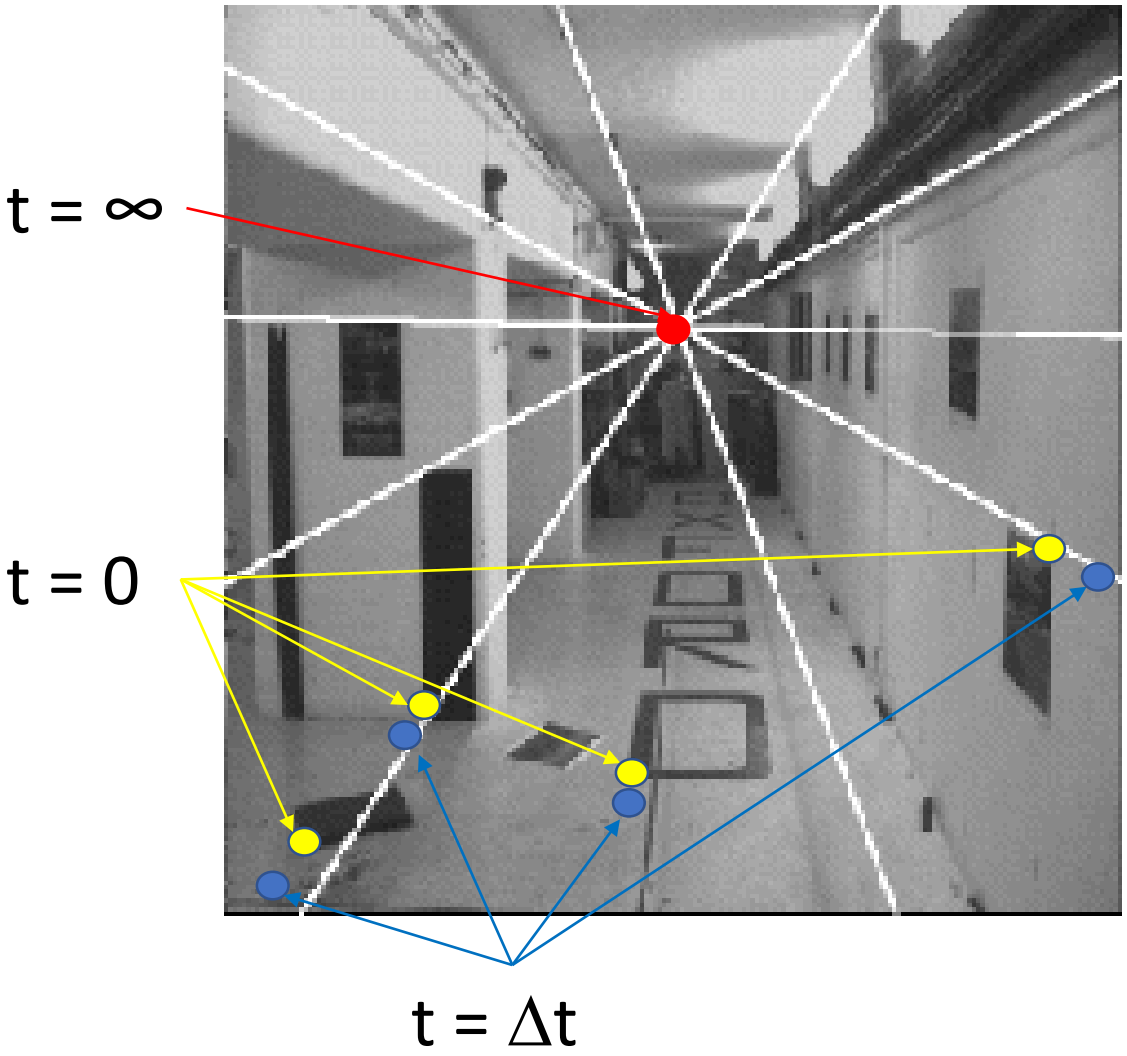


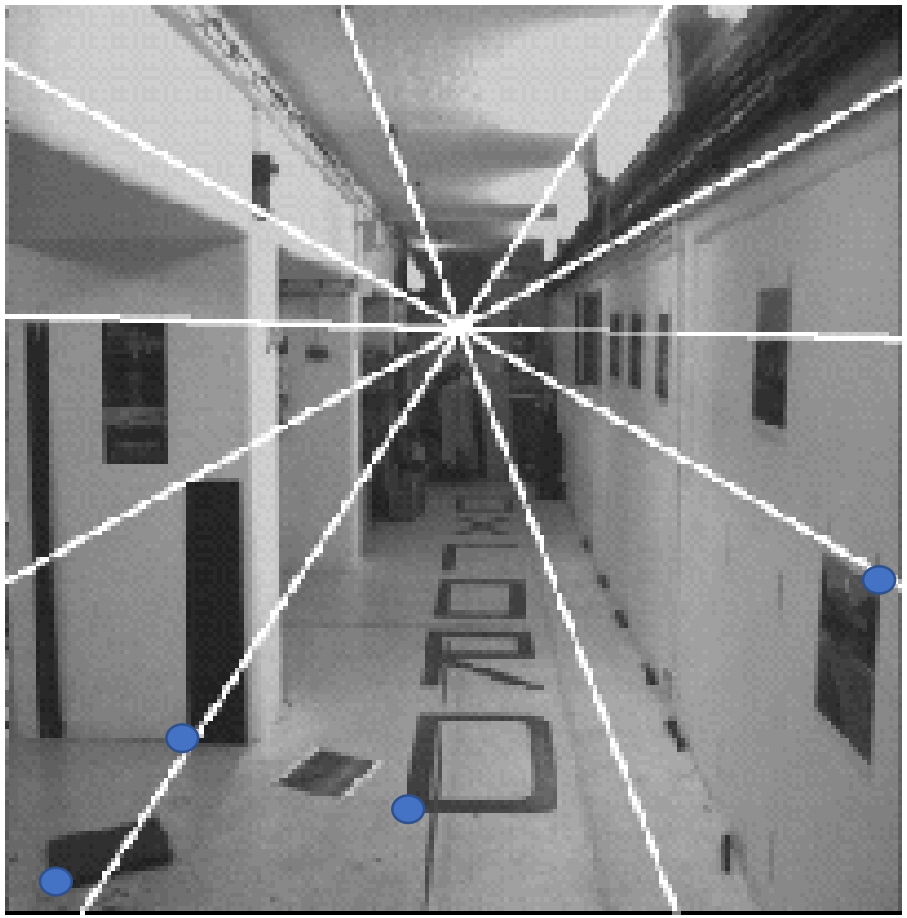


cross ratio of times = cross ratio of image coordinates

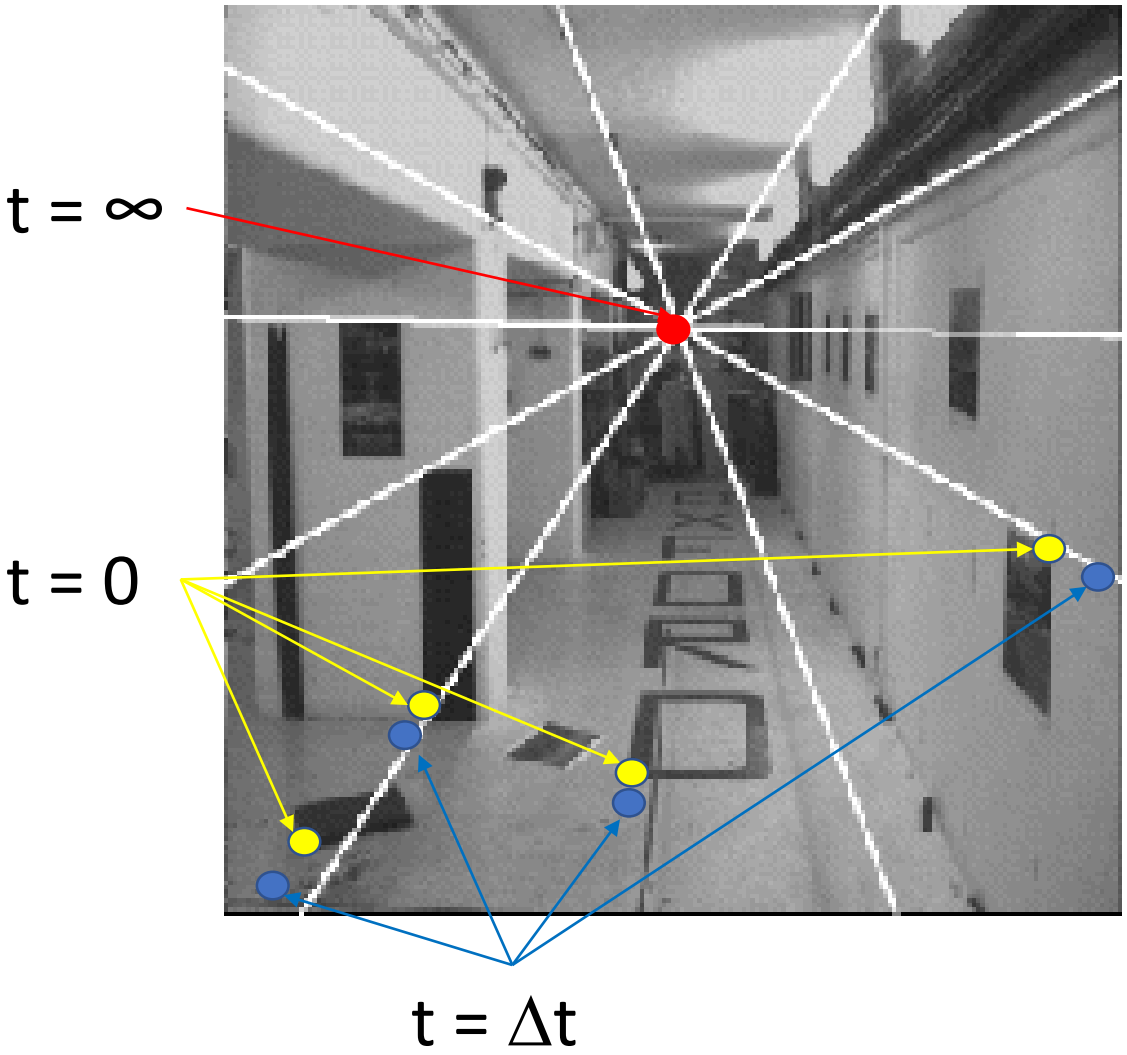
<u>for each moving feature</u>, the cross ratios need 4-tuples, but we only see three:

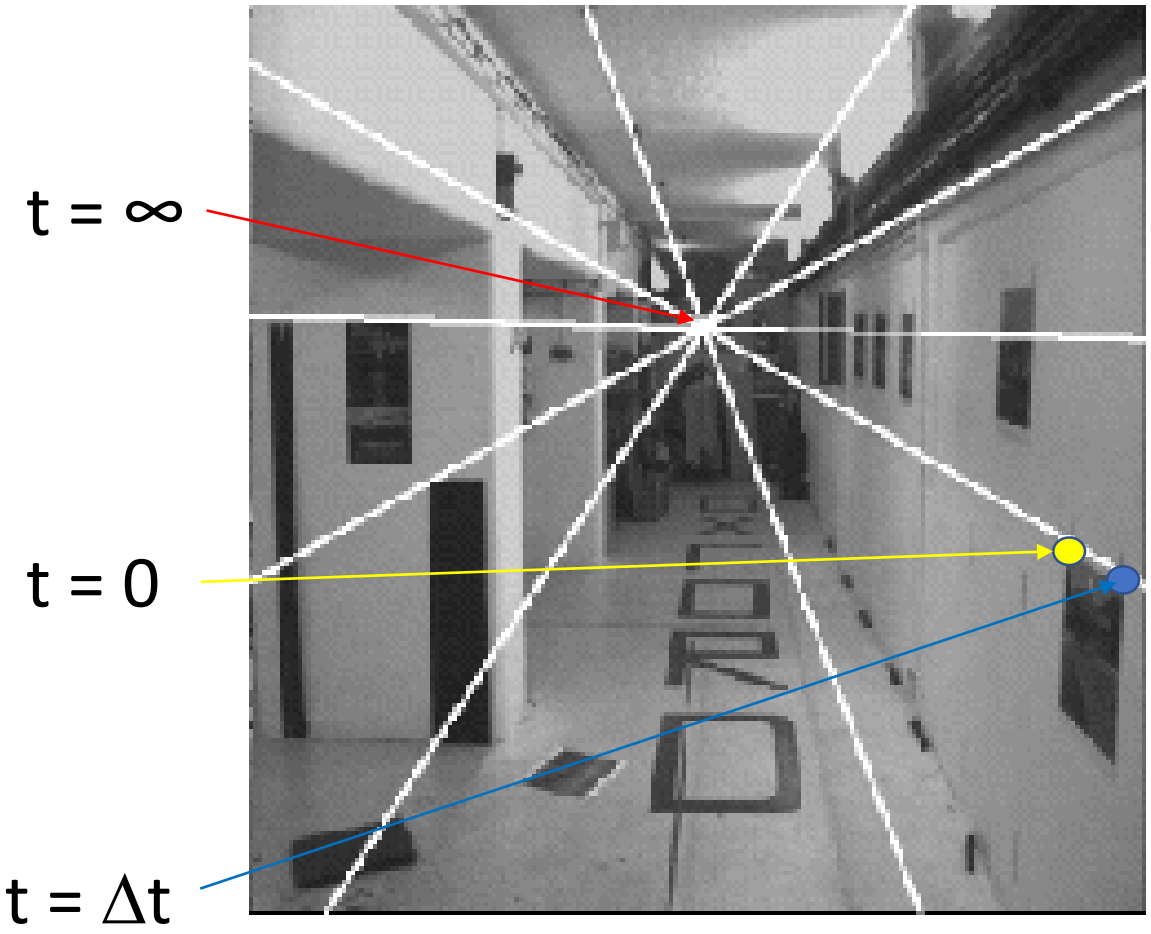
fourth time instant? fourth point in 3D space? its image?





fourth time instant = time of impact *ti* of the feature on the lens plane
fourth real world point = the point-of-impact of the feature on the lens plane
its image???





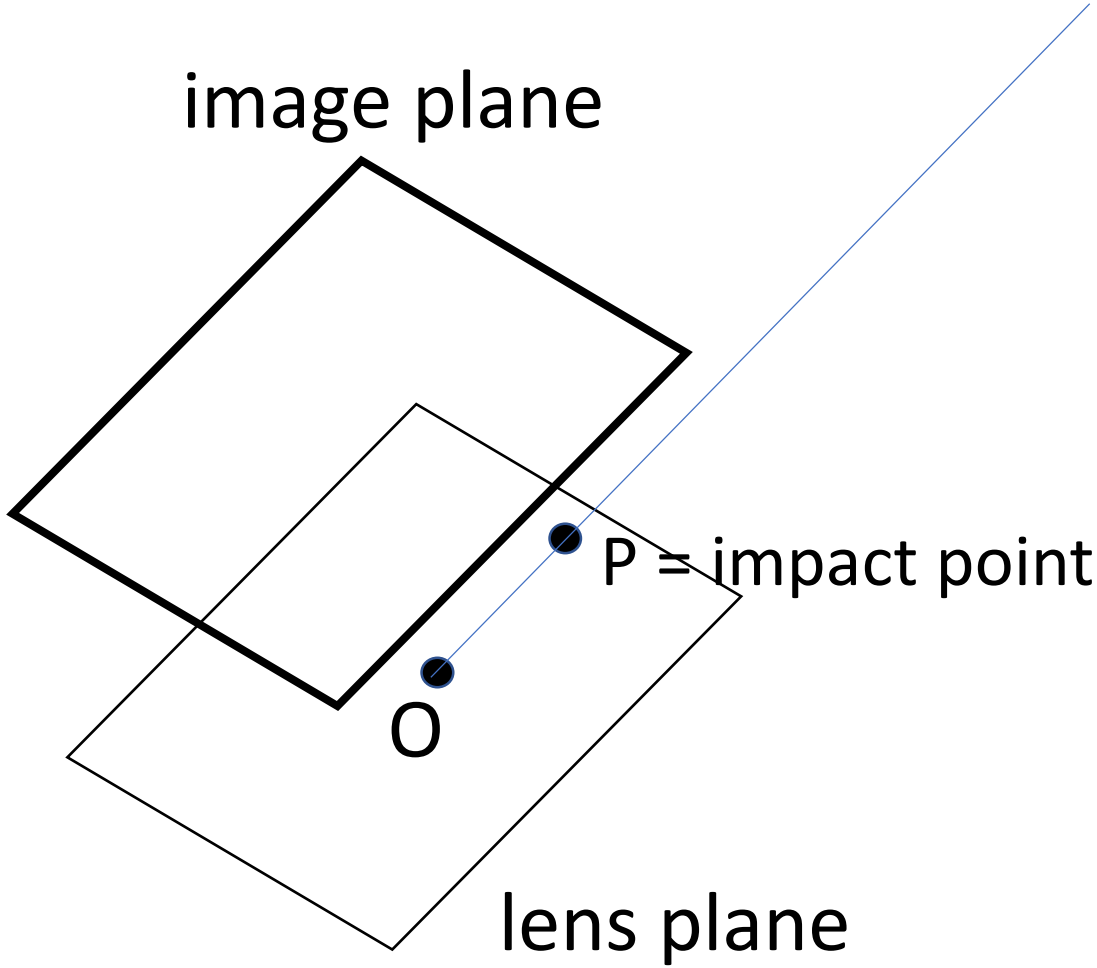


Image of impact point P = intersection
line (P,O) ∩ image plane;
P and O are on lens plane,
lens plane // image plane
→ intersection is at the ∞
→ **x at impact is ∞**

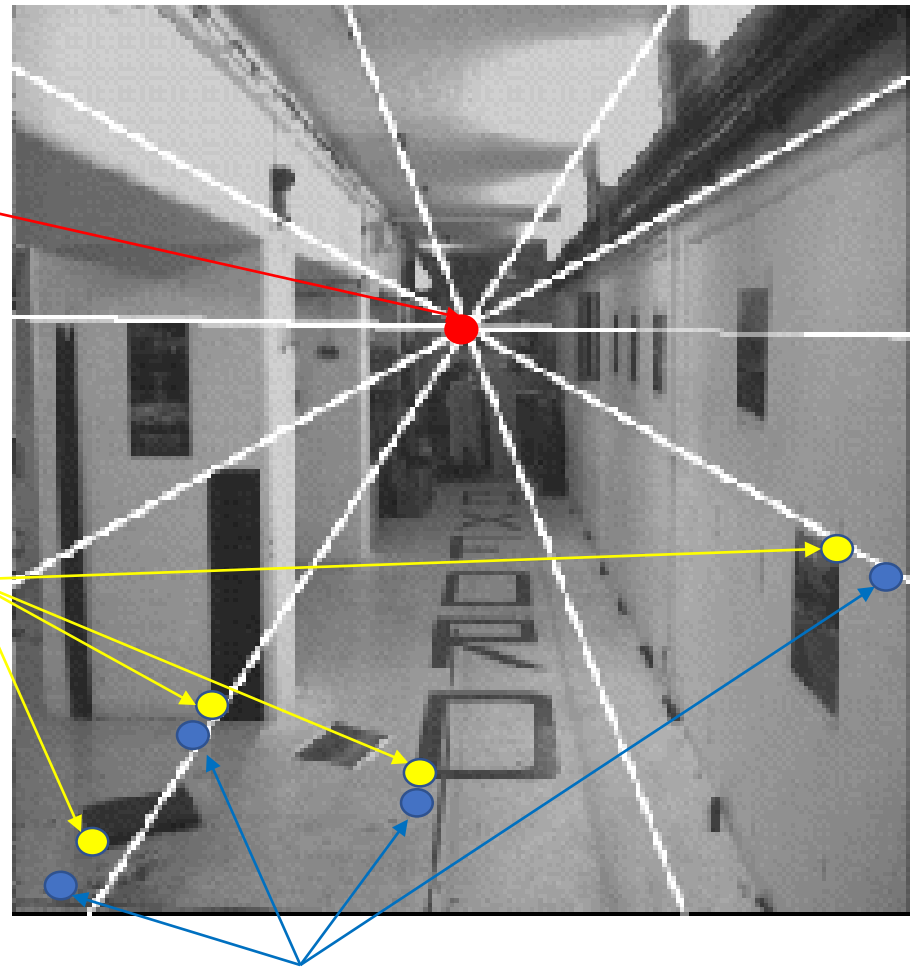
what is the fourth image coordinate?
at time of impact *ti* with lens plane
the image feature is at the ∞

t = 0

t = ∞

t = 0

t = Δt

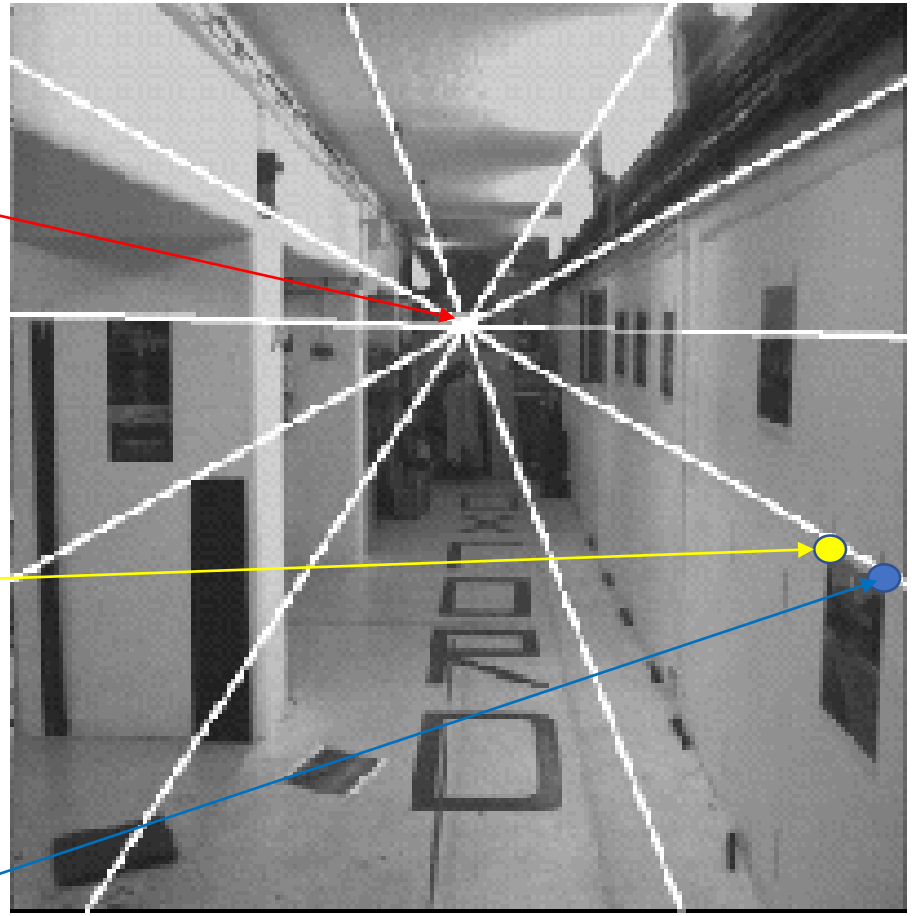


t = Δt

t = ∞

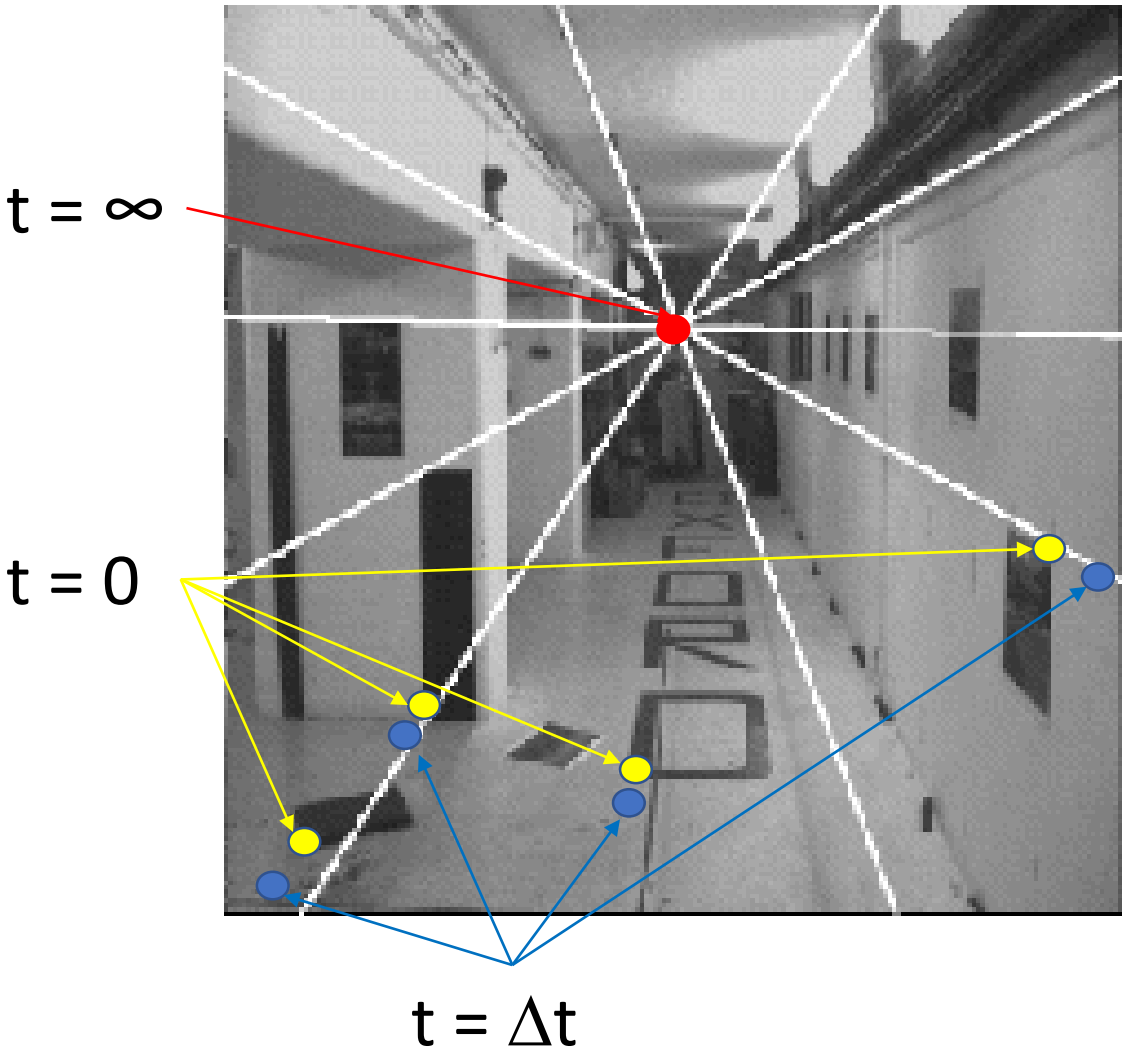
t = 0

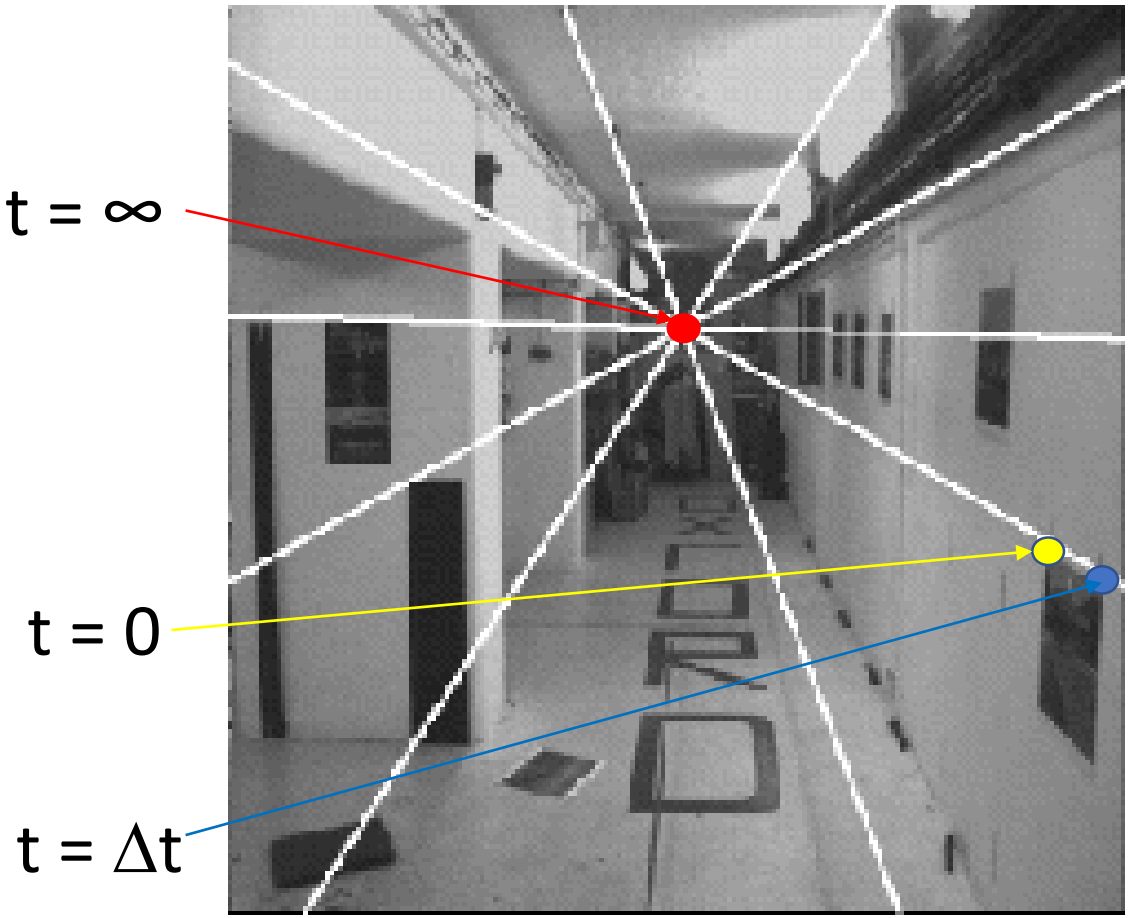
t = Δt



equal cross ratios

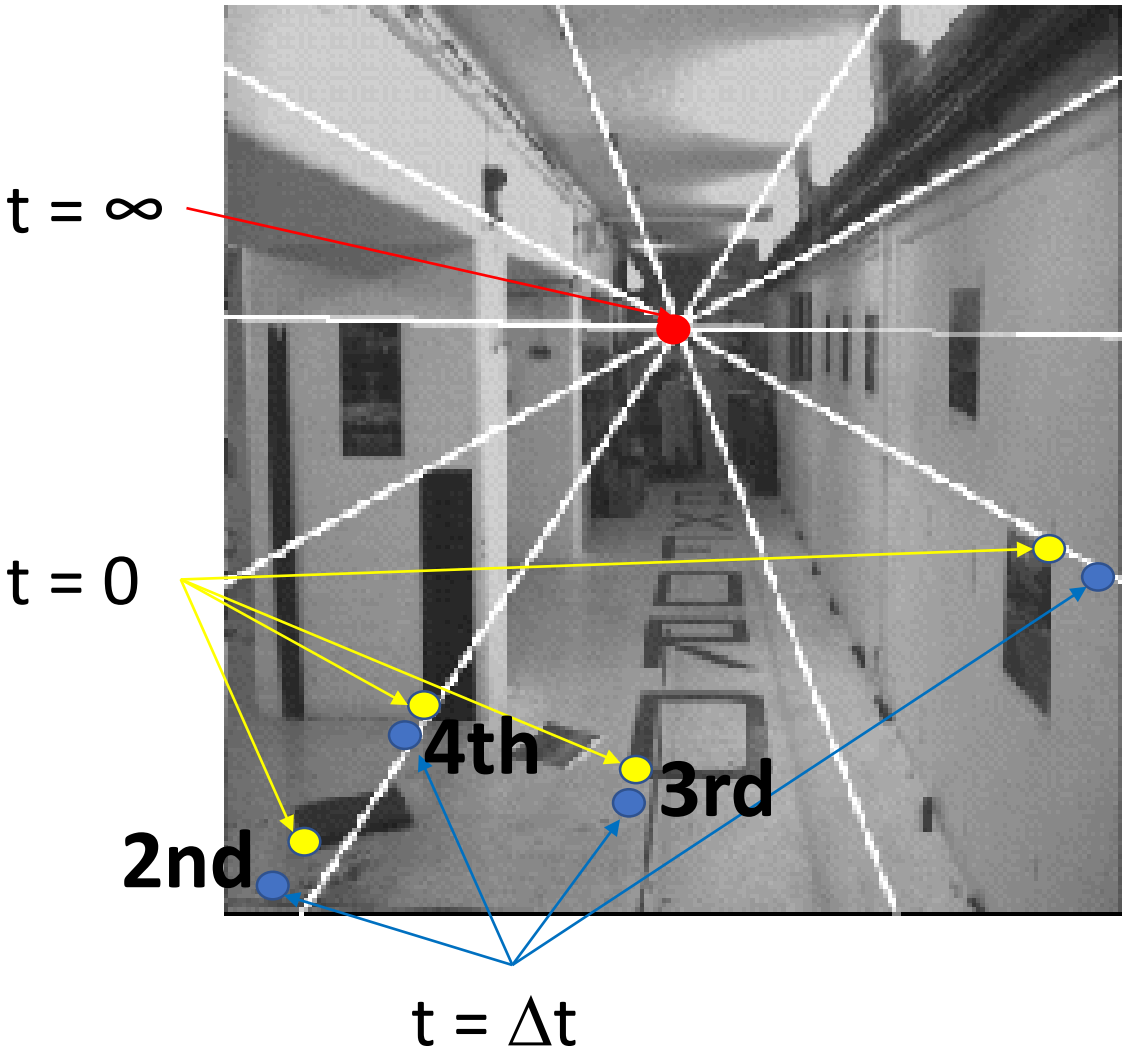
$$\frac{t_i-\Delta t}{t_i-0} \Big/ \frac{\infty-\Delta t}{\infty-0} = \frac{\infty-x'}{\infty-x} \Big/ \frac{x_\infty-x'}{x_\infty-x} \rightarrow \frac{t_i-\Delta t}{t_i-0} = \frac{x_\infty-x}{x_\infty-x'} \rightarrow t_i = \Delta t \frac{x_\infty-x'}{x-x'}$$





equal cross ratios

$$t_i = \Delta t \frac{\mathbf{x}_\infty - \mathbf{x}'}{\mathbf{x} - \mathbf{x}'}$$



sort features by increasing time to impact

Translational motion

time to impact of a feature
proportional to the **depth** of the feature
(i.e. distance from lens plane along motion direction)

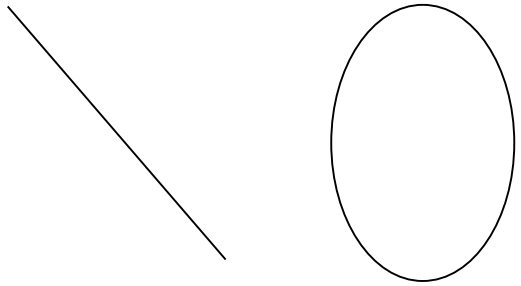
→

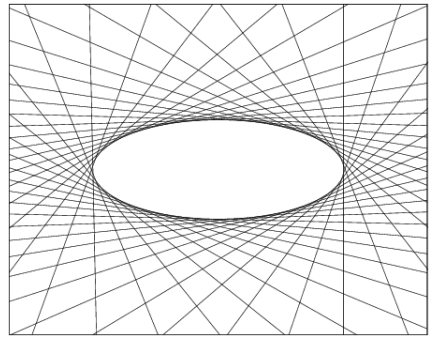
If all feature are moving at the same speed, as in static scene
and camera mounted on translating vehicle,
it allows a «partial 3D reconstruction»

→ Otherwise, e.g., when features translate at different speeds,
(e.g., different vehices driving on a straight road),
time-to-impact is useful «per se»

Planar Projective Geometry

- **Elements**
- Points
- Lines
- **Conics**
- Dual conics

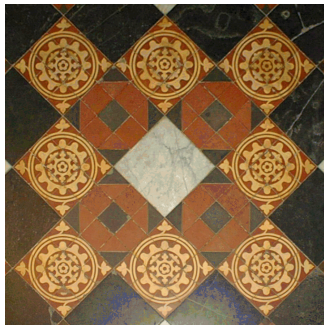




- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities

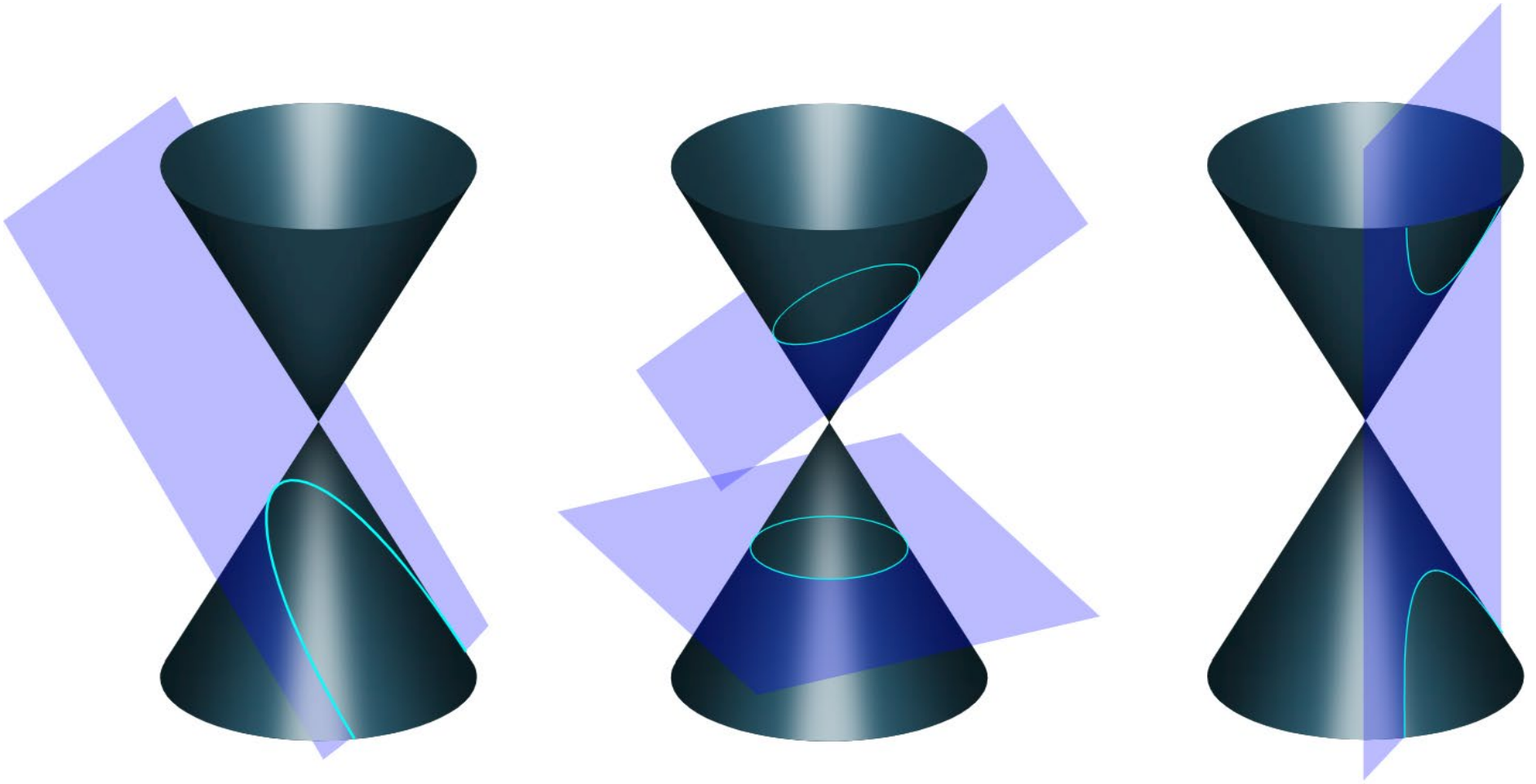






CONICS

Conics name derives from the intersection of cones with planes



PARABOLAE ELLIPSE HYPERBOLAE

Lines: a point $\mathbf{x}$ is on a line $\boldsymbol{l}$ if it satisfies a homogeneous ***linear*** equation, namely

$$\boldsymbol{l}^T\mathbf{x} = \mathbf{0}$$

Conics: a point $\mathbf{x}$ is on a conic $\boldsymbol{C}$ if it satisfies a homogeneous ***quadratic*** equation, namely

$$\mathbf{x}^T\boldsymbol{C}\,\mathbf{x} = \mathbf{0}$$

where $\boldsymbol{C}$ is a 3x3 symmetric matrix.

Conics in $\mathbb{P}^2$

- A conic is a curve described by a second-degree equation in the plane.
- In Euclidean coordinates a conic becomes
- $aX^2 + bXY + cY^2 + dX + eY + f = 0$
- i.e. a polynomial of degree 2. “Homogenizing” this by the replacements:
- X→ x/w, Y→ y/w gives
- $ax^2 + bxy + cy^2 + dxw + eyw + fw^2 = 0$
- or in matrix form $\mathbf{x}^\intercal \boldsymbol{C}\mathbf{x} = \mathbf{0}$

→ $$[x \quad y \quad w]\begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}\begin{bmatrix} x \\ y \\ w \end{bmatrix} = ax^2 + bxy + cy^2 + dxw + eyw + fw^2 = 0$$

- where the conic coefficient matrix $\boldsymbol{C}$ is given by $\boldsymbol{C} = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$

Conics in $\mathbb{P}^2$

- $\mathbf{x}^\intercal \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix} \mathbf{x} = 0$

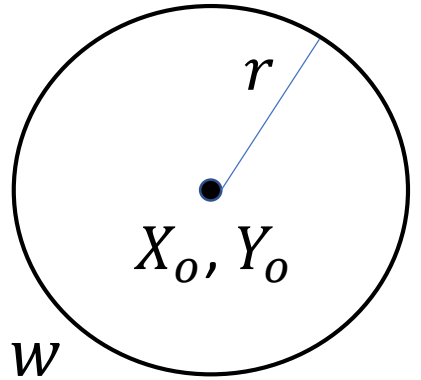
- **Rmk** The conic coefficient matrix is symmetric,
- **Rmk** multiplying $\boldsymbol{C}$ by a non-zero scalar does not change. Only the ratios of the elements in $\boldsymbol{C}$ are important, as for homogeneous points and for lines. → $\boldsymbol{C}$ is a homogeneous matrix
- **Rmk** The conic has five degrees of freedom :
- the ratios $\{a : b : c : d : e : f\}$ or equivalently
- the six elements of a symmetric matrix minus one for scale.

Example: the circumference

First in *cartesian* coordinates:

$$(X - X_o)^2 + (Y - Y_o)^2 - r^2 = 0$$

X_o, Y_o are the center coordinates, r is the radius.



then in *homogeneous* coordinates:

replace X and Y by, respectively, x/w and y/w and multiply by w

$$x^2 - 2X_o xw + X_o^2 w^2 + y^2 - 2Y_o yw + Y_o^2 w^2 - r^2 w^2 = 0$$

reorder the coefficients

$$[x \quad y \quad w] \begin{bmatrix} 1 & 0 & -X_o \\ 0 & 1 & -Y_o \\ -X_o & -Y_o & X_o^2 + Y_o^2 - r^2 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix} = 0$$

$$\mathbf{x}^\mathsf{T} \; \boldsymbol{C} \; \mathbf{x} = 0$$

Intersection of a line and a conic

Intersection of a line and a conic

Conic: quadratic equation on $\mathbf{x}$, namely $\mathbf{x}^{\mathsf{T}}\, \boldsymbol{C}\ \mathbf{x} = 0$

Line: linear equation on $\mathbf{x}$, namely $\boldsymbol{l}^{T}\mathbf{x} = \mathbf{0}$

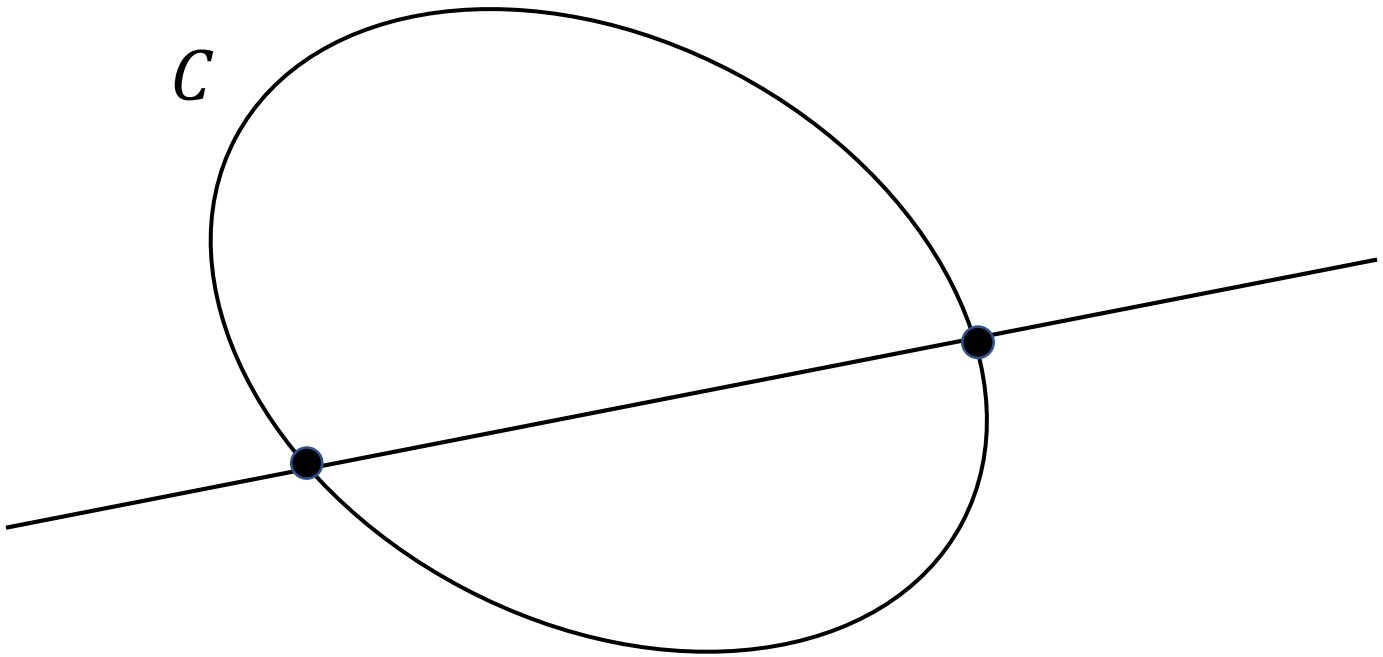
Line-conic intersection leads to a degree 2 equation on $\mathbf{x}$

→

there are always TWO intersection points between a line and a conic:
they can be real, or complex conjugate, distinct or coincident
[Foundamental Theorem of Algebra]

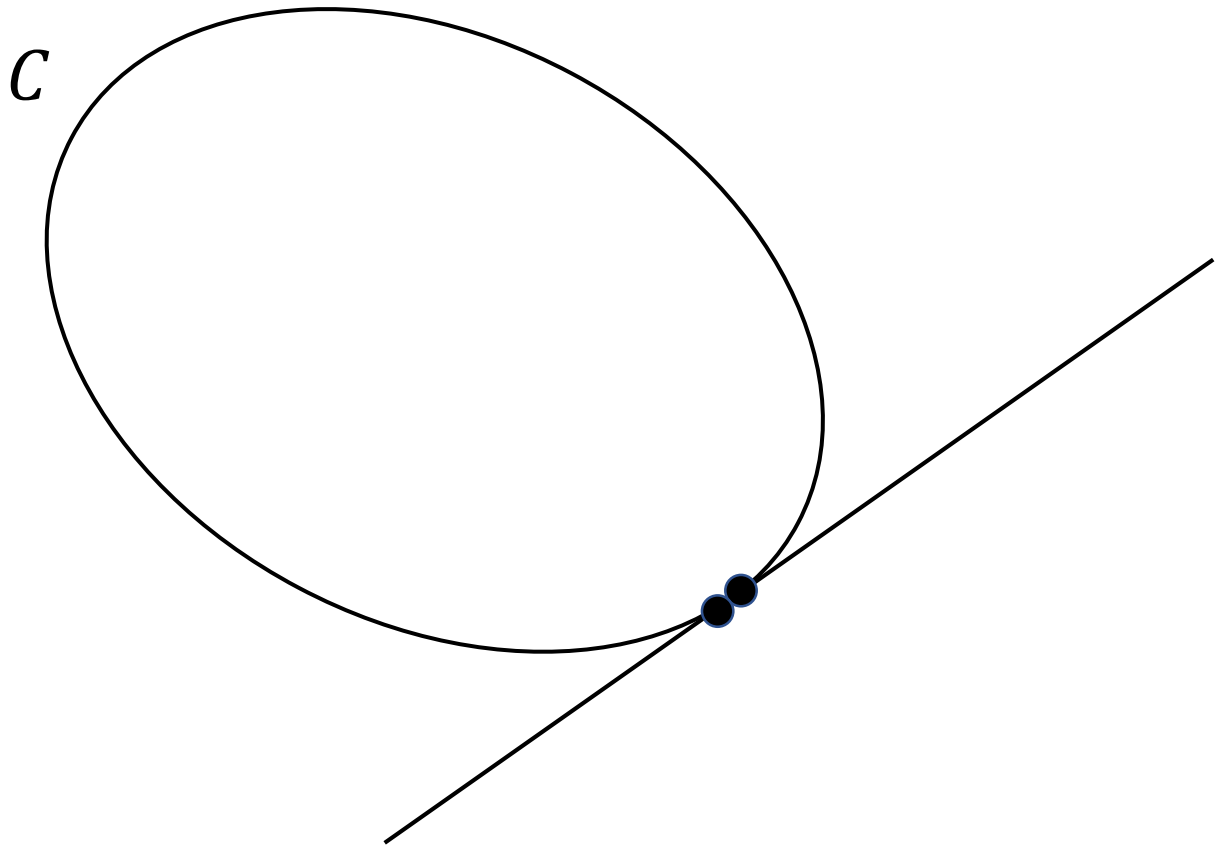
Line – conic intersection: two real distinct solutions

.



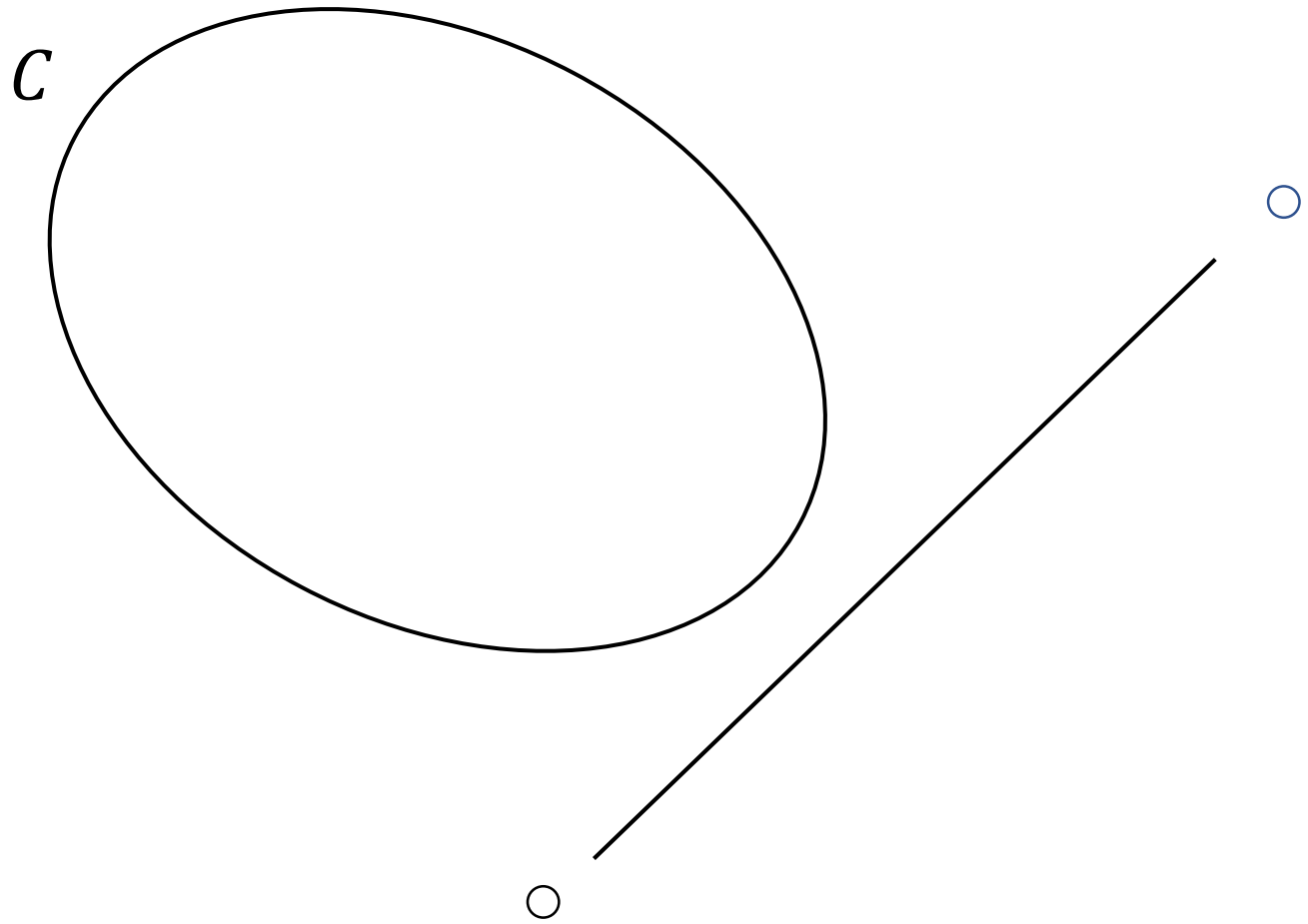
Line – conic intersection:
two real coincident solutions → **tangency**

.



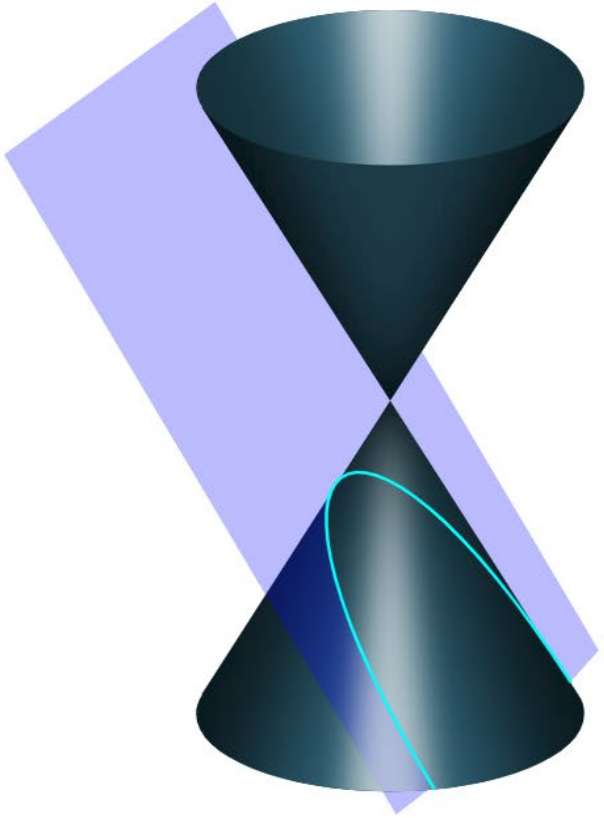
Line – conic intersection:
two complex-conjugate solutions

.



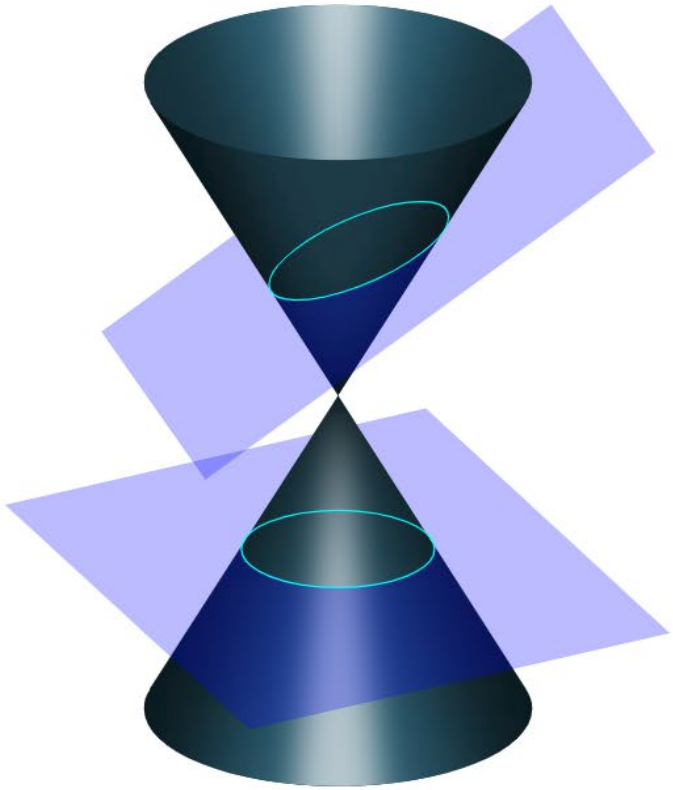
INTERSECTION between the **LINE AT THE ∞** and a CONIC

PARABOLAE



two coincident solutions: point at the ∞ along axis

ELLIPSE



two complex-conjugate solutions: no real solution

HYPERBOLAE



two real distinct solutions: asymptotes

The circular points

- useful in 2D reconstruction

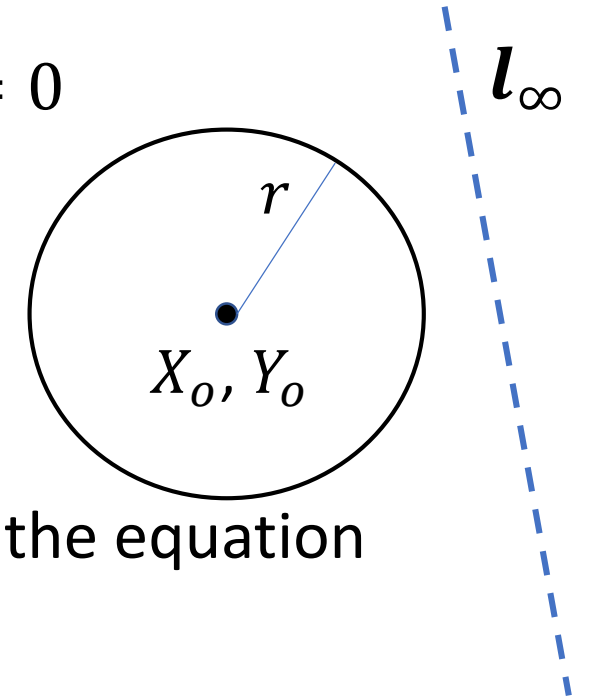
The circular points:

intersecting a circumference and the line at the ∞

$$\begin{cases} x^2 - 2X_o w + X_o^2 w^2 + y^2 - 2Y_o w + Y_o^2 w^2 - r^2 w^2 = 0 \\ w = 0 \end{cases}$$

→

$$\begin{cases} x^2 + y^2 = 0 \\ w = 0 \end{cases}$$



The circumference parameters (center and radius) disappear from the equation

→

the two intersection points are the **same for all** circumferences:

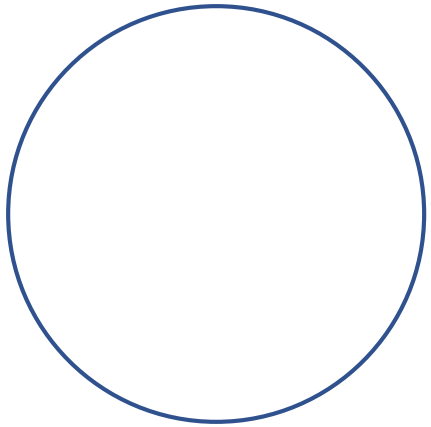
$$\boldsymbol{I} = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} \text{ and } \boldsymbol{J} = \begin{bmatrix} 1 \\ -i \\ 0 \end{bmatrix} \quad (i \text{ is the imaginary unit number})$$

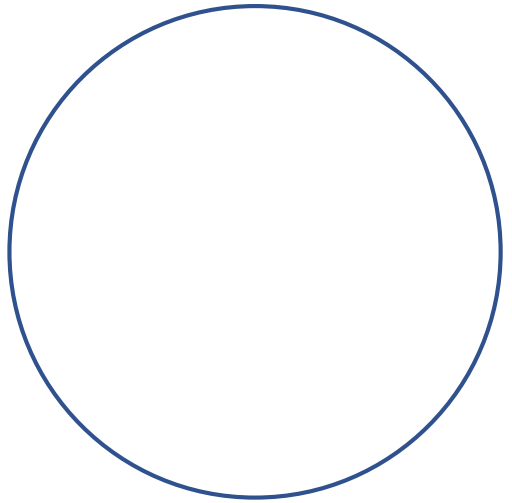
These points deserve a special name: the **CIRCULAR POINTS**

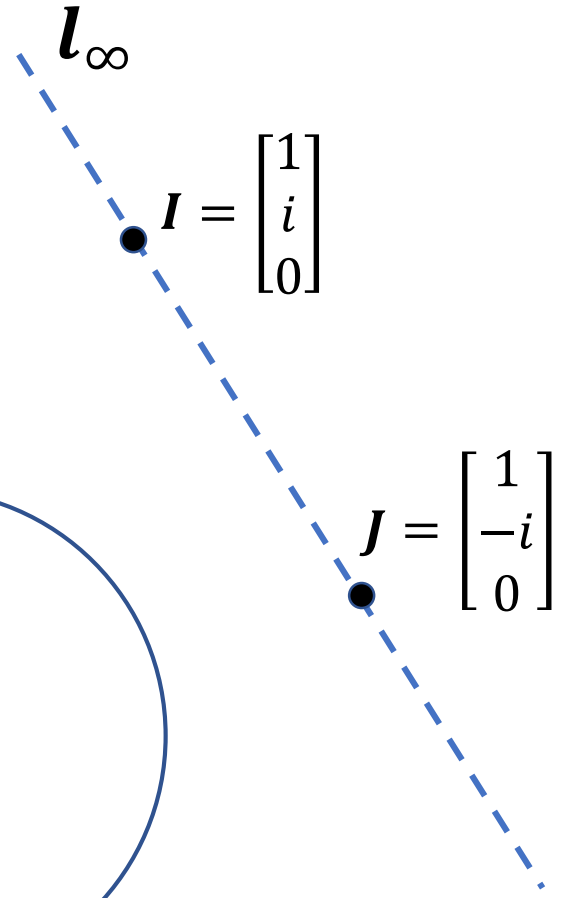
The circular points

Remember their coordinates

$$\boldsymbol{I} = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} \text{ and } \boldsymbol{J} = \begin{bmatrix} 1 \\ -i \\ 0 \end{bmatrix}$$







All circumferences cross the line $\boldsymbol{l}_\infty$ at the same two points $\boldsymbol{I}$ and $\boldsymbol{J}$

All the circumferences contain the two circular points $\boldsymbol{I}$ and $\boldsymbol{J}$

→ **The intersection of any two circumferences contains points $\boldsymbol{I}$ and $\boldsymbol{J}$**

The line tangent to a conic

Consider a point $\mathbf{y}$ on a conic $\boldsymbol{C}$: $\mathbf{y}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{y} = \mathbf{0}$, and consider the line

$\boldsymbol{l} = \boldsymbol{C}\,\mathbf{y}$: the line $\boldsymbol{l}$ goes through $\mathbf{y}$ since $\boldsymbol{l}^{\boldsymbol{T}}\mathbf{y} = \mathbf{y}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{y} = \mathbf{0}$

Find the intersections between line $\boldsymbol{l} = \boldsymbol{C}\,\mathbf{y}$ and the conic C

by solving the system $\begin{cases} \boldsymbol{l}^{\boldsymbol{T}}\mathbf{x} = \mathbf{y}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{x} = \mathbf{0} = \mathbf{x}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{y} \\ \mathbf{x}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{x} = \mathbf{0} \end{cases}$ for $\mathbf{x}$

From the two equalities $\mathbf{y}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{y} = \mathbf{0}$ and $\mathbf{y}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{x} = \mathbf{0}$

we derive $\boldsymbol{C}\,\mathbf{y} = \mathbf{x}$ x $\mathbf{y}$

From the two equalities $\mathbf{x}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{y} = \mathbf{0}$ and $\mathbf{x}^{\boldsymbol{T}}\boldsymbol{C}\,\mathbf{x} = \mathbf{0}$

we derive $\boldsymbol{C}\,\mathbf{x} = \mathbf{x}$ x $\mathbf{y}$

Hence $\boldsymbol{C}\,\mathbf{x} = \boldsymbol{C}\,\mathbf{y} \rightarrow \mathbf{x} = \mathbf{y} \rightarrow$ there is only one solution for $\mathbf{x}$: $\mathbf{x} = \mathbf{y}$

The two solutions are therefore coincident, and thus the **line** $\boldsymbol{l} = \boldsymbol{C}\,\mathbf{y}$ through a point $\mathbf{y}$ on the conic $\boldsymbol{C}$ is **tangent to the conic** $\boldsymbol{C}$

Degenerate conics

Conics: a point $\mathbf{x}$ is on a conic $\boldsymbol{C}$ if it satisfies a homogeneous *quadratic* equation, namely $\mathbf{x}^T \boldsymbol{C}\, \mathbf{x} = \mathbf{0}$, where $\boldsymbol{C}$ is a 3x3 symmetric matrix.

Nondegenerate conics: matrix $\boldsymbol{C}$ is nonsingular, i.e. $|\boldsymbol{C}| \neq 0$, $\mathrm{rank}(\boldsymbol{C}) = 3$

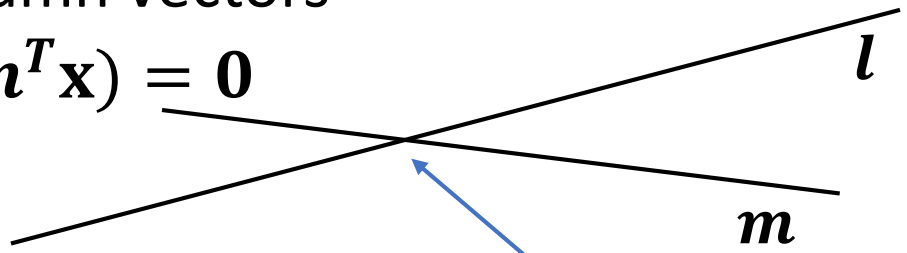
Degenerate conics: matrix $\boldsymbol{C}$ is singular, i.e. $|\boldsymbol{C}| = 0$, $\mathrm{rank}(\boldsymbol{C}) < 3$

two cases:

- rank $(C) = 2$, $C = \boldsymbol{l}\boldsymbol{m}^T + \boldsymbol{m}\boldsymbol{l}^T$ with $\boldsymbol{l}$ and $\boldsymbol{m}$ are column vectors

→ $\mathbf{x}^T \boldsymbol{C}\, \mathbf{x} = \mathbf{x}^T \left(\boldsymbol{l}\boldsymbol{m}^T + \boldsymbol{m}\boldsymbol{l}^T\right)\mathbf{x} = 2\mathbf{x}^T \boldsymbol{l}\boldsymbol{m}^T \mathbf{x} = 2(\mathbf{x}^T \boldsymbol{l})(\boldsymbol{m}^T \mathbf{x}) = \mathbf{0}$

$\mathbf{x}^T \boldsymbol{l} = \mathbf{0}$ OR $\boldsymbol{m}^T \mathbf{x} = \mathbf{0}$

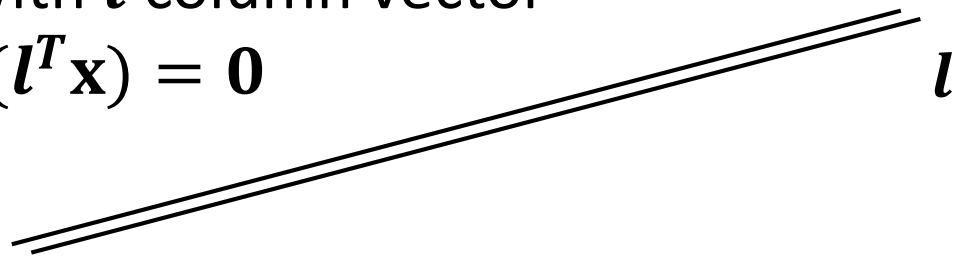


set of points on $\boldsymbol{l}$ or m

- rank $(C) = 1$, $C = \boldsymbol{l}\boldsymbol{l}^T$ with $\boldsymbol{l}$ column vector

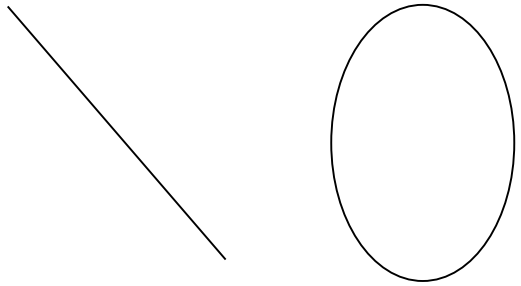
→ $\mathbf{x}^T \boldsymbol{C}\, \mathbf{x} = \mathbf{x}^T \boldsymbol{l}\boldsymbol{l}^T \mathbf{x} = (\mathbf{x}^T \boldsymbol{l})(\boldsymbol{l}^T \mathbf{x}) = \mathbf{0}$

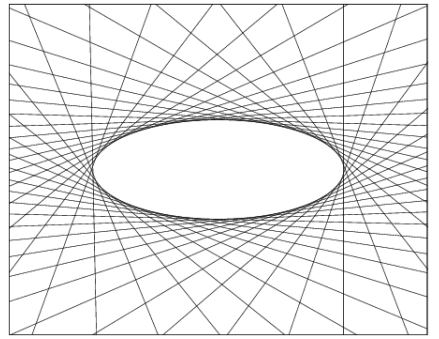
$\mathbf{x}^T \boldsymbol{l} = \mathbf{0}$ counted 2 times



Planar Projective Geometry

- **Elements**
- Points
- Lines
- Conics
- **Dual conics**

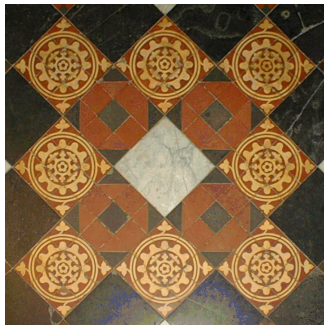




- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities







DUAL CONICS

Conics: a conic is a set of points $\mathbf{x}$ that satisfy equation

$$\mathbf{x}^T \boldsymbol{C}\, \mathbf{x} = \mathbf{0}$$

where $\boldsymbol{C}$ is a 3x3 symmetric matrix.

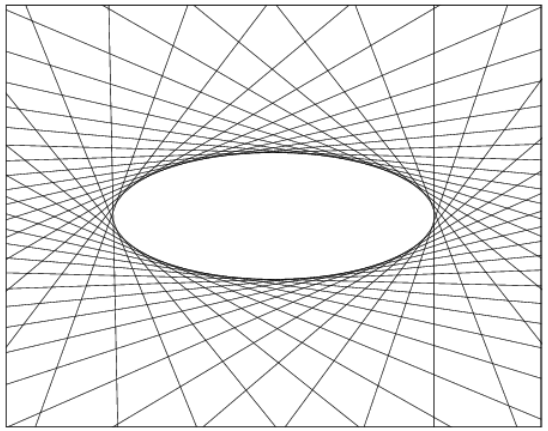
its dual:

Dual conics: a *dual conic* is a set of *lines* $\boldsymbol{l}$ that satisfy equation

$$\boldsymbol{l}^T \boldsymbol{C}^* \boldsymbol{l} = \mathbf{0}$$

where $\boldsymbol{C}^*$ is a 3x3 symmetric matrix.

Nondegenerate dual conics: a nondegenerate dual conic is a dual conic whose matrix $\boldsymbol{C}^*$ is NONSINGULAR, i.e., . $|\boldsymbol{C}^*| \neq 0$, $\text{rank}(\boldsymbol{C}^*) = 3$



A nondegenerate dual conic $\boldsymbol{C}^*$ is the set of lines TANGENT to a nondegenerate conic $\boldsymbol{C}$ with

$$\boldsymbol{C}^* = \boldsymbol{C}^{-1}$$

Degenerate dual conics

Dual Conics: a line $\boldsymbol{l}$ is on a conic $\boldsymbol{C}^*$ if it satisfies a homogeneous *quadratic* equation, namely $\boldsymbol{l}^T\,\boldsymbol{C}^*\boldsymbol{l} = \boldsymbol{0}$, where $\boldsymbol{C}^*$ is a 3x3 symmetric matrix.

Nondegenerate dual conics: matrix $\boldsymbol{C}^*$ is nonsingular, i.e. $|\boldsymbol{C}^*| \neq 0$, $\text{rank}\,\boldsymbol{C}^* = 3$

Degenerate dual conics: matrix $\boldsymbol{C}^*$ is singular, i.e. $|\boldsymbol{C}^*| = 0$, $\text{rank}\,\boldsymbol{C}^* < 3$

two cases:

- rank $\boldsymbol{C}^* = 2$, $\boldsymbol{C}^* = \boldsymbol{pq}^T + \boldsymbol{qp}^T$ with $\boldsymbol{p}$ and $\boldsymbol{q}$ are column vectors

$\rightarrow \boldsymbol{l}^T\,\boldsymbol{C}^*\boldsymbol{l} = \boldsymbol{l}^T(\boldsymbol{pq}^T + \boldsymbol{qp}^T)\boldsymbol{l} = \boldsymbol{2l}^T\boldsymbol{pq}^T\boldsymbol{l} = \boldsymbol{2}(\boldsymbol{l}^T\boldsymbol{p})(\boldsymbol{q}^T\boldsymbol{l}) = \boldsymbol{0}$

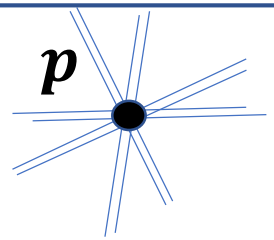
$\boldsymbol{l}^T\boldsymbol{p} = \boldsymbol{0}$ OR $\boldsymbol{q}^T\boldsymbol{l} = \boldsymbol{0}$



- rank $\boldsymbol{C}^* = 1$, $\boldsymbol{C}^* = \boldsymbol{pp}^T$ with $\boldsymbol{p}$ column vector

$\rightarrow \boldsymbol{l}^T\,\boldsymbol{C}^*\boldsymbol{l} = \boldsymbol{l}^T\boldsymbol{pp}^T\boldsymbol{l} = (\boldsymbol{l}^T\boldsymbol{p})(\boldsymbol{p}^T\boldsymbol{l}) = \boldsymbol{0}$

$\boldsymbol{l}^T\boldsymbol{p} = \boldsymbol{0}$ counted 2 times



set of lines through $\boldsymbol{p}$ or $\boldsymbol{q}$

A special (degenerate) dual conic:

The (degenerate) conic-dual to the circular points

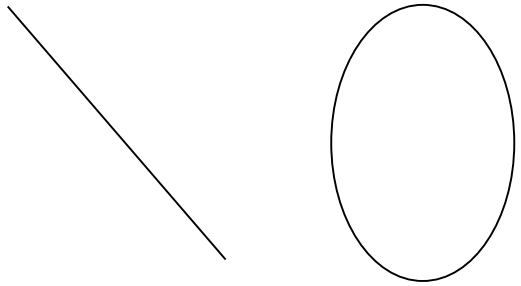
what is the degenerate dual conic $\boldsymbol{C^* = pq^T + qp^T}$
when points $\boldsymbol{p}$ and $\boldsymbol{q}$ are the circular points $\boldsymbol{p = I}$ and $\boldsymbol{q = J}$?

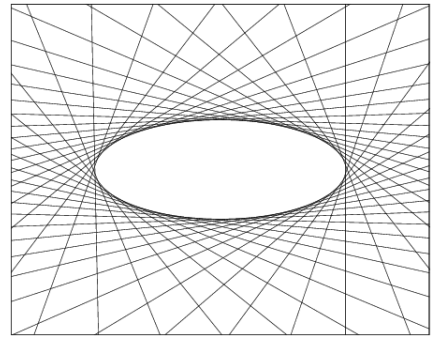
$$\boldsymbol{C_\infty^* = IJ^T + JI^T} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

$\boldsymbol{C_\infty^*}$ is called the conic dual to the circular points: set of all lines going through $\boldsymbol{I}$ or $\boldsymbol{J}$
it will be useful in the 2D reconstruction of planar scenes
(or, **image rectification**)

Planar Projective Geometry

- **Elements**
- Points
- Lines
- Conics
- Dual conics

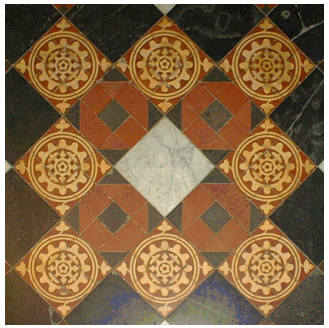




- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities







PROJECTIVE MAPPINGS

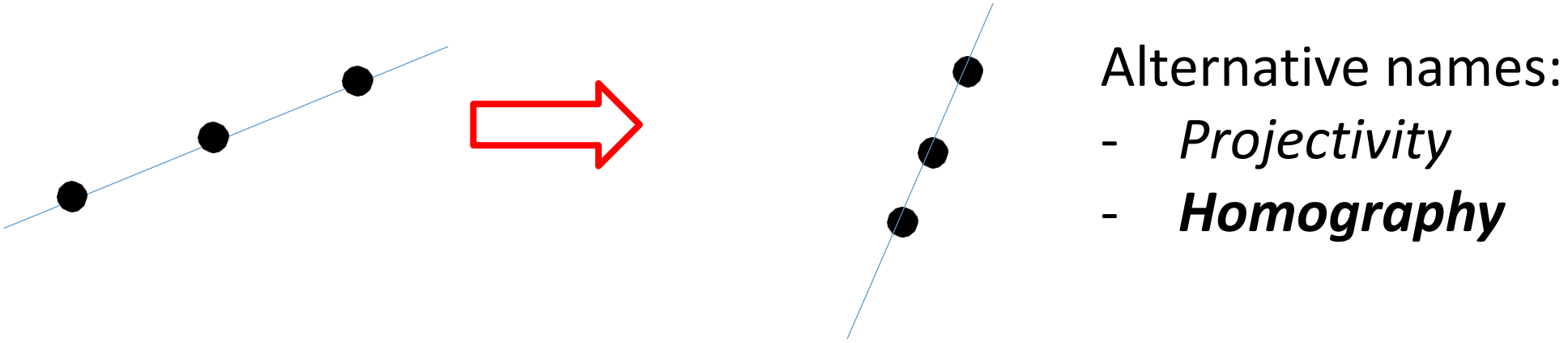
Projective mappings

Def. *A **projective mapping** between a projective plane* $\mathbb{P}^2$ *and an other projective plane* $\mathbb{P}'^2$ *is an **invertible** mapping which **preserves colinearity**:*

$$h\colon \mathbb{P}^2 \rightarrow \mathbb{P}'^2,\ \mathbf{x}' = h(\mathbf{x}),\ \mathbf{x}_1,\ \mathbf{x}_2, \mathbf{x}_3 \text{ are colinear}$$

$$\leftrightarrow$$

$$\mathbf{x}'_1 = h(\mathbf{x}_1),\ \mathbf{x}'_2 = h(\mathbf{x}_2), \mathbf{x}'_3 = h(\mathbf{x}_3) \text{ are colinear}$$



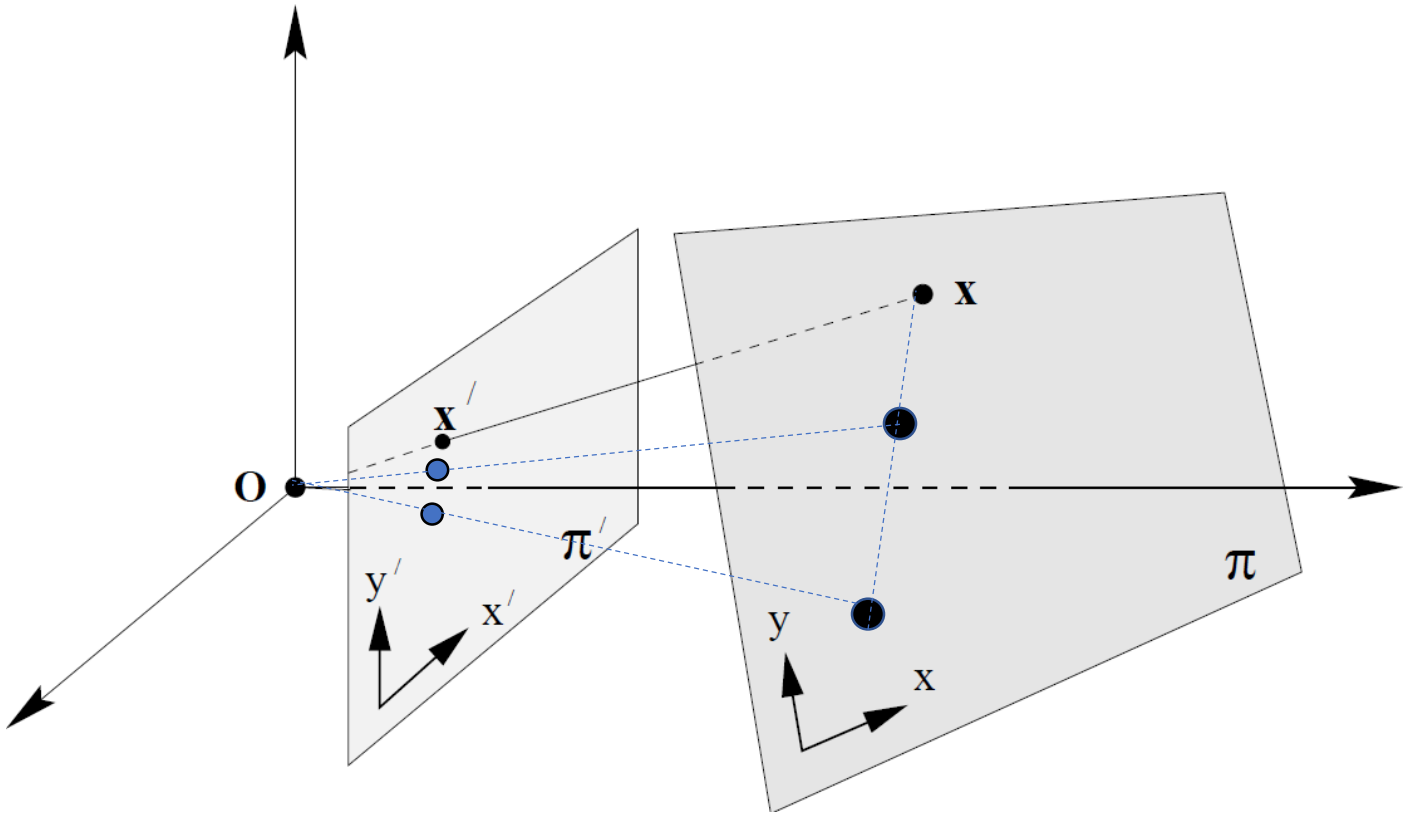
Alternative names:

- *Projectivity*
- ***Homography***

Examples of projective mappings

Mappings between two planes induced by central projection are projective, since they preserve colinearity →

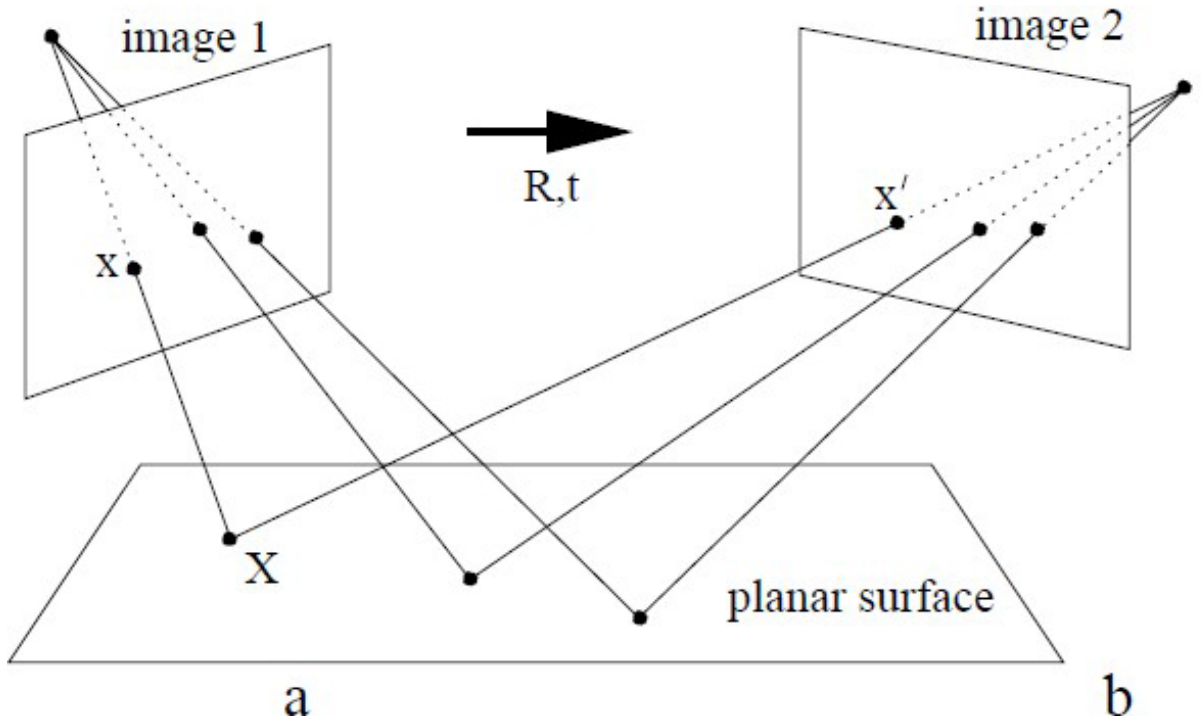
points on a **plane** are **projectively** mapped to their **images** and viceversa



Examples of projective mappings

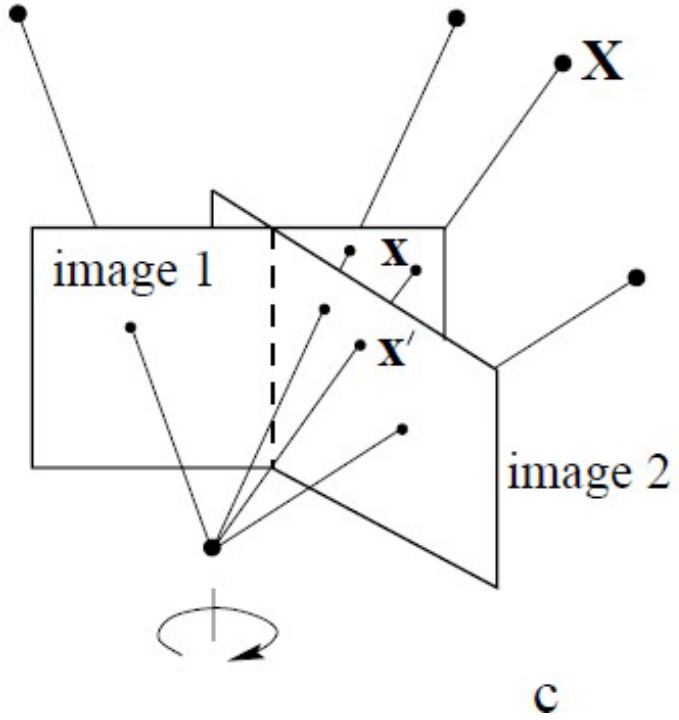
Mapping between two images of a planar scene is a homography

a composition of the 1° image → scene homography and the scene → 2° image homography ⇨ it is a **homography**



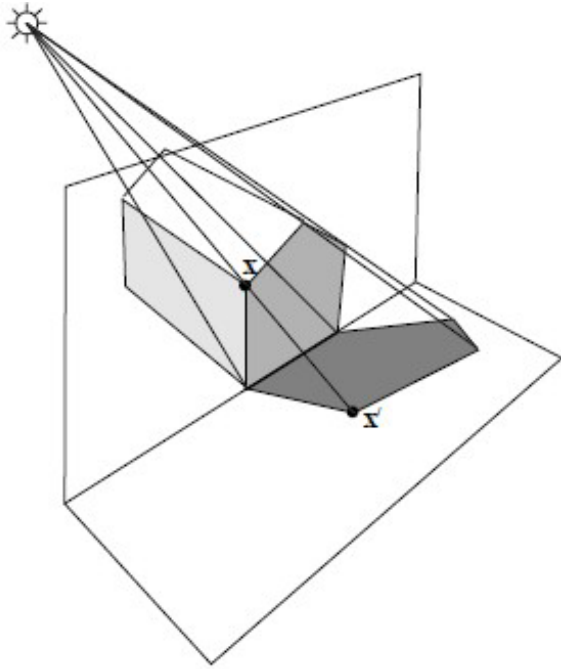
Examples of projective mappings

Two images of a 3D scene, taken by a camera rotating about its center are related by a **homography**, since the 2° image can be regarded as a central projection of the 1° image



Examples of projective mappings

The shadow cast by a **planar** silhouette onto a **ground plane** is a projective transformation of the planar silhouette, since they are related by a central projection



Fundamental Theorem of Projective Geometry

Theorem: *A mapping* $h : \mathbb{P}^2 \rightarrow \mathbb{P}'^2$ *is projective if and only if there exists an invertible 3x3 matrix* H *such that for any point in* $\mathbb{P}^2$ *represented by the vector* $\mathbf{x}$ *, is* $h(\mathbf{x}) = H\,\mathbf{x}$

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

i.e. projective mappings are **linear** in the homogeneous coordinates

(they are not linear in cartesian coordinates)

Homography: 8 degrees of freedom

From the theorem

$$h(\mathbf{x}) = \mathbf{x}' = H\,\mathbf{x}$$

Therefore, if we multiply the matrix H by any nonzero scalar λ, the relation is satisfied by the same points

$$\mathbf{x}' = \lambda H\,\mathbf{x}$$

Thus any nonzero multiple of the matrix H represents the same projective mapping as H.

Hence H is a homogeneous matrix: in spite of its 9 entries,

H has only 8 degrees of freedom, namely the ratios between its elements.

Homography estimation from point correspondences

H has only 8 degrees of freedom, namely the ratios between its elements, e.g.

$$H = \begin{bmatrix} A_{11} & A_{12} & t_1 \\ A_{21} & A_{22} & t_2 \\ v_1 & v_2 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{h_1^T} \\ \mathbf{h_2^T} \\ \mathbf{h_3^T} \end{bmatrix}$$

Therefore, H can be estimated by just FOUR point correspondences,

i.e. four known points in known positions on a plane and their images,

since each point correspondence $\mathbf{x}_i' = H\mathbf{x}_i$ yelds **two** independent equations

DUE to homogeneity, $\mathbf{x}_i' = H\mathbf{x}_i$ actually means that $\mathbf{x}_i' = \lambda H\mathbf{x}_i$ **,**

i.e. the vectors $\mathbf{x}_i'$ and $H\mathbf{x}_i$ are colinear → their vector product $\mathbf{x}_i' \times H\mathbf{x}_i = 0$

Homography estimation by DLT

$\mathbf{x}_i' \times H\mathbf{x}_i = 0$ with $H = \begin{bmatrix} \mathbf{h}_1^{\mathrm{T}} \\ \mathbf{h}_2^{\mathrm{T}} \\ \mathbf{h}_3^{\mathrm{T}} \end{bmatrix}$ becomes, after some manipulations,

$$\begin{bmatrix} \mathbf{0}^T & -w'_i\mathbf{x}_i^{\mathrm{T}} & y'_i\mathbf{x}_i^{\mathrm{T}} \\ w'_i\mathbf{x}_i^{\mathrm{T}} & \mathbf{0}^T & -x'_i\mathbf{x}_i^{\mathrm{T}} \\ -y'_i\mathbf{x}_i^{\mathrm{T}} & x'_i\mathbf{x}_i^{\mathrm{T}} & \mathbf{0}^T \end{bmatrix} \begin{bmatrix} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{bmatrix} = 0$$ but the 3° row linearly depends on the 1° and 2°

Therefore, with FOUR point correspondences, 8 rows are obtained →

$\boldsymbol{A}_{8x9}\mathbf{h} = 0$ where $\mathbf{h} \triangleq \begin{bmatrix} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{bmatrix}$ is a nine-component vector, while $\boldsymbol{A}_{8x9}$ is 8 x 9.

Hence, the solution is $\mathbf{h} = \mathbf{RNS}(\mathbf{A})$ (the trivial solution is discarded)

Homography estimation by DLT

If more than 4 point correspondences are used, then an overdetermined system is obtained, $\boldsymbol{A}_{2nx9}\mathbf{h} = 0$, which doesn't admit a solution in general. Hence the algebraic error is minimized constrained by unit norm of $\mathbf{h}$:

$$\mathbf{h} = \underset{\mathbf{h}}{\operatorname{argmin}} \|\mathbf{A}\mathbf{h}\|_2 \text{ such that } \|\mathbf{h}\|_2 = 1$$

The solution for $\mathbf{h}$ is the eigenvector of $\mathbf{A}^{\boldsymbol{T}}\mathbf{A}$ associated to its minimum eigenvalue.

It can be obtained by the Singular Value Decomposition (SVD) of matrix $\mathbf{A}$:

The output of SVD($\mathbf{A}$) is the product $\mathrm{U}\ \mathrm{D}\ V^T$ where D is diagonal, U and V are orthogonal matrixes.

If the entries of D are arranged in decreasing order down the diagonal, then the solution $\mathbf{h}$ is the last column of V

Projective mapping of points, lines, conics, dual conics

A homography transforms **each point $\mathbf{x}$** into a point $\mathbf{x}'$ such that:

$$\mathbf{x} \rightarrow H\mathbf{x} = \mathbf{x}'$$

A homography transforms **each line $\boldsymbol{l}$** into a line $\boldsymbol{l}'$ such that:

$$\boldsymbol{l} \rightarrow H^{-T}\boldsymbol{l} = \boldsymbol{l}'$$

A homography transforms **each conic C** into a conic C' such that:

$$C \rightarrow H^{-T}CH^{-1} = C'$$

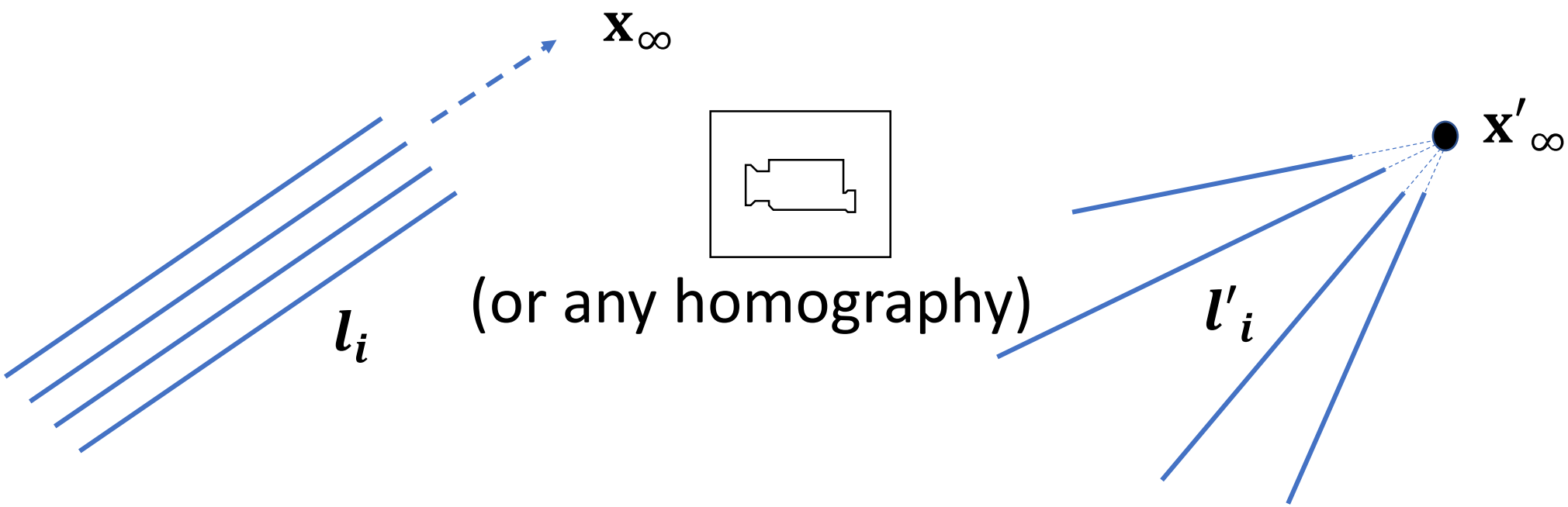
A homography transforms **each dual conic C^*** into a dual conic $C^{*\prime}$

$$C^* \rightarrow HC^*H^T = C^{*\prime}$$

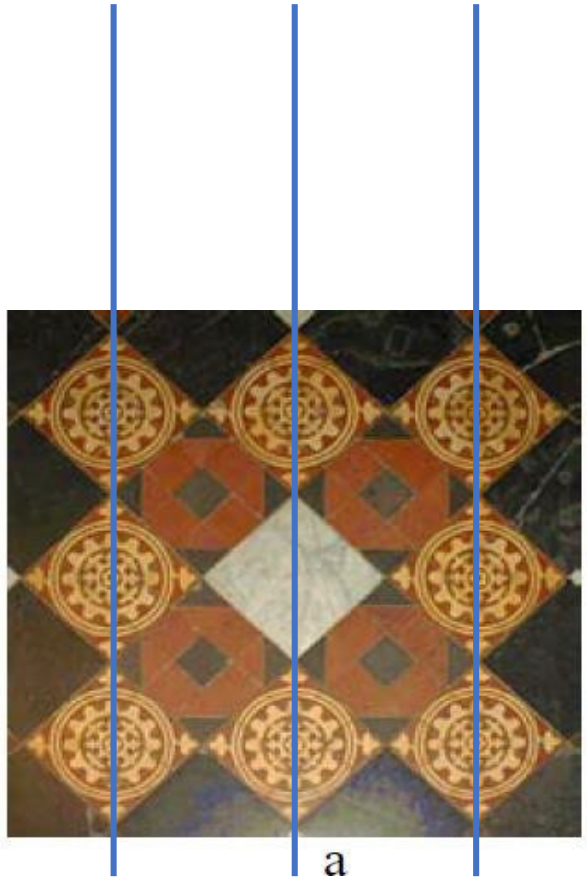
Vanishing points and the vanishing line

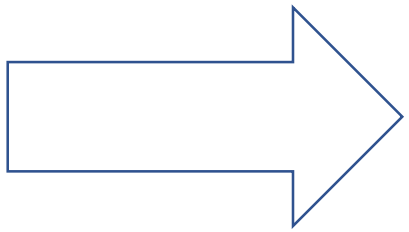
Vanishing point

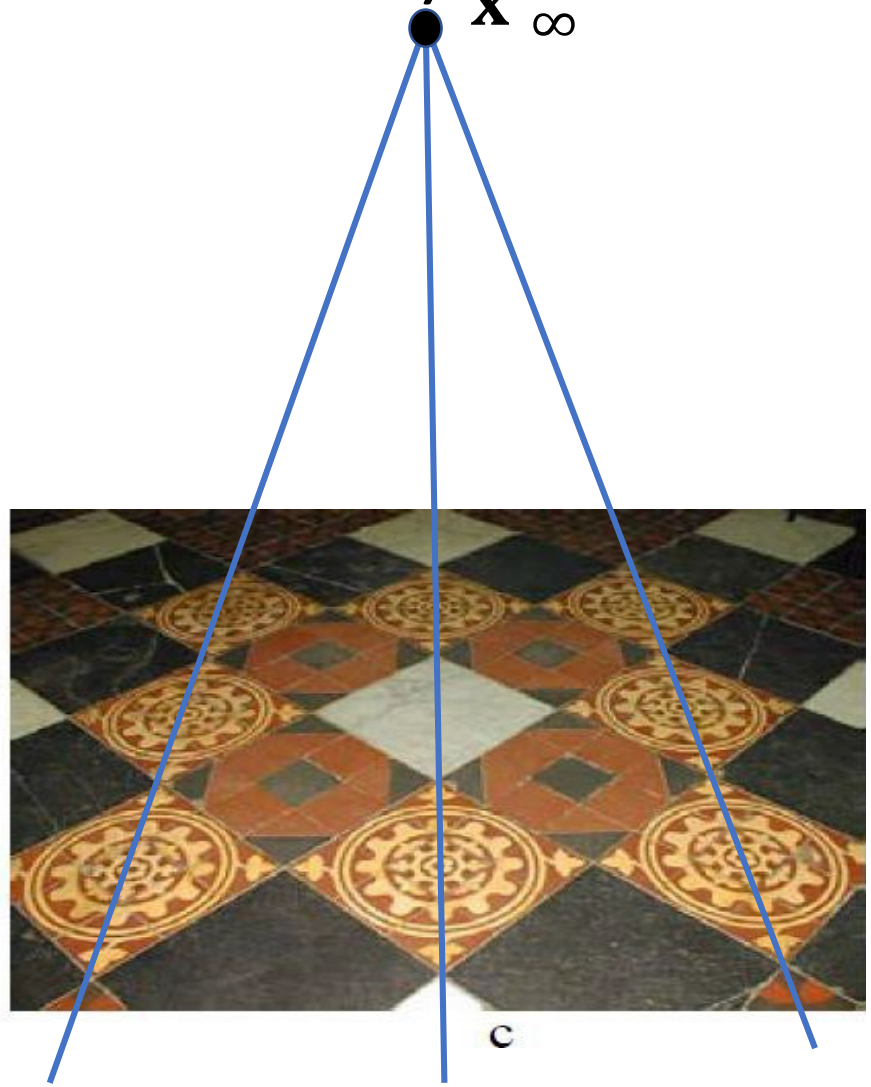
Theorem: *the image of a set of parallel lines* $\boldsymbol{l_i}$ *is a set of lines* $\boldsymbol{l'_i}$ ***concurrent*** *at a common point* $\boldsymbol{x'}$ *called the vanishing point of the direction of lines* $\boldsymbol{l_i}$



Vanishing point: image of a point at the ∞,
(where images of parallel lines concur)

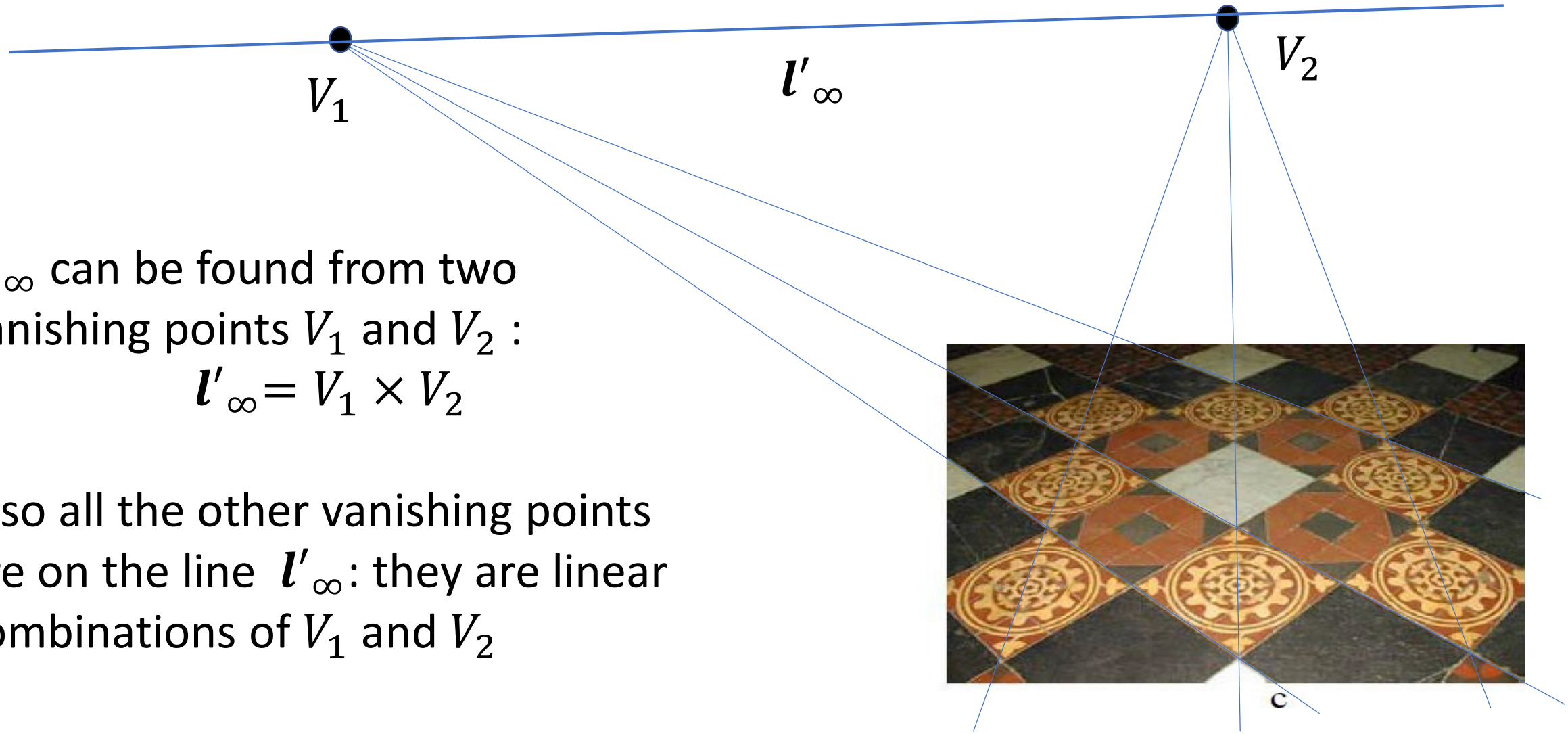






Vanishing <u>line</u> l'_∞ (or horizon) :

image of the set of the points at the ∞ = the image of l_∞

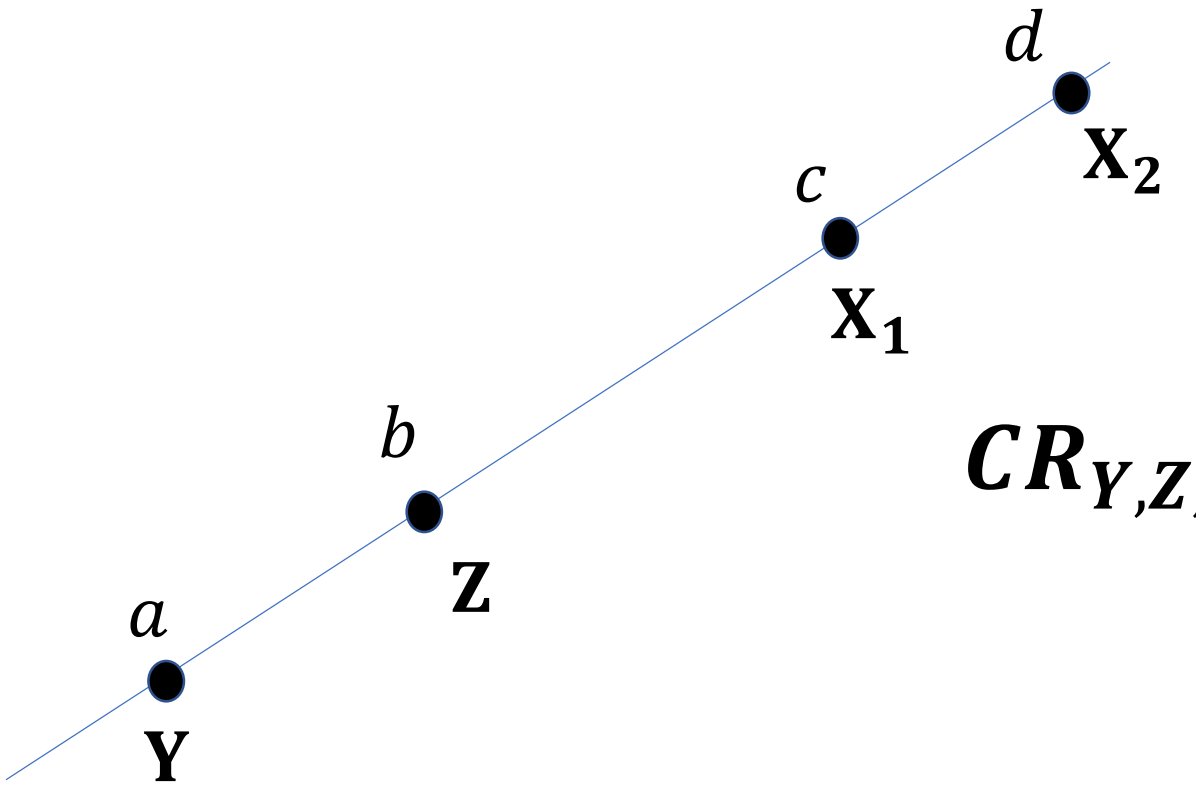


l'_∞ can be found from two vanishing points V_1 and V_2 :

$$l'_\infty = V_1 \times V_2$$

Also all the other vanishing points are on the line l'_∞: they are linear combinations of V_1 and V_2

Cross ratio: a projective invariant



$$CR_{Y,Z,X_1,X_2} = \frac{c-a}{c-b} \Big/ \frac{d-a}{d-b}$$

The Cross Ratio is invariant
under projective mappings

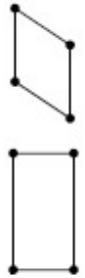
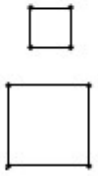
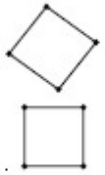


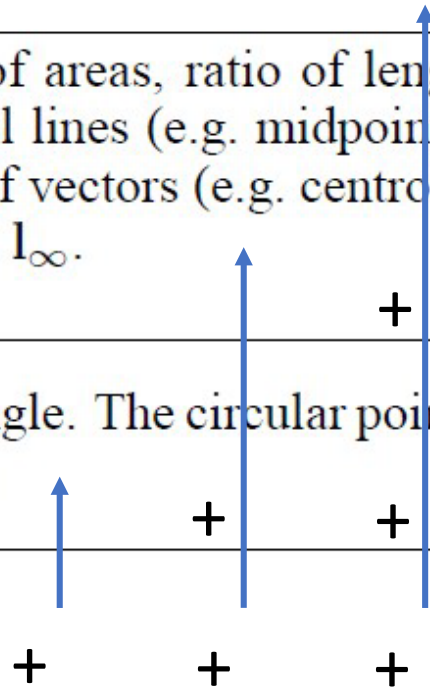


$$CR_{Y,Z,X_1,X_2} = CR_{Y',Z',X'_1,X'_2}$$

Hierarchy of projective transformations

Hierarchy of projective transformations

Group	Matrix	Distortion	Invariant properties
Projective 8 dof	$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$		Concurrency, collinearity, **order of contact**: intersection (1 pt contact); tangency (2 pt contact); inflections (3 pt contact with line); tangent discontinuities and cusps. cross ratio (ratio of ratio of lengths).
Affine 6 dof	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Parallelism, ratio of areas, ratio of lengths on collinear or parallel lines (e.g. midpoints), linear combinations of vectors (e.g. centroids). The line at infinity, $\mathbf{l}_\infty$.
Similarity 4 dof	$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Ratio of lengths, angle. The circular points, $\mathbf{I}, \mathbf{J}$ (see section 2.7.3).
Euclidean 3 dof	$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Length, area



Isometries (or Euclidean mappings)

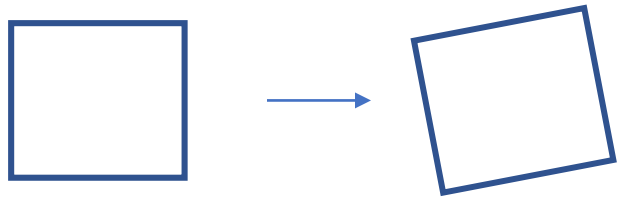
$$H_I = \begin{bmatrix} R_\perp & \boldsymbol{t} \\ 0 \quad 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\vartheta & -\sin\vartheta & t_x \\ \sin\vartheta & \cos\vartheta & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

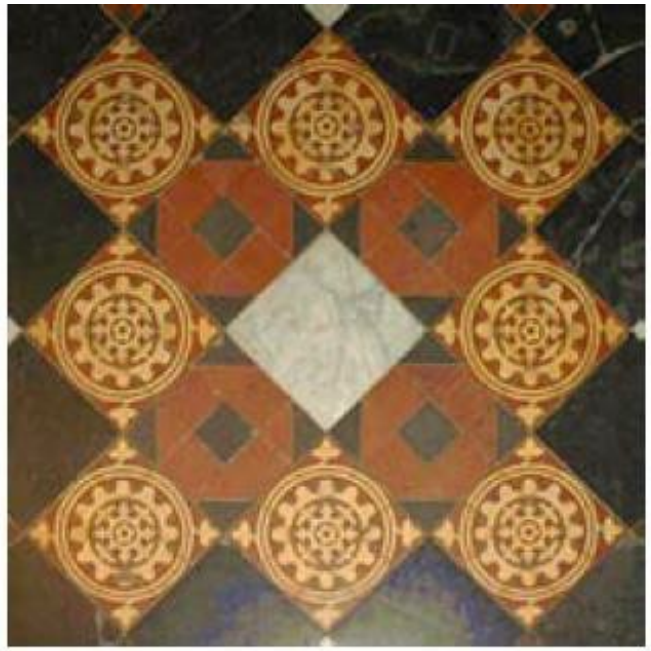
$R_\perp$ is an orthogonal matrix: ${R_\perp}^{-1} = {R_\perp}^{T}$

$\det {R_\perp}^{-1} = 1$ planar rigid displacement (-1 for reflection)

3 dofs: translation $\boldsymbol{t}$ + rotation angle ϑ

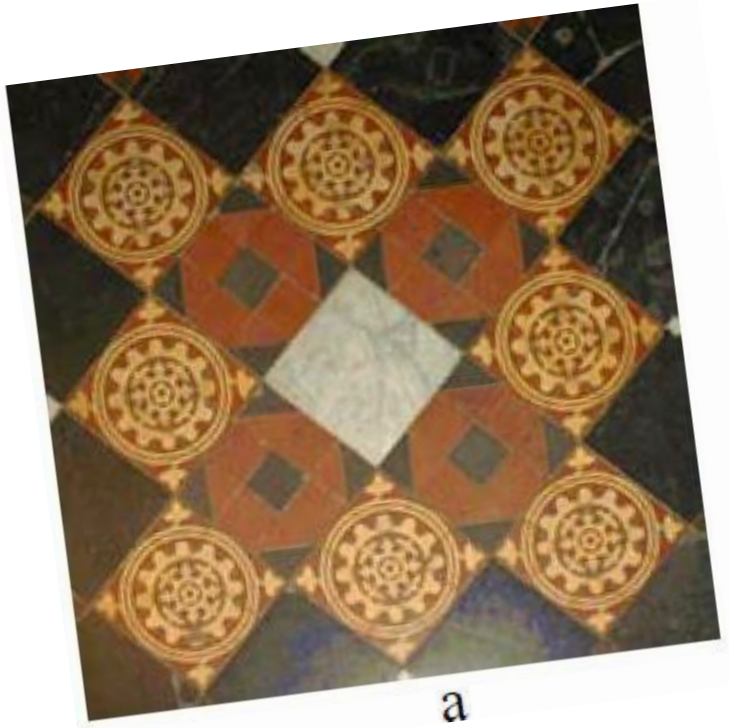
Invariants: lenghts, distances, areas → shape and size → relative positions





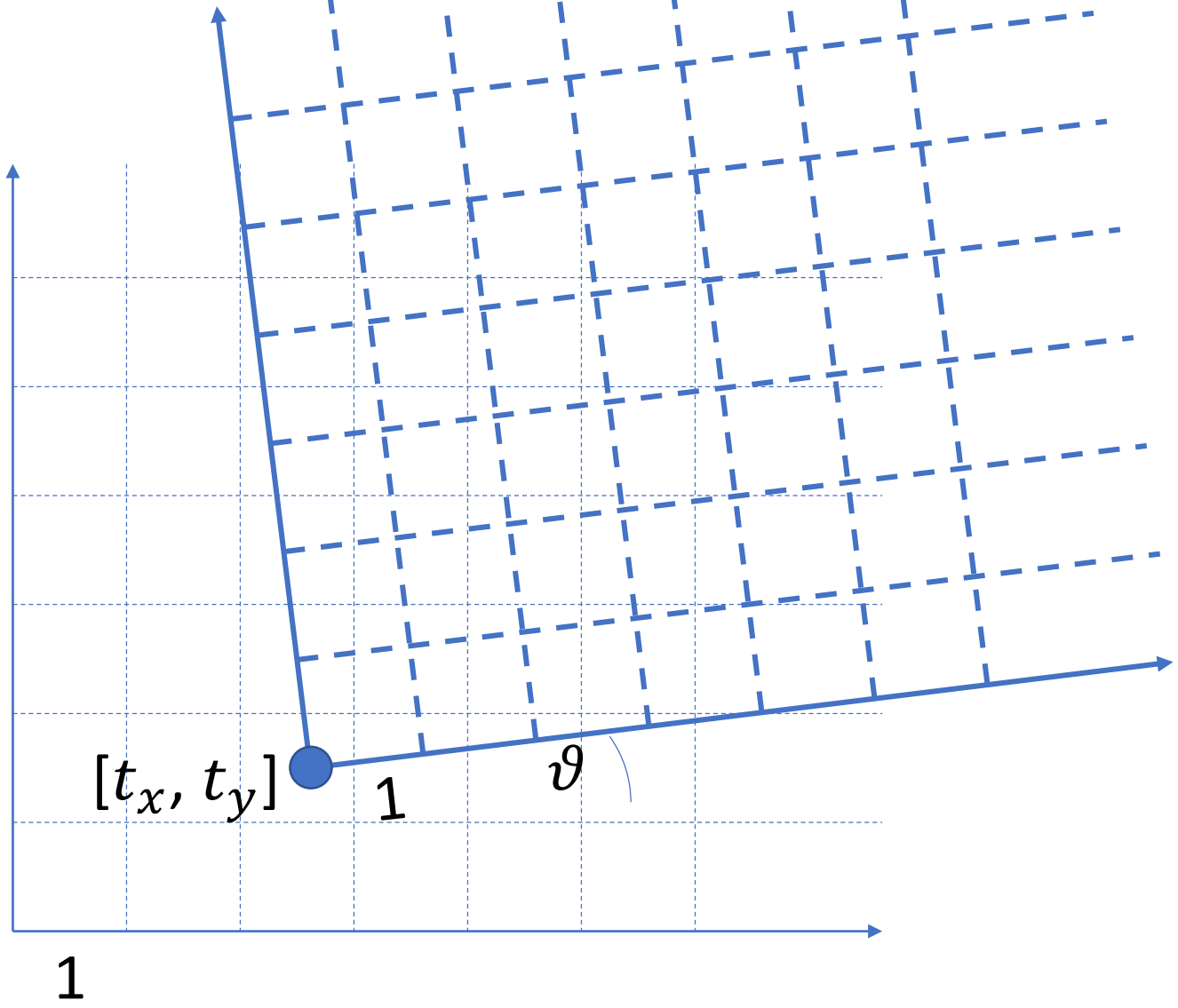
a

isometry



a

Isometry $H_I = \begin{bmatrix} \cos\vartheta & -\sin\vartheta & t_x \\ \sin\vartheta & \cos\vartheta & t_y \\ 0 & 0 & 1 \end{bmatrix}$: 3 degrees of freedom



Similarities

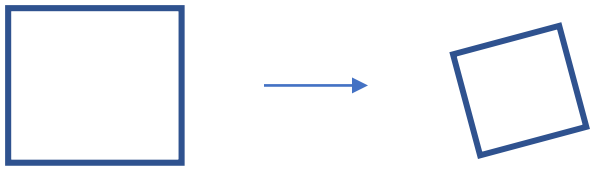
$$H_S = \begin{bmatrix} s\,R_\perp & \boldsymbol{t} \\ 0 \quad 0 & 1 \end{bmatrix} = \begin{bmatrix} s\cos\vartheta & -s\sin\vartheta & t_x \\ s\sin\vartheta & s\cos\vartheta & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

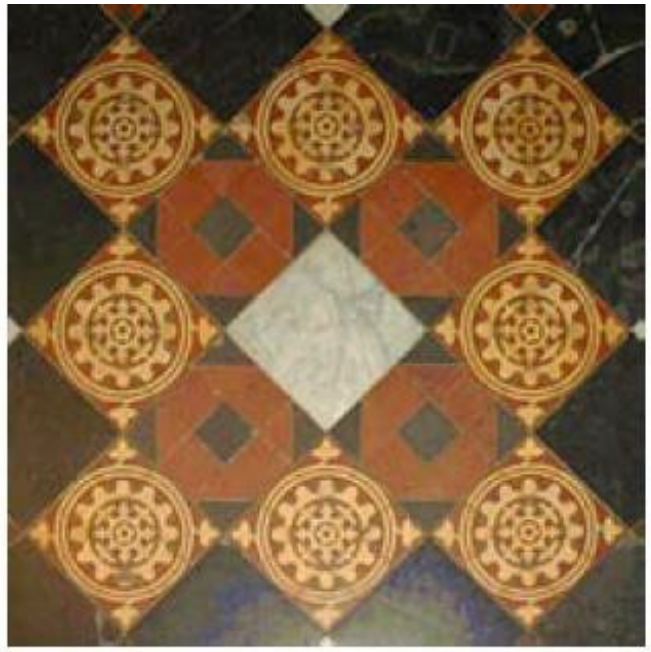
$R_\perp$ is an orthogonal matrix: ${R_\perp}^{-1} = {R_\perp}^{T}$

4 dofs: rigid displacement + *scale*

Invariants: ratio of lenghts, angles → shape (not size)

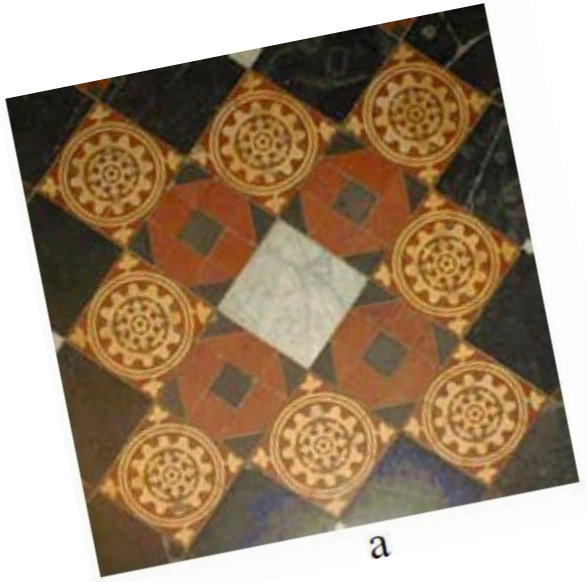
the circular points $\boldsymbol{I} = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$ and $\boldsymbol{J} = \begin{bmatrix} 1 \\ -i \\ 0 \end{bmatrix}$





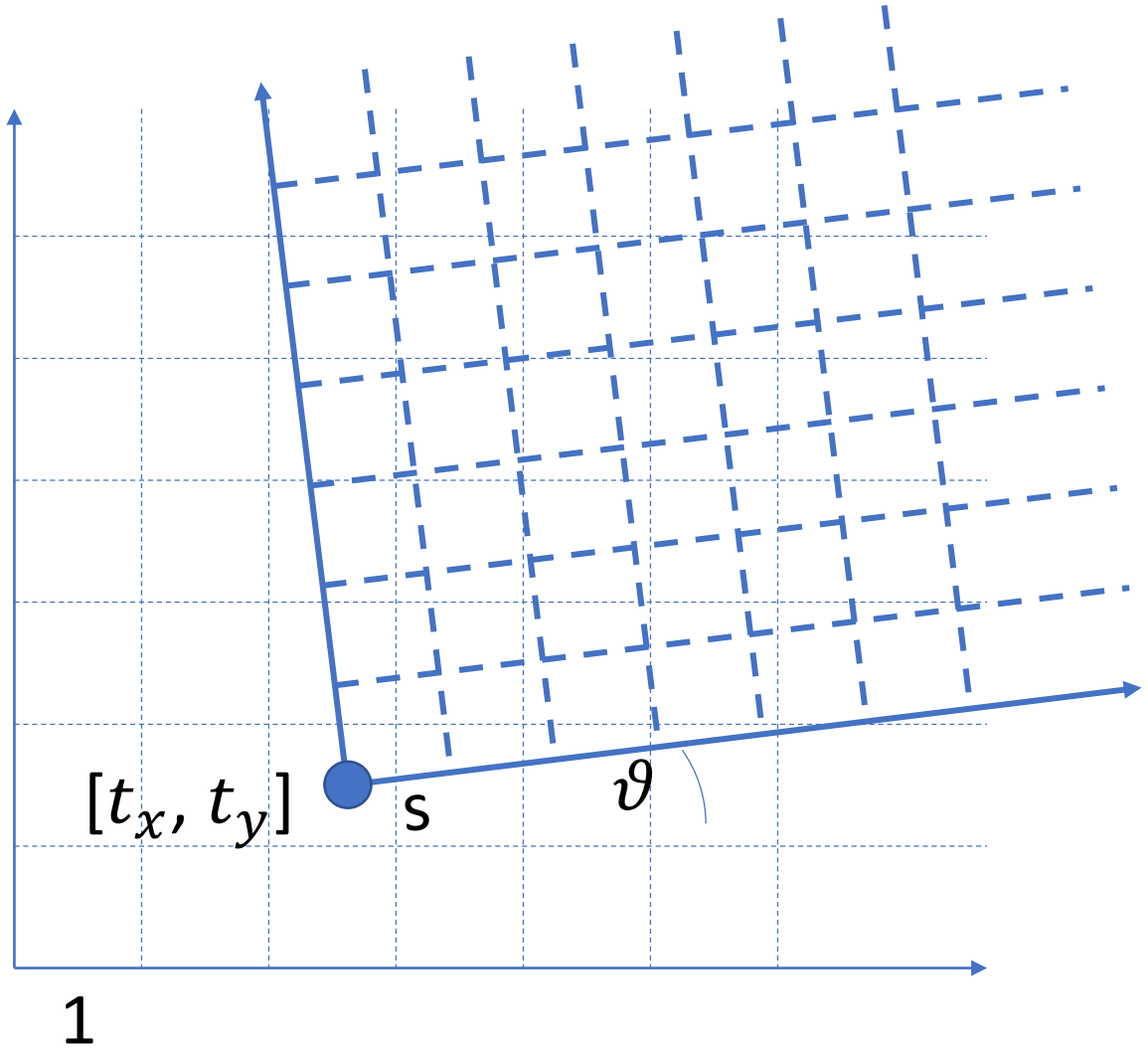
a

similarity



a

Similarity $H_S = \begin{bmatrix} s\cos\vartheta & -s\sin\vartheta & t_x \\ s\sin\vartheta & s\cos\vartheta & t_y \\ 0 & 0 & 1 \end{bmatrix}$: 4 degrees of freedom



Affinities (or affine mappings)

$$H_A = \begin{bmatrix} A & \boldsymbol{t} \\ 0 \quad 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

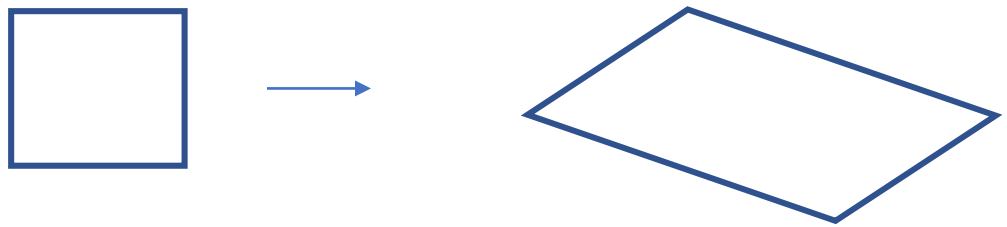
A is any 2x2 rank-2 matrix

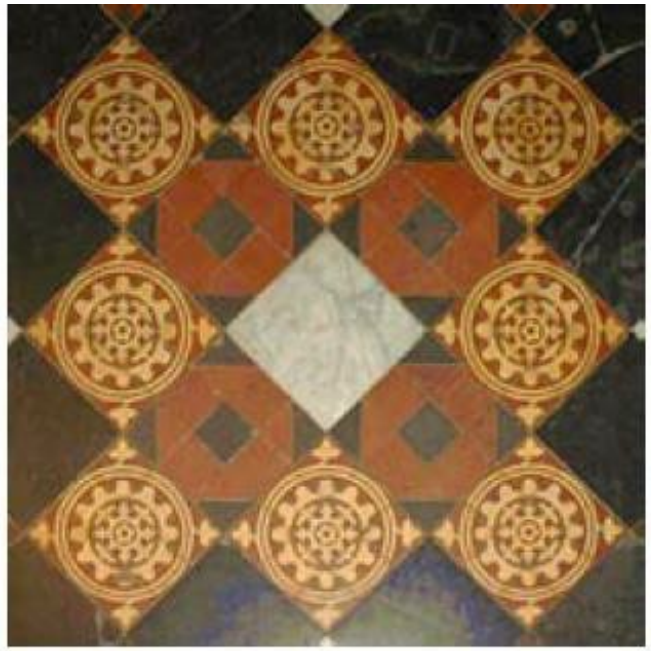
6 dofs: A + $\boldsymbol{t}$

Invariants: parallelism, ratio of parallel lenghts, ratio of areas

the line at the infinity $\boldsymbol{l}_\infty = [\mathbf{0} \quad \mathbf{0} \quad \mathbf{1}]^T$

OBSERVATION: affine mappings are LINEAR in the CARTESIAN COORDINATES





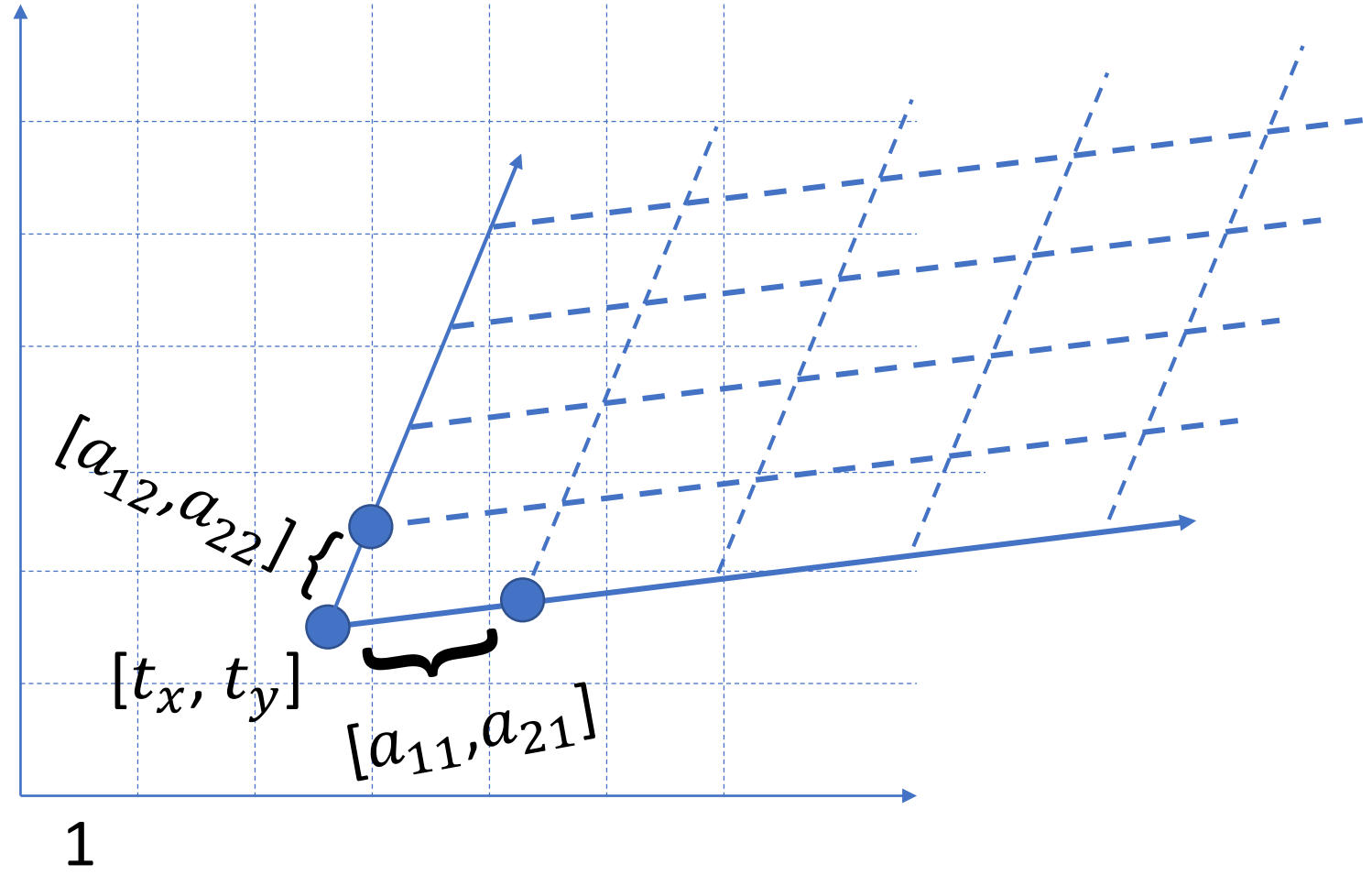
a

affinity



b

Affinity $H_A = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$: 6 degrees of freedom



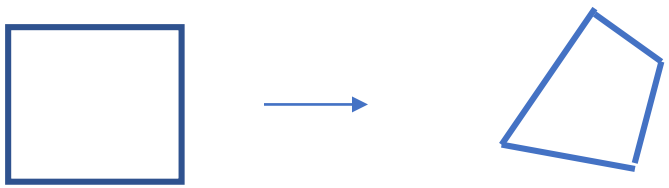
Projectivities (or projective mappings, or **homographies**)

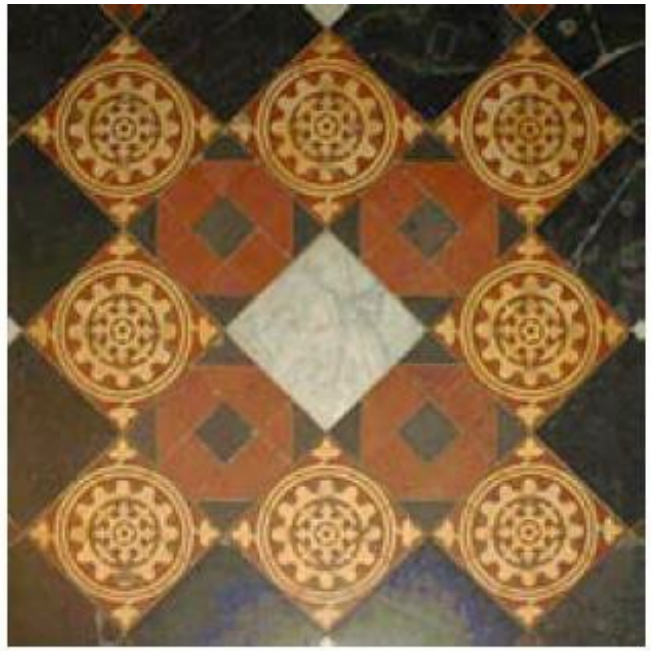
$$H = \begin{bmatrix} A & \boldsymbol{t} \\ \boldsymbol{v}^T & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ v_1 & v_2 & 1 \end{bmatrix}$$

A is any 2x2 rank-2 matrix

8 dofs: $A + \boldsymbol{v} + \boldsymbol{t}$

Invariants: colinearity, incidence, order of contact (crossing, tangency, inflections), the cross ratio





a

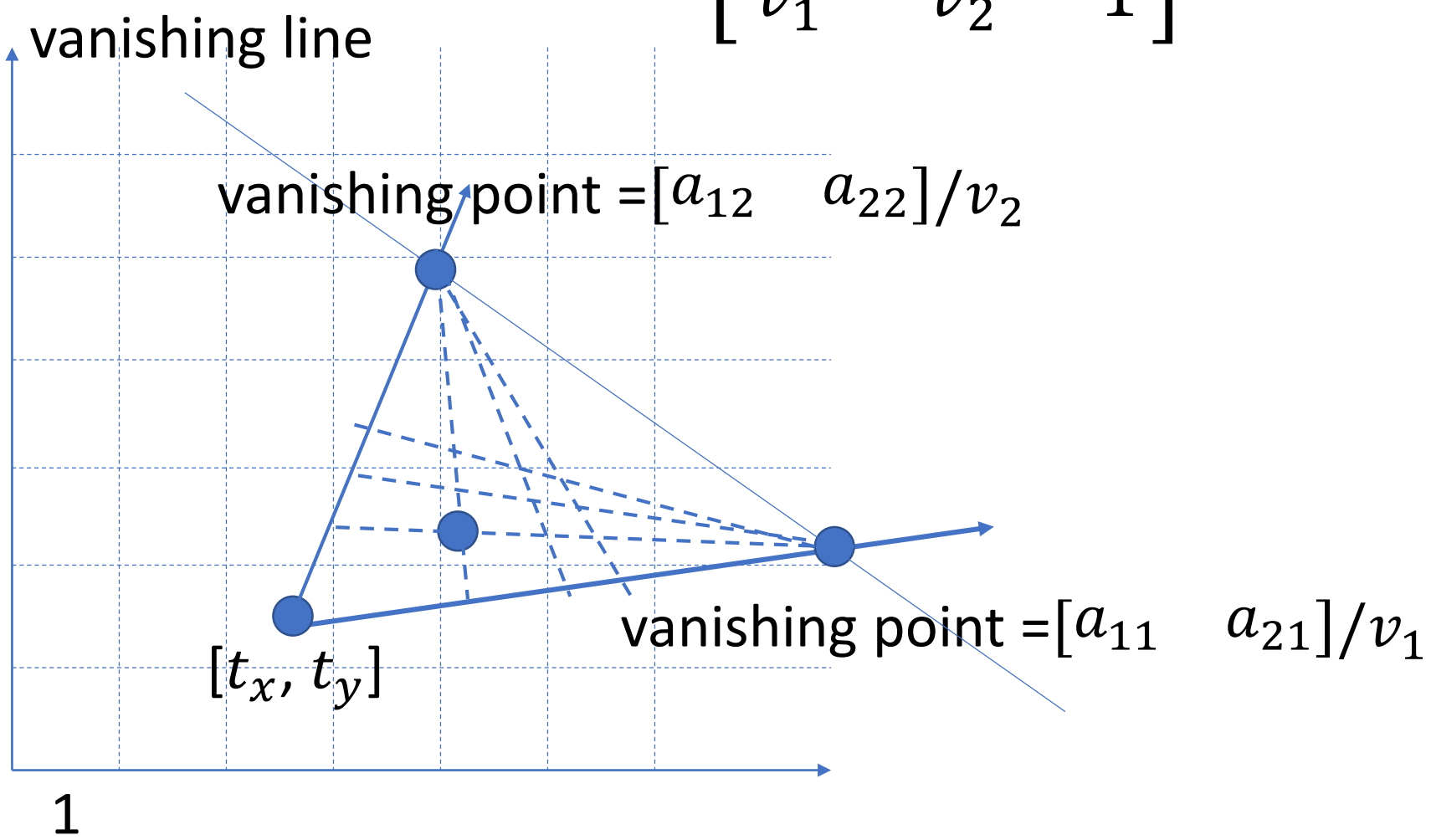
projectivity



c

General Projective mapping

$$H = \begin{bmatrix} A & \boldsymbol{t} \\ \boldsymbol{v}^T & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ v_1 & v_2 & 1 \end{bmatrix}$$



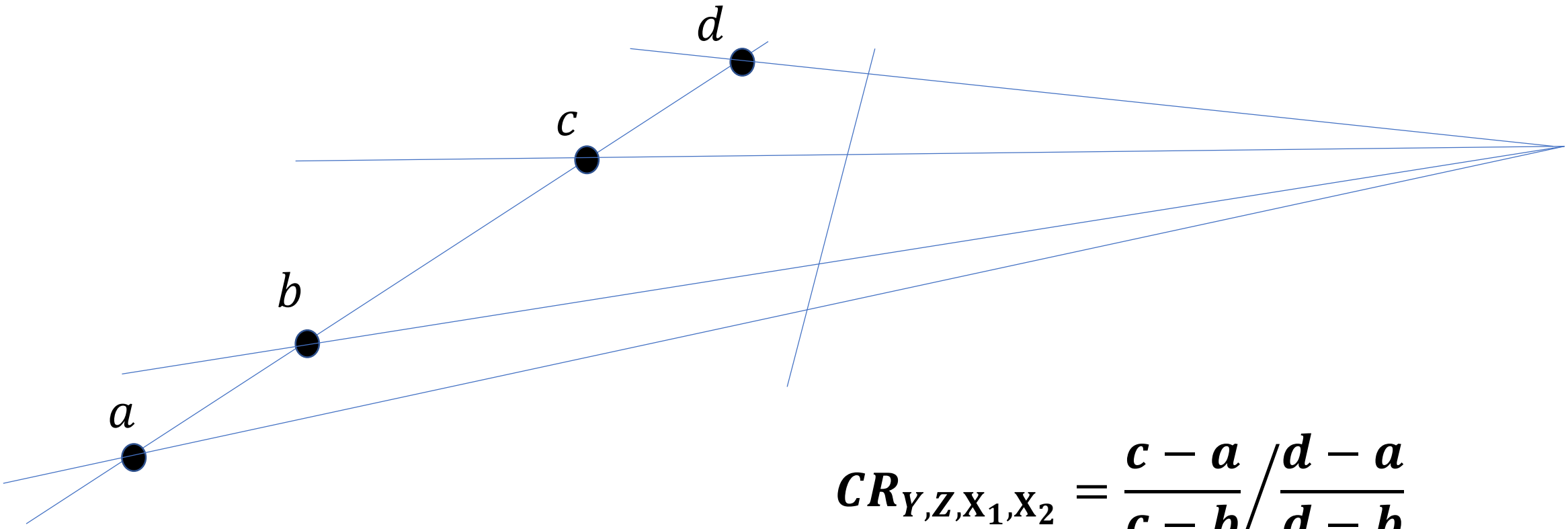
The Cross Ratio is invariant
under projective mappings





$$CR_{Y,Z,X_1,X_2} = CR_{Y',Z',X'_1,X'_2}$$

2D cross ratio of a 4-tuple of coplanar, concurrent lines: the dual of the 1D cross ratio of 4 colinear points. Take any crossing line … compute the 1D cross ratio of intersection points



$$CR_{Y,Z,X_1,X_2} = \frac{c-a}{c-b} \Big/ \frac{d-a}{d-b}$$

2D reconstruction (image rectification): reconstruction of a planar scene

original image



Reconstructed plane (the horizontal plane)

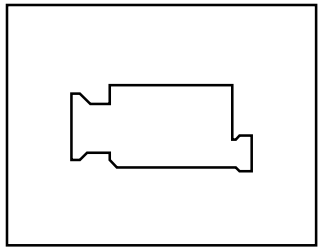


2D reconstruction (image rectification): adopted framework

planar scene



a



projectivity H

image



c

- planar scene: **unknown**
- relative pose of scene and camera: **unknown**
- camera parameters (e.g. focal distance): **unknown**

→

scene-to-image projective **mapping** H : **unknown**

2D reconstruction (or image rectification) problem formulation

given: an image of an **unknown** planar scene

recover a ***model*** of the scene, that satisfies some requirements

Scene: a set of **unknown** points $\mathbf{x}_i$ on a plane

Image: a projective transformation of the scene → $\mathbf{x}'_i = H\mathbf{x}_i$

the image points $\mathbf{x}'_i$ are **known**, but H is **unknown**

Difficulty: the projective mapping H is **unknown**

→ we can not simply invert the mapping H

2D reconstruction problem: utopistic formulation

unknown projective map $H??$



unknown planar scene



known image

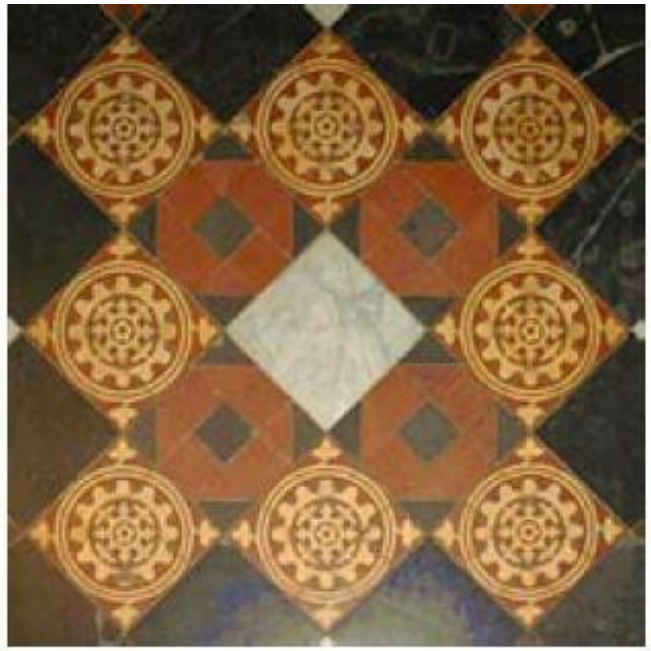
design a mapping H_R



recovered (reconstructed) scene ***equal*** *to the original scene*

combined mapping: an identity $\mathbf{I_3}$

if only the image is given → too many unknowns (8)



a

?



c

Combined strategy:

1. reduce unknowns

2. add constraints

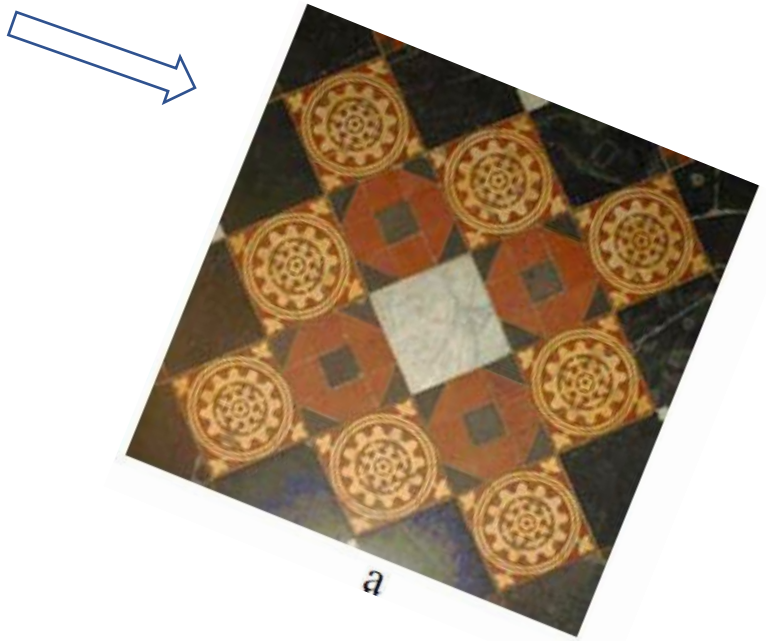
1. Reduce unknowns: relax requirements

shape reconstruction: 4 unkns = 8 - 4



c

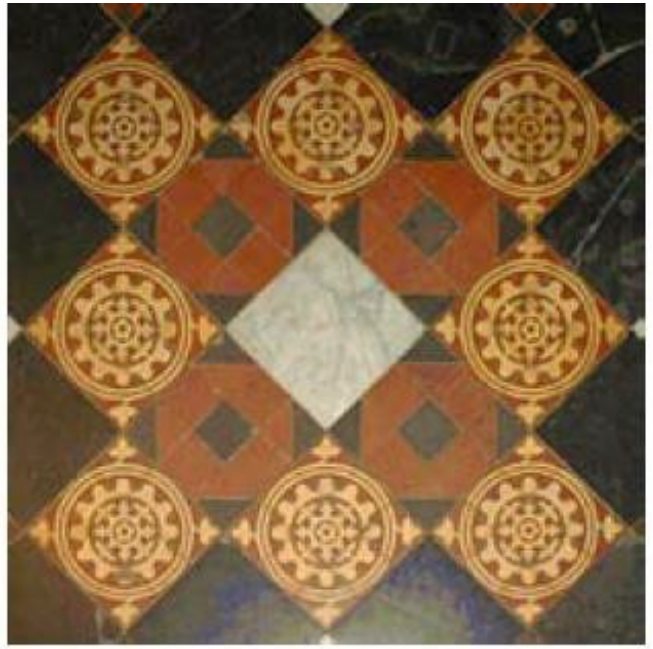
given image



a

similarity

reconstructed scene:
same shape as the original,
but different size and pose



a

original scene

2D reconstruction: 1.a. reduce unknowns 8 → 4

unknown
projective
map $H??$

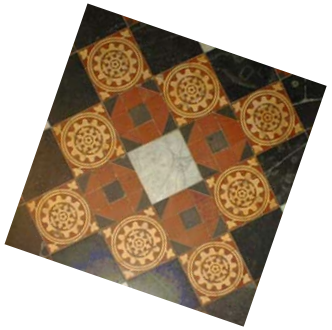


unknown
original scene



known image

design a
mapping
H_R



recovered
(reconstructed) scene:
similar *to the original*
→
same ***shape*** *but*
different size and pose

combined mapping:
a **similarity** H_S
not an identity

1.b Further reduce unknowns

affine reconstruction (less satisfactory): often used as an intermediate step towards shape reconstruction

affine reconstruction: 2 unkns = 8 - 6



c

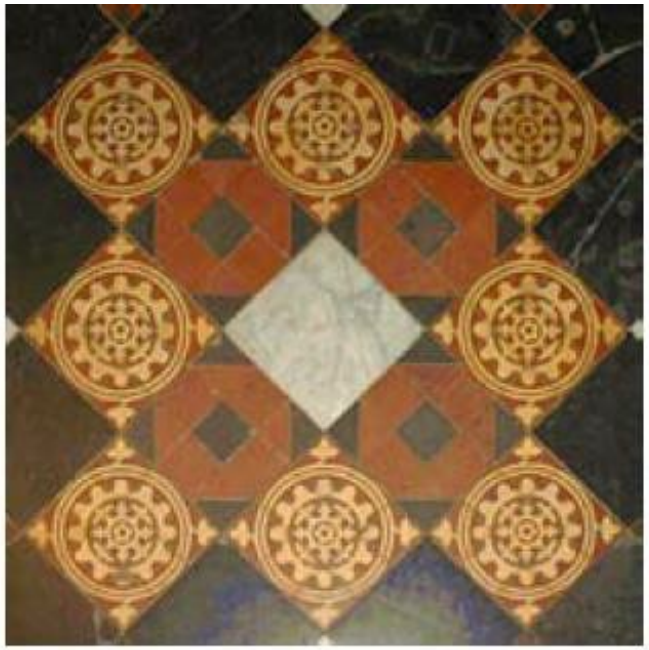
given image



b

affinity

reconstructed scene:
parallelism preserved
different shape, pose and size



a

original scene

2D reconstruction: 1.b. reduce unknowns

unknown
projective
map $H??$



known image

design a
mapping
H_{AR}



b

recovered
(reconstructed) scene:
affine to the original
but different shape



unknown
original scene

combined mapping:
an affinity H_A

2. Add constraints by additional knownledge

Add constraints (from additional information): two cases

- **affine** reconstruction

  (reconstructed scene is an affine mapping of the original one)

- **shape** reconstruction

  (reconstructed scene is a similarity mapping of the original one)

affine reconstruction

2D affine reconstruction problem

unknown projective map $H??$



known image

design a mapping H_{AR}



unknown original scene



recovered (reconstructed) scene: *affine to the original*

combined mapping: an affinity H_A

A theorem on an affine invariant

Theorem. *A projective transformation H maps the line at the infinity $\boldsymbol{l}_\infty$ onto itself (i.e., $\boldsymbol{l}_\infty$ is invariant under the projective transformation H)*

$$\updownarrow$$

*H is **affine***

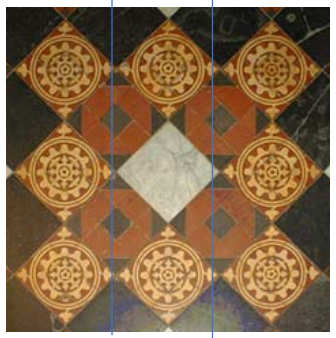
Proof: Any point at the infinity $\mathbf{x}_\infty = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$ is mapped onto a point $\mathbf{x}' = H\mathbf{x}_\infty$ also at the infinity if and only if the third coordinate of $\mathbf{x}'$ is = 0, i.e.,

$$\forall (x, y)\ [v_1 \quad v_2 \quad 1] \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = 0 \rightarrow [v_1 \quad v_2 \quad 1] = [0 \quad 0 \quad 1]$$

namely,

$$H = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ v_1 & v_2 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \text{ is } \textbf{affine}$$

l_∞



unknown
original scene

unknown
projective
map H



known image

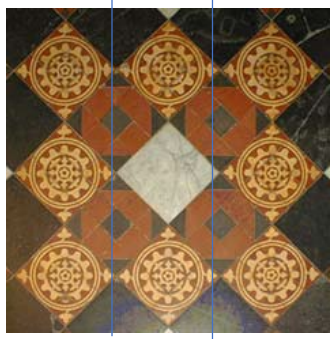
design a
mapping
H_{AR}

l_∞



recovered
(reconstructed)
scene: *affine to the original*

combined mapping: an affinity H_A



unknown
original scene

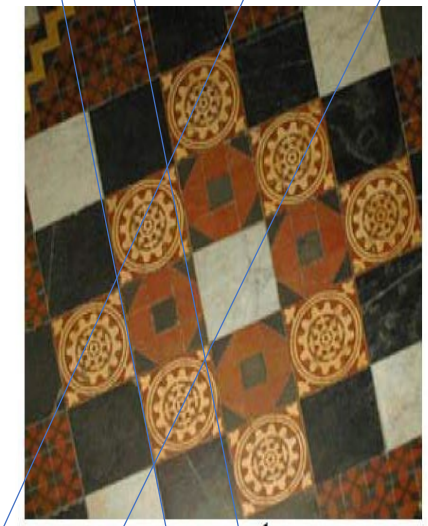
unknown H



known: image
+ l'_∞

additional information

design a mapping H_{AR}



recovered (reconstructed) scene model: affine to the original

combined mapping: affine mapping H_A

l_∞ H l'_∞ H_{AR} l_∞

affine reconstruction: use a H_{AR} that maps l'_∞ back to l_∞

The given image is a general projective mapping of the original scene

→

the vanishing line l'_∞ is the image of the line at the ∞ $l_\infty = [0 \quad 0 \quad 1]^T$ of the plane containing the scene: In general $l'_\infty \neq l_\infty$!!

Use l'_∞ as **additional information**: if we apply to the image a new projective mapping H_{AR} that maps l'_∞ back to l_∞, we obtain a new, modified image

The (new) image of the line at the infinity l_∞ is again l_∞ (itself)

→

From the theorem, the obtained model (i.e. the new image) is an **affine** mapping of the original scene

→

The obtained model is an **affine reconstruction** of the scene

Affine rectification from the image $\boldsymbol{l'_\infty}$ of the vanishing line $\boldsymbol{l_\infty}$

- Image of points at the infinity = vanishing points $\mathbf{v_1}$, $\mathbf{v_2}$ → vanishing line $\boldsymbol{l'_\infty}$

$$\boldsymbol{l'_\infty} = \mathbf{v_1} \times \mathbf{v_2} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

- Affine rectification matrix

$$H_{AR} = \begin{bmatrix} * & * & * \\ * & * & * \\ & \boldsymbol{l'^T_\infty} & \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ a & b & c \end{bmatrix}$$

WHY?

- Affine reconstructed model $M_A = H_{AR} * \text{Image}$

Shape (Euclidean) reconstruction theory

2D shape reconstruction problem

unknown projective map $H??$

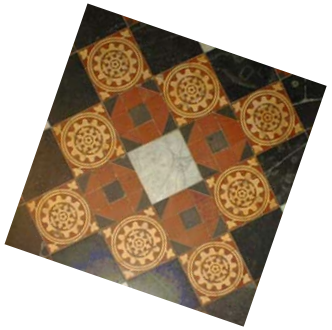


known image

design a mapping H_{SR}



unknown original scene



recovered (reconstructed) scene model: same shape as original

combined mapping: a **similarity** H_S

A theorem on an invariant under similarities

Theorem.

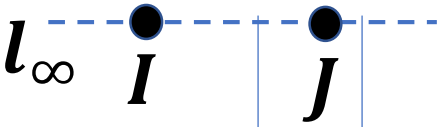
*A projective transformation H maps the **circular points $\boldsymbol{I}$** and $\boldsymbol{J}$ onto themselves (i.e., $\boldsymbol{I}$ and $\boldsymbol{J}$ are invariant under the projective transformation H)*

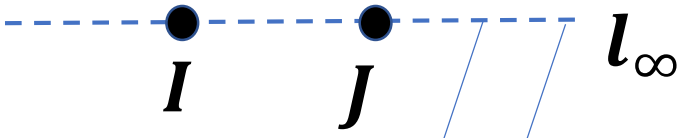
$\updownarrow$

*H is **a similarity***

Proof:

$\{\boldsymbol{I}, \boldsymbol{J}\}$ = circle $\cap$ $\boldsymbol{l}_\infty$ ------- H_S ------> other circle $\cap$ $\boldsymbol{l}_\infty$ = $\{\boldsymbol{I}, \boldsymbol{J}\}$



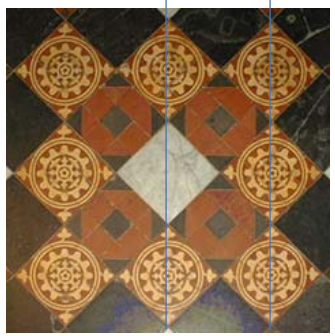


unknown H

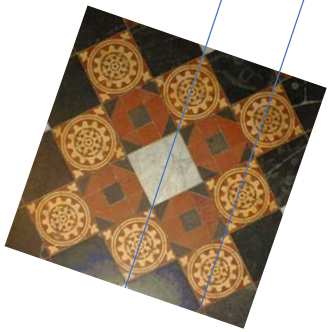


known: image

design a mapping H_{SR}

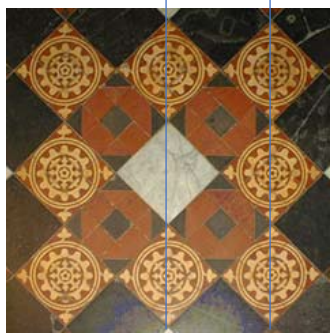


unknown
original scene



reconstructed scene model: same shape as original

combined mapping: a **similarity** H_S



unknown
original scene

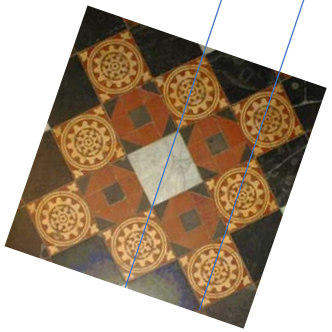
unknown H



known: image
+ $(\boldsymbol{I}', \boldsymbol{J}')$

additional information

design a
mapping
H_{SR}



reconstructed
scene model:
same shape as
original

combined mapping: a **similarity** H_S

2D shape reconstruction:
use H_{SR} that maps $(\boldsymbol{I}', \boldsymbol{J}')$ back to $(\boldsymbol{I}, \boldsymbol{J})$

The given image is a general projective mapping of the original scene

→

the image $(\boldsymbol{I}', \boldsymbol{J}')$ of circular points $(\boldsymbol{I}, \boldsymbol{J})$ of the scene plane is in general $\neq (\boldsymbol{I}, \boldsymbol{J})$.

Use $(\boldsymbol{I}', \boldsymbol{J}')$ as **additional information**: if we apply to the image a new projective mapping H_{SR} that maps $(\boldsymbol{I}', \boldsymbol{J}')$ back to $(\boldsymbol{I}, \boldsymbol{J})$, we obtain a new, modified image

The (new) image of the circular points $(\boldsymbol{I}, \boldsymbol{J})$ is again $(\boldsymbol{I}, \boldsymbol{J})$ (themselves)

→

From the theorem, the obtained model (new image) is *similar* to the original scene

→

The obtained model is a **shape reconstruction** of the scene

Shape (Euclidean) reconstruction
via Singular Value Decomposition

Shape (Euclidean) reconstruction

Instead of using the image $\boldsymbol{I'}$ and $\boldsymbol{J'}$ of the circular points, one can use the image $C_\infty^{*\prime} = \boldsymbol{I'}\,\boldsymbol{J'}^{\boldsymbol{T}} + \boldsymbol{J'}\boldsymbol{I'}^{\boldsymbol{T}}$ of the conic dual to the circular points C_∞^*

$$C_\infty^* = \boldsymbol{I}\boldsymbol{J}^{\boldsymbol{T}} + \boldsymbol{J}\boldsymbol{I}^{\boldsymbol{T}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$C_\infty^{*\prime} = \boldsymbol{I'}\boldsymbol{J'}^{\boldsymbol{T}} + \boldsymbol{J'}\boldsymbol{I'}^{\boldsymbol{T}} \neq C_\infty^*$$

Additional information $C_\infty^{*\prime}$ is equivalent to $\boldsymbol{I'}$ and $\boldsymbol{J'}$

→ 2D shape reconstruction:

use H_{SR} that maps $C_\infty^{*\prime}$ back to C_∞^*

Singular value decomposition

Why $\mathbf{C}_\infty^{*\prime} = \mathbf{U}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}\mathbf{U}^T$?

Normally: SVD (Singular Value Decomposition)

$$\mathbf{C}_\infty^{*\prime} = \mathbf{V}\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}\mathbf{U}^T$$ with **U** and **V** orthogonal

But $\mathbf{C}_\infty^{*\prime}$ is symmetric → $\mathbf{C}_\infty^{*\prime T} = \mathbf{U}\mathbf{D}\mathbf{V}^T = \mathbf{V}\mathbf{D}\mathbf{U}^T = \mathbf{C}_\infty^{*\prime}$

and SVD is unique → $\mathbf{U} = \mathbf{V}$

Observation : **H=U** orthogonal (3x3): not a P^2 isometry

Use of $C_\infty^{*\prime}$ in shape reconstruction

finding a projectivity H_{SR} that maps $(\boldsymbol{I}', \boldsymbol{J}')$ back to $(\boldsymbol{I}, \boldsymbol{J})$

reduces to finding a projectivity H_{SR} that maps $C_\infty^{*\prime}$ back to C_∞^*

Using the transformation rule for dual conics under projective mappings we obtain

$$C_\infty^* = H_{SR}\, C_\infty^{*\prime}\, {H_{SR}}^T.$$

Solving this for $C_\infty^{*\prime}$ yields

$$C_\infty^{*\prime} = {H_{SR}}^{-1} C_\infty^* \, {H_{SR}}^{-T} = {H_{SR}}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} ({H_{SR}}^{-1})^T$$

but, from SVD applied to the symmetric matrix $C_\infty^{*\prime}$

$$(SVD) C_\infty^{*\prime} = U_\perp \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} {U_\perp}^T \rightarrow {H_{SR}}^{-1} = U_\perp \rightarrow H_{SR} = {U_\perp}^{-1} = {U_\perp}^T$$

i.e., one of the possible ∞^4 solutions

Up to now: image rectification from the image $(\boldsymbol{I}', \boldsymbol{J}')$ of the circular points $(\boldsymbol{I}, \boldsymbol{J})$

- Image of the circular points → image of the conic dual to the circular points

$$C_{\infty}^{*\prime} = \boldsymbol{I}'\boldsymbol{J}'^{\boldsymbol{T}} + \boldsymbol{J}'\boldsymbol{I}'^{\boldsymbol{T}}$$

- Singular value decomposition

$$\text{svd}(C_{\infty}^{*\prime}) = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{SR}^{-1} C_{\infty}^{*} H_{SR}^{-T}$$

- Rectifying transformation (from svd output U)

$$H_{SR} = U^T$$

- Euclidean reconstructed model $M_S = H_{SR} \ * \text{Image}$

A numerical **accuracy issue** in image rectification

- In principle, given $C_\infty^{*\prime}$

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{rect}^{-1} C_\infty^* H_{rect}^{-T}$$

- But, due to noise and numerical errors, SVD gives

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{rect}^{-1} C_\infty^* H_{rect}^{-T}$$

- In principle, given $C_\infty^{*\prime}$

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{rect}^{-1} C_\infty^* H_{rect}^{-T}$$

- But, due to noise and numerical errors, SVD output is

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{rect}^{-1} C_\infty^* H_{rect}^{-T}$$

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = U \begin{bmatrix} \sqrt{a} & 0 & 0 \\ 0 & \sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{a} & 0 & 0 \\ 0 & \sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

$$\rightarrow H_{rect}^{-1} = U \begin{bmatrix} \sqrt{a} & 0 & 0 \\ 0 & \sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow H_{rect} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

To sum up: image rectification from the image $(\boldsymbol{I}', \boldsymbol{J}')$ of the circular points $(\boldsymbol{I}, \boldsymbol{J})$

- Image of the circular points → image of the conic dual to the circular points

$$C_\infty^{*\prime} = \boldsymbol{I}'\boldsymbol{J}'^{\boldsymbol{T}} + \boldsymbol{J}'\boldsymbol{I}'^{\boldsymbol{T}}$$

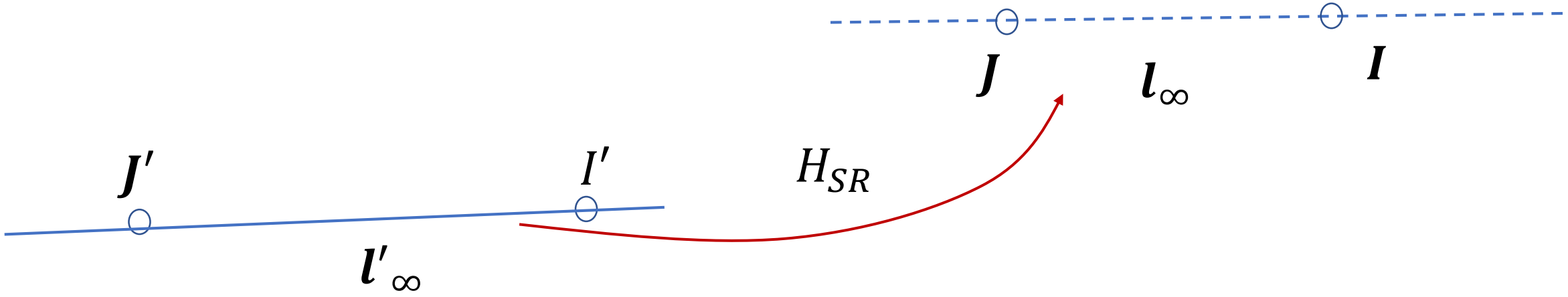
- Singular value decomposition

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{SR}^{-1} C_\infty^* H_{SR}^{-T}$$

- Rectifying transformation (from svd output U)

$$H_{SR} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

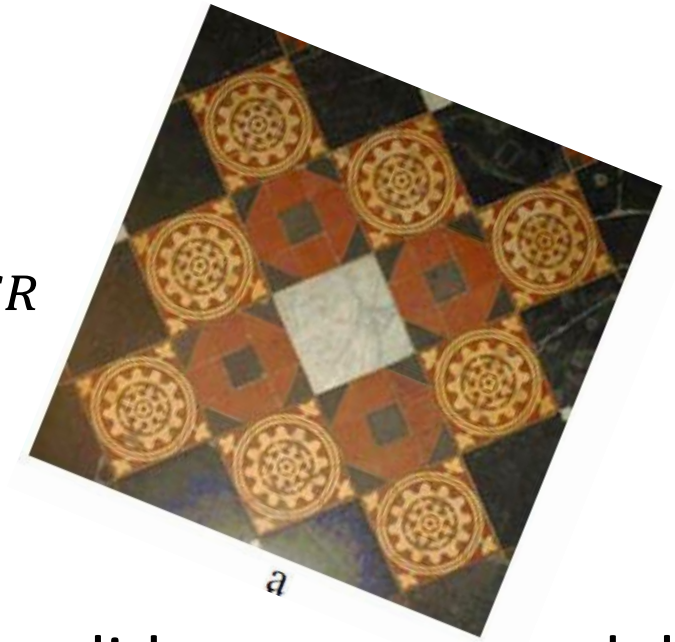
- Euclidean reconstructed model $M_S = H_{SR} \ * \text{Image}$





image

shape reconstruction H_{SR}



euclidean scene model

Shape (Euclidean) reconstruction

practice

How to find $(\boldsymbol{I}', \boldsymbol{J}')$, or $C_{\infty}^{*\prime}$, in practical cases?

How to find $(\boldsymbol{I}', \boldsymbol{J}')$, or $C_{\infty}^{*\prime}$, in practical cases?

In 2D rectification we use $(\boldsymbol{I}', \boldsymbol{J}')$, or equivalently $\boldsymbol{C}_{\infty}^{*\prime}$,

as **additional information**

How can we find $(\boldsymbol{I}', \boldsymbol{J}')$ (or $C_{\infty}^{*\prime}$) **?**

from information on the observed scene we derive constraints on $C_{\infty}^{*\prime}$

a. known angles between lines

b. known shape of objects, e.g., circumferences

c. combinations of a. and b.

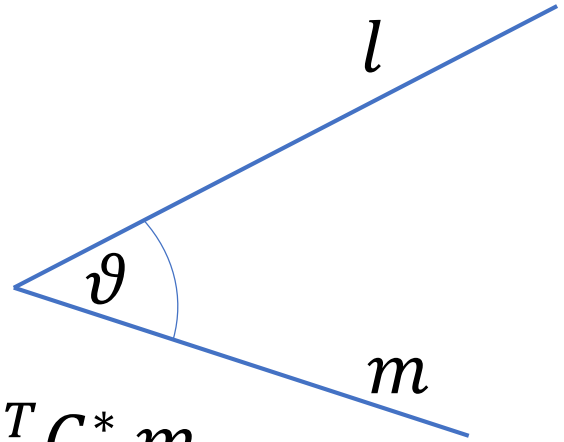
c. observation of rigid planar motion

a. known angles between lines

RESULT: **The angle between two lines** via C_∞^*

The angle ϑ between $l = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix}$ and $m = \begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix}$ is the angle between their normals $[a_1 \quad b_1]$ and $[a_2 \quad b_2]$:

$$\cos\vartheta = \frac{a_1 a_2 + b_1 b_2}{\sqrt{(a_1{}^2 + b_1{}^2)(a_2{}^2 + b_2{}^2)}}$$



But, e.g., $a_1 a_2 + b_1 b_2 = [a_1 \quad b_1 \quad c_1] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix} = l^T C_\infty^* m$

$$\rightarrow \cos\vartheta = \frac{l^T C_\infty^* m}{\sqrt{(l^T C_\infty^* l)(m^T C_\infty^* m)}}$$

Express the angle ϑ between two lines **in the scene** in terms of elements **in the image**: l', m', $C_\infty^{*\prime}$

$$\cos\vartheta = \frac{a_1 a_2 + b_1 b_2}{\sqrt{(a_1^2 + b_1^2)(a_2^2 + b_2^2)}} = \frac{l^T C_\infty^* m}{\sqrt{(l^T C_\infty^* l)(m^T C_\infty^* m)}}$$

from transformation rules for lines and dual conics

e.g., $l^T C_\infty^* m = l'^T H H^{-1} C_\infty^{*\prime} H^{-T} H^T m' = l'^T C_\infty^{*\prime} m'$

→

$$\cos\vartheta = \frac{l'^T C_\infty^{*\prime} m'}{\sqrt{(l'^T C_\infty^{*\prime} l')(m'^T C_\infty^{*\prime} m')}}$$

Express the angle ϑ between two lines **in the scene** in terms of elements **in the image**: l', m', $C_\infty^{*\prime}$

$$\cos\vartheta = \frac{l'^T C_\infty^{*\prime} m'}{\sqrt{(l'^T C_\infty^{*\prime} l')(m'^T C_\infty^{*\prime} m')}}$$

Here, l' and m' are extracted from the image,
whereas $\boldsymbol{C}_\infty^{*\prime}$ is the **unknown** matrix we want to estimate

Known angle ϑ between two **scene lines** → nonlinear eqn on $C_\infty^{*\prime}$

if the scene lines are perpendicular, $\cos\vartheta = 0$ → $l'^T C_\infty^{*\prime} m' = 0$ **linear**

Knowledge of l'_∞ → constraints on $C^{*\prime}_\infty$

$$C^{*\prime}_\infty l'_\infty = (I'J'^T + J'I'^T)l'_\infty = I'(J'^T l'_\infty) + \mathrm{J}'(I'^T l'_\infty) = \mathbf{0}$$

$J'^T l'_\infty$ → 0

$I'^T l'_\infty$ → 0

$$C^{*\prime}_\infty l'_\infty = 0$$

2 constraints

Image of line at the infinity l'_∞ = RNS($C^{*\prime}_\infty$)

How many constraints are needed to find $C_\infty^{*\prime}$?

Matrix $C_\infty^{*\prime}$ is symmetric, homogeneous, and **singular**

→ 4 degrees of freedom (e.g. $\boldsymbol{I}'$, $\boldsymbol{J}'$)

→ 4 **independent** constraints are needed

however

singularity is in general a nonlinear constraint

→use 5 linear constraints (orthogonal lines)

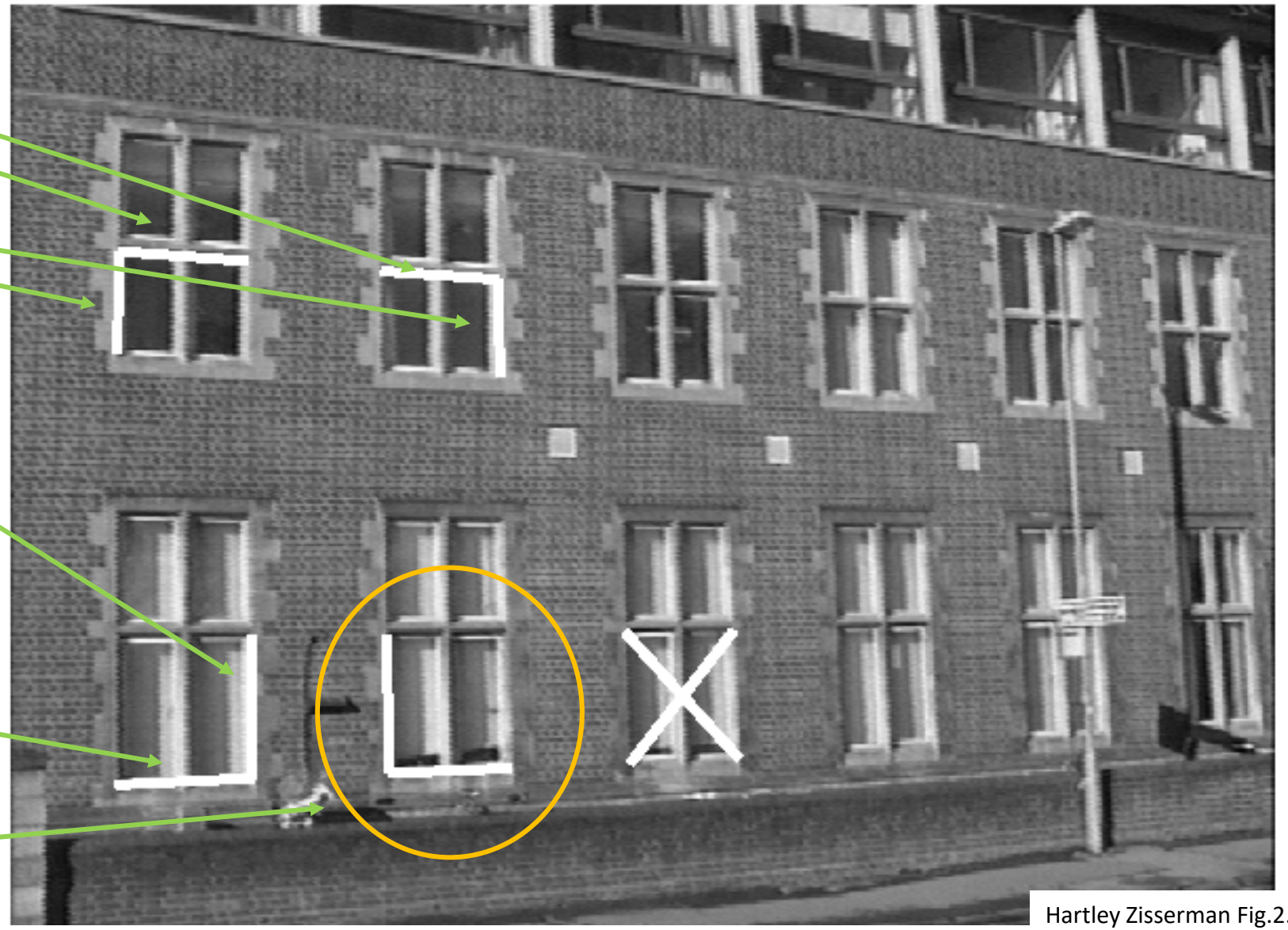
or

if l'_∞ can be derived from parallelism (or pairs of orthogonal lines with one line in common), use

$l'_\infty = RNS(C_\infty^{*\prime})$: 2 linear constraints which imply singularity

+ 2 linear constraints from pairs of orthogonal lines

Are the constraints associated to the 5 indicated pairs of orthogonal lines independent? NO: l'_∞ can be derived

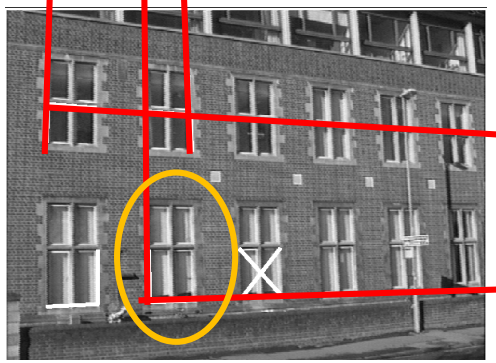


Hartley Zisserman Fig.2.17

Left-low window lines are constrained by the others:

- vertical line must concur where other vertical lines do
- horizontal line must be colinear to l_5

- → four pairs are sufficient



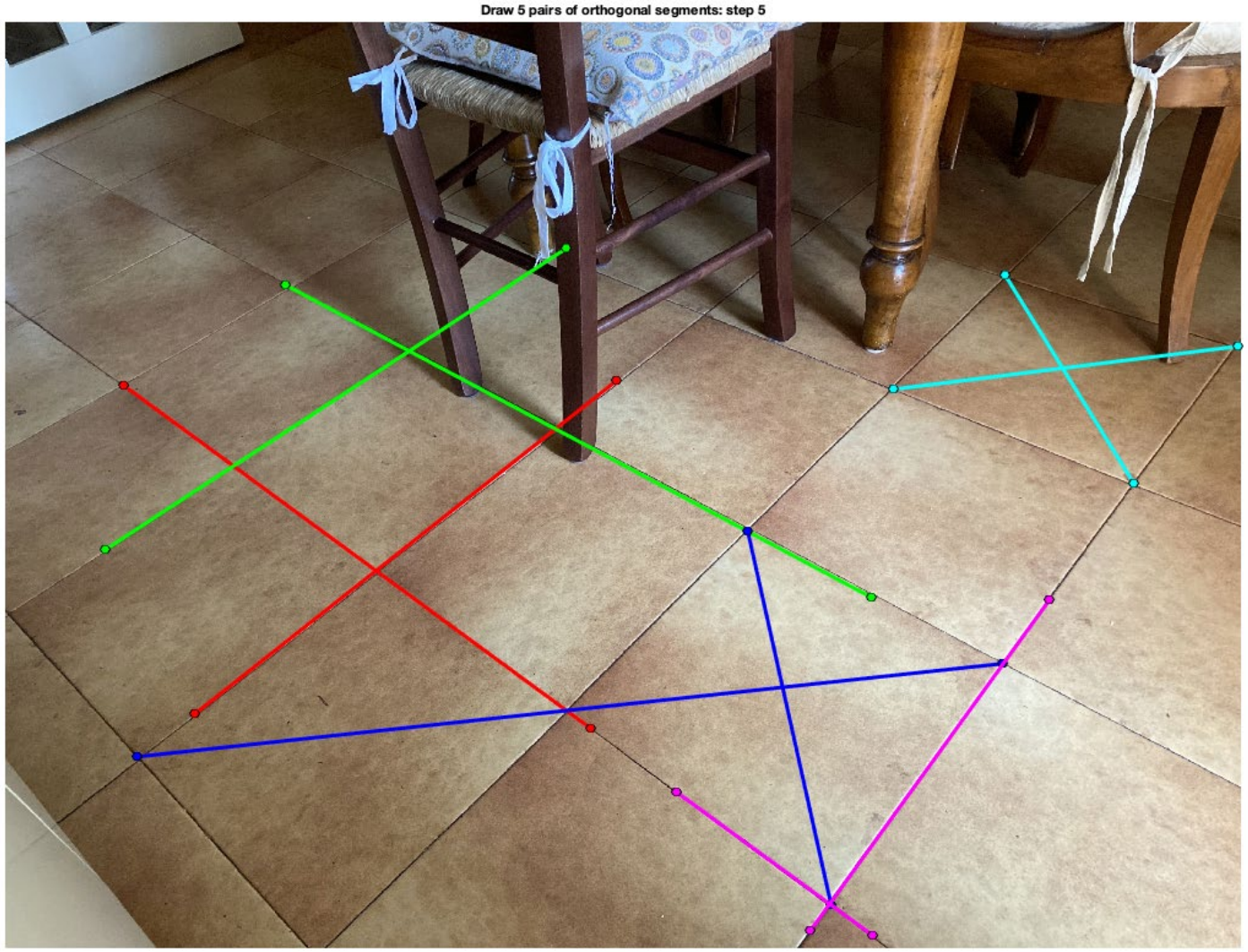
l_1 l_2 l_3 l_4 l_5

Rectified image from pairs of orthogonal lines



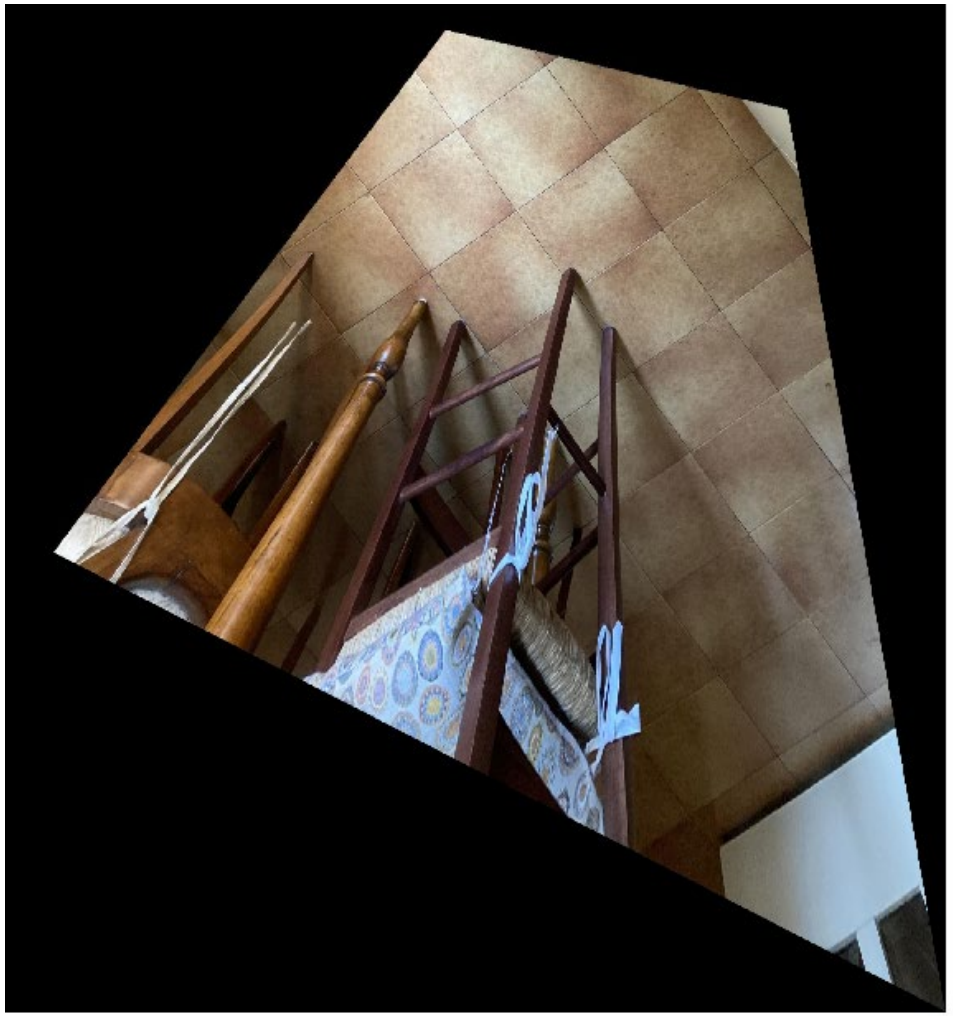
Hartley Zisserman Fig.2.17

Are they all independent?



rectified image

Caveat: just one plane (+ all planes parallel to it) is rectified, not the other planes present in the scene!

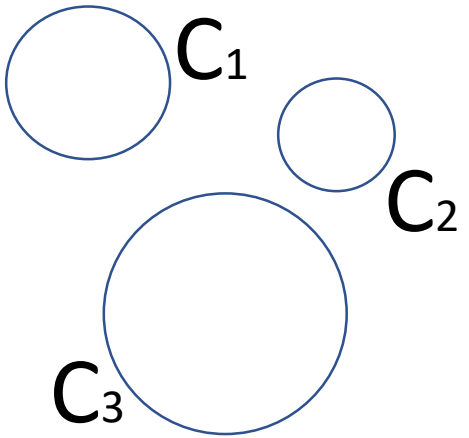


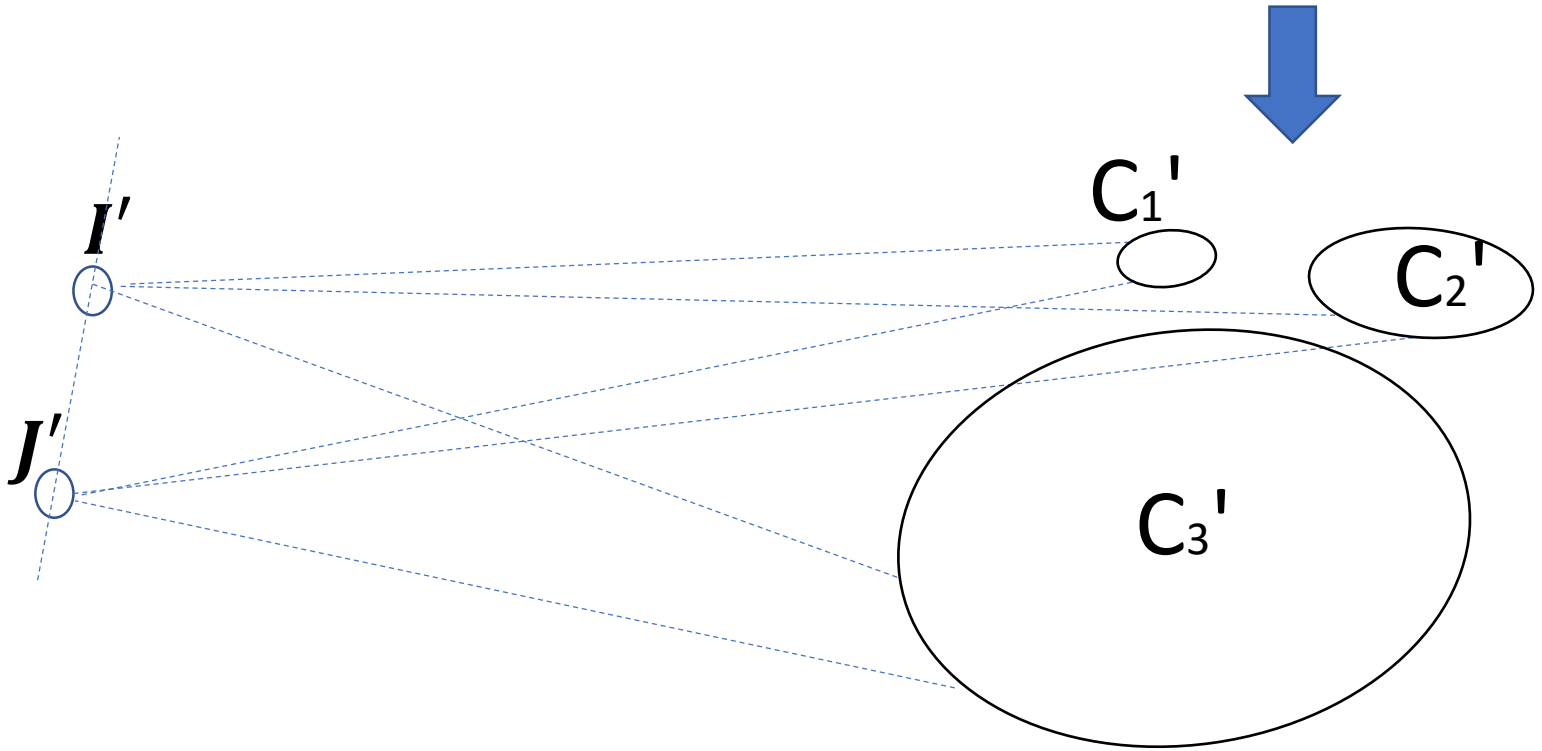
credits: Luca Magri

b. Example: image of coplanar circumferences



intersect the images of circumferences
→ image of circular points





Observation: also parallel, but non coplanar, circumferences intersect in the circular points



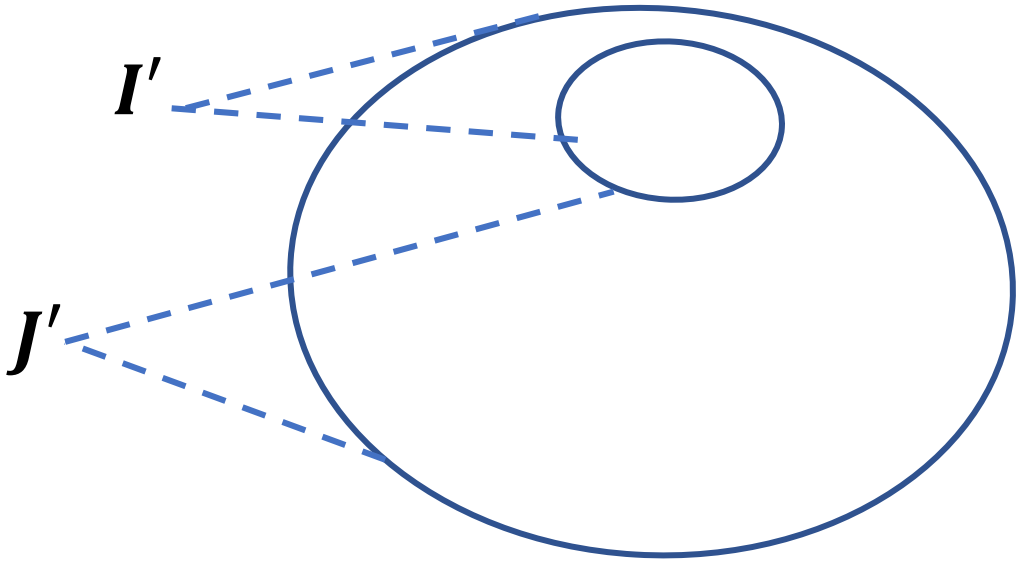


Image rectification from the image of circular points: $\{I', J'\} = C_1' \cap C_2' \cap C_3'$

- Image of the circular points → image of the conic dual to the circular points

$$C_\infty^{*\prime} = \boldsymbol{I'J'^T} + \boldsymbol{J'I'^T}$$

- Singular value decomposition

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{SR}^{-1} C_\infty^* H_{SR}^{-T}$$

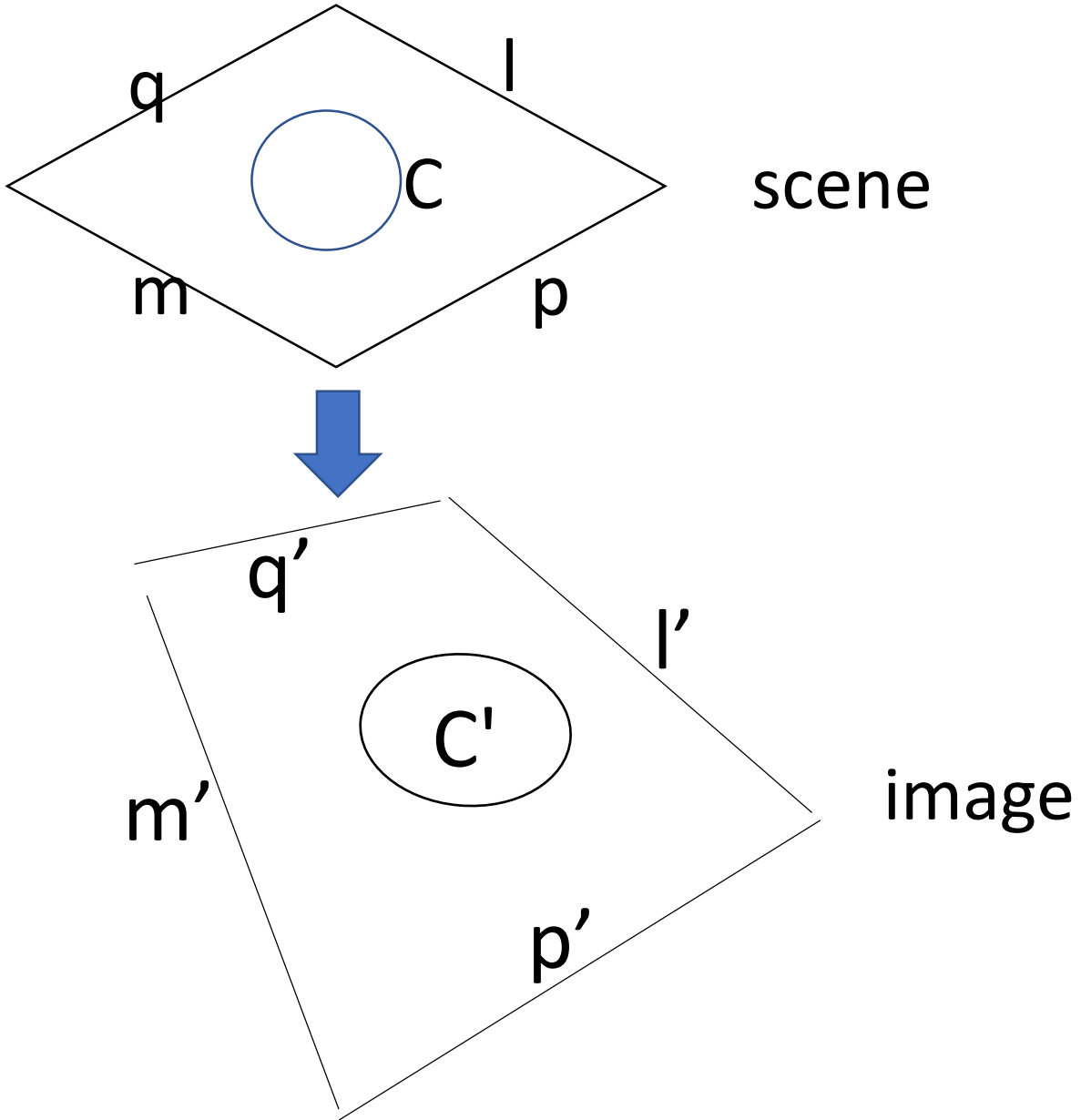
- Rectifying transformation (from svd output U)

$$H_{SR} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

- Euclidean reconstructed model $M_S = H_{SR} * \text{Image}$

c. Example: a circumference + a parallelogram





c. Example: a circumference + a parallelogram

$\mathbf{v}_2 = \mathrm{l}' \times m'$

$\mathbf{v}_1 = p' \times q'$

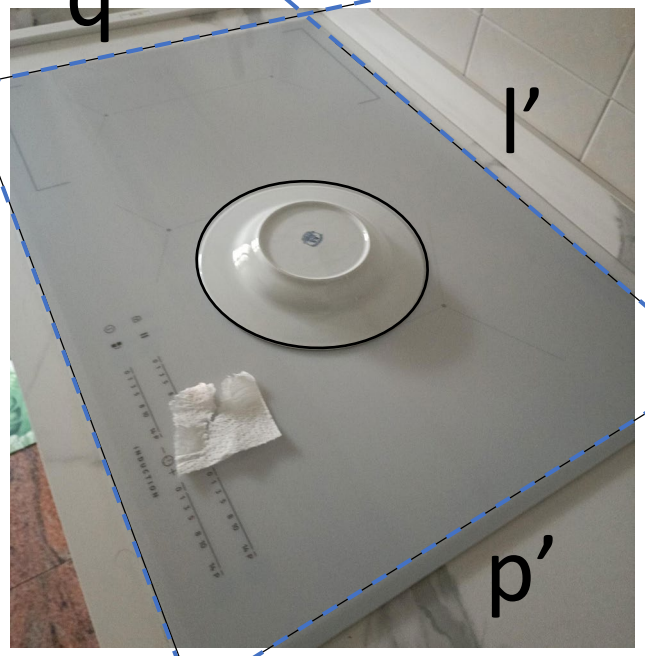


image of the line at the infinity

$\mathbf{v}_1 = p' \times q'$

$\mathbf{v}_2 = \mathrm{l}' \times m'$

$l'_\infty = \mathbf{v}_1 \times \mathbf{v}_2$

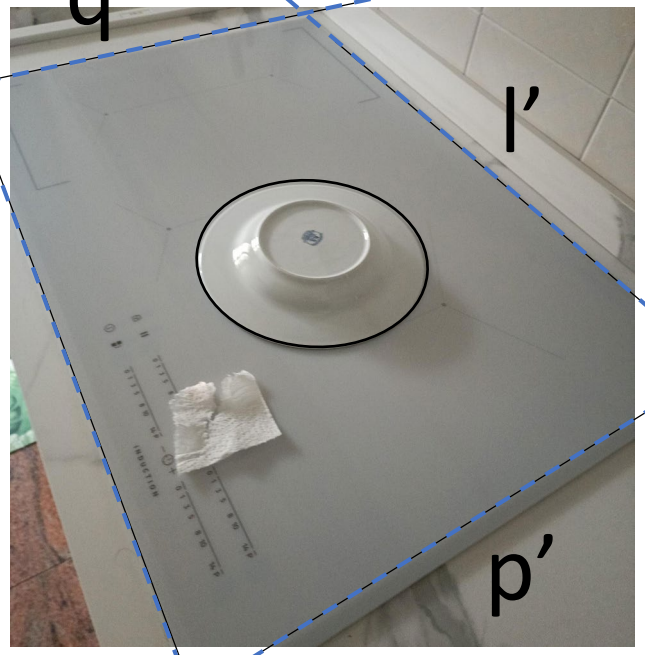
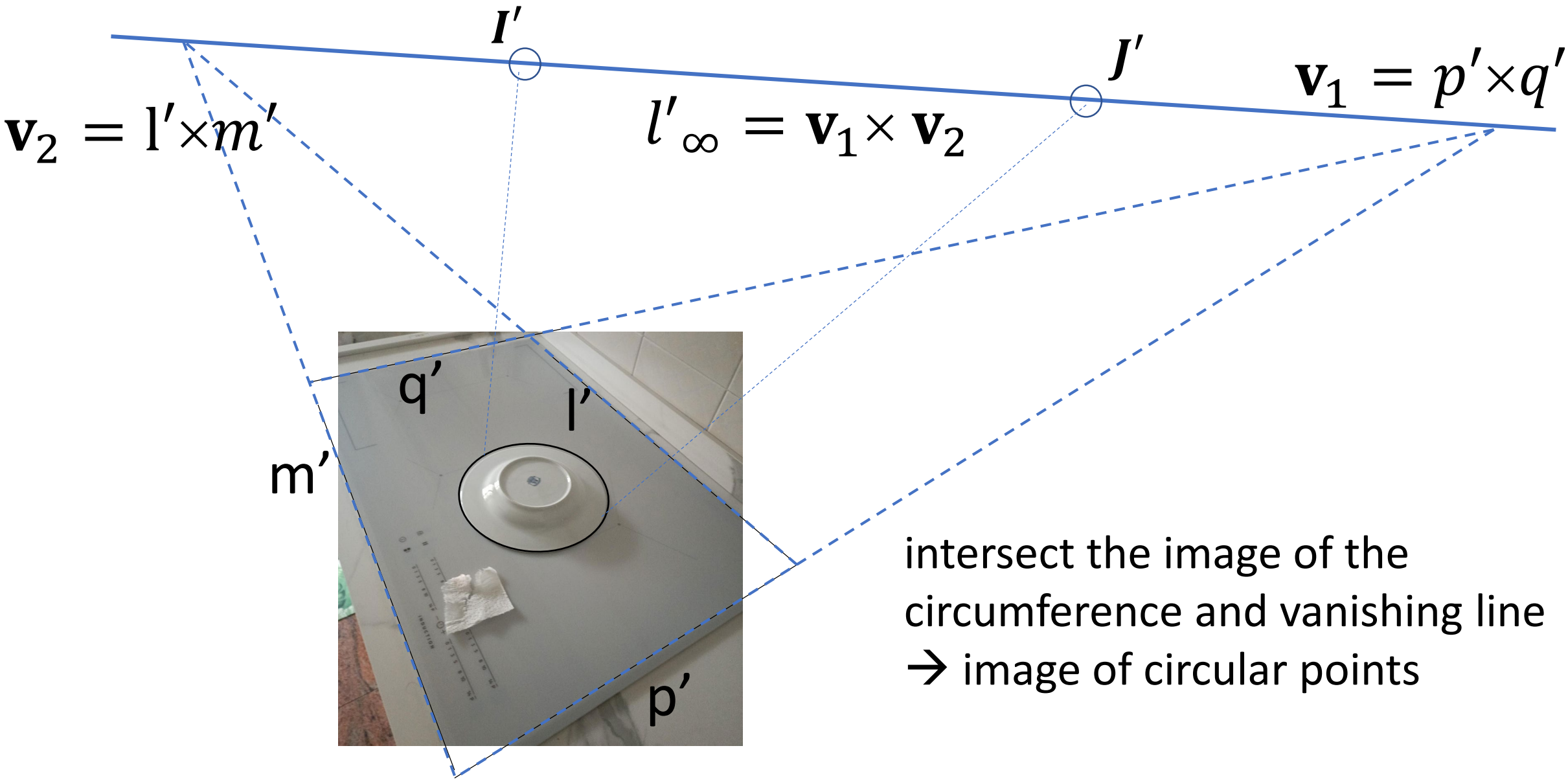


Image of circular points: $\{l', J'\} = l'_\infty \cap C'$



intersect the image of the circumference and vanishing line
→ image of circular points

Image rectification from the image of circular points: $\{I', J'\} = l'_\infty \cap C'$

- Image of the circular points → image of the conic dual to the circular points

$$C_\infty^{*\prime} = \boldsymbol{I'J'^T} + \boldsymbol{J'I'^T}$$

- Singular value decomposition

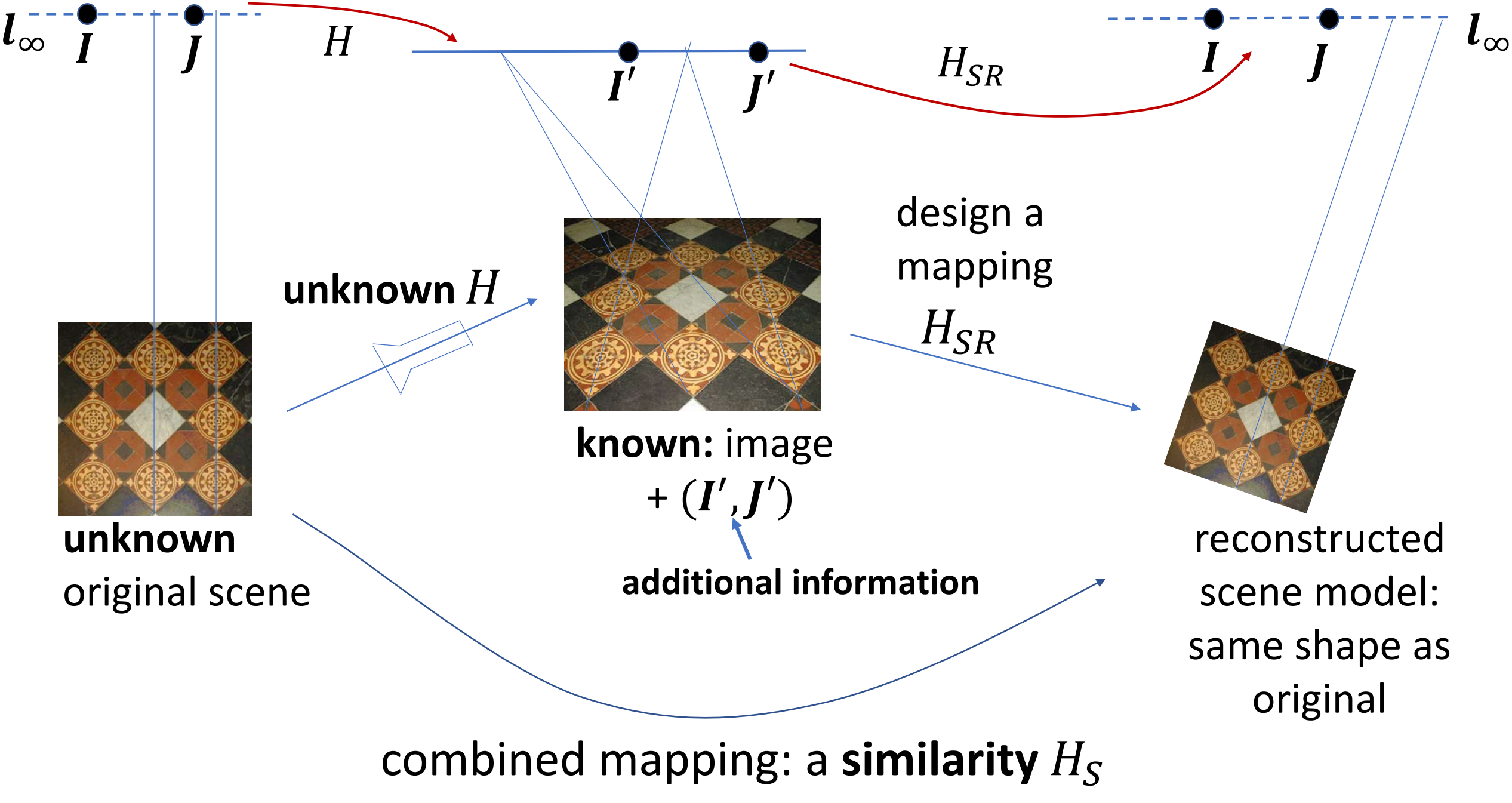
$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{SR}^{-1} C_\infty^* H_{SR}^{-T}$$

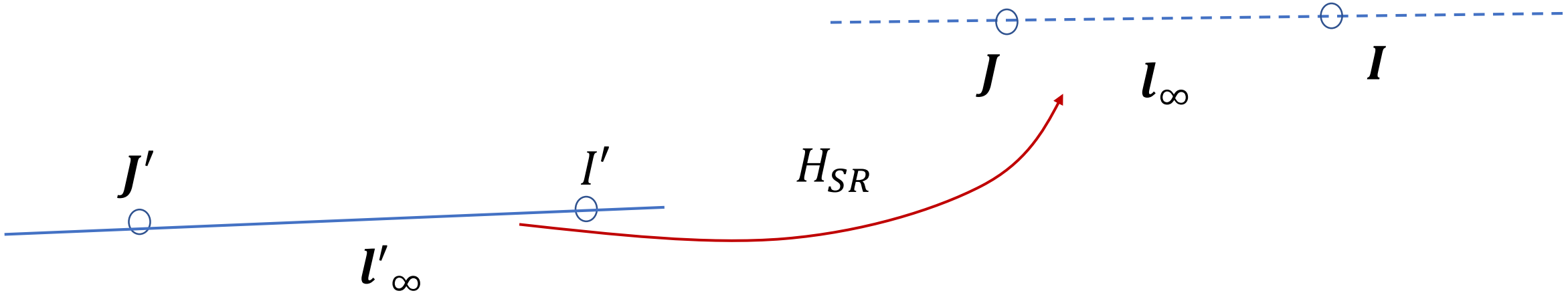
- Rectifying transformation (from svd output U)

$$H_{SR} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

- Euclidean reconstructed model $M_S = H_{SR} * \text{Image}$

Summary on 2D Reconstruction

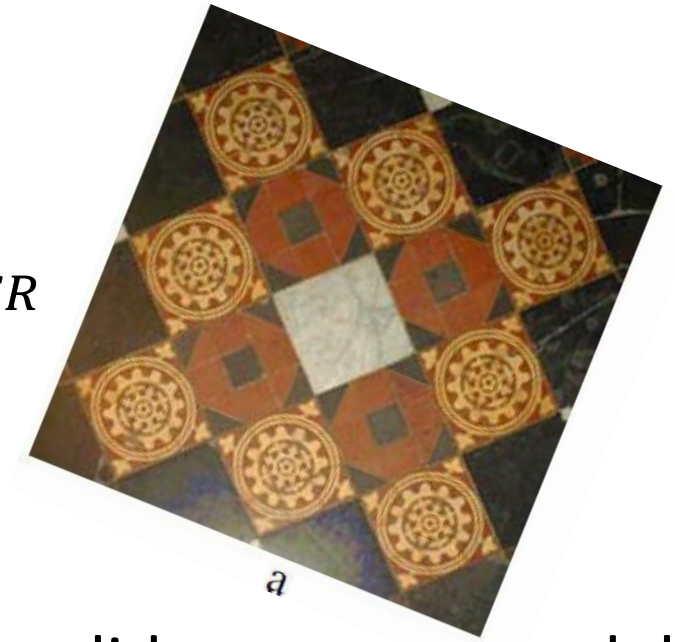

l_∞
I
J
H
I'
J'
H_{SR}
I
J
l_∞
unknown H
design a mapping H_{SR}
known: image + (I', J')
unknown
original scene
additional information
reconstructed scene model: same shape as original
combined mapping: a **similarity** H_S





image

shape reconstruction H_{SR}



euclidean scene model

Theory: image rectification from the image $(\boldsymbol{I}', \boldsymbol{J}')$ of the circular points $(\boldsymbol{I}, \boldsymbol{J})$

- Image of the circular points → image of the conic dual to the circular points

$$C_\infty^{*\prime} = \boldsymbol{I}'\boldsymbol{J}'^{\boldsymbol{T}} + \boldsymbol{J}'\boldsymbol{I}'^{\boldsymbol{T}}$$

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$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{SR}^{-1} C_\infty^* H_{SR}^{-T} = H_{SR}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} H_{SR}^{-T}$$

- Rectifying transformation (from svd output U)

$$H_{SR} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

transformation rule for dual conic

- Euclidean reconstructed model $M_S = H_{SR} * \text{Image}$

How to find $(\boldsymbol{I}', \boldsymbol{J}')$, or $C_{\infty}^{*\prime}$, in practical cases?

In 2D rectification we use $(\boldsymbol{I}', \boldsymbol{J}')$, or equivalently $\boldsymbol{C}_{\infty}^{*\prime}$,

as **additional information**

How can we find $(\boldsymbol{I}', \boldsymbol{J}')$ (or $C_{\infty}^{*\prime}$) **?**

from information on the observed scene we derive constraints on $C_{\infty}^{*\prime}$

a. known angles between lines

b. known shape of objects, e.g., circumferences

c. combinations of a. and b.

c. observation of rigid planar motion 

d. Example: rectification from planar displacement

Image of a rigid planar motion

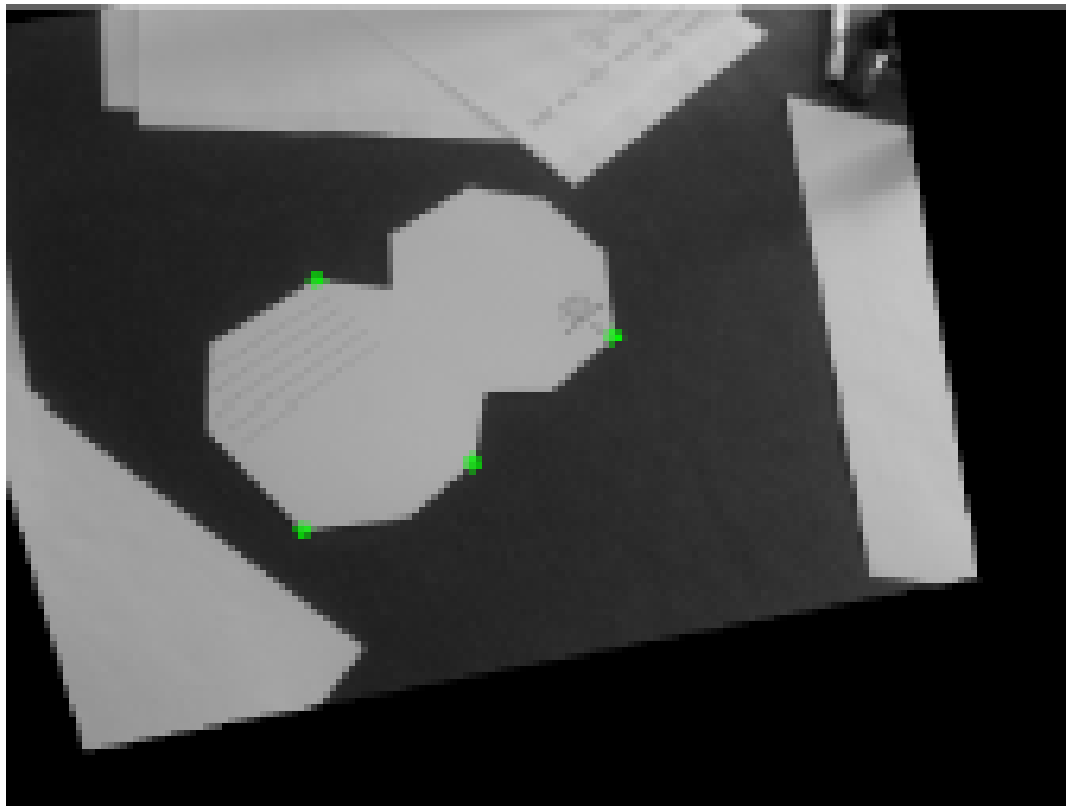


image before displacement

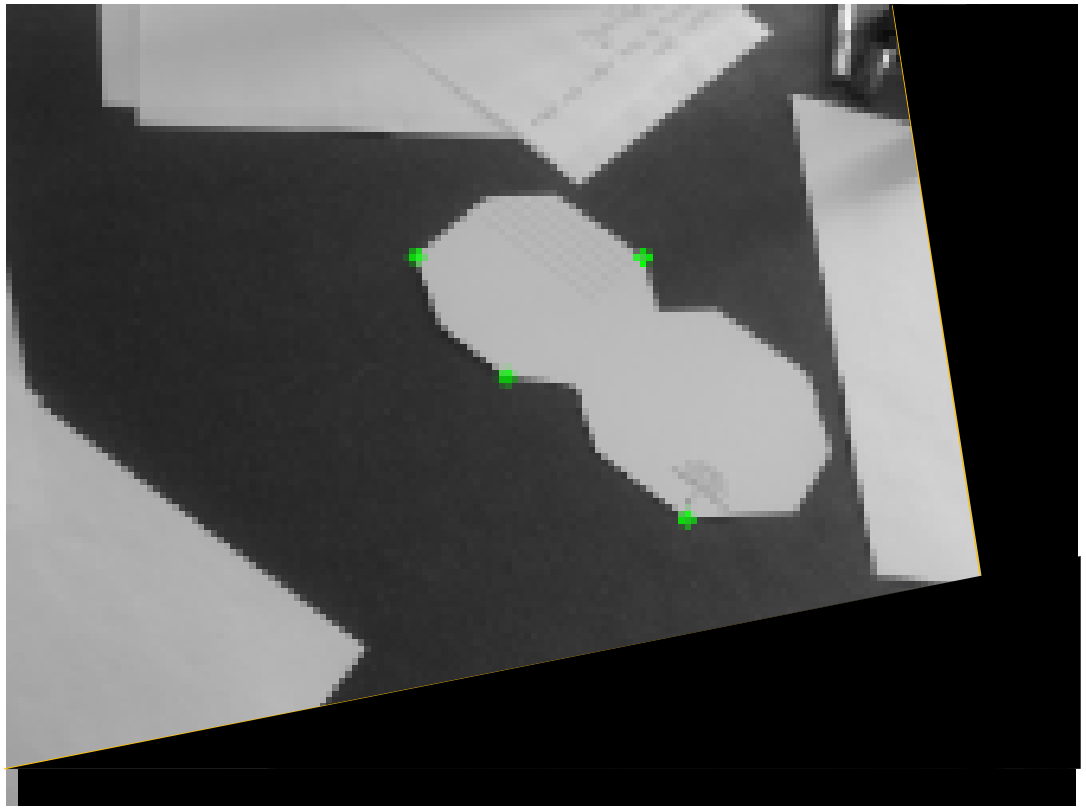
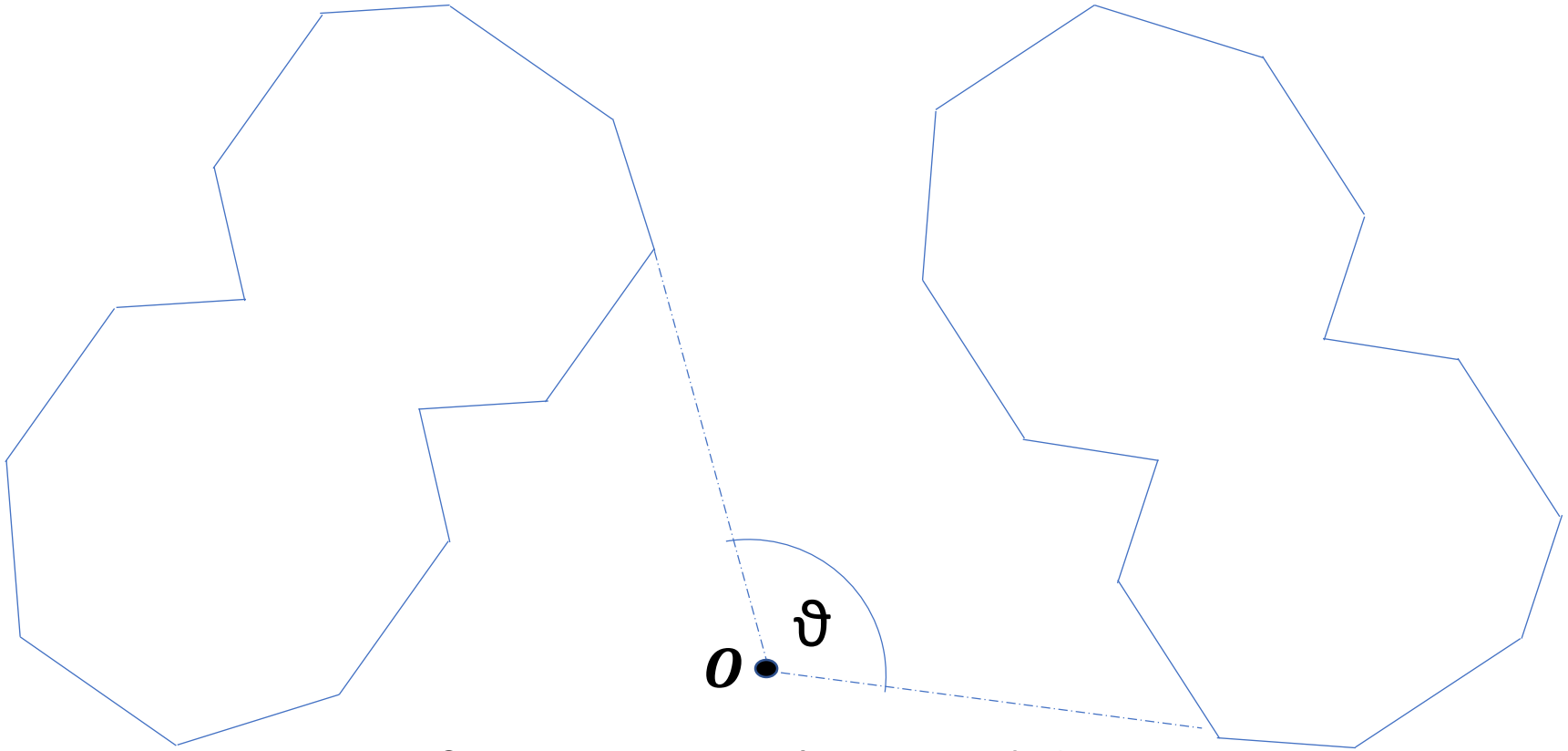


image after displacement

Property of rigid planar motion: **any roto-translation** is a **pure rotation**



ϑ : rotation angle around O

rigid planar motion

- Planar rigid motion: isometry → is a similarity → **invariant** circ. pts $\boldsymbol{I}, \boldsymbol{J}$
- any (planar) rigid displacement is a pure rotation: 3 dofs X_o, Y_o, ϑ

→ Center of rotation $\boldsymbol{O}$ is also invariant

→ 3 **invariants**: circular points $\boldsymbol{I}, \boldsymbol{J}$ and center of rotation $\boldsymbol{O}$

these 3 points, namely, $\boldsymbol{I}, \boldsymbol{J}$ and $\boldsymbol{O}$ remain fixed during motion

⇨

also their images $\boldsymbol{I}'$, $\boldsymbol{J}'$ and $\boldsymbol{O}$' remain fixed during motion

image points $\boldsymbol{I}'$, $\boldsymbol{J}'$ and $\boldsymbol{O}'$ remain fixed
corresponding features → estimate homography H
$\boldsymbol{I}'$, $\boldsymbol{J}'$ and $\boldsymbol{O}'$ are invariant under mapping H

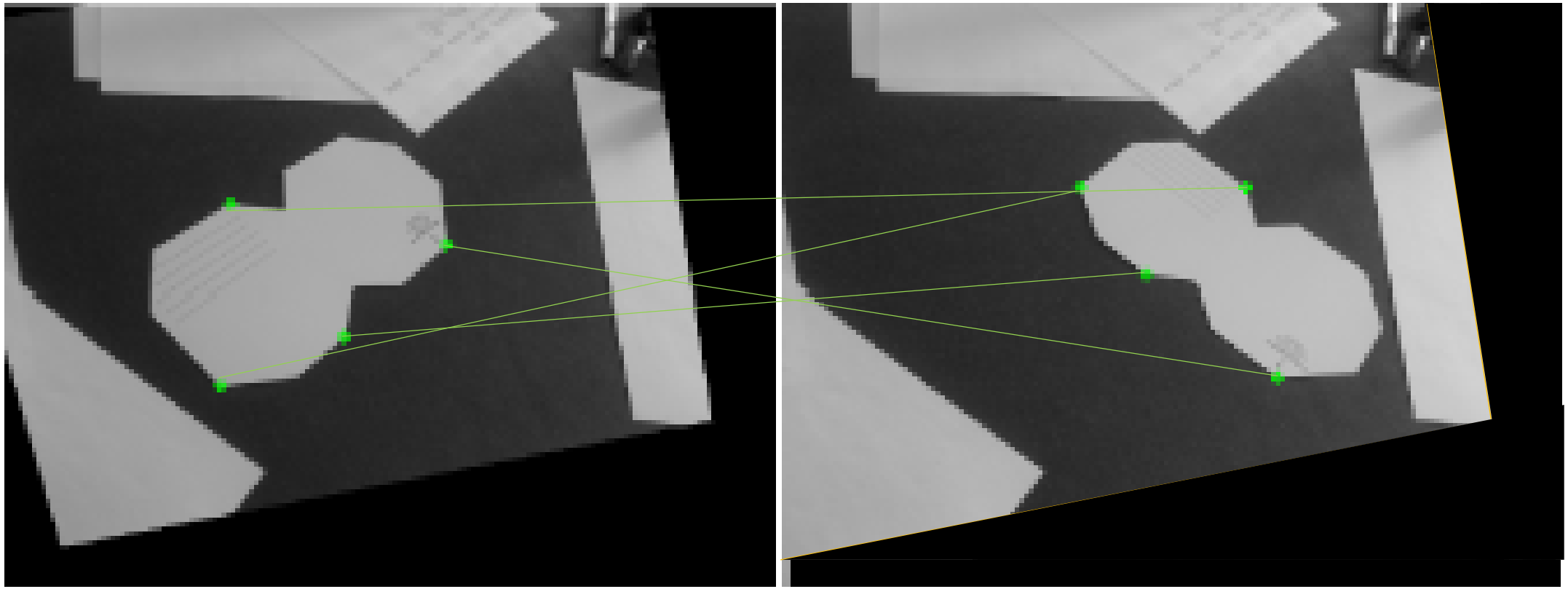


image before displacement

image after displacement

Invariants under a projective mapping H

Homogeneous coordinates:

invariants $He = \lambda e$ ←→ **eigenvectors**

it can be shown that:

- eigenvectors I', J' correspond to complex eigenvalues
- the phase of their eigenvalues is the rotation angle
- eigenvector O' corresponds to the real eigenvalue

Rectification from planar motion

- Fit homography H between the image before and after displacement
- find eigenvector-eigenvalues of H :
- eigenvalues are proportional to $\lambda' = 1, \lambda'' = e^{i\theta}, \lambda''' = e^{-i\theta}$
- eigenvector e' associated to $\lambda' = 1$ is the image of the C.O.R. $\boldsymbol{O}$
- angle θ is the rotation angle
- eigenvectors e'' and e''' associated to λ'' and λ''' are the images $\boldsymbol{I}', \boldsymbol{J}'$ of the circular points $\boldsymbol{I}, \boldsymbol{J}$
- thus $C_\infty^{*\prime} = \boldsymbol{I}'\boldsymbol{J}'^{\boldsymbol{T}} + \boldsymbol{J}'\boldsymbol{I}'^{\boldsymbol{T}}$
- apply singular value decomposition svd$(C_\infty^{*\prime}) = UC_\infty^* U^T$
- we obtain the rectification matrix $H_{SR} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$

Example 1.
rectification from two planar rectangles

Remember: angle between scene lines expressed in terms of image lines

- From transformation rules for lines and dual conics:

$$\cos\theta = \frac{l^T C_\infty^* m}{\sqrt{(l^T C_\infty^* l)\ (m^T C_\infty^* m)}} = \frac{l'^T C_\infty^{*\prime} m'}{\sqrt{(l'^T C_\infty^{*\prime} l')\ (m'^T C'^*_\infty m')}}$$

known angle → equation on $C_\infty^{*\prime}$

e.g., for perpendicular lines $l'^T C_\infty^{*\prime} m' = 0$ **linear**

Image rectification from two coplanar rectangles



or … rectification from vanishing points



Direct method

Direct method

1. Find $C_{\infty}^{*\prime}$
2. Compute H_{rect}

- 4 dof for $C_{\infty}^{*\prime}$:

  9 elements minus

  - 3x3 symmetric → 3 constraints;
  - homogeneous matrix → 1 constraint;
  - singular matrix → 1 constraint

direct method to reconstruct the upper face

vanishing points

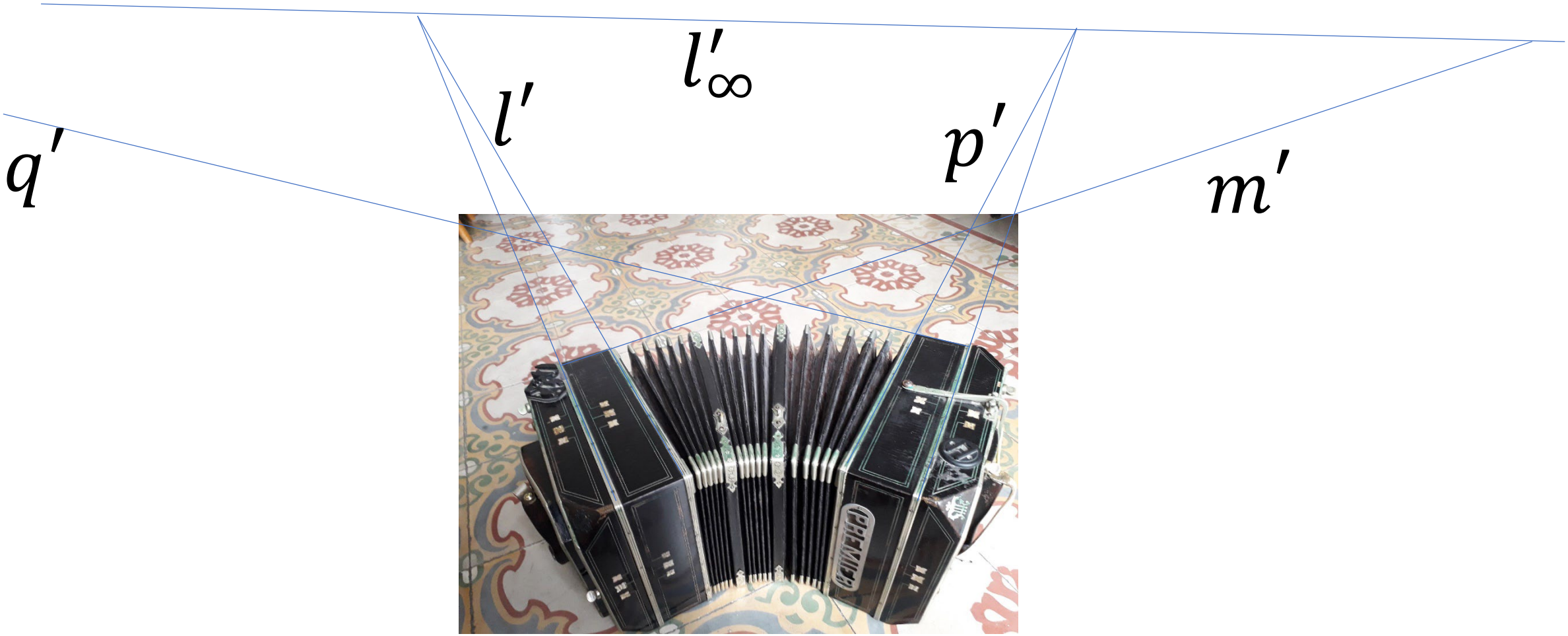


Vanishing line: image of the line at the infinity

l'_∞



known angles (pairs of perpendicular lines)



4 linear constraints on $C_\infty^{*\prime}$

$l'^T C_\infty^{*\prime} m' = 0$ (from orthogonal pair: 1 constr)

$p'^T C_\infty^{*\prime} q' = 0$ (from orthogonal pair: 1 constr)

$$0 = C_\infty^{*\prime} l'_\infty = (I' J'^T + J' I'^T) l'_\infty =$$
$$= I'(J'^T l'_\infty) + \mathrm{J}'(I'^T l'_\infty) = \mathbf{0} \quad (2 \text{ constr})$$

0 0

line at the infinity l'_∞ = RNS($C_\infty^{*\prime}$)

Image rectification

- From the above constraints → find $C_\infty^{*\prime} = \boldsymbol{I'J'^T} + \boldsymbol{J'I'^T}$
- Singular value decomposition

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{SR}^{-1} C_\infty^* H_{SR}^{-T}$$

- Rectifying transformation (from svd output U)

$$H_{SR} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

- Euclidean reconstructed model $M_S = H_{SR} \ * \text{Image}$

reconstructed upper face
(+ all planes parallel to it, e.g. the ground floor)



2D reconstruction: stratified method

Stratified method

1. First step: affine reconstruction - from projective to affine

then

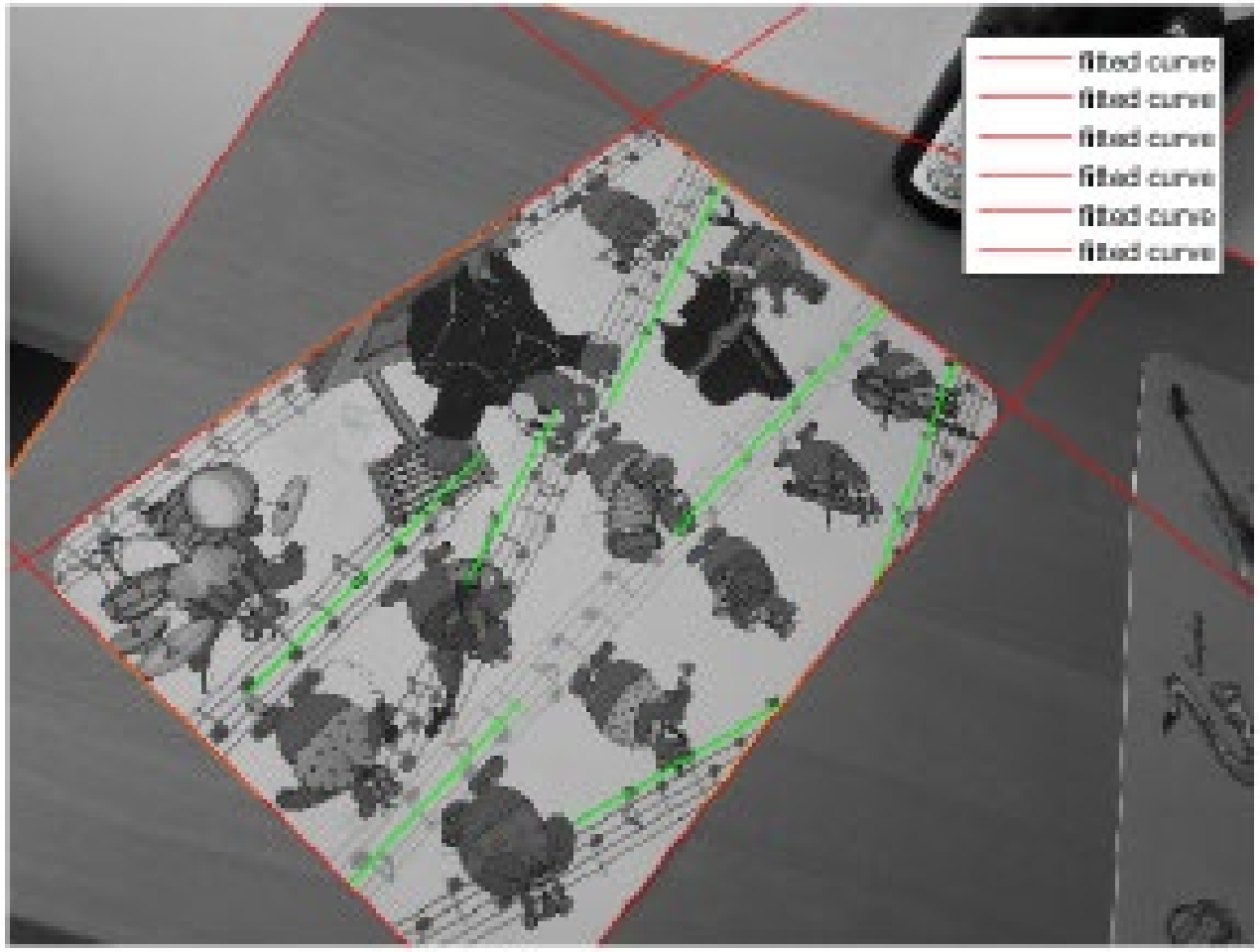
2. Second step: shape reconstruction - from affine to metric
(= image rectification or metric or euclidean reconstruction)

- sometimes reduces numerical errors

Stratified method

1. affine reconstruction - from projective to affine

Affine rectification



Vanishing line l'_∞ from vanishing points

$\mathbf{v}_1 = l_1 \times l_2$

$l'_\infty = \mathbf{v}_1 \times \mathbf{v}_2$

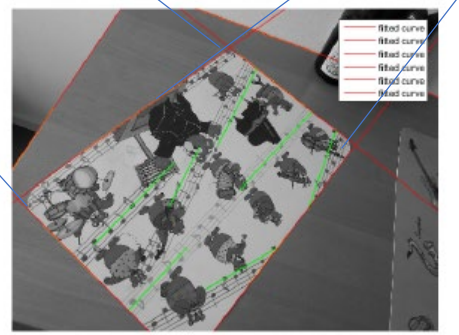
$\mathbf{v}_2 = l_3 \times l_4$

l_1

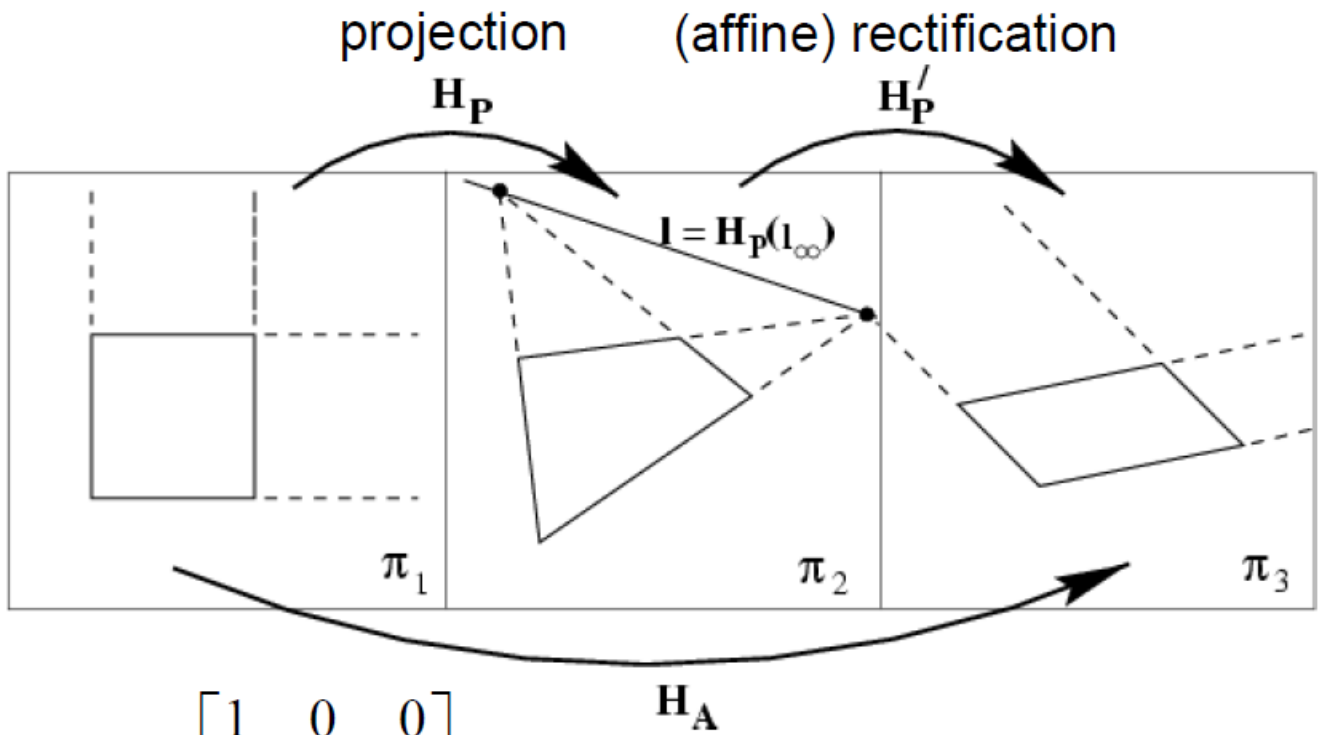
l_2

l_3

l_4



Affine properties from images



$$\mathbf{H'}_P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix} \mathbf{H}_A \qquad \mathbf{l'}_\infty = \begin{bmatrix} l_1 & l_2 & l_3 \end{bmatrix}^T, l_3 \neq 0$$

in fact, any point x on l'_∞ is mapped to a point at the ∞

Affine rectification

- Apply the above mapping $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{vmatrix}$ to image lines and points

- while following transformation rules for lines and points respectively

In this phase, no need to use information about conic dual to circular points

Affinely rectified image



direct rectification versus stratified rectification

Metric properties from images

$$
\begin{aligned}
\mathbf{C}_\infty^{*\prime} &= (\mathbf{H}_P\mathbf{H}_A\mathbf{H}_S)\mathbf{C}_\infty^{*}(\mathbf{H}_P\mathbf{H}_A\mathbf{H}_S)^{\mathsf{T}} \\
&= (\mathbf{H}_P\mathbf{H}_A)\mathbf{H}_S\mathbf{C}_\infty^{*}\mathbf{H}_S^{\mathsf{T}}(\mathbf{H}_P\mathbf{H}_A)^{\mathsf{T}} \\
&= (\mathbf{H}_P\mathbf{H}_A)\mathbf{C}_\infty^{*}(\mathbf{H}_P\mathbf{H}_A)^{\mathsf{T}} \\
&= \begin{bmatrix} \mathbf{K}\mathbf{K}^{\mathsf{T}} & \mathbf{K}^{\mathsf{T}}\mathrm{v} \\ \mathrm{v}^{\mathsf{T}}\mathbf{K} & \mathrm{v}^{\mathsf{T}}\mathrm{v} \end{bmatrix}
\end{aligned}
$$

Rectifying transformation from SVD

$$
\mathbf{C}_\infty^{*\prime} = \mathbf{U}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}\mathbf{U}^{\mathsf{T}} \qquad \mathbf{H} = \mathbf{U}^{\mathsf{T}}
$$

after affine rectification

- Relationship between

original conic dual to the circular points $C_\infty^* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

and

affine(ly rectified) image $C_\infty^{*\prime}$ of C_∞^* is **affine**

$$C_\infty^{*\prime} = H_A C_\infty^* H_A^T = \begin{bmatrix} G & \boldsymbol{t} \\ \mathbf{0^T} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} G^T & \mathbf{0} \\ \boldsymbol{t}^T & 1 \end{bmatrix} = \begin{bmatrix} GG^T & 0 \\ \mathbf{0^T} & 0 \end{bmatrix}$$

G **: unknown** → first estimate GG^T

after affine rectification

- **Important Property** after an affine rectification, $C_\infty^{*\prime}$ can be written as

$$C_\infty^{*\prime} = \begin{bmatrix} GG^\intercal & \mathbf{0} \\ \mathbf{0}^\intercal & 0 \end{bmatrix}$$

Where $GG^\intercal$ is a 2×2 invertible matrix (cfr HZ example 2.26)

- this is the image of the canonical C_∞^* through an affine transformation since similarities do not change C_∞^*, and projective transformation has been removed by the affine rectification
- We can take G upper triangular: $G = \begin{bmatrix} g_{11} & g_{12} \\ 0 & g_{22} \end{bmatrix}$

$$\rightarrow GG^\intercal = \begin{bmatrix} g_{11}^2 + g_{12}^2 & g_{12}g_{22} \\ g_{12}g_{22} & g_{22}^2 \end{bmatrix}$$

Stratified method

1. First step: affine reconstruction - from projective to affine

then

2. Shape reconstruction - from affine to metric

Remember: costraints from known angles

$$\cos\theta = \frac{l^T C_\infty^* m}{\sqrt{l^T C_\infty^* l \;\; m^T C_\infty^* m}} = \frac{l'^T C_\infty^{*\prime} m'}{\sqrt{l'^T C_\infty^{*\prime} l' \;\; m'^T C'^*_\infty m'}}$$

known angle → equation on $C_\infty^{*\prime}$

and for perpendicular lines $l'^T C_\infty^{*\prime} m' = 0$ **linear**

image rectification from orthogonal lines

- If $\boldsymbol{l}'$ and $\boldsymbol{m}'$ are images two orthogonal lines $\boldsymbol{l}$ and $\boldsymbol{m}$ (in the 3D world), then

$$\boldsymbol{l}'^{\mathsf{T}} C_\infty^{*\prime} \boldsymbol{m}' = 0$$

- Let us use this information to compute $C_\infty^{*\prime}$, thus compute H
- Remember that, after an affine rectification

$$C_\infty^{*\prime} = \begin{bmatrix} GG^{\mathsf{T}} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & 0 \end{bmatrix}$$

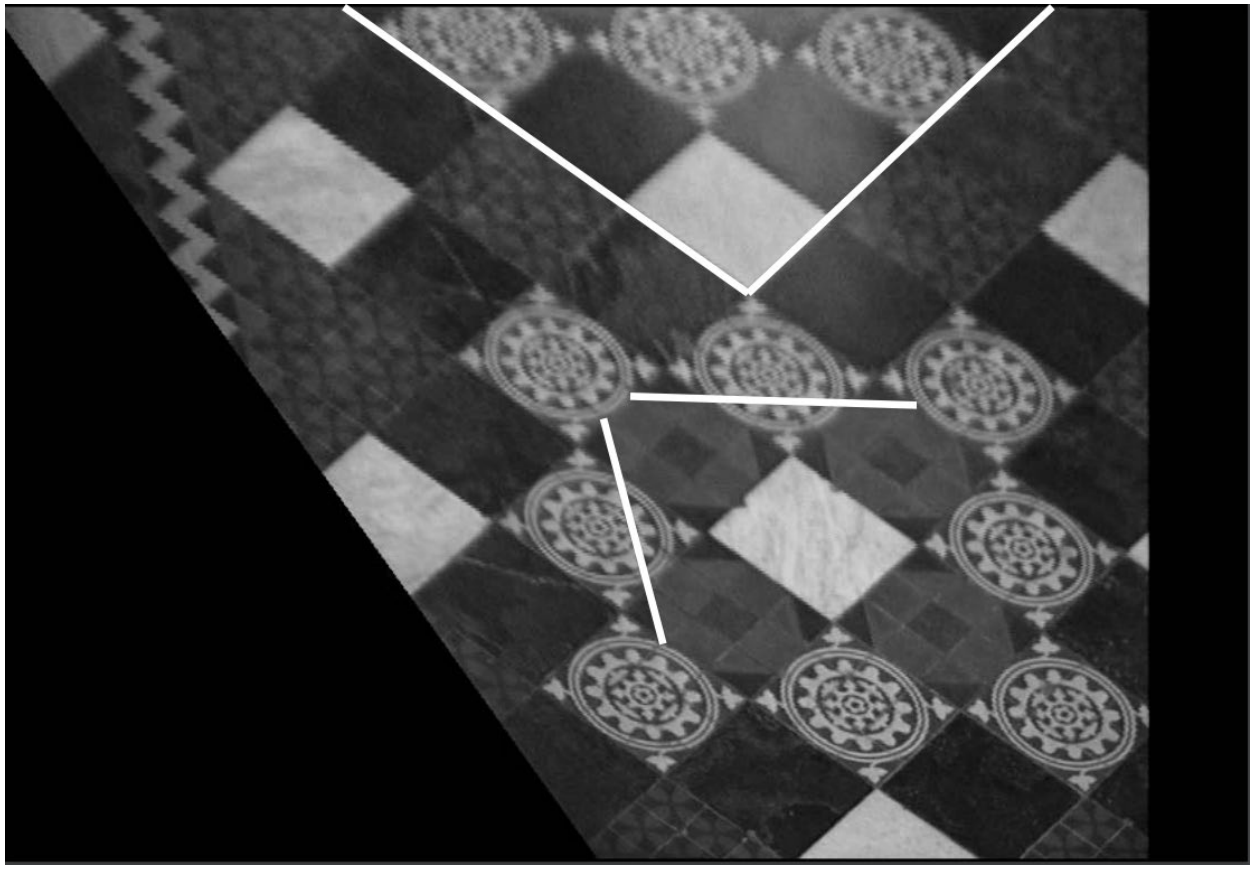
- Since GG^{T} **is a symmetric homogeneous matrix** $(GG^{\mathsf{T}} = G^{\mathsf{T}}G)$, there are only 2 unknowns to identify $C_\infty^{*\prime}$
- Each pair of orthogonal lines yields a single equation $\boldsymbol{l}'(1{:}2)^{\mathsf{T}} GG^{\mathsf{T}}\, \boldsymbol{m}'(1{:}2) = 0$

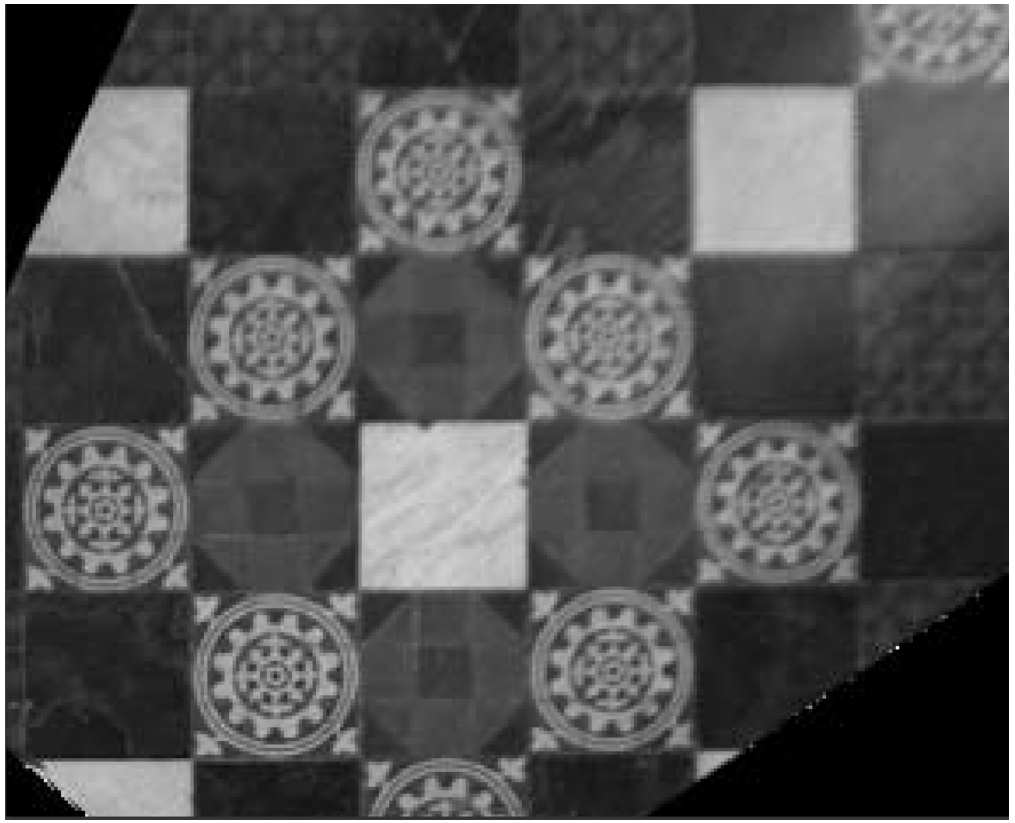
$$= l'_1 m'_1 (g_{11}^2 + g_{12}^2) + (l'_1 m'_2 + l'_2 m'_1) g_{12} g_{22} + l'_2 m'_2 g_{22}^2 = 0$$

Two pairs of orthogonal lines are enough to identify $\boldsymbol{C}_\infty^{*\prime}$

Stratified Rectification from Orthogonal Lines

- $\boldsymbol{l}'(1:2)^{\mathsf{T}} S\, \boldsymbol{m}'(1:2) = 0$
- Two (different, i.e. not parallel) pairs of orthogonal lines are enough to estimate GG^{T}, (thus G through Choleski factorisation)





Hartley Zisserman Fig.2.17

$$l'^T \begin{vmatrix} GG^T & 0 \\ 0 & 0 \end{vmatrix} m' = 0$$



$$l'_1 m'_1 (g_{11}^2 + g_{12}^2) + (l'_1 m'_2 + l'_2 m'_1) g_{12} g_{22} + l'_2 m'_2 g_{22}^2 = 0$$

$$l'^T \begin{vmatrix} GG^T & 0 \\ 0 & 0 \end{vmatrix} m' = 0$$



$$l'_1 m'_1 (g_{11}^2 + g_{12}^2) + (l'_1 m'_2 + l'_2 m'_1) g_{12} g_{22} + l'_2 m'_2 g_{22}^2 = 0$$

$$p'^T \begin{vmatrix} GG^T & 0 \\ 0 & 0 \end{vmatrix} q' = 0$$



$$p'_1 q'_1 (g_{11}^2 + g_{12}^2) + (p'_1 q'_2 + p_2 q'_1) g_{12} g_{22} + p'_2 q'_2 g_{22}^2 = 0$$

From affine reconstruction to metric reconstruction

After affine reconstruction, the recovered conic dual to the circular point is in the form

$$C_\infty^{*\prime} = \begin{bmatrix} GG^T & 0 \\ 0 & 1 \end{bmatrix} \text{ i.e, } C_\infty^{*\prime} = H_A C_\infty^* H_A^T = \begin{bmatrix} G & \boldsymbol{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} G^T & \mathbf{0} \\ \boldsymbol{t}^T & 1 \end{bmatrix}$$

$$\rightarrow H_{rect} = H_A^{-1} = \begin{bmatrix} G & t \\ \mathbf{0}^T & 1 \end{bmatrix}$$

where GG^T is estimated from two pairs of orthogonal lines, and G is derived from Cholesky Factorisation of GG^T.

Then the image is rectified by applying $H_{rect} = \begin{bmatrix} G & t \\ \mathbf{0}^T & 1 \end{bmatrix}^{-1}$

NOTE: no SVD is needed

Result of shape reconstruction



same stratified approach for

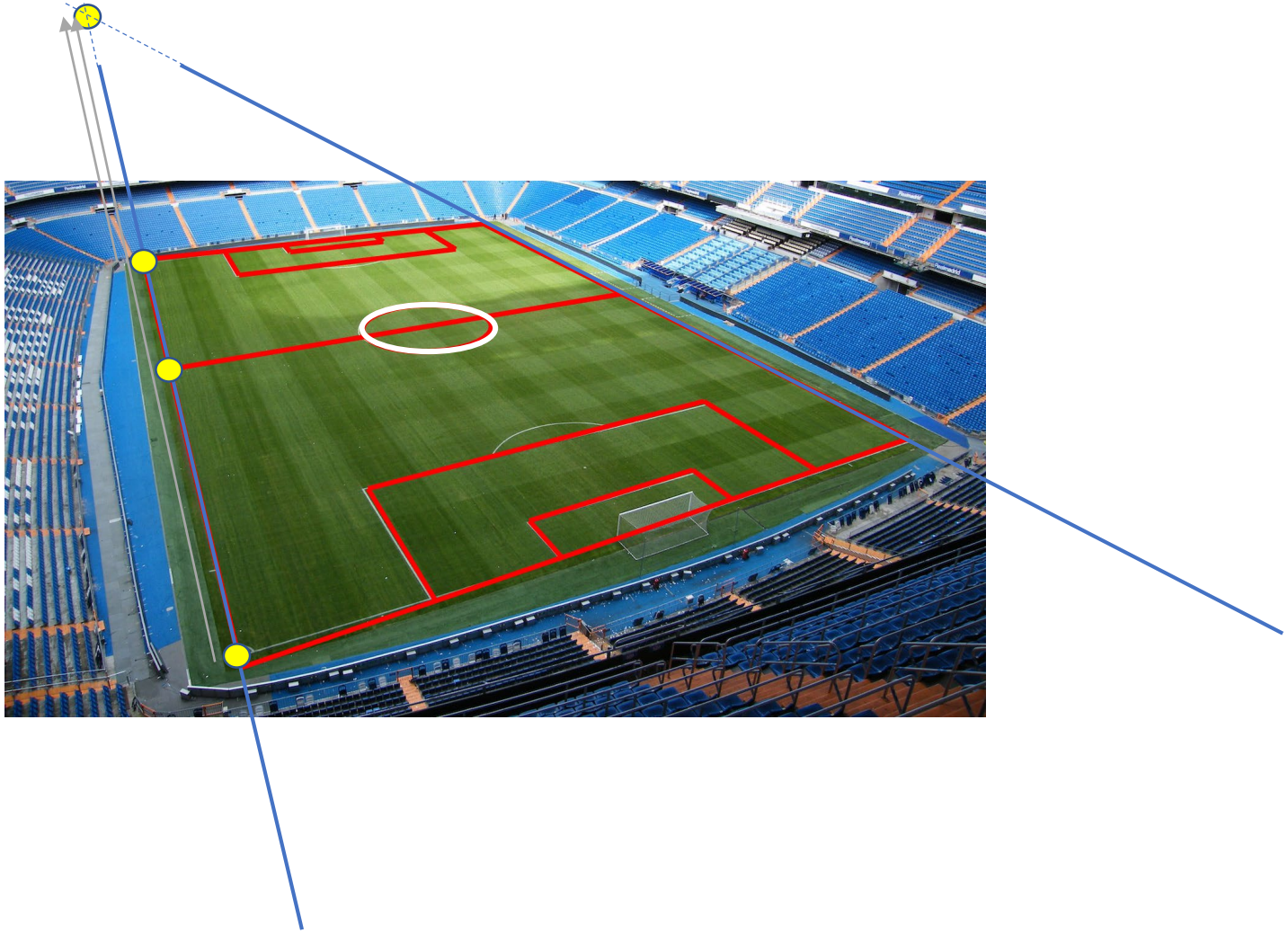


reconstructed upper face

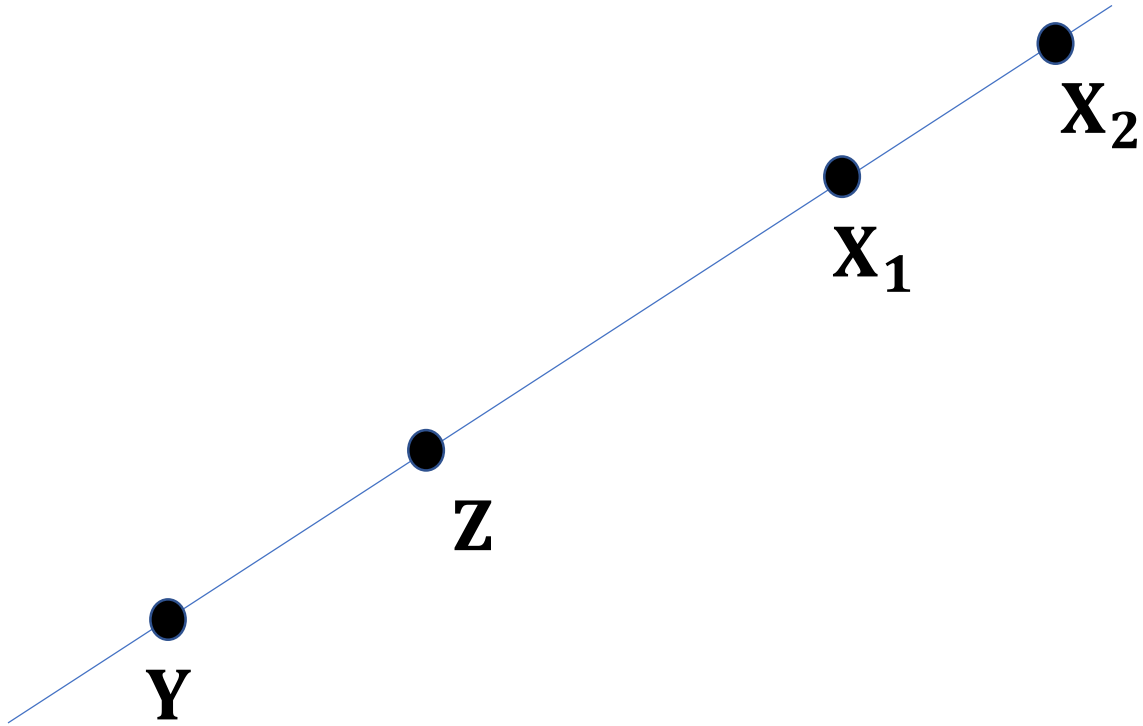


APPENDIX:
the cross ratio

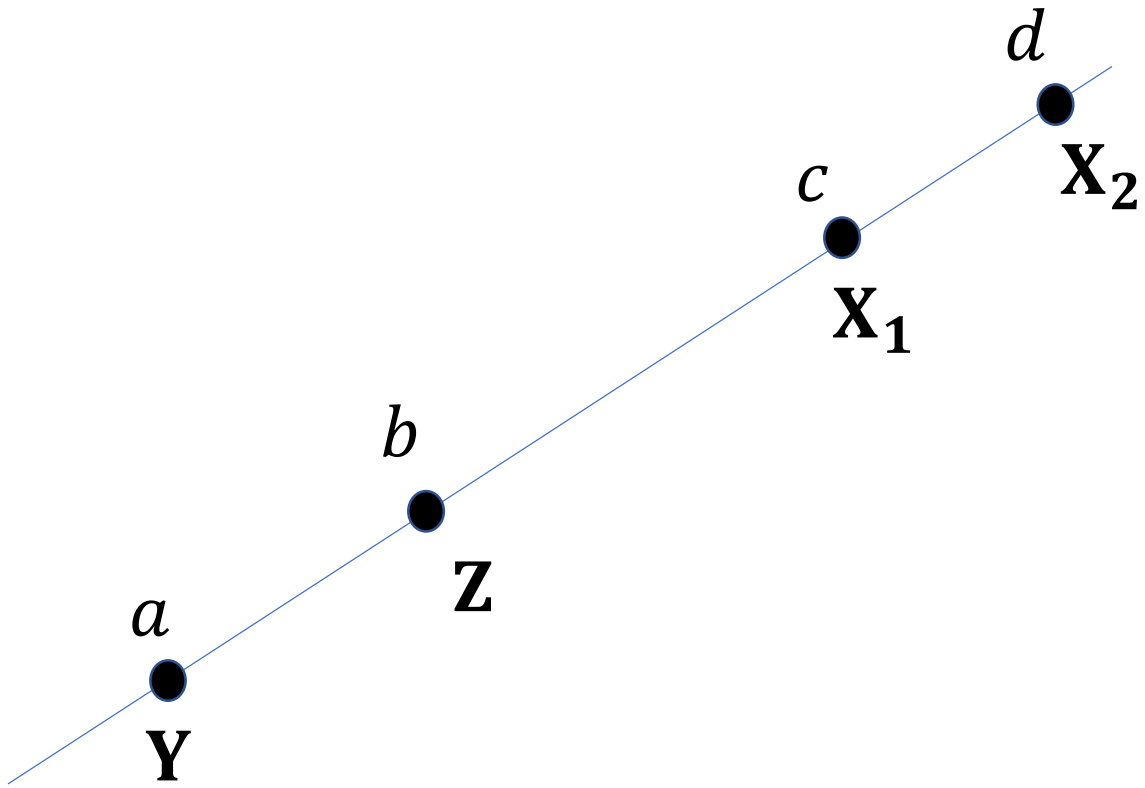
Let’s study the ratio of ratios (for 4 colinear points) → **cross ratio**



Cross ratio
of 4 colinear points

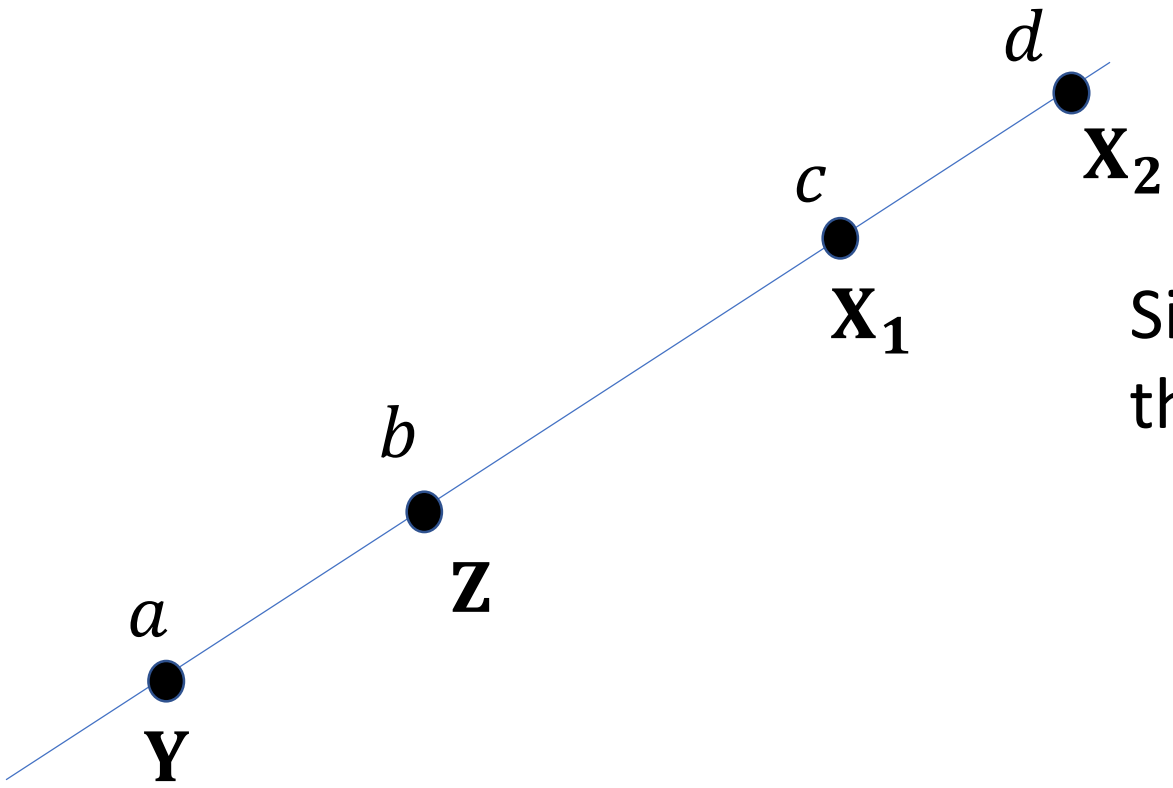


Define a coordinate (abscissa) along the line



Cross ratio of the 4-tuple of colinear points

$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{c-a}}{\boldsymbol{c-b}} \Big/ \frac{\boldsymbol{d-a}}{\boldsymbol{d-b}}$$



Since $\mathbf{X_1}$ and $\mathbf{X_2}$ are colinear with $\mathbf{Y}, \mathbf{Z}$ they are linear combinations of $\mathbf{Y}$ and $\mathbf{Z}$

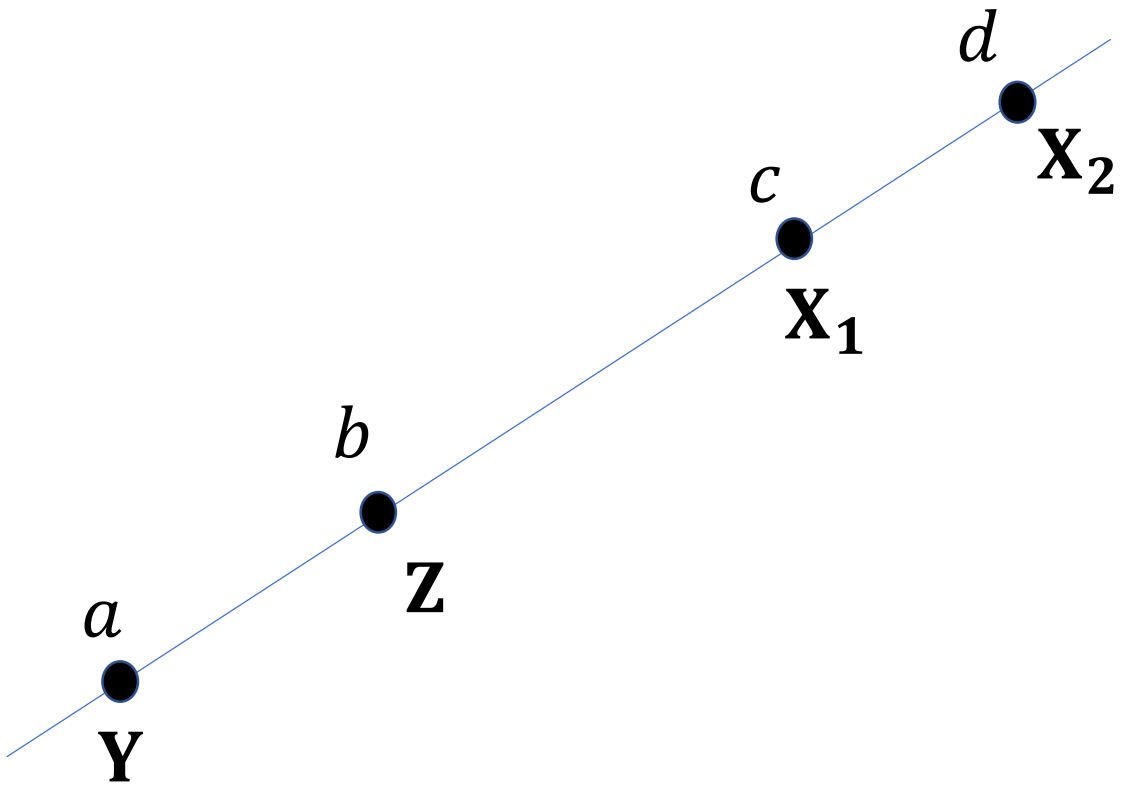
$$\mathbf{X_1} = \alpha_1 \mathbf{Y} + \beta_1 \mathbf{Z}$$

and

$$\mathbf{X_2} = \alpha_2 \mathbf{Y} + \beta_2 \mathbf{Z}$$

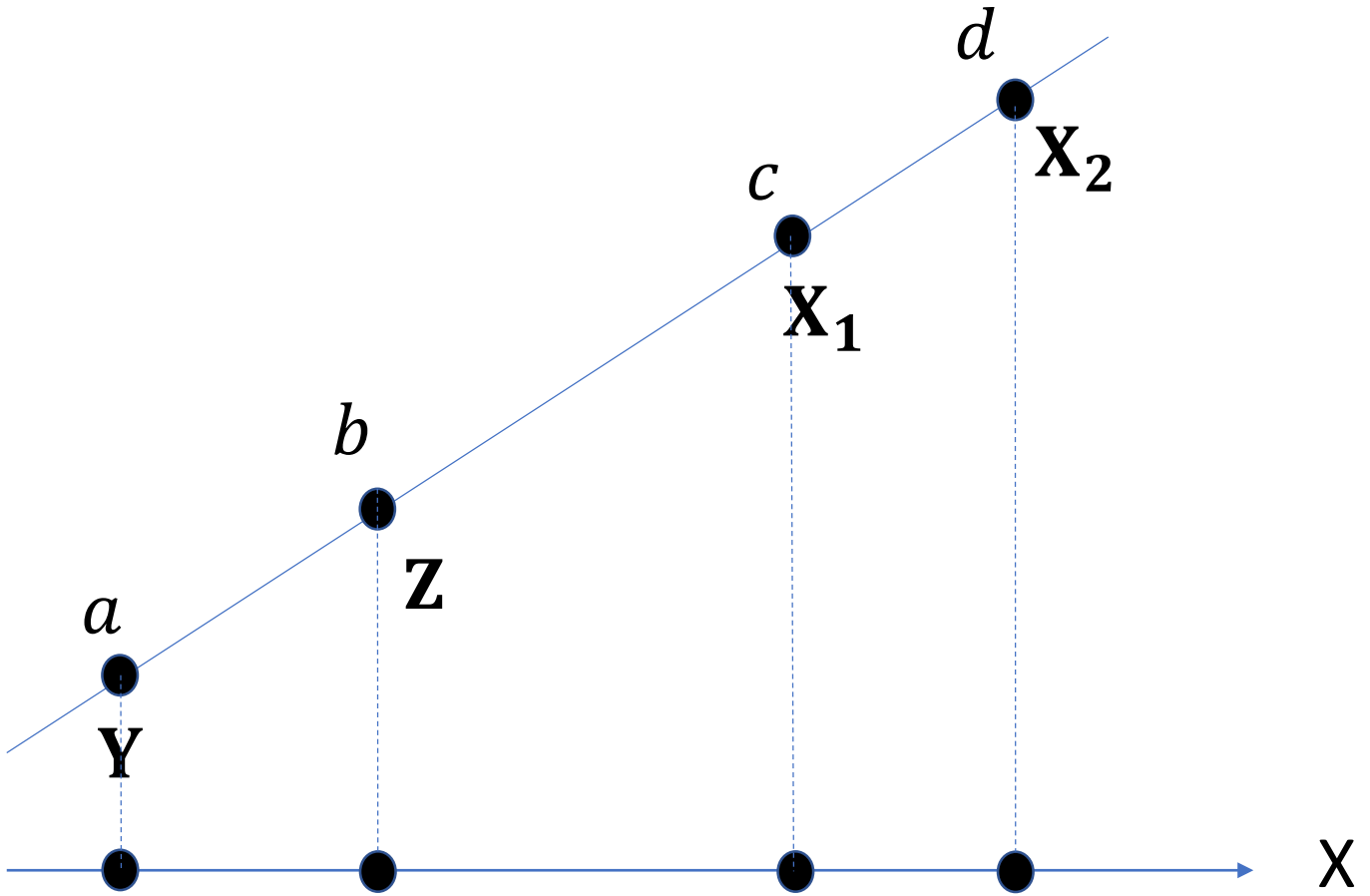
Cross ratio: A result

$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{c-a}}{\boldsymbol{c-b}} \Big/ \frac{\boldsymbol{d-a}}{\boldsymbol{d-b}} = \frac{\beta_1/\alpha_1}{\beta_2/\alpha_2}$$



Proof: since the abscissae are proportional to, e.g., the X cartesian coordinates, we can replace the abscissae by the X coordinates (let us reuse the same names)

$$CR_{Y,Z,X_1,X_2} = \frac{c-a}{c-b} \Big/ \frac{d-a}{d-b}$$



Proof: since the abscissae are proportional to, e.g., the X cartesian coordinates, we can replace the abscissae by the X coordinates (let us the same names …)

Let $\mathbf{Y} = \begin{bmatrix} \boldsymbol{y} \\ * \\ \boldsymbol{v} \end{bmatrix}, \mathbf{Z} = \begin{bmatrix} \boldsymbol{z} \\ * \\ \boldsymbol{w} \end{bmatrix}$: then $\boldsymbol{X_1} = \begin{bmatrix} \boldsymbol{\alpha_1 y + \beta_1 z} \\ * \\ \boldsymbol{\alpha_1 v + \beta_1 w} \end{bmatrix}$ and $\boldsymbol{X_2} = \begin{bmatrix} \boldsymbol{\alpha_2 y + \beta_2 z} \\ * \\ \boldsymbol{\alpha_2 v + \beta_2 w} \end{bmatrix}$

The difference between the X coordinates of $\boldsymbol{X_1}$ and $\mathbf{Y}$ is

$$c - a = \frac{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v} - (\boldsymbol{\alpha_1 v + \beta_1 w})\boldsymbol{y}}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v}} = \frac{\boldsymbol{\beta_1}(\boldsymbol{zv - yw})}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v}}$$

The difference between the X coordinates of $\boldsymbol{X_1}$ and $\mathbf{Z}$ is

$$c - b = \frac{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w} - (\boldsymbol{\alpha_1 v + \beta_1 w})\boldsymbol{z}}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w}} = \frac{-\boldsymbol{\alpha_1}(\boldsymbol{zv - yw})}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w}}$$

The difference between the X coordinates of $\boldsymbol{X_1}$ and $\mathbf{Y}$ is

$$c - a = \frac{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v} - (\boldsymbol{\alpha_1 v + \beta_1 w})\boldsymbol{y}}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v}} = \frac{\boldsymbol{\beta_1}(\boldsymbol{zv - yw})}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v}}$$

The difference between the X coordinates of $\boldsymbol{X_1}$ and $\mathbf{Z}$ is

$$c - b = \frac{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w} - (\boldsymbol{\alpha_1 v + \beta_1 w})\boldsymbol{z}}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w}} = \frac{-\boldsymbol{\alpha_1}(\boldsymbol{zv - yw})}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w}}$$

The ratio between these differences is $\frac{c-a}{c-b} = -\frac{\boldsymbol{\beta_1}}{\boldsymbol{\alpha_1}}\frac{w}{v}$ and, similarly,

$$\frac{d-a}{d-b} = -\frac{\boldsymbol{\beta_2}}{\boldsymbol{\alpha_2}}\frac{w}{v}$$

Thus the cross ratio of the four colinear points $\boldsymbol{Y, Z, X_1, X_2}$ is

$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{c-a}}{\boldsymbol{c-b}} \Big/ \frac{\boldsymbol{d-a}}{\boldsymbol{d-b}} = \frac{\boldsymbol{\beta_1}}{\boldsymbol{\alpha_1}} \Big/ \frac{\boldsymbol{\beta_2}}{\boldsymbol{\alpha_2}}$$

Q.E.D.

Cross ratio invariance
under projective mappings





Cross ratio of four colinear points:

$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{\beta_1}}{\boldsymbol{\alpha_1}} \Big/ \frac{\boldsymbol{\beta_2}}{\boldsymbol{\alpha_2}}$$

where the third and fourth points $\mathbf{X_1}$ and $\mathbf{X_2}$ are linear combinations of the first two points $\mathbf{Y}$ and $\mathbf{Z}$:

$$\mathbf{X_1} = \propto_1 \mathbf{Y} + \beta_1 \mathbf{Z}$$
$$\mathbf{X_2} = \propto_2 \mathbf{Y} + \beta_2 \mathbf{Z}$$

Apply a projective transformation H to the four points:

$$\mathbf{Y'} = \mathbf{HY}$$
$$\boldsymbol{Z'} = \mathbf{HZ}$$
$$\mathbf{X'_1} = \mathbf{HX_1}$$
$$\mathbf{X'_2} = \mathbf{HX_2}$$

Cross ratio of four colinear points:

$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{\beta_1}}{\boldsymbol{\alpha_1}} \Big/ \frac{\boldsymbol{\beta_2}}{\boldsymbol{\alpha_2}}$$

where the third and fourth points $\mathbf{X_1}$ and $\mathbf{X_2}$ are linear combinations of the first two points $\mathbf{Y}$ and $\mathbf{Z}$:

$$\mathbf{X_1} = \propto_1 \mathbf{Y} + \beta_1 \mathbf{Z}$$
$$\mathbf{X_2} = \propto_2 \mathbf{Y} + \beta_2 \mathbf{Z}$$

Apply a projective transformation H to the four points:

$$\mathbf{Y'} = \mathbf{HY}$$
$$\boldsymbol{Z'} = \mathbf{HZ}$$
$$\mathbf{X'_1} = \boldsymbol{H}\mathbf{X_1} = \boldsymbol{H}(\propto_1 \mathbf{Y} + \beta_1 \mathbf{Z}) = \propto_1 \mathbf{HY} + \beta_1 \mathbf{HZ} = \propto_1 \mathbf{Y'} + \beta_1 \mathbf{Z'}$$
$$\mathbf{X'_2} = \boldsymbol{H}\mathbf{X_2} = \boldsymbol{H}(\propto_2 \mathbf{Y} + \beta_2 \mathbf{Z}) = \propto_2 \mathbf{HY} + \beta_2 \mathbf{HZ} = \propto_2 \mathbf{Y'} + \beta_2 \mathbf{Z'}$$

The linear combination coefficients are the **same** . Therefore their CR is

$$\boldsymbol{CR_{Y',Z',X'_1,X'_2}} = \frac{\boldsymbol{\beta_1}}{\boldsymbol{\alpha_1}} \Big/ \frac{\boldsymbol{\beta_2}}{\boldsymbol{\alpha_2}} = \boldsymbol{CR_{Y,Z,X_1,X_2}}$$

APPENDIX to 2D reconstruction:
accuracy issues in 2D shape reconstruction

accuracy issues in image rectification

1. Noise and numerical errors (already seen)

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T$$

- In principle, given $C_\infty^{*\prime}$

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{rect}^{-1} C_\infty^* H_{rect}^{-T}$$

- But, due to noise and numerical errors, SVD output is

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{rect}^{-1} C_\infty^* H_{rect}^{-T}$$

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = U \begin{bmatrix} \sqrt{a} & 0 & 0 \\ 0 & \sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{a} & 0 & 0 \\ 0 & \sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

$$\rightarrow H_{rect}^{-1} = U \begin{bmatrix} \sqrt{a} & 0 & 0 \\ 0 & \sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow H_{rect} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

accuracy issues in image rectification

2. Poor estimation:

e.g., intersection of two lines, that are too close to eachother



intersection of two lines, that are too close to eachother

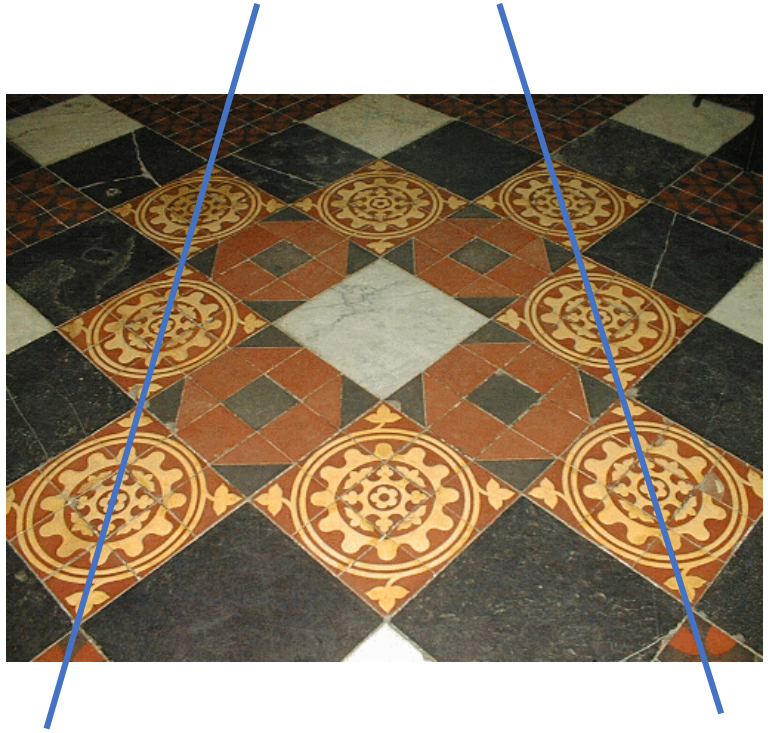
Uncertainty in the intersection point



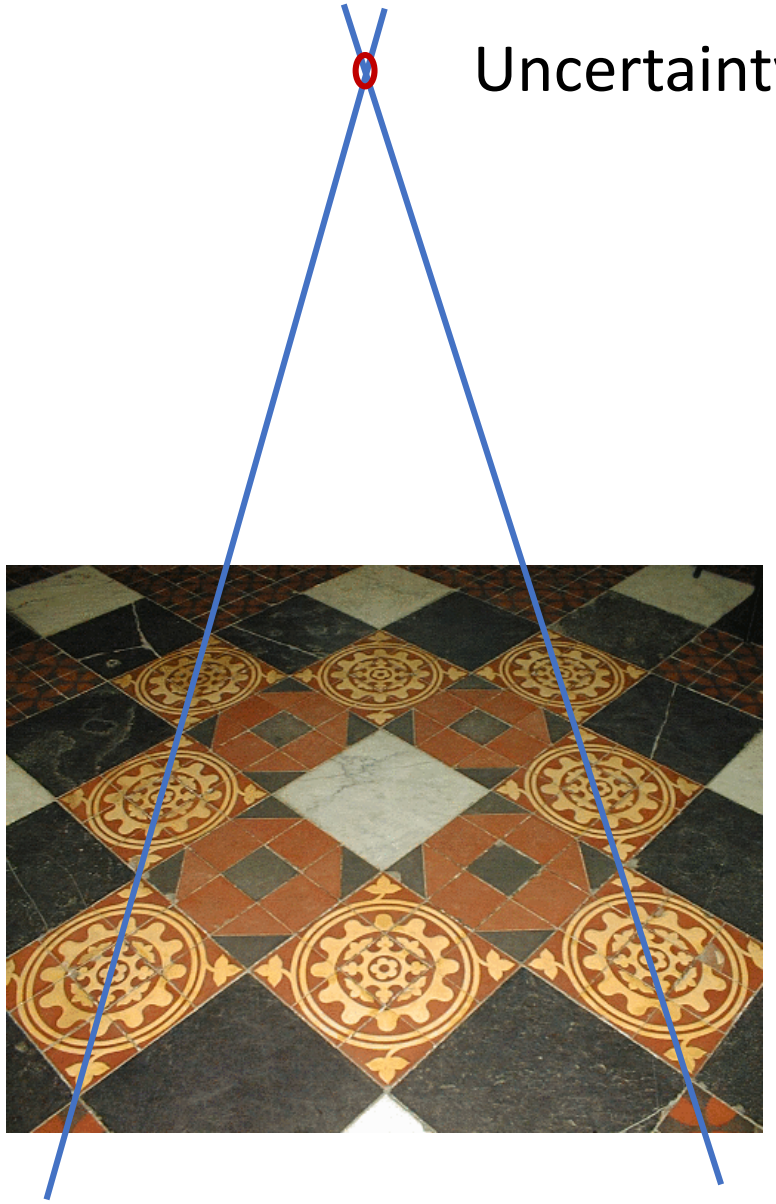
accuracy issues in image rectification

2. Poor estimation:

e.g., intersection of two lines, better **use lines that are far from eachother**



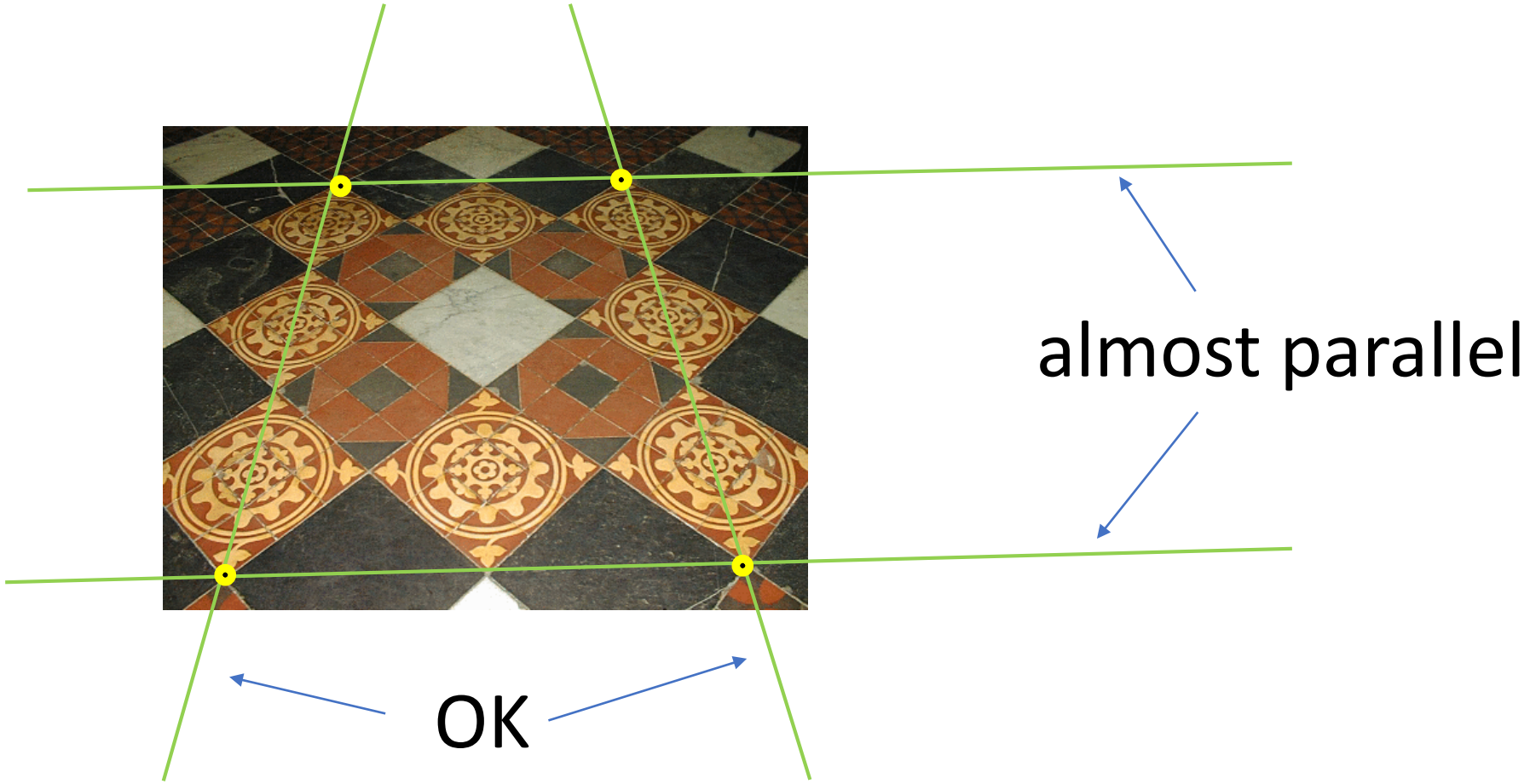
… better **use lines that are far from eachother**



Uncertainty in the intersection point

accuracy issues in image rectification

3. Vanishing point almost at the ∞



Vanishing points towards the ∞

When the images of parallel lines are almost parallel

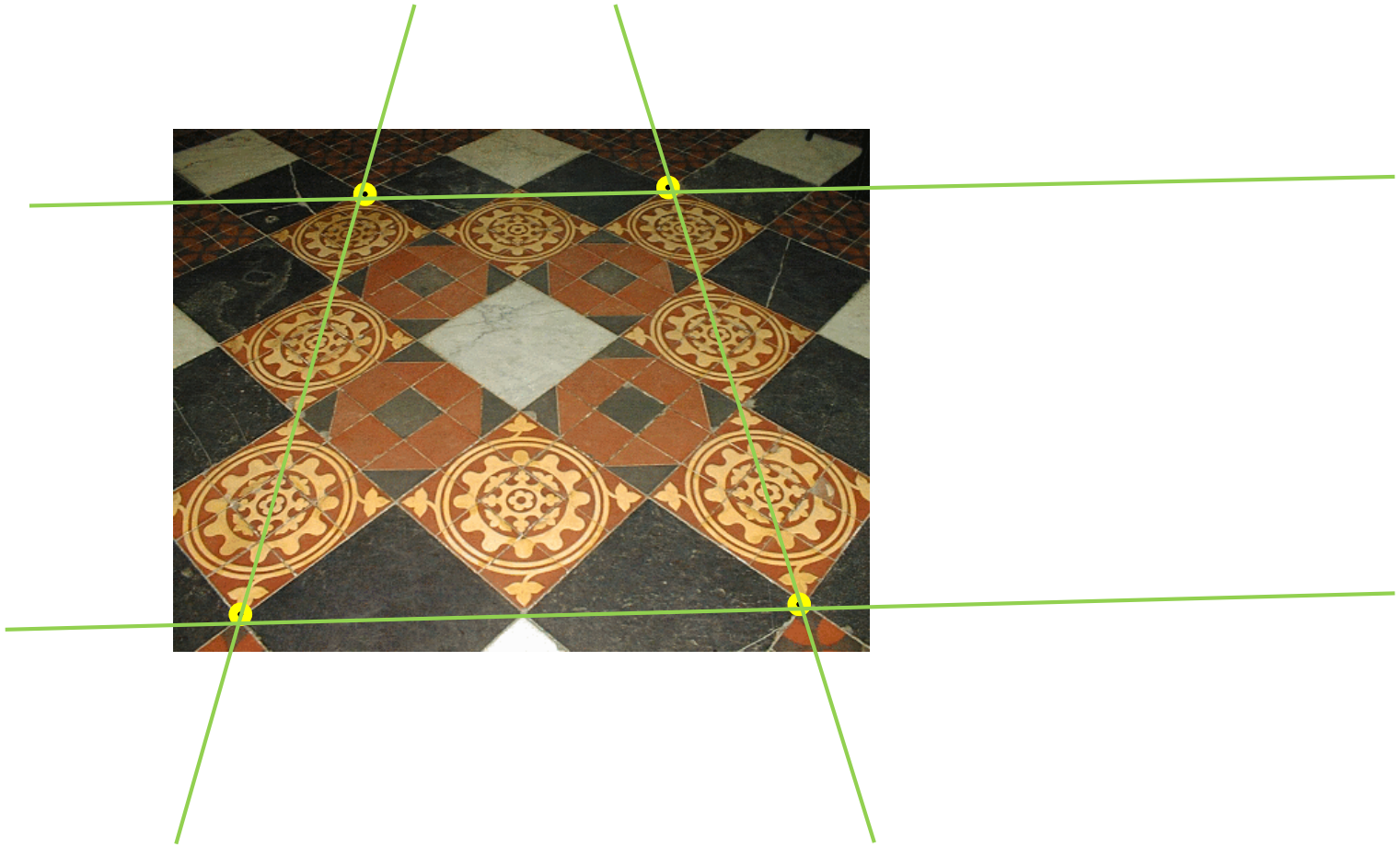
→ Estimate of vanishing points: not accurate

→ estimate of l'_∞ (image of l_∞): not accurate

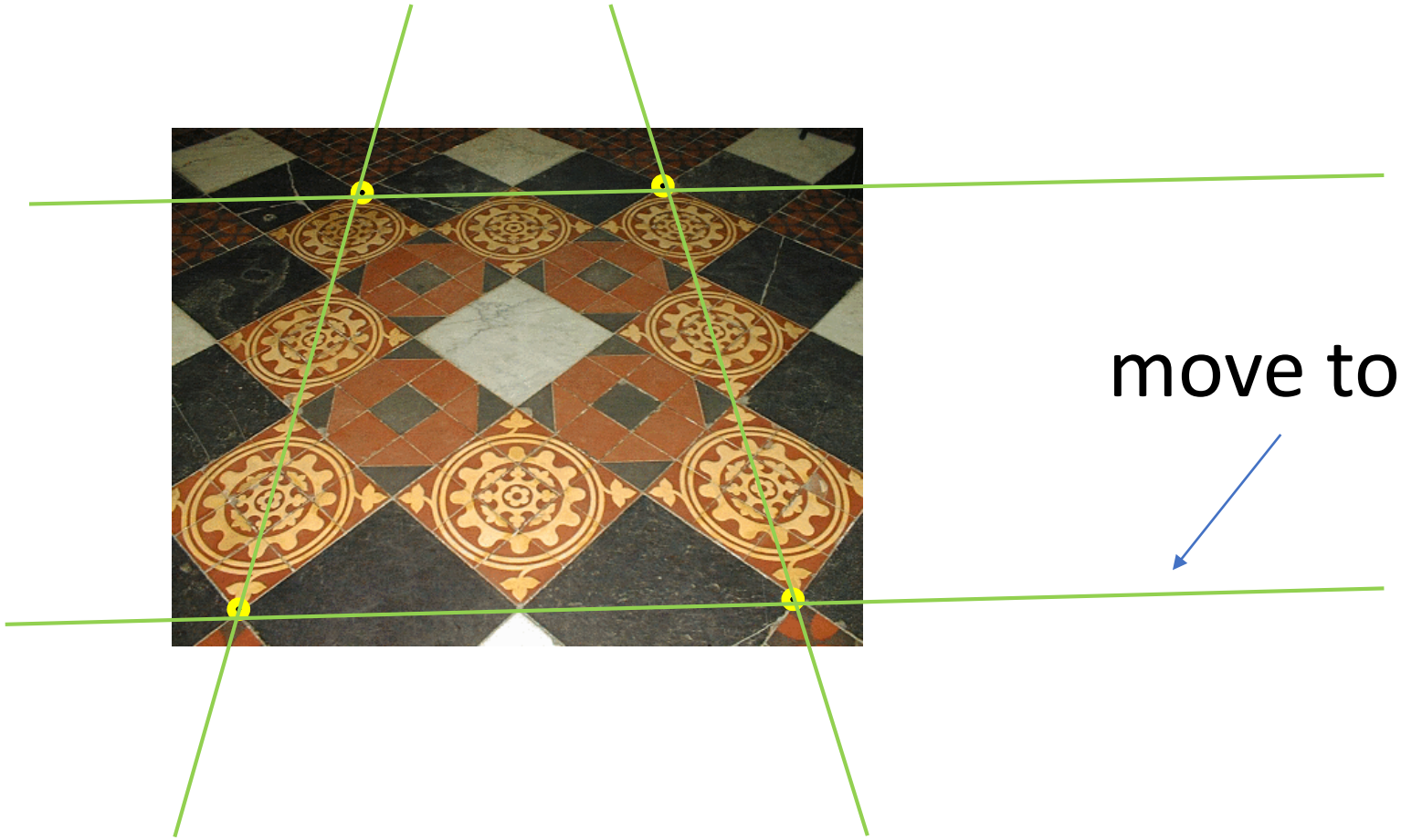
→ affine rectification: not accurate

Stratified «geometric» reconstruction

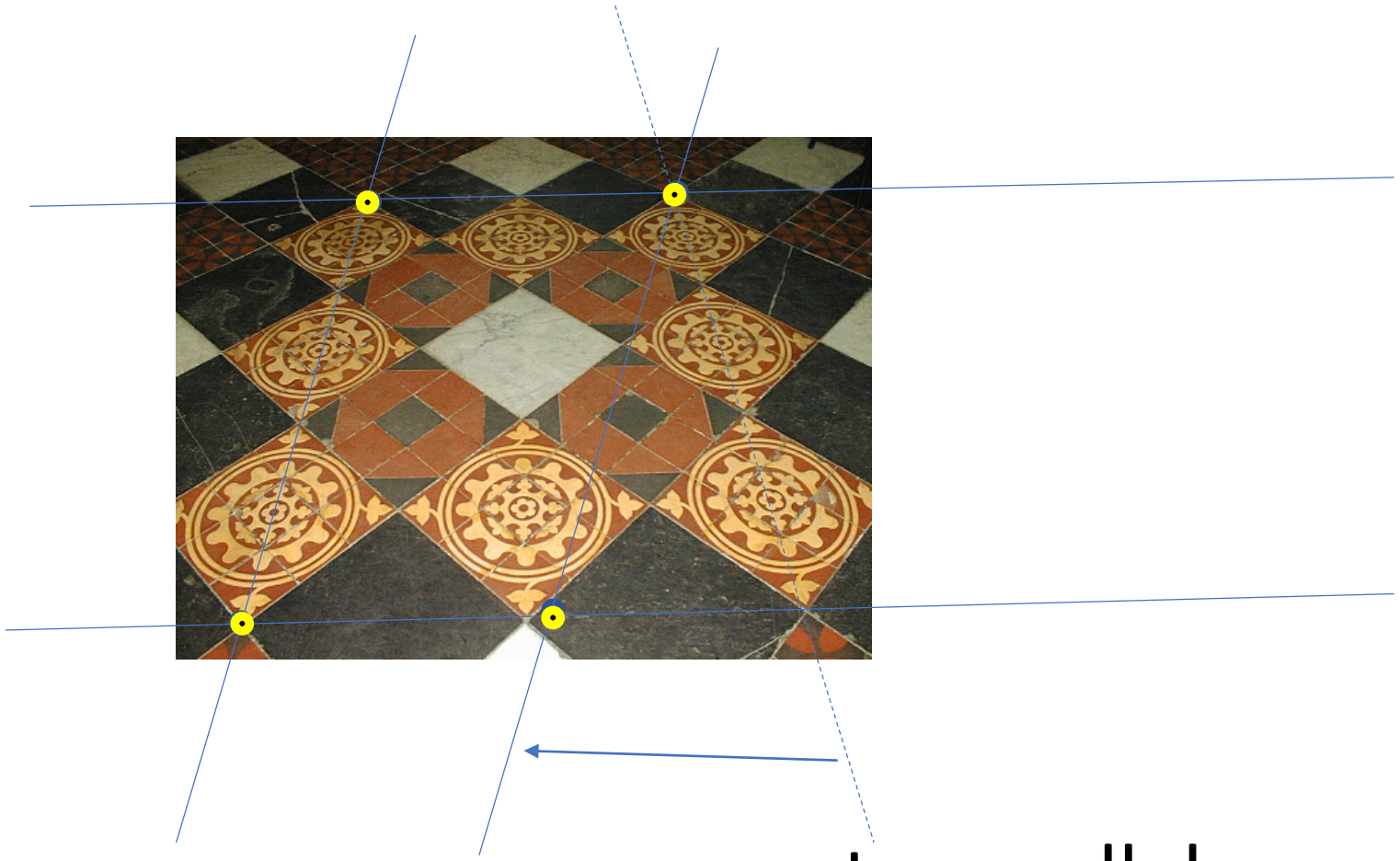
Stratified «geometric» method



1° step: affine rectification

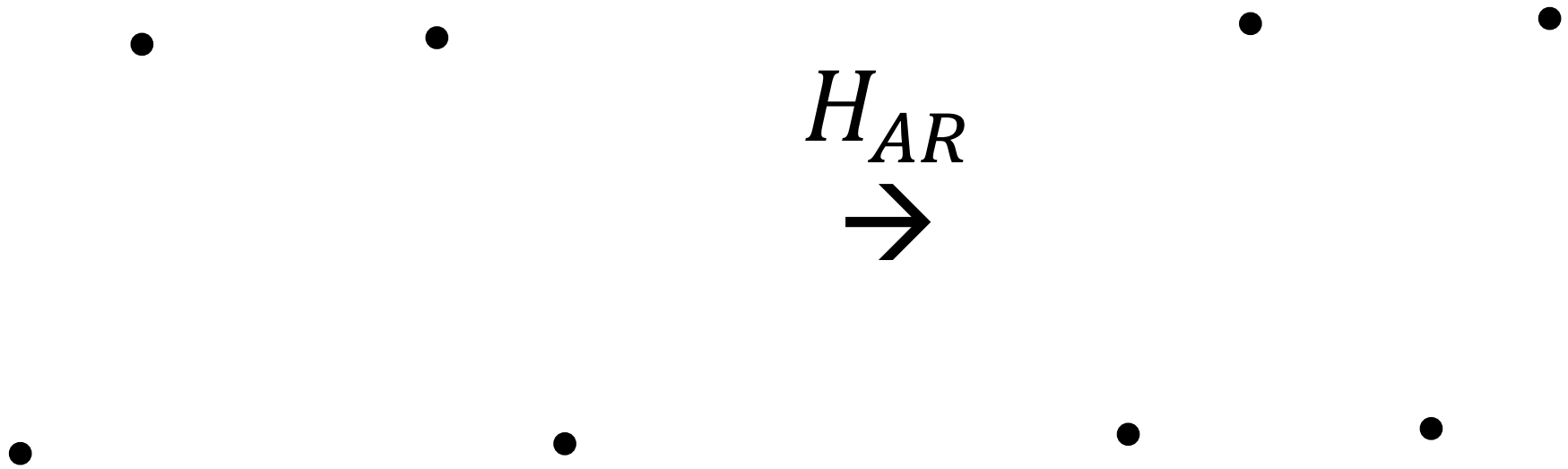


1° step: affine rectification

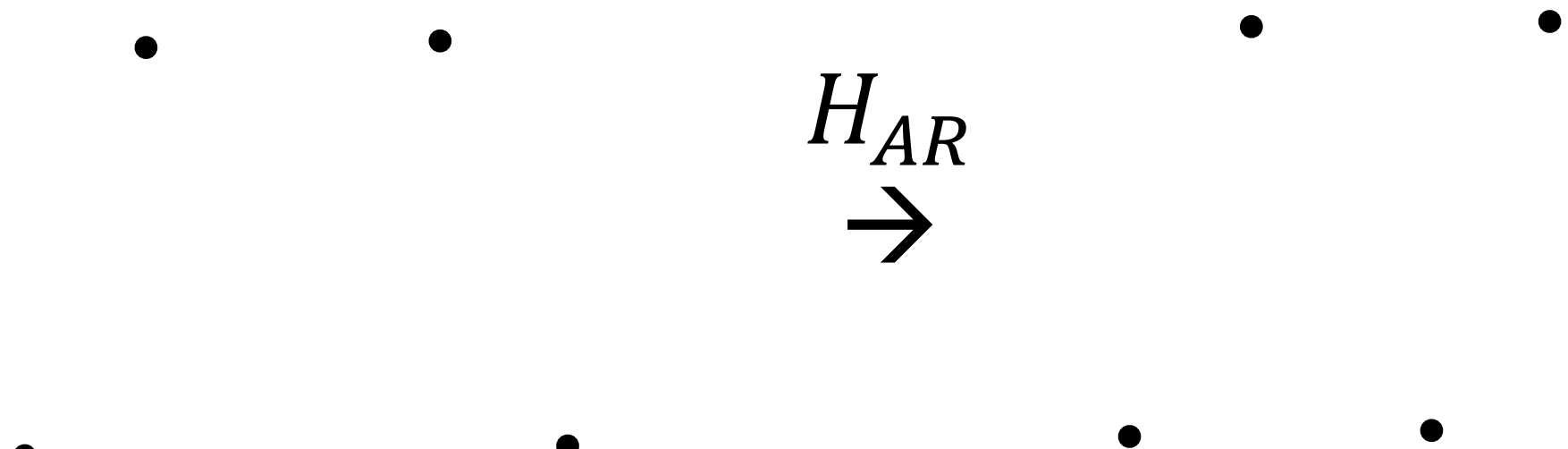


move to parallel

4 points → 4 points HOMOGRAPHY



4 points → 4 points HOMOGRAPHY

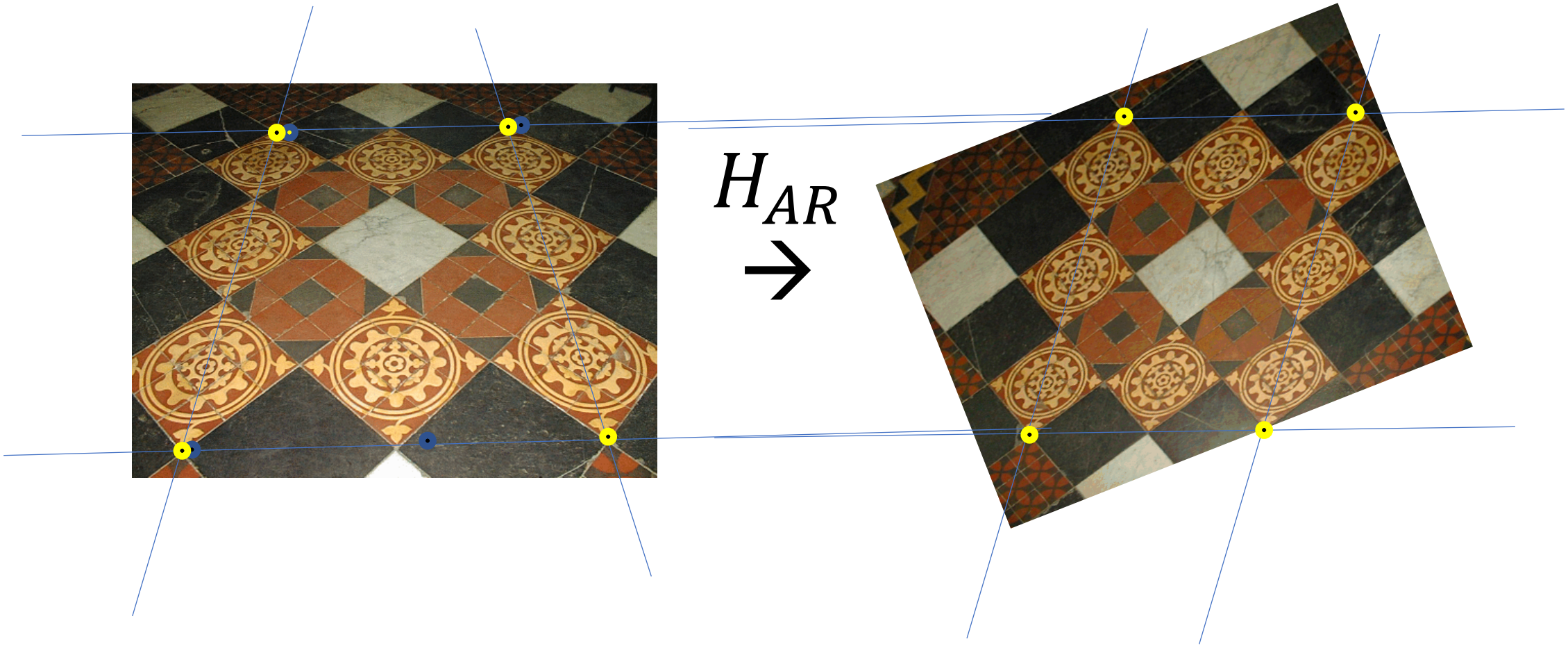


where H_{AR} is the solution of

- $A = H_{AR}\mathrm{A}$
- $B = H_{AR}\mathrm{B}$
- $C = H_{AR}\mathrm{C}$
- $D' = H_{AR}\mathrm{D}$

(homogeneous coordinates)

affine reconstruction



2° step: affine → metric (Euclidean)

two INDEPENDENT constraints:

- two pairs of image lines, that are images of orthogonal lines

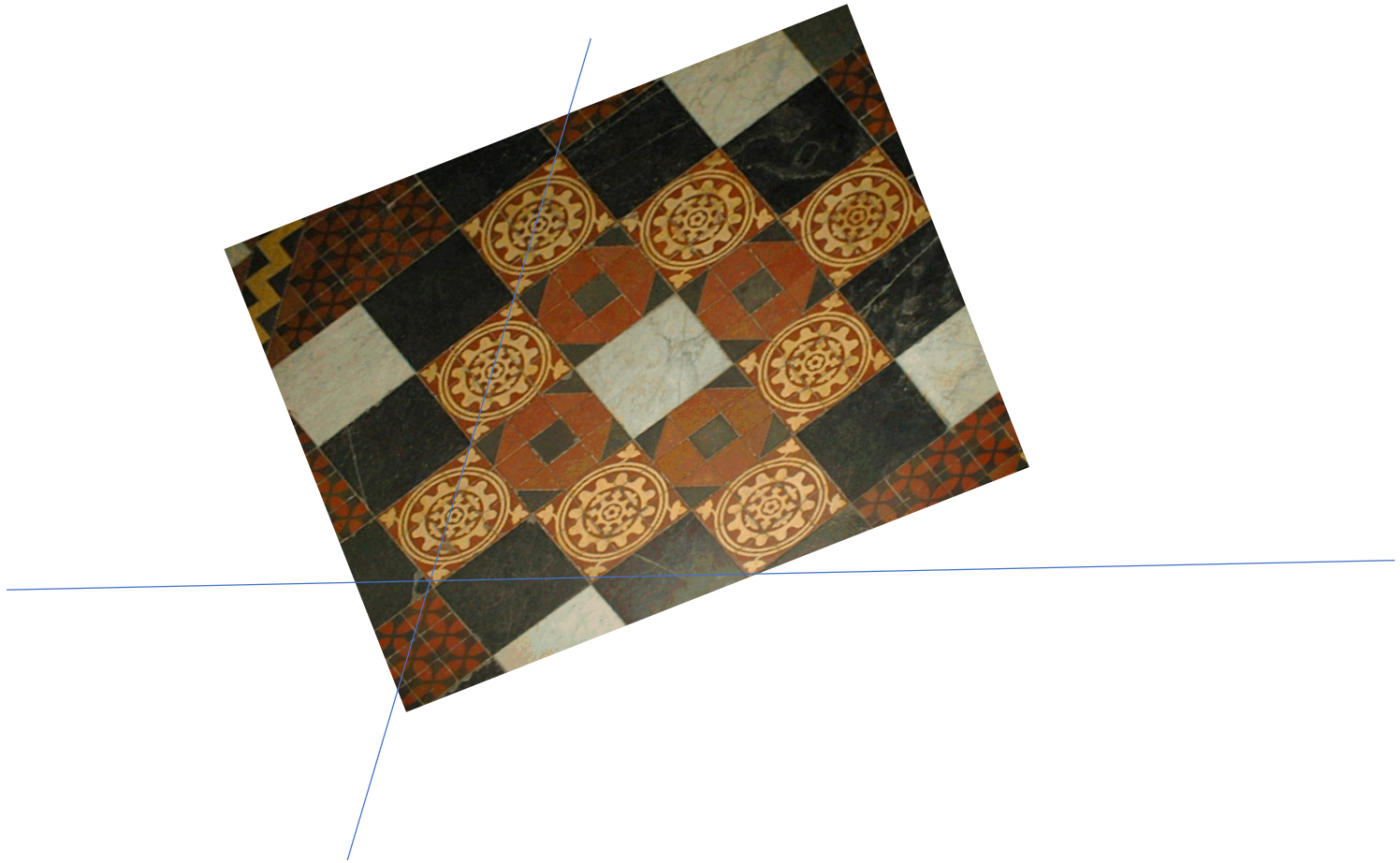
$$l'^T C_\infty^{*\prime} m' = 0$$

$$p'^T C_\infty^{*\prime} q' = 0$$

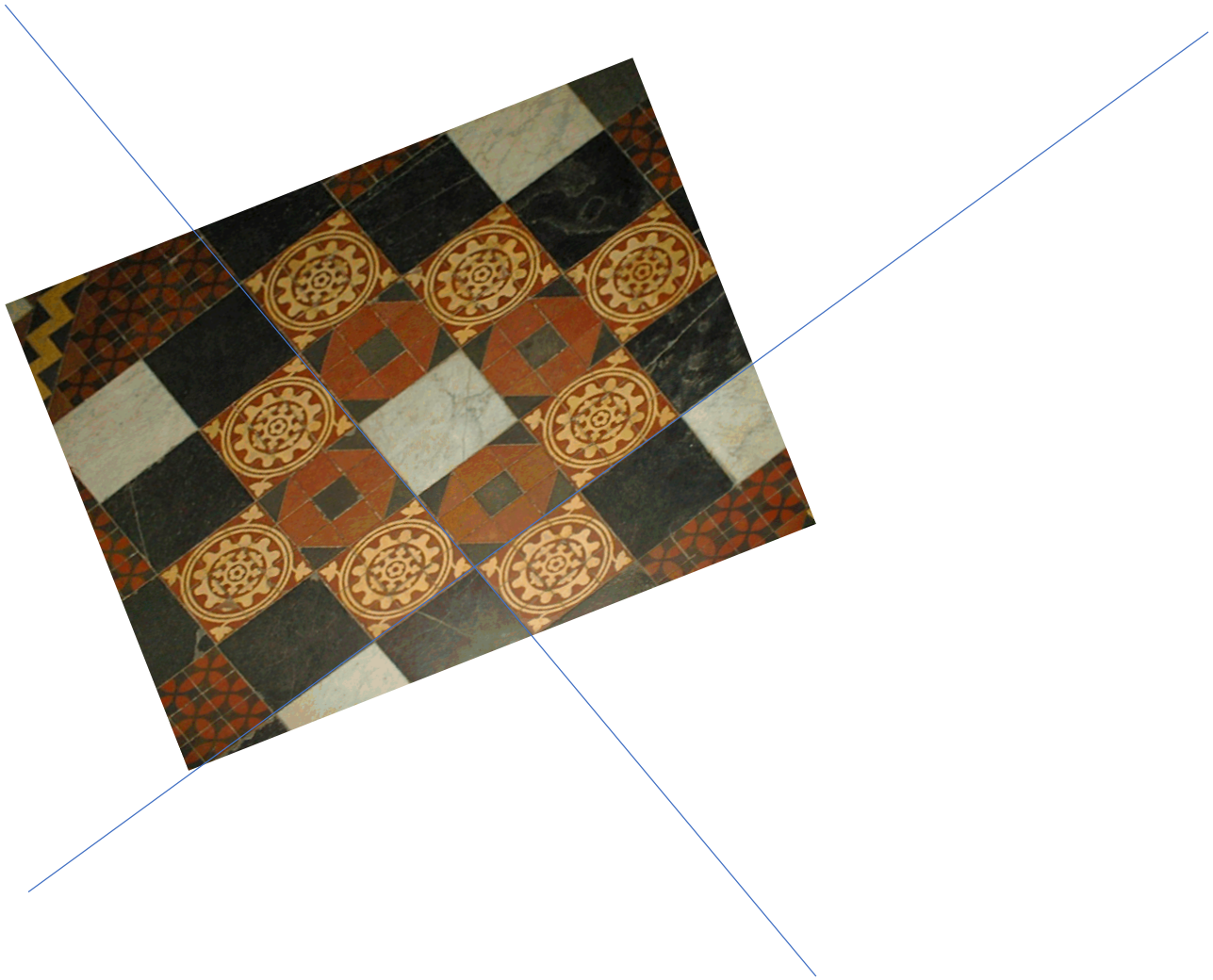
these two pairs lead to the same constraint as *they are not independent*



→ use only one of the two pairs as first pair



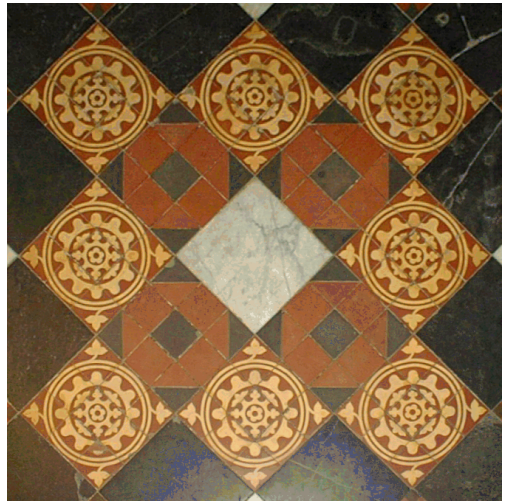
→ use a new pair as a second pair



Metric rectification from affine



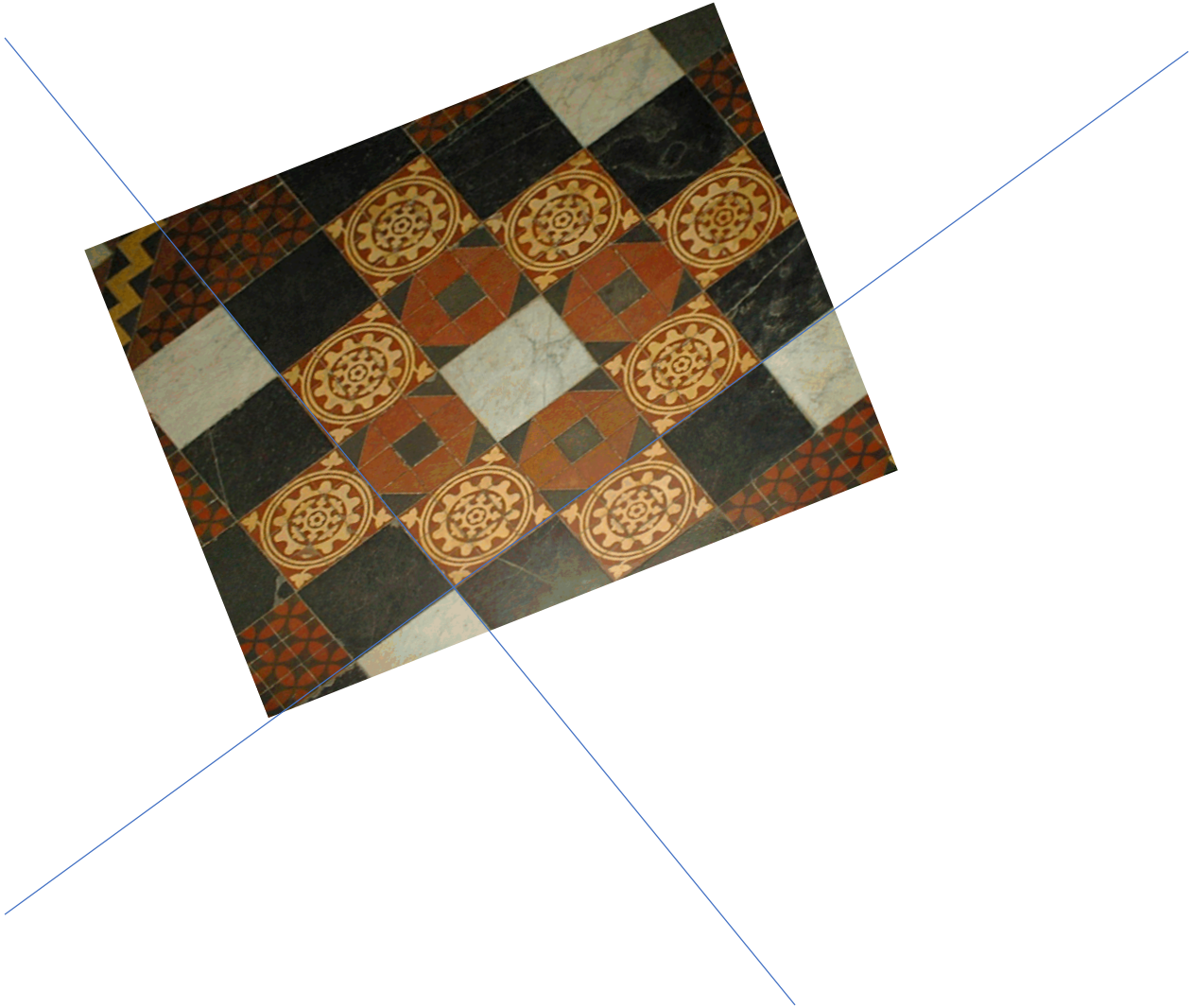
H_{rect}



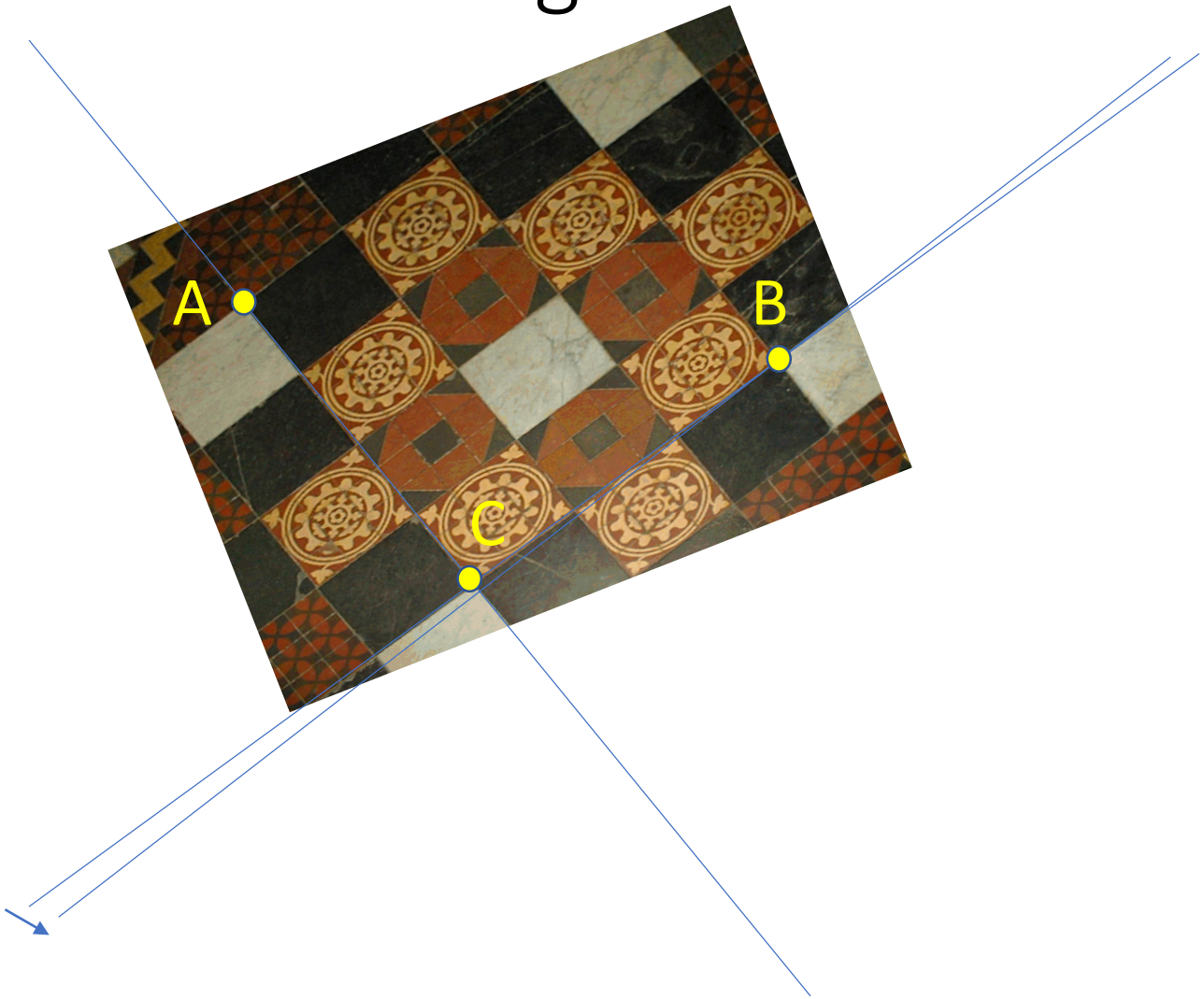
affine rectification

metric rectification

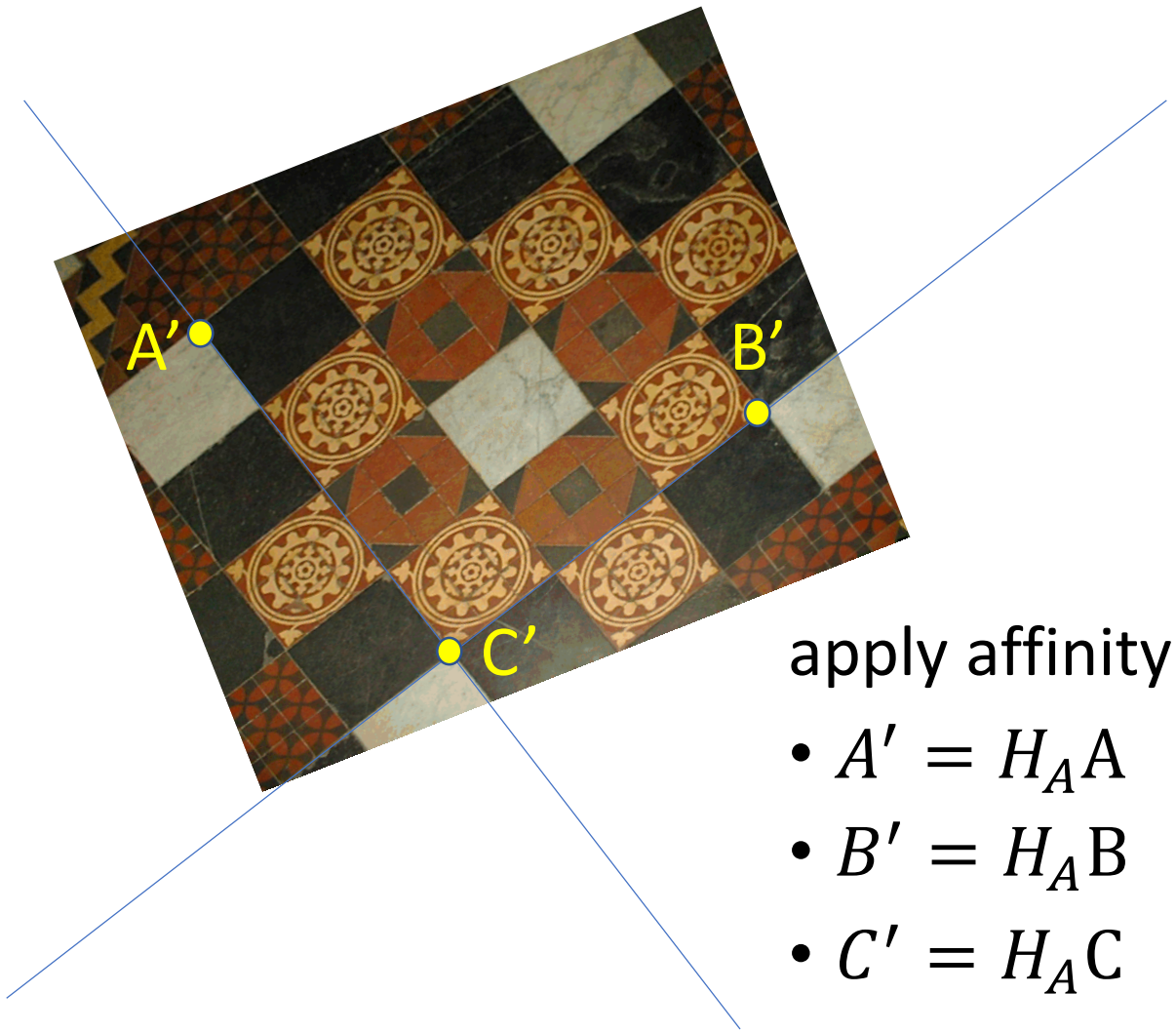
from affine to metric: **geometric** method with two pairs of orthogonal lines



First pair of orthogonal lines: apply a homography (in fact, an **affinity**) that maps the second line onto a line orthogonal to the first line



First pair of orthogonal lines: apply an affinity that maps the second line onto a line orthogonal to the first line



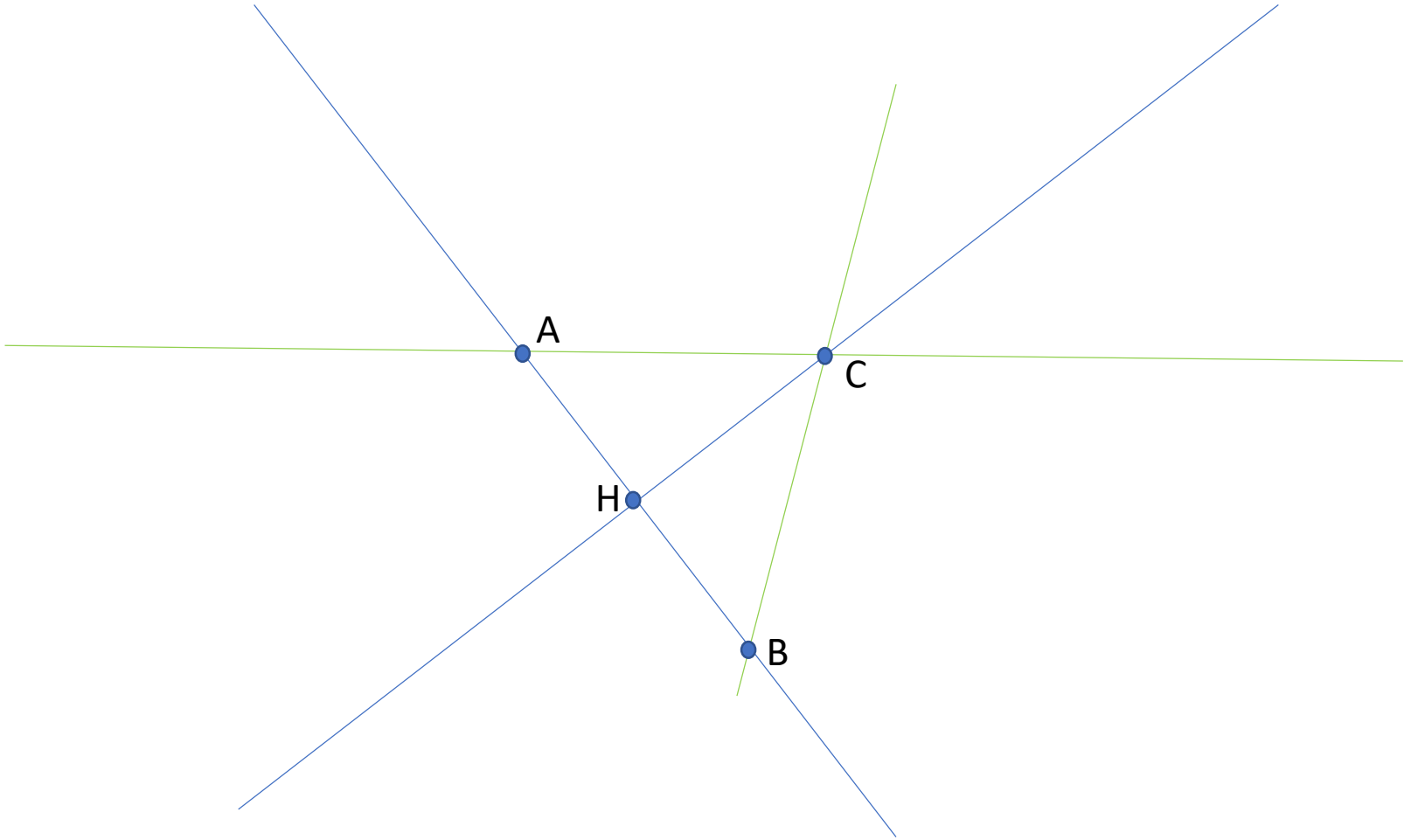
apply affinity H_A, which solves

- $A' = H_A \mathrm{A}$
- $B' = H_A \mathrm{B}$
- $C' = H_A \mathrm{C}$

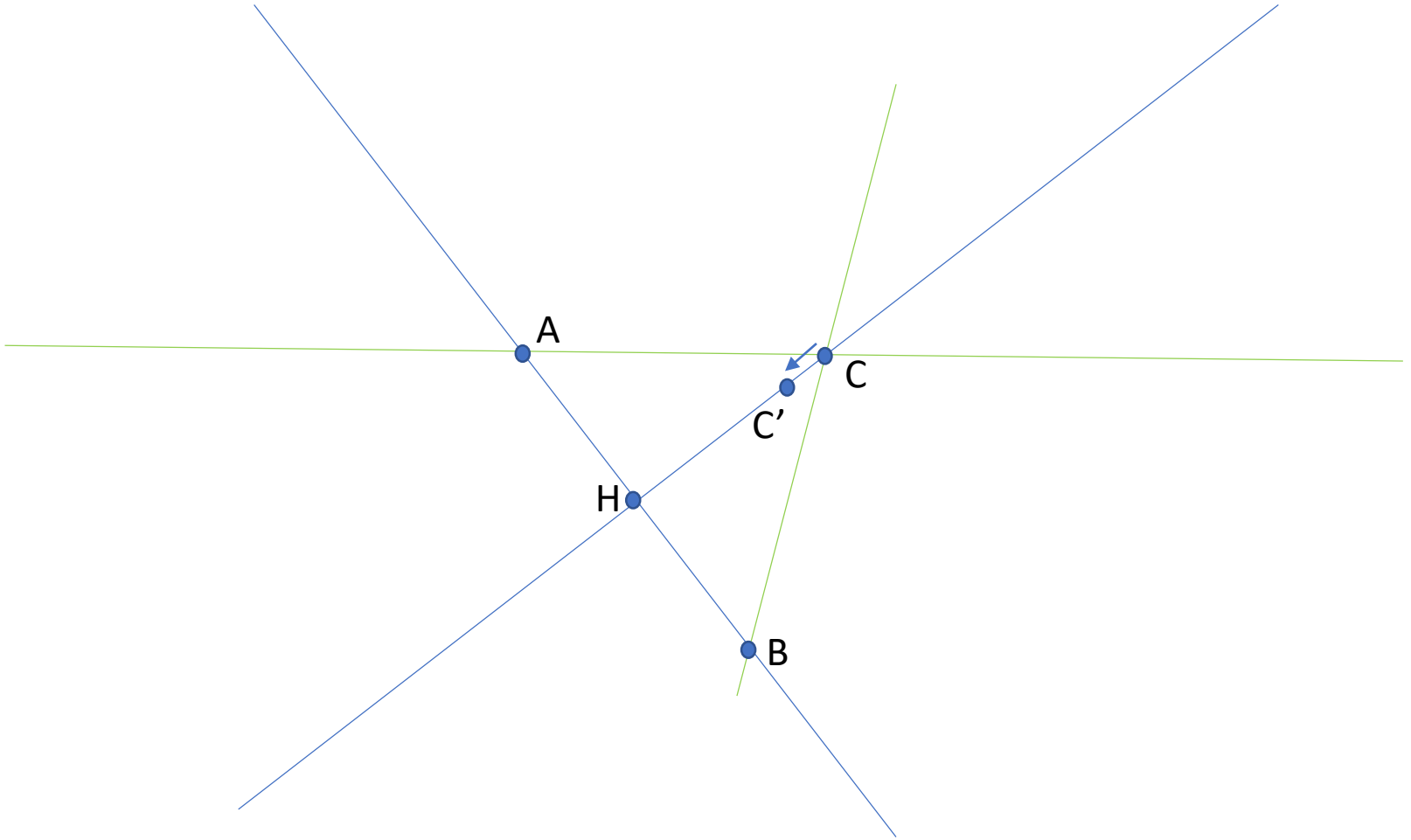
Second pair of images of orthogonal lines



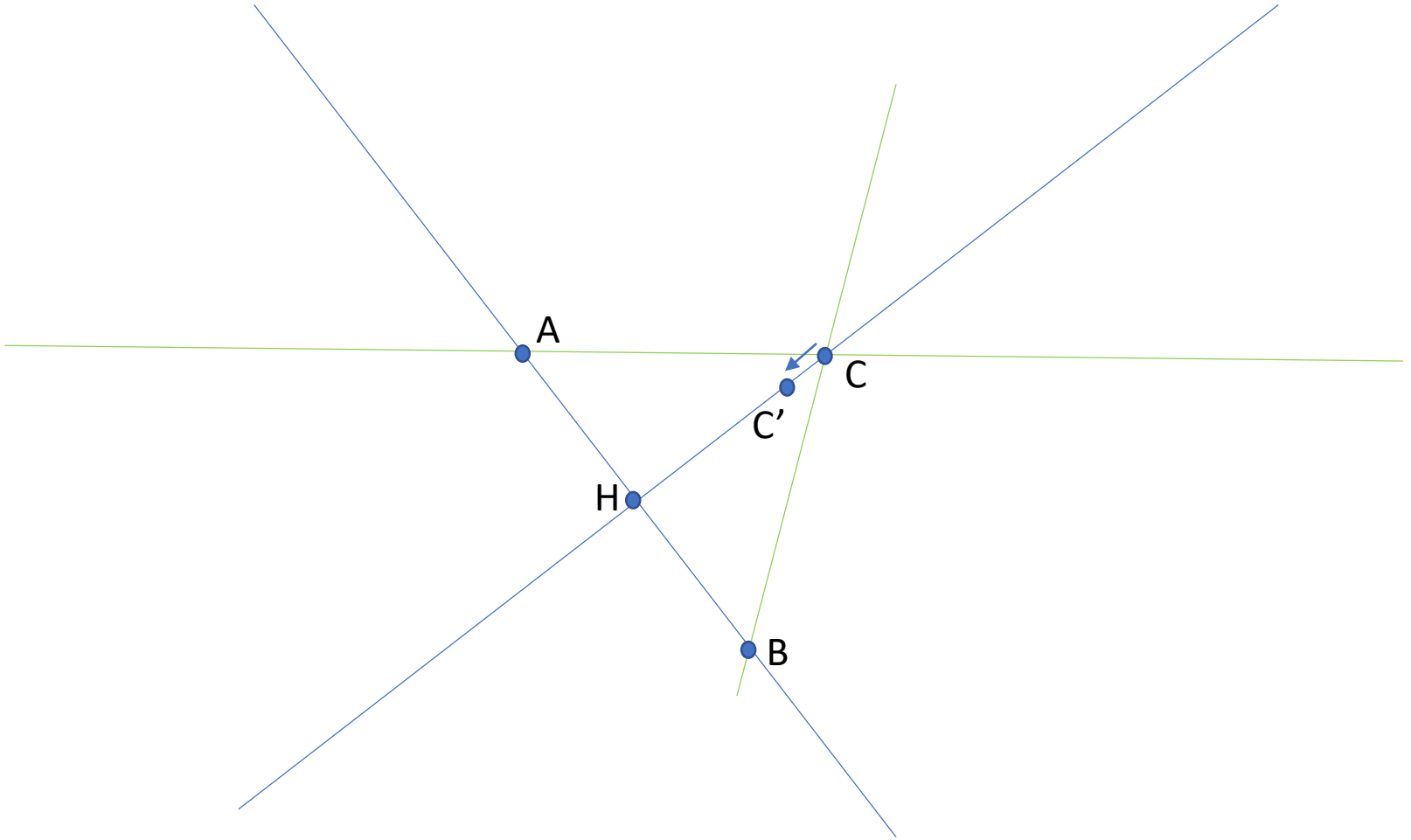
Second pair of image lines must be mapped onto orthogonal lines **while keeping the first pair orthogonal**



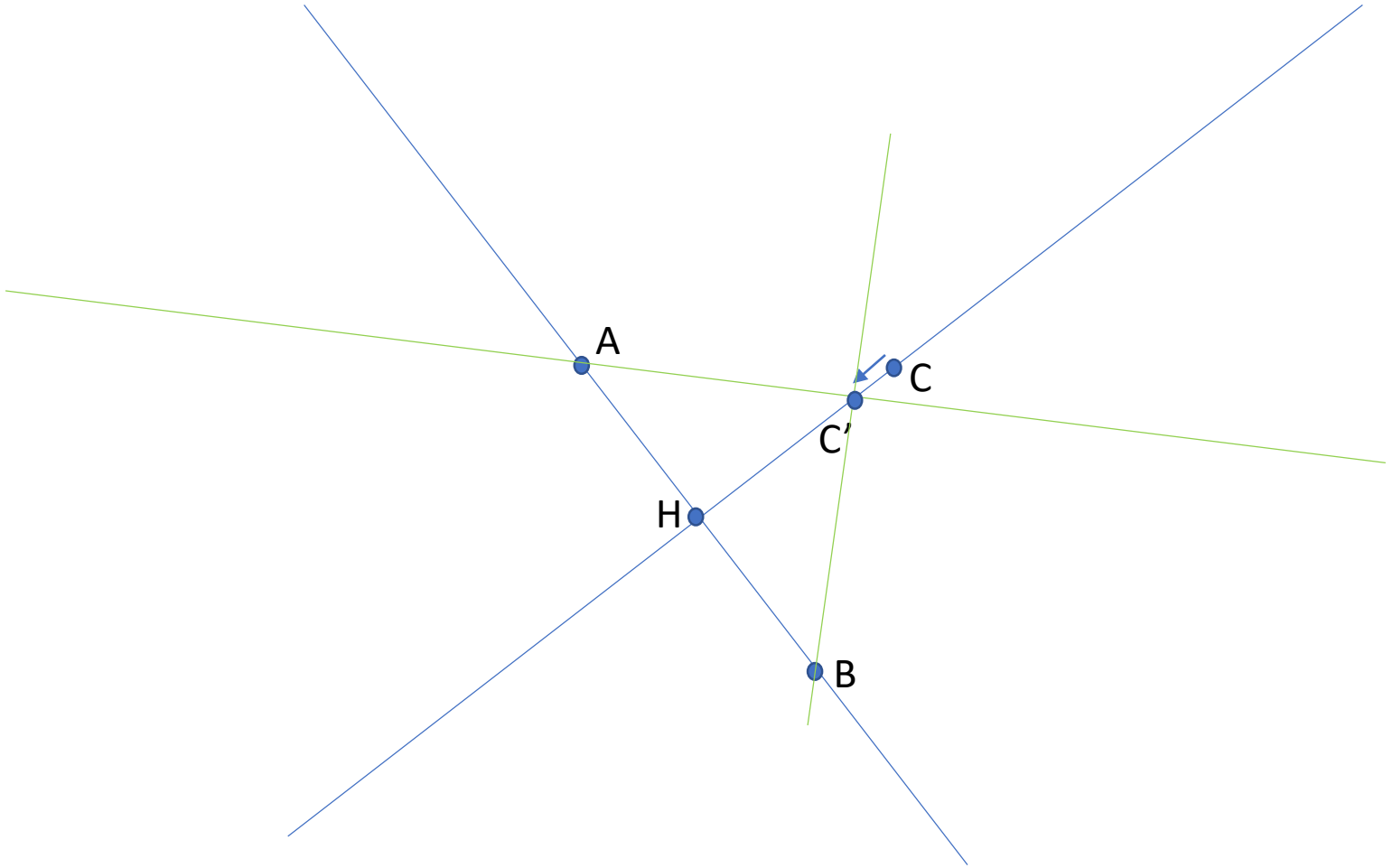
Map point C onto a point C’ still on the line HC such that line HC’ is orthogonal to line AB



Euclides theorem: ABC’ is a rectangular triangle iff HC’ = $\sqrt{AH\ HB}$

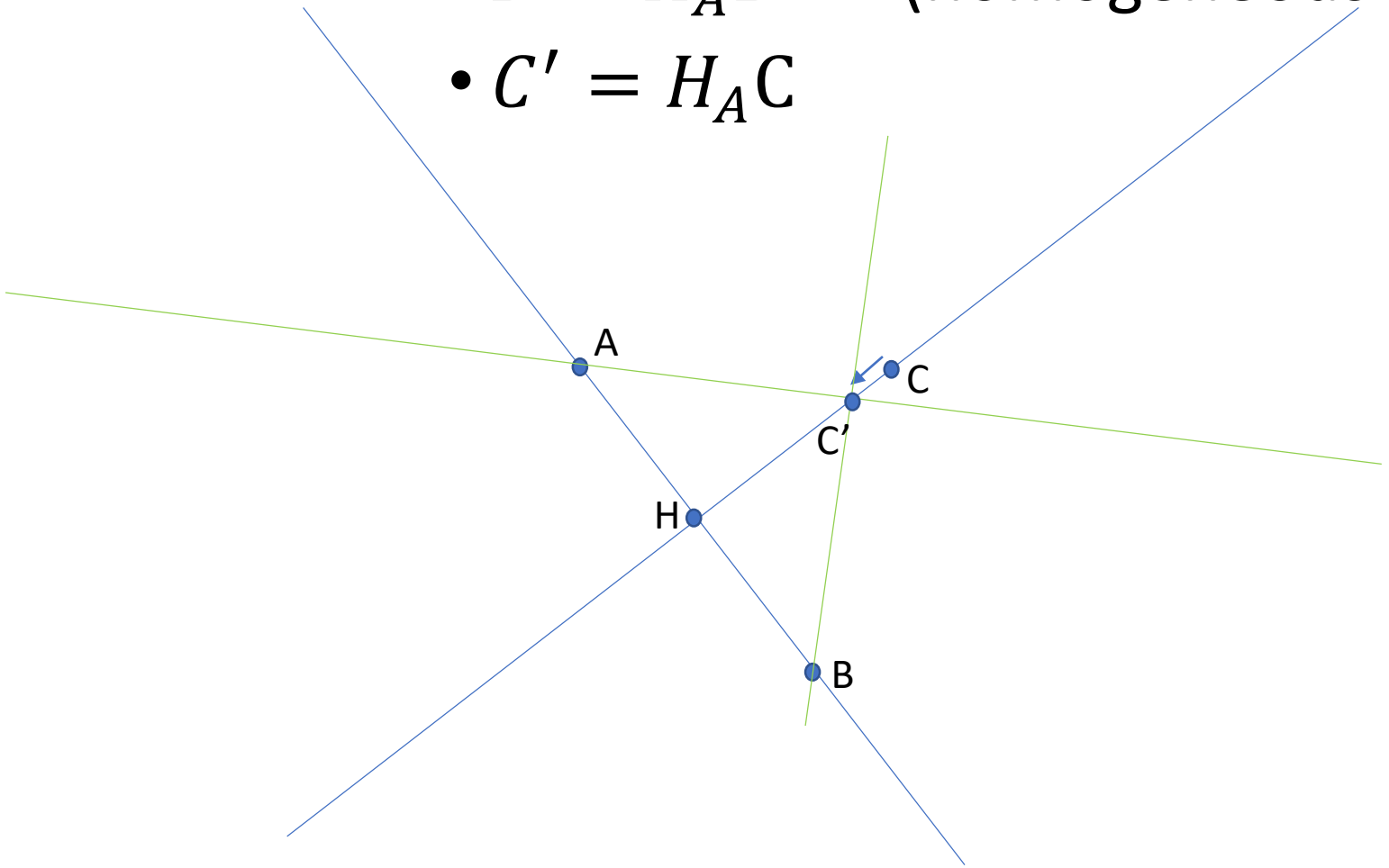


Euclides theorem: ABC’ is a rectangular triangle iff HC’ = $\sqrt{AH\ HB}$

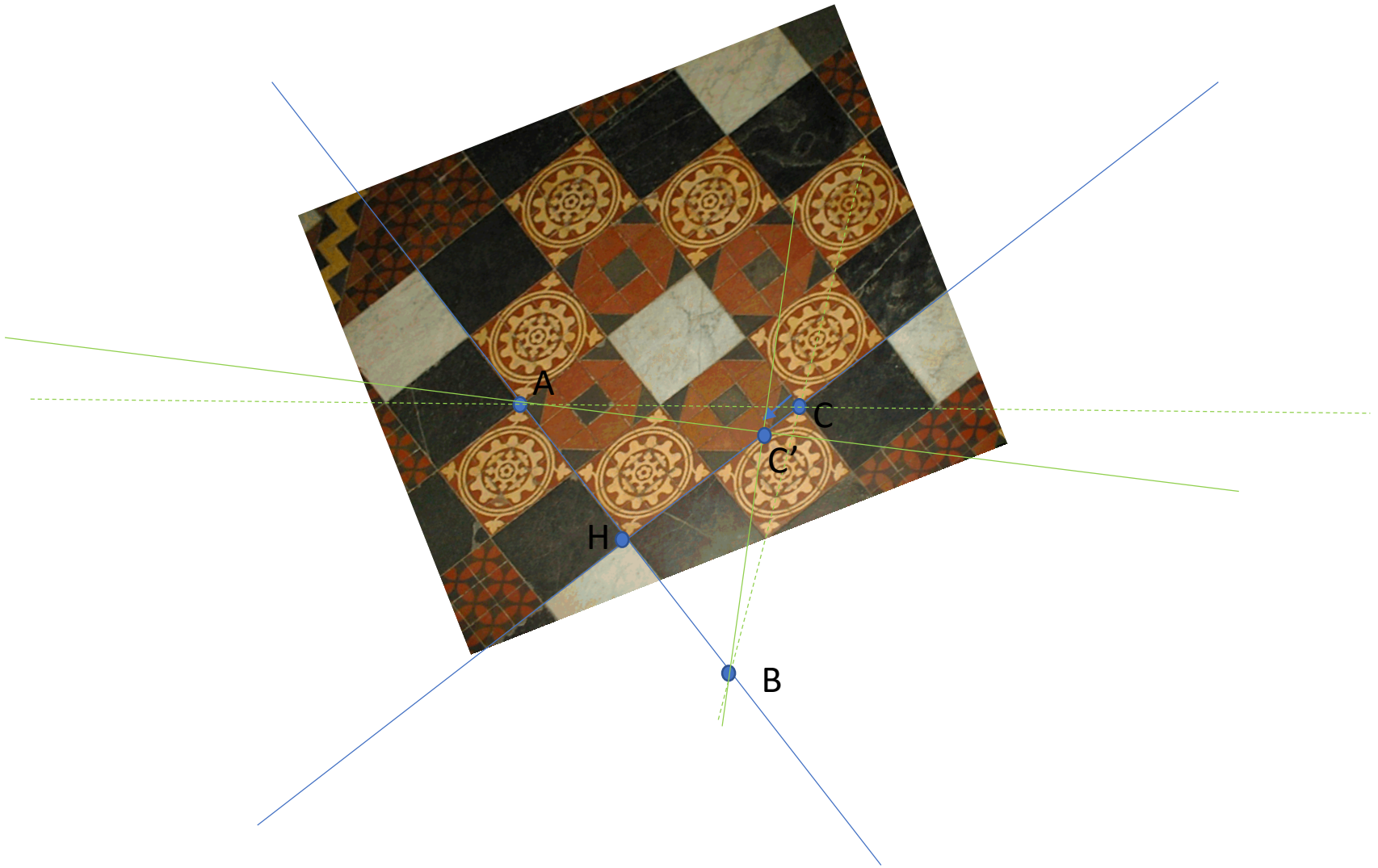


apply affinity H_A, where H_A is the solution of

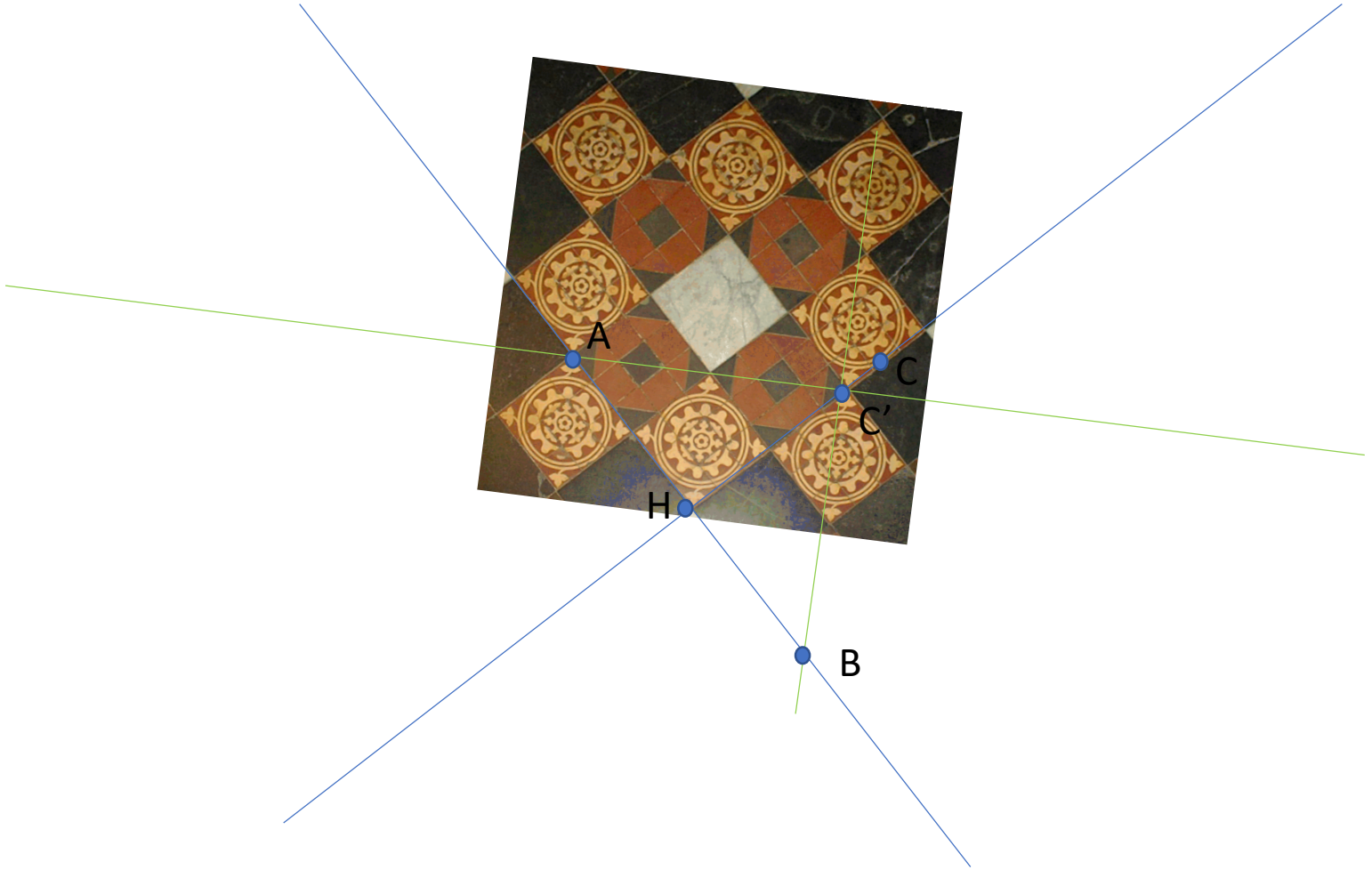
- $A = H_A \mathrm{A}$
- $B = H_A \mathrm{B}$ (homogeneous coordinates)
- $C' = H_A \mathrm{C}$



Apply affinity H_A



Apply affinity H_A
→ reconstructed shape



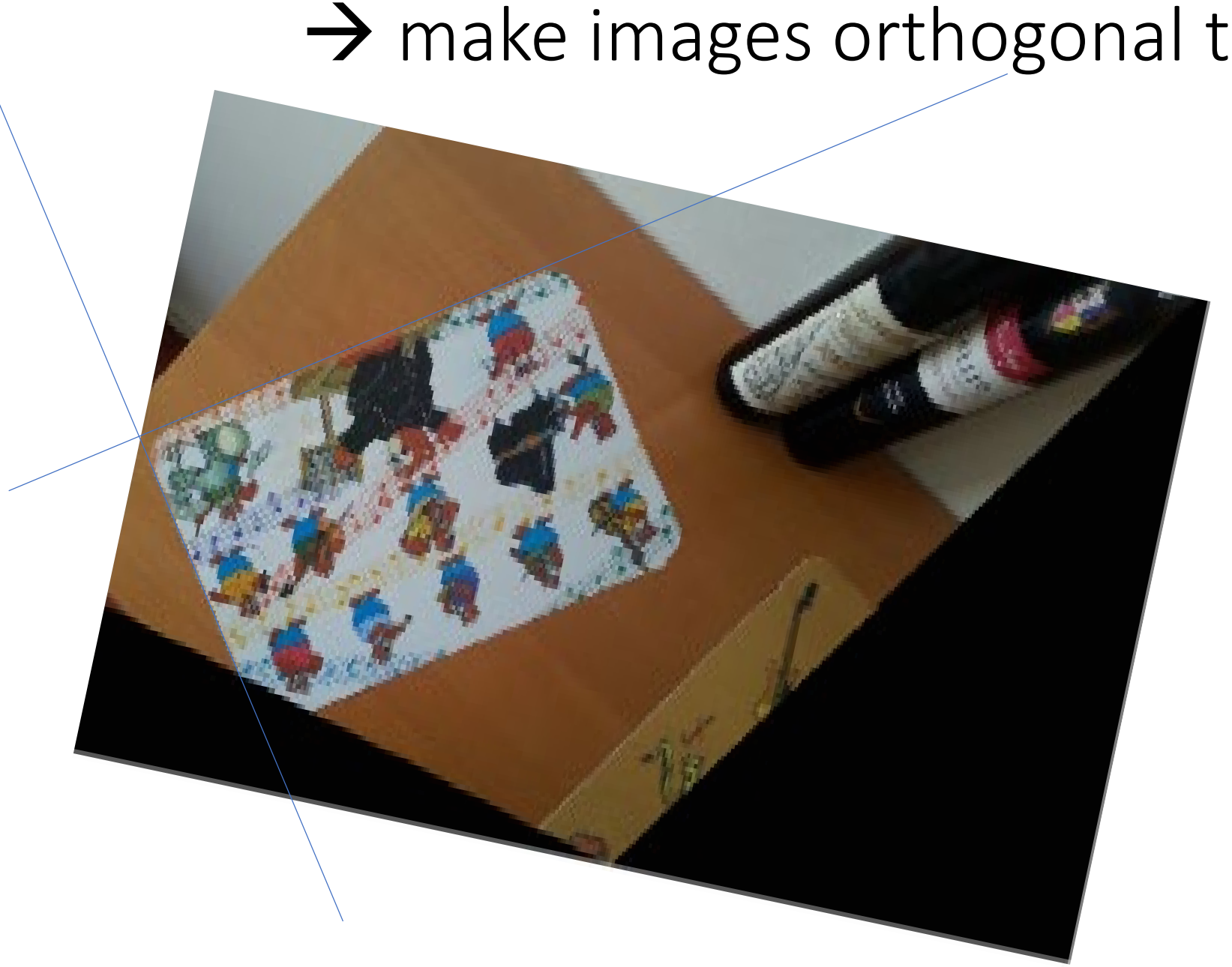
metric from affine: other example



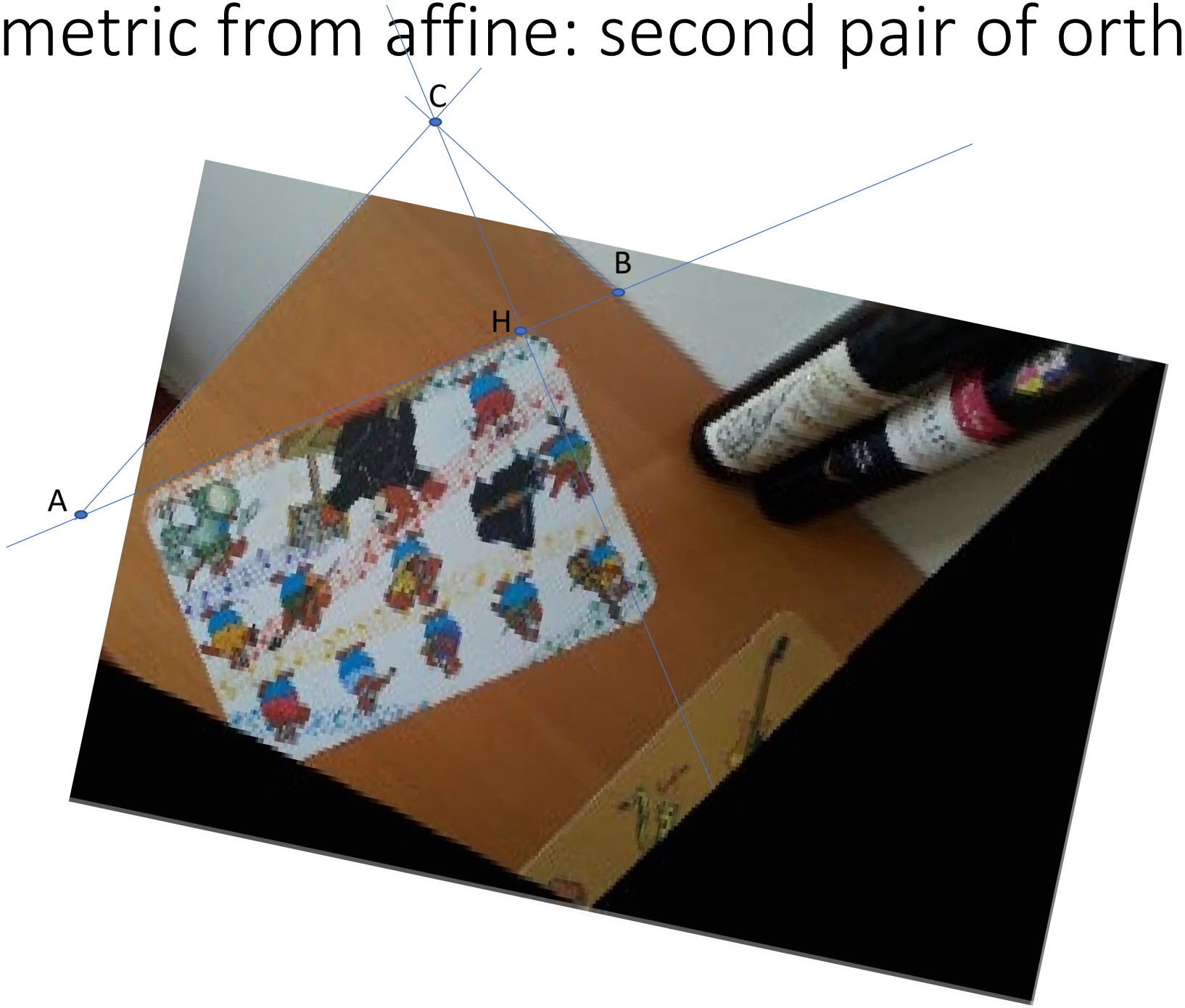
metric from affine: other example



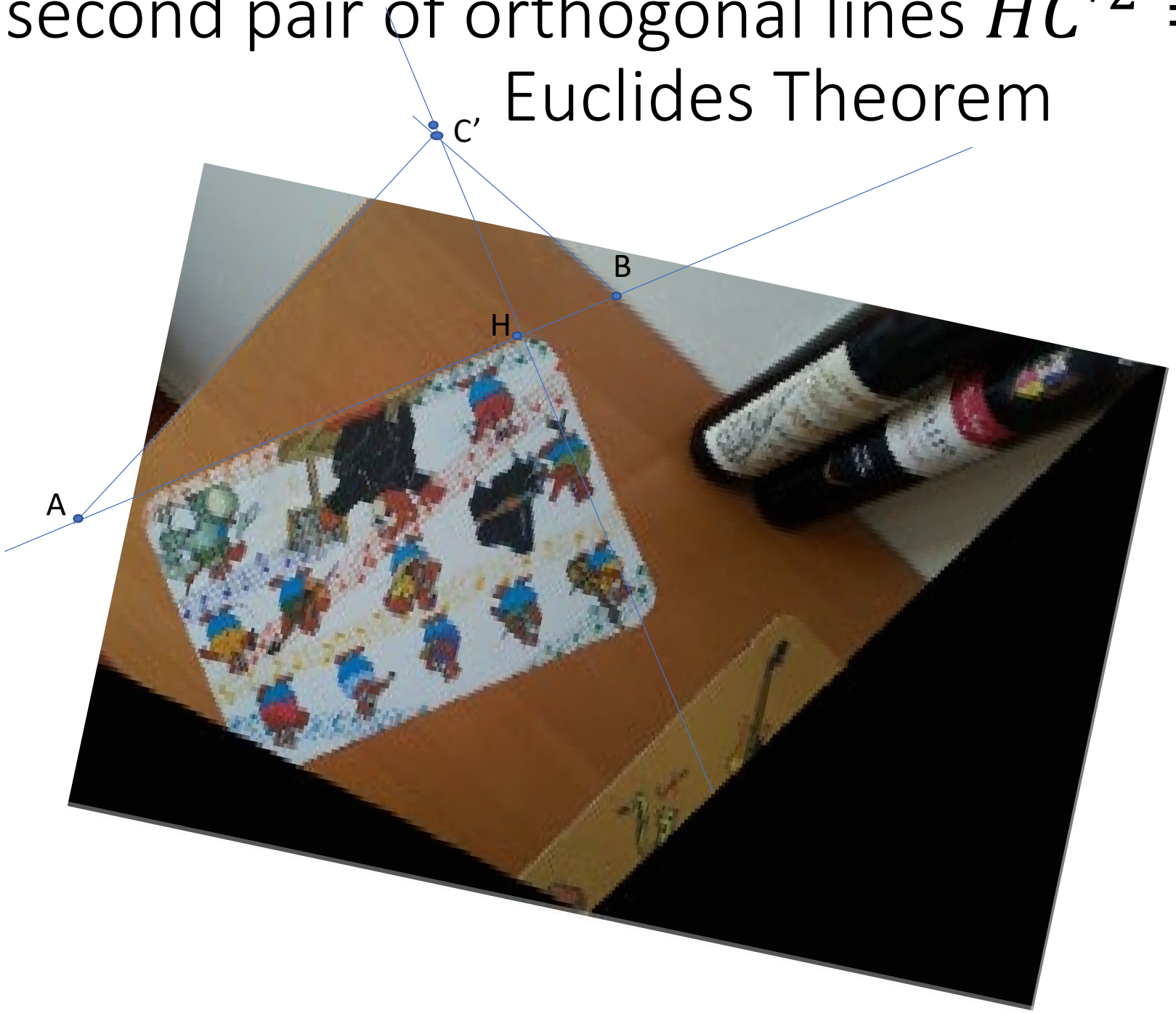
metric from affine: first pair of orthogonal lines → make images orthogonal too



metric from affine: second pair of orthogonal lines



second pair of orthogonal lines $HC'^2 = AH \times BH$ Euclides Theorem



rectified image (2D shape reconstructed)

